

Basic Mathematics

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# Mathematics

## Lecture Notes

A friendly path through algebra, geometry, and calculus.

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# Chapter 1

## Introduction

### 1.1 Purpose and Philosophy

These notes are an invitation to look at mathematics in a different way. For many students, mathematics has long appeared as a collection of formulas, rules, and mnemonic patterns: one recognises the type of problem, chooses a remembered procedure, performs the required operations, and moves on. Such skills have their place, but they are not the heart of mathematics. Mathematics begins when we stop for a moment and ask what we are really looking at, what kind of object has appeared, what question is being asked, what is known, what is unknown, and why a proposed step is justified.

The purpose of this course is therefore not only to practise calculations. We will learn how to read mathematical expressions, formulate precise questions, distinguish an object from its notation, notice structure, and build simple but meaningful chains of reasoning. A solved problem is valuable not only because it produces an answer, but also because it can reveal a method of thought: how to begin, what to observe, which assumptions matter, what can be transformed, what remains unchanged, and how a conclusion follows from earlier facts.

The course is organised into three connected parts: linear algebra, analytic geometry, and calculus. In the first part, we focus on matrix methods: matrices, determinants, inverse matrices, and systems of linear equations. The second part develops the language of coordinates and vectors and uses it to describe points, lines, planes, directions, distances, and angles. The third part introduces sequences and functions, limits, continuity, derivatives, integrals, change, and accumulation. These parts are not isolated collections of techniques. Together they form a basic mathematical language for describing structure, space, and change.

We will try to study this language from the inside. A definition should not be treated as a sentence to memorise, but as a way of introducing a new kind of object. A theorem should not be treated as a decorative statement, but as an answer to a clear question under precise assumptions. An example should not be treated as a template to copy blindly, but as a small experiment in reasoning. Even a simple calculation can show something important if we ask why it works and what it tells us.

For this reason, some habits of thought will deliberately return in different places. The same question of what is fixed and what is variable appears when we study functions, vectors, equations, and transformations. The same need for precision appears when we define a line, compute a derivative, solve a system of equations, or interpret an integral. A mathematical question can often be seen algebraically, geometrically, and conceptually. This repetition is intentional: it helps the reader recognise patterns of reasoning rather than only patterns of exercises.

The number of answers produced is not, by itself, a measure of progress. What matters is the quality of the work. Did we understand the question? Did we identify the object correctly? Did we choose notation consciously? Did we know why each important step was valid? Did we interpret or verify the result? These questions are not an addition to mathematics; they are mathematics being done carefully.

Mathematics is useful in physics, engineering, computer science, data analysis, modelling, and many other fields, but its value is not only practical. It is also a way of seeing. It teaches precision, discipline, imagination, and the ability to move from simple observations to general conclusions. These notes are meant to show a piece of that beauty, strength, and richness. The

goal is not merely to survive mathematical techniques, but to begin to appreciate mathematics as the art of asking good questions and giving well-reasoned answers.

## 1.2 How the Course Works

This book is part of a weekly learning process rather than a collection of material to be read only before an examination. Each of the twelve thematic chapters is designed to support one week of study. A chapter provides the mathematical structure for the lecture, a complete narrative to which the student can return, and a starting point for individual problem solving and classroom discussion.

In a fifteen-week semester, the twelve thematic weeks establish the main rhythm of the course. The remaining time can be used to introduce the method of work, connect topics, revise important ideas, and present more substantial solutions.

### 1.2.1 The Weekly Learning Cycle

A student should not leave a topic behind when the lecture ends. A typical week consists of the following connected stages:

1. **Attend the lecture.** The lecture introduces the topic, motivates the central questions, and demonstrates how the relevant mathematical objects and methods should be read.
2. **Return to the chapter.** Read the corresponding chapter carefully after the lecture. The PDF is not merely a summary of slides. It is an organised narrative containing definitions, explanations, examples, interpretations, and mathematical arguments that can be examined more than once.
3. **Ask questions and clarify the material.** Mark every place where a definition, transition, calculation, or conclusion is not yet clear. Use the text, your own examples, classroom discussion, and appropriate digital tools to resolve these questions.
4. **Solve the complete assigned problem set.** Move from following an explanation to constructing arguments independently. Work on every assigned problem and prepare a separate, self-contained note for each one.
5. **Prepare professional mathematical notes.** Use precise notation, explanatory sentences, justified transitions, and a clear conclusion. The note should show not only the answer, but also how it follows from the given information, definitions, and earlier results.
6. **Present and discuss your work.** During exercise classes, be prepared to explain a solution, answer questions about its steps, and describe what you learned while preparing it.

These stages form one continuous process. Each stage prepares the next, and the presentation and feedback lead back to further reading, revision, and improved work.

### 1.2.2 Exercise Classes: Presentation, Feedback, and Revision

Presenting mathematics is part of learning mathematics. It reveals whether an argument has genuinely been understood and whether the student can communicate it clearly. During a presentation, the student should be able to read the notation, explain the role of each important formula, justify the method, and respond to questions without relying on AI to speak on the student's behalf.

Feedback given during exercise classes is intended for the whole group, not only for the student whose work is currently being presented. Everyone should follow the argument and

listen carefully to the discussion. By comparing different examples, students learn what a complete solution looks like, which explanations make an argument clear, which steps require justification, and which weaknesses require revision.

Students should not focus only on comments addressed directly to their own work. Listening to feedback on the work of others provides a broader set of examples and develops an operational understanding of the course standards: what makes mathematical work clear, justified, and complete, and what still needs improvement.

The feedback concerns the mathematical work and its results, not the personal worth of the student. What is examined is the quality of the note, the understanding demonstrated in the argument, and the clarity of the presentation. Feedback should identify what already works and what should be changed in order to produce a stronger solution.

The first submitted version does not always have to be the final version. Students should learn to receive comments, reconsider their choices, correct errors, add missing explanations, and improve both the written note and its presentation. Revision is not evidence of failure. It is a normal part of serious intellectual work.

This process is important throughout university study and later in professional life. Students and professionals are given tasks, asked to solve problems, expected to present results, and required to respond to feedback and critical review. Learning to improve an initial solution is therefore not an additional classroom exercise. It is a fundamental habit of responsible academic and professional work.

### 1.2.3 Progress and Assessment

Progress is demonstrated by the quality and increasing independence of the student's work. By the end of a weekly cycle, the student should be able to identify the main mathematical objects, state the essential definitions, solve representative problems, justify the principal steps, verify or interpret the result, and communicate the argument clearly.

Assessment in the exercise component focuses primarily on how the student works throughout the semester. Evidence of serious engagement includes:

- regular preparation and completion of the assigned problem sets,
- documented attempts to solve problems rather than bare answers,
- active participation in presentations and classroom discussion,
- separate, self-contained notes for individual problems,
- careful revision of work after feedback,
- increasing clarity, independence, and mathematical precision.

A student is not defined by the quality of the first solution or presentation. Some students may initially find mathematical notation, formal writing, or public explanation difficult. A weak beginning does not prevent a very good final grade for the exercise component. A student who works conscientiously, responds to feedback, corrects earlier weaknesses, and demonstrates substantial development may finish the exercise component with a very good grade.

This approach does not lower the standard of mathematical work. It recognises that reaching the standard through sustained effort and revision is itself a meaningful achievement. The exercise-component grade reflects a process of learning, not merely a snapshot taken at the beginning of the semester.

Systematic work according to these guidelines provides the basis for earning a passing grade for the exercise component. The examination is assessed separately and tests the student's mathematical knowledge and ability to use it independently. Regular work does not replace the examination, but it develops the understanding and skills that substantially increase the likelihood of successfully completing the course.

## 1.3 How to Prepare Mathematical Notes

A mathematical note is not merely a place where an answer is recorded. Its purpose is to show how a problem is understood, how relevant information is selected, how a method is chosen, and how a conclusion follows from definitions, assumptions, and earlier results.

Mathematical expressions are compact. A single formula may contain several ideas at once: objects, operations, conditions, and logical relationships. Reading a formula therefore means more than pronouncing its symbols. A student should be able to explain what the objects represent, what operation is being performed, why the expression is meaningful, and what information the formula provides.

The same principle applies to writing. A sequence of formulas without explanatory sentences rarely communicates a complete mathematical argument. Formulas should be connected by words that explain their role. A good solution tells the reader why a calculation is being performed, which definition or theorem is being used, whether its assumptions are satisfied, and how the result answers the original question.

### 1.3.1 Make the Note Self-Contained

Prepare one separate, self-contained note for each assigned problem, including all of its subparts. Begin by writing the problem or a complete and faithful formulation of it. The reader should not need to search through another document to discover what is being solved. After several weeks, you should still be able to open the note and understand both the question and the solution.

Identify what is given and what must be found, proved, or explained. Introduce the relevant notation and determine what kinds of mathematical objects appear in the problem. Before calculating, ask what information is available and what would count as a satisfactory answer.

### 1.3.2 Make the Reasoning Visible

Present the principal idea or method before, or while, carrying out the calculation. Important transitions should be explained with complete sentences. For example:

- Because the determinant is non-zero, the matrix is invertible.
- We may therefore multiply both sides by the inverse matrix.
- By the definition of the derivative, we consider the following limit.
- Since the direction vector is non-zero, it determines a line.
- Substituting the result into the original equation verifies the solution.

Such sentences are not decoration around the mathematics. They are part of the mathematics because they show the logical relationship between successive formulas.

It is not necessary to describe every elementary arithmetic operation. The amount of explanation should be proportional to the problem. A simple exercise may require only a few sentences, while a more substantial problem may require a longer argument. The goal is not to make every note long. The goal is to make every essential step understandable and justified.

Write as briefly as possible, but as completely as necessary.

### 1.3.3 A Complete Solution

A complete note should make the following five stages visible:

1. **Understand the problem.** Record the problem, identify the given information, and state clearly what must be found, proved, or explained.

2. **Choose a method.** Select the definitions, theorems, or techniques that connect the available information with the goal.
3. **Construct the argument.** Carry out the necessary steps and connect formulas with sentences that explain why each important transition is valid.
4. **Verify the result.** Check calculations, assumptions, domain restrictions, or geometric meaning whenever appropriate.
5. **State the conclusion.** Answer the original question clearly and in its proper context.

A result without its context is not a complete solution. A bare final answer, an unexplained sequence of formulas, or a few unconnected calculations will be treated as incomplete work and should be revised.

Quality cannot be replaced by quantity, but quality does not remove the obligation to complete the assigned work. Students are expected to prepare separate, complete solutions for the entire assigned problem set. Producing many unexplained answers is insufficient, but preparing one polished solution while ignoring the remaining problems is also insufficient.

### 1.3.4 Working with AI

We are studying in an age in which AI tools are widely available. The free versions of these tools are sufficient for the kinds of support expected in this course: asking questions, clarifying notation, generating simpler examples, checking reasoning, identifying missing steps, and improving the organisation of mathematical writing. Access to expensive software is not required.

AI should be used as an individual learning assistant. It may help transform rough calculations into clear mathematical prose, but it does not replace the student's mathematical work. Receiving a well-written answer is not the same as understanding it. The student remains responsible for checking every definition, calculation, assumption, conclusion, and formula included in the submitted note.

Use AI with a focused context. Work on one problem at a time and prepare one complete note for that problem. Do not paste an entire problem set into a single conversation and request all answers at once. Handling many unrelated tasks together encourages brief, generic, and mathematically weak notes. Focused work makes it easier to question each step, identify missing justifications, and produce a solution that can be understood and presented.

The one-problem-at-a-time principle does not mean that only one problem from the set should be prepared. Students are expected to repeat this focused process for every problem in the complete assigned set.

A polished AI-generated note is insufficient if the student cannot read its formulas, explain its terminology, justify its transitions, or present the argument independently. When using AI, ask not only for a solution, but also questions that deepen understanding:

- What does this formula mean?
- What mathematical object does each symbol represent?
- Why is this step valid?
- Which definition or theorem is being used?
- Are the assumptions of the theorem satisfied?
- Is there a simpler way to explain this transition?
- How can the result be checked?

- What should I be able to explain during class?

Preparing notes in this way is part of learning mathematics. It develops the ability to read notation, recognise mathematical objects, organise information, construct arguments, and communicate ideas precisely. Repeating this process for the complete problem set builds the habits of systematic and conscientious work that should accompany the student throughout university study and later professional life.

### 1.3.5 Student GitHub Workspace

The current instructions, problem templates, and technical workflow for student work are maintained in the public repository [dchorazkiewicz/Math\\_Problems\\_Repo](#). Students should create a public fork of this repository and commit their notes to their own fork. The course repository is the common template; it is not the place where individual student solutions should be submitted.

The repository contains a short map of the working process:

- `README.md` explains the purpose and structure of the workspace and provides the weekly index;
- `SOURCE_MATERIAL.md` links the current lecture PDF and maps each week to the corresponding LaTeX problem source used by an AI assistant;
- `NOTE_GUIDELINES.md` defines the standard of a complete mathematical note;
- `AI_WORKFLOW.md` describes the one-problem-at-a-time workflow with AI, review, commits, presentation, feedback, and revision;
- `AGENTS.md` gives a connected AI assistant the operational rules for working safely in the student's fork;
- the `docs/` directory contains twelve weekly folders and one Markdown file for each problem.

The workspace uses Markdown for student notes, Git and GitHub for version history and public forks, MathJax for mathematical typesetting, MkDocs for the web presentation, and GitHub Actions for automatic validation and deployment. These instructions and templates may be updated independently of this PDF, so students should consult the repository for the current working procedure. The repository links back to this PDF and to its LaTeX sources. Students use the PDF as the readable course material containing the theory, definitions, examples, and problem sets. A connected AI assistant uses the current mapped LaTeX file as the exact machine-readable source of a selected problem statement.

# Chapter 2

## Matrices

### 2.1 What matrices are

A matrix is a rectangular arrangement of mathematical objects, usually numbers. This description is correct, but it is not yet very informative. If we stop at this description, a matrix looks like a table, and matrix algebra looks like a collection of rules for moving numbers around. In this chapter we will try to see more than that. A matrix is a way of organizing information. It can describe data, relations between quantities, coefficients of equations, or a rule that transforms one vector into another.

The first skill is therefore not calculation, but reading. When we see a matrix, we should ask what its shape is, what its entries represent, and how its rows and columns are meant to be interpreted. The same array of numbers may be only a table in one problem, but in another problem it may describe a transformation, a system of equations, or a geometric operation.

#### Definition

A matrix with  $m$  rows and  $n$  columns is called an  $m \times n$  matrix. It has the form

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

The entry  $a_{ij}$  is located in row  $i$  and column  $j$ .

This notation is simple, but it contains an important discipline. The first index tells us where to look vertically, and the second index tells us where to look horizontally. A student who confuses these indices will often make mistakes later when multiplying matrices or describing systems of equations.

#### Example

Consider the matrix

$$M = \begin{pmatrix} 4 & -2 & 0 \\ 7 & 1 & 5 \end{pmatrix}.$$

This is a  $2 \times 3$  matrix. The entry  $m_{11}$  is 4, the entry  $m_{12}$  is  $-2$ , and the entry  $m_{23}$  is 5. We are not doing any operation yet. We are only learning to read the object.

The size of a matrix is part of its identity. A row matrix, a column matrix, and a square matrix may contain similar numbers, but they play different roles. A column matrix can represent a vector. A square matrix can sometimes represent a transformation from a space to itself. A rectangular matrix may describe a relation between two collections of quantities of different sizes.

### Definition

A matrix is called square if it has the same number of rows and columns. An  $n \times n$  matrix is a square matrix of order  $n$ .

We will often begin with square matrices because many basic ideas are easiest to see there. However, rectangular matrices are just as important. The point is not to memorize a list of shapes, but to develop the habit of asking: what kind of object am I holding, and what operations make sense for this object?

## 2.2 Basic matrix operations

Once we know how to read a matrix, we can begin to operate on it. The simplest operations are addition, subtraction, and multiplication by a number. These operations are deliberately simple because they preserve the shape of the matrix. They do not change a table into a different kind of object; they change its entries while keeping the same positions.

### Definition

Let  $A = (a_{ij})$  and  $B = (b_{ij})$  be matrices of the same size  $m \times n$ . Their sum is the matrix

$$A + B = (a_{ij} + b_{ij}),$$

and their difference is the matrix

$$A - B = (a_{ij} - b_{ij}).$$

If  $\lambda$  is a number, then

$$\lambda A = (\lambda a_{ij}).$$

The phrase “of the same size” is not a technical detail. It is the reason the operation is meaningful. When two matrices have the same size, each entry of one matrix has a partner in the other matrix. Addition and subtraction then happen position by position.

### Example

Let

$$A = \begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 5 & 4 \\ -2 & 1 \end{pmatrix}.$$

Then

$$A + B = \begin{pmatrix} 2+5 & -1+4 \\ 0+(-2) & 3+1 \end{pmatrix} = \begin{pmatrix} 7 & 3 \\ -2 & 4 \end{pmatrix}.$$

Also,

$$A - B = \begin{pmatrix} 2-5 & -1-4 \\ 0-(-2) & 3-1 \end{pmatrix} = \begin{pmatrix} -3 & -5 \\ 2 & 2 \end{pmatrix}.$$

Finally,

$$-3A = \begin{pmatrix} -6 & 3 \\ 0 & -9 \end{pmatrix}.$$

The calculation is not difficult. The important thing is the discipline of positions. We do not add rows to columns, and we do not rearrange entries because it looks convenient. We respect the shape of the object.

These operations satisfy the familiar rules one expects from ordinary arithmetic. For matrices of the same size,

$$A + B = B + A, \quad (A + B) + C = A + (B + C).$$

This should not be surprising: addition is performed entry by entry, and ordinary addition of numbers is commutative and associative.

#### Remark

Whenever a matrix operation is defined entry by entry, many properties of ordinary arithmetic pass directly to matrices. But this principle must be used carefully. Matrix multiplication will not be entry-by-entry multiplication, and therefore it will behave differently.

## 2.3 Matrix multiplication

Matrix multiplication is the first operation in this chapter that is not simply performed position by position. This is why it deserves careful attention. Many mistakes with matrices come from treating multiplication as if it were only another version of addition. It is not. Addition compares matching positions. Multiplication combines rows of the first matrix with columns of the second matrix.

#### Definition

Let  $A = (a_{ij})$  be an  $m \times n$  matrix and let  $B = (b_{ij})$  be an  $n \times p$  matrix. The product  $AB$  is the  $m \times p$  matrix  $C = (c_{ij})$ , where

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}.$$

In words, the entry  $c_{ij}$  is obtained by multiplying row  $i$  of  $A$  by column  $j$  of  $B$ .

The condition on sizes is built into the definition. If  $A$  has size  $m \times n$ , then each row of  $A$  has  $n$  entries. To multiply this row by a column of  $B$ , each column of  $B$  must also have  $n$  entries. Therefore  $B$  must have  $n$  rows. We summarize this by writing

$$(m \times n)(n \times p) = m \times p.$$

The inner dimensions must match; the outer dimensions describe the size of the result.

#### Example

Let

$$A = \begin{pmatrix} 1 & 3 \\ -2 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ 5 & -1 \end{pmatrix}.$$

The first entry of  $AB$  comes from the first row of  $A$  and the first column of  $B$ :

$$1 \cdot 2 + 3 \cdot 5 = 17.$$

The first row and second column give

$$1 \cdot 0 + 3 \cdot (-1) = -3.$$

Continuing in the same way,

$$AB = \begin{pmatrix} 17 & -3 \\ 16 & -4 \end{pmatrix}.$$

The example is small, but the point is general. Each entry of the product answers one local question: how does a chosen row of the first matrix interact with a chosen column of the second matrix?

### Why the order matters

For ordinary numbers,  $ab = ba$ . For matrices this is not generally true. The reason is conceptual, not only computational. The product  $AB$  asks rows of  $A$  to interact with columns of  $B$ . The product  $BA$  asks a different question.

Using the matrices from the previous example,

$$BA = \begin{pmatrix} 2 & 0 \\ 5 & -1 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ -2 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 6 \\ 7 & 11 \end{pmatrix}.$$

Thus  $AB \neq BA$ . The matrices have not changed, but the order has changed, and the question asked by the multiplication has changed.

#### Remark

The statement  $AB \neq BA$  does not mean that two matrices can never commute. Some matrices do commute. The point is that commutativity is not automatic. If it matters, it must be checked or justified from the structure of the matrices.

Matrix multiplication is associative:

$$(AB)C = A(BC),$$

whenever the products are defined. This means that if several compatible matrices are multiplied in a fixed order, parentheses do not change the result. But the order of the matrices themselves still matters. Associativity tells us how to group operations. It does not allow us to swap them.

## 2.4 Special matrices and visible structure

Some matrices are important because their structure is easy to see. They help us learn a useful lesson: a matrix is not only a collection of entries. The position of the entries matters, and certain patterns can make an operation simple or meaningful.

### The zero matrix and the identity matrix

The zero matrix is the matrix whose entries are all zero. It behaves like zero for matrix addition. If  $A$  is an  $m \times n$  matrix and  $0$  is the zero matrix of the same size, then

$$A + 0 = A.$$

An identity matrix is always square. The size needed in a product is determined by the dimensions of the other matrix.

#### Definition

The  $n \times n$  identity matrix is

$$I_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

If  $A \in \mathbb{R}^{m \times n}$ , then

$$I_m A = A, \quad A I_n = A.$$

The two identity matrices in these products need not have the same size. The left identity must have  $m$  rows and columns, while the right identity must have  $n$  rows and columns.

The identity matrix is important because it represents doing nothing. This phrase is informal, but helpful. Multiplying by the correctly sized identity preserves the matrix, just as multiplying a number by 1 preserves the number.

## Transpose

The transpose interchanges rows and columns. If

$$A = (a_{ij}) \in \mathbb{R}^{m \times n},$$

then its transpose is the  $n \times m$  matrix

$$A^T = (a_{ji}).$$

### Definition

For

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix},$$

the transpose is

$$A^T = \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{pmatrix}.$$

For example,

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}.$$

A square matrix is called symmetric if

$$A^T = A.$$

This means that its entries are mirrored across the main diagonal.

## Diagonal matrices

A diagonal matrix is a square matrix whose non-zero entries, if any, lie only on the main diagonal.

### Definition

A square matrix  $D = (d_{ij})$  is diagonal if

$$d_{ij} = 0 \quad \text{whenever } i \neq j.$$

We often write

$$D = \text{diag}(d_1, d_2, \dots, d_n).$$

Diagonal matrices are useful because they act independently on different positions. There is no mixing between coordinates. This is why many calculations with diagonal matrices are transparent.

### Example

Let

$$D = \text{diag}(3, -1, 2).$$

Then

$$D^2 = \text{diag}(9, 1, 4), \quad D^3 = \text{diag}(27, -1, 8).$$

The power is computed by taking powers of the diagonal entries. This works because all off-diagonal entries are zero, so the coordinates do not mix.

If two diagonal matrices have the same size, then they commute. Indeed,

$$\text{diag}(a_1, \dots, a_n) \text{diag}(b_1, \dots, b_n) = \text{diag}(a_1 b_1, \dots, a_n b_n),$$

and the same result is obtained in the opposite order because ordinary numbers commute. This is a good example of structure explaining an algebraic fact.

### Patterns worth noticing

When a matrix appears in a problem, it is useful to ask whether it has visible structure. Is it diagonal? Does it contain many zeros? Are two rows equal? Are rows or columns multiples of one another? Is it symmetric? Such observations can simplify computations, but more importantly they help us understand the object.

### Remark

A calculation often becomes easier after we notice structure. But noticing structure is not a trick; it is part of mathematical understanding. We learn not only to compute with objects, but also to see them.

## 2.5 Row operations and problem solving

Matrices are often used because they allow us to transform a problem without rewriting every sentence of the problem. One of the most important examples is the use of row operations. Row operations will become essential when we solve systems of linear equations, but it is useful to meet them already here.

A row operation changes a matrix in a controlled way. We do not change entries randomly. We perform an operation that has a clear description and a clear purpose.

### Definition

The basic elementary row operations are:

- swapping two rows,
- multiplying a row by a non-zero number,
- adding a multiple of one row to another row.

The value of row operations is not that they make matrices look different. Their value is that they help reveal structure. For example, they can help us simplify a matrix, detect whether

equations are dependent, or prepare a system for solving.

### Example

Consider the matrix

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 4 & 3 \\ 0 & 1 & 5 \end{pmatrix}.$$

Suppose we replace the second row by the second row minus twice the first row. Then

$$(2, 4, 3) - 2(1, 2, -1) = (0, 0, 5).$$

The new matrix is

$$\begin{pmatrix} 1 & 2 & -1 \\ 0 & 0 & 5 \\ 0 & 1 & 5 \end{pmatrix}.$$

This operation did not merely produce new numbers. It revealed that part of the second row was already explained by the first row, except for the last entry.

This example should be read slowly. The operation has three layers. First, there is arithmetic. Second, there is a transformation of the row. Third, there is interpretation: the new row tells us something about dependence between rows.

### Column vectors as matrices

A column vector may be viewed as a matrix with one column. This viewpoint will be used constantly in linear algebra. It allows us to write vector equations and matrix equations in the same language.

For instance,

$$u = \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}$$

is a  $3 \times 1$  matrix. Its transpose is the row matrix

$$u^\top = (2 \quad -1 \quad 4).$$

The distinction between a row and a column is not cosmetic. The products  $u^\top u$  and  $uu^\top$  have different sizes and different meanings:

$$u^\top u$$

is a  $1 \times 1$  matrix, while

$$uu^\top$$

is a  $3 \times 3$  matrix.

This is another example of a recurring principle: before calculating, read the shapes.

### How to approach matrix exercises

When solving matrix exercises, the aim is not to rush toward the answer. The answer is only the last line of a process. A good solution should make the process visible.

## Summary

When working with matrices, ask:

- What is the size of each matrix?
- Which operations are defined?
- Am I adding entries position by position, or multiplying rows by columns?
- Does the order of multiplication matter here?
- Is there a visible structure that simplifies the problem?
- What does the result tell me about the original objects?

A student who asks these questions is not merely applying a recipe. They are beginning to use matrices as mathematical objects. That is the real purpose of this chapter.

# Chapter 3

## Determinants

### 3.1 A new function on square matrices

In the previous chapter we learned how to read matrices and how to perform basic operations on them. We now introduce a new object associated with a square matrix: a single number called its *determinant*.

For each size  $n$ , the determinant is a function

$$\det: \mathbb{R}^{n \times n} \longrightarrow \mathbb{R}.$$

It takes an  $n \times n$  matrix as input and returns one real number.

At this point we should not pretend that the formula is forced by ideas that have not yet been developed. We will begin by defining determinants explicitly for matrices of size  $1 \times 1$ ,  $2 \times 2$ , and  $3 \times 3$ . The definition may initially appear somewhat artificial. Its usefulness will become visible through the properties that follow from it.

This is a common order in mathematics:

$$\text{definition} \longrightarrow \text{calculation} \longrightarrow \text{properties} \longrightarrow \text{interpretation}.$$

We first agree what the object is. We then investigate what the definition allows us to do.

#### Remark

The determinant is defined only for square matrices in this course. Thus expressions such as

$$\det \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$$

are not defined.

The chapter will follow a controlled progression.

- We define determinants of small matrices and practise direct calculation.
- We examine how the value changes when rows or columns are modified.
- We study what can be concluded when the determinant is zero.
- We extend the definition to arbitrary square matrices using minors, cofactors, and Laplace expansion.
- We collect general properties that make determinant calculations efficient.

The main question is therefore not yet “What deep object does the determinant represent?” The immediate question is more elementary and more honest:

#### Summary

How is the determinant defined, and what follows from that definition?

## 3.2 Determinants of small matrices

We begin with explicit formulas. These formulas are the definitions for the small sizes considered in this section.

### 3.2.1 The determinant of a $1 \times 1$ matrix

For a matrix containing one entry, we define

$$\det(a) = a.$$

The determinant is simply the entry itself.

This convention will later be needed when larger determinants are reduced to smaller ones.

### 3.2.2 The determinant of a $2 \times 2$ matrix

#### Definition

For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

the determinant of  $A$  is

$$\det A = ad - bc.$$

We also write

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

The products  $ad$  and  $bc$  use the two diagonals of the matrix. The order matters: the second product is subtracted from the first.

#### Example

Let

$$A = \begin{pmatrix} 3 & 1 \\ 2 & 5 \end{pmatrix}.$$

Then

$$\det A = 3 \cdot 5 - 1 \cdot 2 = 15 - 2 = 13.$$

#### Example

Let

$$B = \begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix}.$$

Then

$$\det B = 1 \cdot 6 - 2 \cdot 3 = 6 - 6 = 0.$$

For the moment, this is simply the result of applying the definition. In the next section we will examine why proportional rows force this cancellation.

A determinant is sensitive to the positions of the entries. For example,

$$\det \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = 4 - 6 = -2,$$

whereas

$$\det \begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix} = 6 - 4 = 2.$$

The same four numbers appear, but the rows have been exchanged and the sign has changed.

### 3.2.3 The determinant of a $3 \times 3$ matrix

For a  $3 \times 3$  matrix we again begin with an explicit formula.

#### Definition

For

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix},$$

we define

$$\det A = aei + bfg + cdh - ceg - bdi - afh.$$

One way to remember this formula is Sarrus' rule. Repeat the first two columns:

$$\begin{array}{ccc|cc} a & b & c & a & b \\ d & e & f & d & e \\ g & h & i & g & h \end{array}$$

Add the products along the three downward diagonals and subtract the products along the three upward diagonals:

$$\det A = (aei + bfg + cdh) - (ceg + bdi + afh).$$

#### Remark

Sarrus' rule is a mnemonic for the  $3 \times 3$  formula. It is not a rule for determinants of size  $4 \times 4$  or larger.

#### Example

Let

$$A = \begin{pmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \\ 2 & 0 & 5 \end{pmatrix}.$$

Then

$$\begin{aligned} \det A &= (1 \cdot 3 \cdot 5 + 2 \cdot 4 \cdot 2 + 0 \cdot (-1) \cdot 0) \\ &\quad - (0 \cdot 3 \cdot 2 + 2 \cdot (-1) \cdot 5 + 1 \cdot 4 \cdot 0) \\ &= (15 + 16 + 0) - (0 - 10 + 0) \\ &= 41. \end{aligned}$$

#### Example

Consider

$$C = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 4 & 0 \\ 3 & 1 & 5 \end{pmatrix}.$$

Using the formula,

$$\det C = 1 \cdot 4 \cdot 5 + 2 \cdot 0 \cdot 3 + 0 \cdot 2 \cdot 1 - 0 \cdot 4 \cdot 3 - 2 \cdot 2 \cdot 5 - 1 \cdot 0 \cdot 1 = 0.$$

The first two rows are proportional. The cancellation is not an accident; it is one of the basic properties of determinants.

### 3.2.4 What should be learned from the formulas?

At this stage there are two separate skills.

- Apply the correct formula without losing signs or terms.
- Compare examples and look for regular behaviour.

The second skill leads to the next section. Instead of computing many unrelated examples, we will use the formulas to establish rules that hold for whole families of matrices.

## 3.3 Properties discovered from the formulas

The formulas for  $2 \times 2$  and  $3 \times 3$  determinants are now known. We will not treat the next properties as a new list of rules to memorize. We will obtain them from the formulas.

For each property we follow the same pattern:

1. perform a small numerical experiment;
2. describe what seems to happen;
3. repeat the calculation with general entries;
4. only then write the result as a theorem.

The numerical examples are not proofs. They show us what question to ask. The calculation with general entries explains why the observation is always true.

### 3.3.1 Changing one row at a time

We begin with two matrices that have the same second row:

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix}, \quad B = \begin{pmatrix} 4 & -1 \\ 3 & 5 \end{pmatrix}.$$

Their determinants are

$$\det A = 1 \cdot 5 - 2 \cdot 3 = 5 - 6 = -1$$

and

$$\det B = 4 \cdot 5 - (-1) \cdot 3 = 20 + 3 = 23.$$

Now add only the first rows:

$$(1, 2) + (4, -1) = (5, 1).$$

Keeping the second row unchanged gives

$$C = \begin{pmatrix} 5 & 1 \\ 3 & 5 \end{pmatrix}.$$

Its determinant is

$$\det C = 5 \cdot 5 - 1 \cdot 3 = 25 - 3 = 22.$$

We notice that

$$22 = (-1) + 23,$$

so in this example

$$\det C = \det A + \det B.$$

The example suggests that adding two possible first rows may add the corresponding determinants. We now check this from the formula.

Take the two possible first rows

$$(a, b) \quad \text{and} \quad (p, q),$$

and keep the second row

$$(c, d)$$

fixed. After adding the first rows, the matrix is

$$\begin{pmatrix} a+p & b+q \\ c & d \end{pmatrix}.$$

Using the definition of a  $2 \times 2$  determinant,

$$\begin{aligned} \det \begin{pmatrix} a+p & b+q \\ c & d \end{pmatrix} &= (a+p)d - (b+q)c \\ &= ad + pd - bc - qc \\ &= (ad - bc) + (pd - qc) \\ &= \det \begin{pmatrix} a & b \\ c & d \end{pmatrix} + \det \begin{pmatrix} p & q \\ c & d \end{pmatrix}. \end{aligned}$$

Every line has a specific purpose. The first line applies the determinant formula. The second expands the brackets. The third groups together the terms coming from the first row  $(a, b)$  and the terms coming from the second possible row  $(p, q)$ . The last line recognizes the two determinants.

### 3.3.2 Multiplying one row by a number

Consider

$$A = \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}.$$

We first calculate

$$\det A = 2 \cdot 4 - 1 \cdot 3 = 8 - 3 = 5.$$

Multiply only the first row by 3:

$$(2, 1) \longrightarrow (6, 3).$$

The new matrix is

$$B = \begin{pmatrix} 6 & 3 \\ 3 & 4 \end{pmatrix}.$$

Its determinant is

$$\det B = 6 \cdot 4 - 3 \cdot 3 = 24 - 9 = 15.$$

Since

$$15 = 3 \cdot 5,$$

the determinant was also multiplied by 3.

Now replace the numerical factor 3 by an arbitrary number  $k$ . For

$$\begin{pmatrix} ka & kb \\ c & d \end{pmatrix}$$

we obtain

$$\begin{aligned} \det \begin{pmatrix} ka & kb \\ c & d \end{pmatrix} &= (ka)d - (kb)c \\ &= kad - kbc \\ &= k(ad - bc) \\ &= k \det \begin{pmatrix} a & b \\ c & d \end{pmatrix}. \end{aligned}$$

The factor  $k$  appears in both products, so it can be taken in front of the determinant.

### Theorem

For a  $2 \times 2$  determinant, addition in one row becomes addition of determinants, and multiplication of one row by a number multiplies the determinant by the same number. The other row must remain fixed.

This property is called *linearity in one row*. The phrase “in one row” matters. It does not mean that

$$\det(A + B) = \det A + \det B$$

for arbitrary matrices.

For example, let

$$A = B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Then

$$\det A + \det B = 1 + 1 = 2,$$

but

$$A + B = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

and

$$\det(A + B) = 2 \cdot 2 - 0 \cdot 0 = 4.$$

Thus linearity applies to one selected row while the other rows are fixed. It is not linearity of the whole matrix.

### 3.3.3 A check for a $3 \times 3$ determinant

The same mechanism can already be seen in the  $3 \times 3$  formula. We check multiplication of the first row by  $k$ .

Start with

$$\begin{pmatrix} ka & kb & kc \\ d & e & f \\ g & h & i \end{pmatrix}.$$

Applying the six-term formula gives

$$\begin{aligned}
& \det \begin{pmatrix} ka & kb & kc \\ d & e & f \\ g & h & i \end{pmatrix} \\
&= (ka)ei + (kb)fg + (kc)dh - (kc)eg - (kb)di - (ka)fh \\
&= k(aei) + k(bfg) + k(cdh) - k(ceg) - k(bdi) - k(afh) \\
&= k(aei + bfg + cdh - ceg - bdi - afh) \\
&= k \det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}.
\end{aligned}$$

The calculation is longer than in the  $2 \times 2$  case, but the idea is unchanged: every term contains exactly one entry from the first row, so multiplying that row by  $k$  multiplies every term by  $k$ .

We can also test addition in the first row without introducing another large collection of letters. Keep the last two rows fixed and consider

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 1 & 0 & 2 \\ 0 & 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & -1 & 1 \\ 1 & 0 & 2 \\ 0 & 1 & 1 \end{pmatrix}.$$

Using the  $3 \times 3$  formula,

$$\det A = -4, \quad \det B = -2.$$

Adding the first rows gives

$$(1, 2, 0) + (2, -1, 1) = (3, 1, 1),$$

so

$$C = \begin{pmatrix} 3 & 1 & 1 \\ 1 & 0 & 2 \\ 0 & 1 & 1 \end{pmatrix}.$$

A direct calculation gives

$$\det C = -6 = -4 + (-2).$$

This numerical calculation is evidence of the same pattern. Later, after the general definition by Laplace expansion, we will prove linearity for every size without writing an increasingly long explicit formula.

### 3.3.4 Exchanging two rows

Take

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

Then

$$\det A = 1 \cdot 4 - 2 \cdot 3 = 4 - 6 = -2.$$

Exchange the rows:

$$B = \begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix}.$$

Now

$$\det B = 3 \cdot 2 - 4 \cdot 1 = 6 - 4 = 2.$$

The value changed from  $-2$  to  $2$ . The sign changed.

To see why this always happens, start with

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Its determinant is

$$\det A = ad - bc.$$

After exchanging the rows,

$$B = \begin{pmatrix} c & d \\ a & b \end{pmatrix}.$$

Therefore

$$\begin{aligned} \det B &= cb - da \\ &= bc - ad \\ &= -(ad - bc) \\ &= -\det A. \end{aligned}$$

The equality  $cb = bc$  uses ordinary multiplication of numbers. The third line takes a minus sign in front of the whole expression.

### Theorem

Exchanging the two rows of a  $2 \times 2$  determinant changes its sign.

### 3.3.5 Equal rows

Suppose both rows are equal:

$$A = \begin{pmatrix} a & b \\ a & b \end{pmatrix}.$$

The determinant can be calculated directly:

$$\begin{aligned} \det A &= a \cdot b - b \cdot a \\ &= ab - ab \\ &= 0. \end{aligned}$$

There is also a useful reasoning argument. If we exchange two equal rows, the matrix does not change. Therefore its determinant should still be the same number. But we have just proved that exchanging two rows changes the sign. Hence the same determinant must satisfy

$$\det A = -\det A.$$

Adding  $\det A$  to both sides gives

$$2 \det A = 0,$$

and therefore

$$\det A = 0.$$

The direct calculation checks the result from the formula. The second argument shows how one proved property can be used to obtain another.

### 3.3.6 Proportional rows

Now suppose the second row is  $k$  times the first row:

$$A = \begin{pmatrix} a & b \\ ka & kb \end{pmatrix}.$$

Applying the determinant formula,

$$\begin{aligned} \det A &= a(kb) - b(ka) \\ &= kab - kab \\ &= 0. \end{aligned}$$

The two products are equal, so they cancel.

We can see the same cancellation directly in the  $3 \times 3$  formula. Let the second row be  $k$  times the first:

$$A = \begin{pmatrix} a & b & c \\ ka & kb & kc \\ g & h & i \end{pmatrix}.$$

Then

$$\begin{aligned} \det A &= a(kb)i + b(kc)g + c(ka)h \\ &\quad - c(kb)g - b(ka)i - a(kc)h \\ &= kabi + kbcg + kach - kbcg - kabi - kach \\ &= 0. \end{aligned}$$

We have written all six terms so that the cancellation is visible.

#### Theorem

For the  $2 \times 2$  and  $3 \times 3$  determinants already defined, equal or proportional rows give determinant zero.

### 3.3.7 Adding a multiple of one row to another

Consider

$$A = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}.$$

Its determinant is

$$\det A = 2 \cdot 3 - 1 \cdot 5 = 6 - 5 = 1.$$

Replace the second row by the second row minus twice the first:

$$(5, 3) - 2(2, 1) = (5, 3) - (4, 2) = (1, 1).$$

The new matrix is

$$B = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

Its determinant is

$$\det B = 2 \cdot 1 - 1 \cdot 1 = 2 - 1 = 1.$$

The determinant did not change.

Now check the general calculation. Start with

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

and replace the second row by

$$(c, d) + k(a, b) = (c + ka, d + kb).$$

The new determinant is

$$\begin{aligned} \det \begin{pmatrix} a & b \\ c + ka & d + kb \end{pmatrix} &= a(d + kb) - b(c + ka) \\ &= ad + kab - bc - kab \\ &= ad - bc \\ &= \det \begin{pmatrix} a & b \\ c & d \end{pmatrix}. \end{aligned}$$

The new terms  $kab$  and  $-kab$  cancel. This is why the determinant remains unchanged.

### Theorem

Adding a multiple of one row to another row does not change a  $2 \times 2$  determinant.

This property can also be reconstructed from the earlier discoveries:

- addition in one row separates into a sum of determinants;
- the added part is a multiple of the other row;
- a determinant with proportional rows is zero.

The direct calculation above is still important because it shows the cancellation explicitly.

### 3.3.8 Rows and columns

So far we have worked mainly with rows. Columns satisfy the same properties. For  $2 \times 2$  matrices this can be checked by transposition.

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad A^T = \begin{pmatrix} a & c \\ b & d \end{pmatrix}.$$

Then

$$\det A = ad - bc,$$

while

$$\begin{aligned} \det(A^T) &= ad - cb \\ &= ad - bc \\ &= \det A. \end{aligned}$$

The middle equality again uses  $cb = bc$ .

Transposition changes rows into columns without changing the determinant. Therefore a row calculation can be transferred to columns. For example, proportional columns become proportional rows after transposition, so their determinant is also zero.

### Theorem

For the small determinants considered here,

$$\det(A^T) = \det A.$$

Consequently, the corresponding row properties also hold for columns.

### What has been learned

## Summary

The formulas were the starting point, not the final lesson.

We first performed concrete calculations. They suggested patterns. We then returned to the definitions and showed the patterns by expanding, regrouping, and cancelling terms. Only after this work did we state the properties:

- one row may be added or multiplied by a number while the other rows remain fixed;
- exchanging two rows changes the sign;
- equal or proportional rows give determinant zero;
- adding a multiple of one row to another does not change the determinant;
- the corresponding statements hold for columns.

A formula summarizes a piece of reasoning. In a complete solution, the intermediate steps must remain visible so that the reader can see why the conclusion follows.

## 3.4 What a zero determinant can reveal

A determinant equal to zero is often easier to understand than a general determinant value. In small dimensions we can connect the equation  $\det A = 0$  directly with relations among rows or columns.

### 3.4.1 The $2 \times 2$ case

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Then

$$\det A = 0 \iff ad = bc.$$

If the first row is non-zero, this equation implies that the second row is proportional to the first. To see this, suppose first that  $a \neq 0$ . Then

$$d = \frac{bc}{a},$$

so

$$(c, d) = \frac{c}{a}(a, b).$$

If  $a = 0$ , then  $b \neq 0$ , and the equation  $ad = bc$  gives  $c = 0$ . Hence

$$(c, d) = \frac{d}{b}(0, b).$$

If the first row is the zero row, proportionality is immediate.

Thus, for  $2 \times 2$  matrices, determinant zero is equivalent to the rows being proportional. The same is true for the columns.

### 3.4.2 A first observation from matrix–vector multiplication

The previous chapter introduced multiplication of a matrix by a column vector. Consider a matrix whose second row is  $\lambda$  times its first row:

$$A = \begin{pmatrix} a & b \\ \lambda a & \lambda b \end{pmatrix}.$$

For an arbitrary column vector

$$x = \begin{pmatrix} u \\ v \end{pmatrix},$$

we obtain

$$Ax = \begin{pmatrix} au + bv \\ \lambda au + \lambda bv \end{pmatrix} = \begin{pmatrix} t \\ \lambda t \end{pmatrix}, \quad t = au + bv.$$

No matter which values of  $u$  and  $v$  are chosen, the two entries of  $Ax$  satisfy the fixed relation

$$(Ax)_2 = \lambda(Ax)_1.$$

### Example

Let

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}.$$

Then  $\det A = 0$ , and for every  $x = (u, v)^T$ ,

$$Ax = \begin{pmatrix} u + 2v \\ 2u + 4v \end{pmatrix} = \begin{pmatrix} t \\ 2t \end{pmatrix}.$$

The resulting columns are restricted to a single pattern: the second entry is always twice the first.

This does not yet require a general theory of linear mappings. It is simply an algebraic observation obtained from matrix multiplication. It gives a first indication that determinant zero records a restriction in the possible outputs.

A similar phenomenon appears for three rows. If

$$r_3 = r_1 + r_2,$$

then for every column vector  $x$ , the third entry of  $Ax$  is the sum of the first two entries. The output again satisfies a fixed relation.

### 3.4.3 Parameters

When a matrix contains a parameter, its determinant is an expression in that parameter. The equation

$$\det A(t) = 0$$

identifies the parameter values for which a row or column relation appears.

### Example

Let

$$A(t) = \begin{pmatrix} t & 1 \\ 4 & t \end{pmatrix}.$$

Then

$$\det A(t) = t^2 - 4 = (t - 2)(t + 2).$$

Therefore the determinant is zero for

$$t = 2 \quad \text{or} \quad t = -2.$$

For  $t = 2$ ,

$$A(2) = \begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix},$$

and the second row is twice the first. For  $t = -2$ ,

$$A(-2) = \begin{pmatrix} -2 & 1 \\ 4 & -2 \end{pmatrix},$$

and the second row is  $-2$  times the first.

Here the roots of the determinant are not merely numbers obtained from an equation. They identify exactly when proportionality occurs.

### Example

Consider

$$B(s) = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & s & 2 \end{pmatrix}.$$

Using the  $3 \times 3$  formula,

$$\det B(s) = 1 - s.$$

Thus  $\det B(s) = 0$  when  $s = 1$ . At that value,

$$(1, 1, 2) = (1, 0, 1) + (0, 1, 1),$$

so the third row is the sum of the first two rows.

### Summary

When a determinant is zero, first look for a relation among rows or columns. With parameters, solve  $\det A = 0$  and then inspect what relation appears at the exceptional values.

## 3.5 Determinants of larger matrices

The formulas for  $2 \times 2$  and  $3 \times 3$  matrices are explicit, but they do not provide a practical definition for every size. We now extend the determinant recursively: a determinant of size  $n \times n$  will be expressed in terms of determinants of size  $(n - 1) \times (n - 1)$ .

### 3.5.1 Minors

Let  $A = (a_{ij})$  be an  $n \times n$  matrix. Remove row  $i$  and column  $j$ . The remaining matrix has size  $(n - 1) \times (n - 1)$ .

#### Definition

The determinant of the matrix obtained by deleting row  $i$  and column  $j$  is called the minor of the entry  $a_{ij}$ . It is denoted by

$$M_{ij}.$$

### Example

Let

$$A = \begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 4 \\ 5 & 2 & 1 \end{pmatrix}.$$

To find  $M_{12}$ , delete row 1 and column 2:

$$M_{12} = \det \begin{pmatrix} 0 & 4 \\ 5 & 1 \end{pmatrix} = 0 \cdot 1 - 4 \cdot 5 = -20.$$

### 3.5.2 Cofactors

The signs used in the recursive formula alternate in a checkerboard pattern:

$$\begin{pmatrix} + & - & + & - & \cdots \\ - & + & - & + & \cdots \\ + & - & + & - & \cdots \\ - & + & - & + & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

The sign in position  $(i, j)$  is  $(-1)^{i+j}$ .

### Definition

The cofactor of the entry  $a_{ij}$  is

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

For the previous example,

$$M_{12} = -20, \quad C_{12} = (-1)^{1+2} M_{12} = 20.$$

### 3.5.3 The recursive definition

We already defined the determinant of a  $1 \times 1$  matrix by

$$\det(a) = a.$$

For  $n \geq 2$ , we define the determinant by expanding along the first row.

### Definition

Let  $A = (a_{ij})$  be an  $n \times n$  matrix. Its determinant is

$$\det A = a_{11}C_{11} + a_{12}C_{12} + \cdots + a_{1n}C_{1n}.$$

Equivalently,

$$\det A = \sum_{j=1}^n a_{1j}C_{1j}.$$

This definition reduces the size by one at every step. Eventually every computation reaches determinants of size  $1 \times 1$ .

For a  $2 \times 2$  matrix,

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

the recursive definition gives

$$\det A = a \det \begin{pmatrix} d \end{pmatrix} - b \det \begin{pmatrix} c \end{pmatrix} = ad - bc.$$

Thus it agrees with the earlier definition.

For a  $3 \times 3$  matrix,

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix},$$

expansion along the first row gives

$$\begin{aligned} \det A &= a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix} \\ &= a(ei - fh) - b(di - fg) + c(dh - eg) \\ &= aei + bfg + cdh - ceg - bdi - afh. \end{aligned}$$

Thus the recursive definition also agrees with the earlier  $3 \times 3$  formula.

### 3.5.4 Expansion along any row or column

Although the determinant was defined using the first row, it can be proved that the same value is obtained by expanding along any row or any column.

#### Theorem

Expansion along row  $i$  gives

$$\det A = \sum_{j=1}^n a_{ij} C_{ij}.$$

Expansion along column  $j$  gives

$$\det A = \sum_{i=1}^n a_{ij} C_{ij}.$$

This freedom is computationally important. A row or column containing many zeros usually gives the shortest expansion.

#### Example

Consider

$$B = \begin{pmatrix} 4 & 0 & 1 & 2 \\ 0 & 3 & 0 & -1 \\ 2 & 0 & 5 & 0 \\ 1 & 0 & 0 & 6 \end{pmatrix}.$$

The second column contains only one non-zero entry. Expanding along that column,

$$\det B = 3C_{22}.$$

Since the sign in position  $(2, 2)$  is positive,

$$\det B = 3 \det \begin{pmatrix} 4 & 1 & 2 \\ 2 & 5 & 0 \\ 1 & 0 & 6 \end{pmatrix}.$$

Expanding the remaining determinant along its second row or using the  $3 \times 3$  formula gives

$$\det \begin{pmatrix} 4 & 1 & 2 \\ 2 & 5 & 0 \\ 1 & 0 & 6 \end{pmatrix} = 98.$$

Therefore

$$\det B = 294.$$

### 3.5.5 Using Laplace expansion intelligently

Laplace expansion is a definition and a general method, but it should not be applied blindly. Before expanding:

- inspect the matrix for zeros;
- choose a row or column that produces few non-zero terms;
- write the cofactor signs carefully;
- keep the smaller determinants visible;
- use row or column properties when they shorten the calculation.

## 3.6 General properties and efficient calculation

The recursive definition extends the determinant to every square size. The properties observed for  $2 \times 2$  and  $3 \times 3$  matrices continue to hold in arbitrary dimension.

### 3.6.1 Multilinearity and alternating behaviour

#### Theorem

The determinant is linear in each row separately when all other rows are fixed. It is also linear in each column separately.

If the  $i$ -th row is written as a sum  $r_i = u + v$ , then

$$\det(r_1, \dots, u + v, \dots, r_n) = \det(r_1, \dots, u, \dots, r_n) + \det(r_1, \dots, v, \dots, r_n).$$

If the row is multiplied by  $\lambda$ , then the determinant is multiplied by  $\lambda$ .

#### Theorem

Exchanging two rows changes the sign of the determinant. Consequently, a matrix with two equal rows has determinant zero.

Combining multilinearity with the equal-row property gives a stronger conclusion. If one row is a linear combination of the other rows, then the determinant is zero.

For example, if

$$r_3 = 2r_1 - r_2,$$

then

$$\begin{aligned}\det(r_1, r_2, r_3) &= \det(r_1, r_2, 2r_1 - r_2) \\ &= 2 \det(r_1, r_2, r_1) - \det(r_1, r_2, r_2) \\ &= 0.\end{aligned}$$

Both terms contain repeated rows.

Because  $\det(A^T) = \det A$ , all corresponding column statements also hold.

### 3.6.2 Elementary row operations

The effect of the three elementary row operations can now be summarized.

| Row operation                        | Effect on the determinant |
|--------------------------------------|---------------------------|
| Exchange two rows                    | multiply by $-1$          |
| Multiply one row by $\lambda$        | multiply by $\lambda$     |
| Add a multiple of one row to another | no change                 |

These rules allow us to simplify a matrix while keeping track of the determinant.

#### Example

Compute

$$\det \begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 1 & 0 & 4 \end{pmatrix}.$$

Apply

$$R_2 \leftarrow R_2 - 2R_1, \quad R_3 \leftarrow R_3 - R_1.$$

These operations do not change the determinant:

$$\begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 1 & 0 & 4 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & -2 & 3 \end{pmatrix}.$$

Next use

$$R_3 \leftarrow R_3 + 2R_2,$$

which again leaves the determinant unchanged:

$$\begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & -2 & 3 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 5 \end{pmatrix}.$$

The final matrix is upper triangular, so

$$\det A = 1 \cdot 1 \cdot 5 = 5.$$

### 3.6.3 Triangular matrices

#### Theorem

The determinant of an upper or lower triangular matrix is the product of its diagonal entries:

$$\det A = a_{11}a_{22} \cdots a_{nn}.$$

This follows from Laplace expansion: the zeros force the calculation to continue through smaller triangular matrices until only the diagonal entries remain.

A useful consequence is that a triangular matrix has determinant zero exactly when at least one diagonal entry is zero.

### 3.6.4 Scaling the whole matrix

If every row of an  $n \times n$  matrix is multiplied by  $\lambda$ , multilinearity contributes one factor of  $\lambda$  from each row. Therefore

$$\det(\lambda A) = \lambda^n \det A.$$

This is why the determinant is not a linear function of the whole matrix.

### 3.6.5 The determinant of a product

A further fundamental property is

$$\det(AB) = \det A \det B$$

for square matrices  $A$  and  $B$  of the same size.

The full proof is not needed for the present calculations, but the statement is consistent with the rules already established. For diagonal matrices it is immediate: the diagonal entries multiply position by position, and so do their products.

#### Remark

The formula  $\det(AB) = \det A \det B$  concerns matrix multiplication. It is different from entry-by-entry multiplication and should not be confused with linearity in rows or columns.

### 3.6.6 A structured example

Consider the Vandermonde-type determinant

$$V = \begin{pmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{pmatrix}.$$

Apply the determinant-preserving column operations

$$C_2 \leftarrow C_2 - C_1, \quad C_3 \leftarrow C_3 - C_1.$$

Then

$$\det V = \det \begin{pmatrix} 1 & 0 & 0 \\ x & y-x & z-x \\ x^2 & y^2-x^2 & z^2-x^2 \end{pmatrix}.$$

Expanding along the first row gives

$$\det V = \det \begin{pmatrix} y-x & z-x \\ y^2-x^2 & z^2-x^2 \end{pmatrix}.$$

Factor the two columns:

$$y^2 - x^2 = (y - x)(y + x), \quad z^2 - x^2 = (z - x)(z + x).$$

Therefore

$$\begin{aligned} \det V &= (y - x)(z - x) \det \begin{pmatrix} 1 & 1 \\ y + x & z + x \end{pmatrix} \\ &= (y - x)(z - x)(z - y). \end{aligned}$$

The factorization immediately shows when the determinant is zero: whenever two of  $x, y, z$  are equal, two columns of the original matrix are equal.

## What to remember

### Summary

Before computing a determinant:

- check whether rows or columns are equal, proportional, or related by a simple linear combination;
- look for zeros and triangular structure;
- decide whether row or column operations will simplify the matrix;
- track sign changes and extracted factors;
- use Laplace expansion only after choosing a useful row or column.

The determinant begins as a definition. Its mathematical content is revealed by the algebraic properties that the definition satisfies.

# Chapter 4

## Matrix Inversion

### 4.1 The meaning of an inverse matrix

In the previous chapter we learned that the determinant can distinguish between singular and non-singular square matrices. The condition

$$\det A \neq 0$$

means that the matrix has not collapsed its structure. This raises a natural question: if a matrix has not collapsed information, can we reverse its action?

This question leads to the idea of an inverse matrix.

For ordinary numbers, the inverse of a non-zero number  $a$  is the number  $a^{-1}$  such that

$$aa^{-1} = 1.$$

For example, the inverse of 5 is  $\frac{1}{5}$ , because

$$5 \cdot \frac{1}{5} = 1.$$

The number 1 is special because multiplication by 1 changes nothing. In matrix algebra, the identity matrix plays the same role.

#### Definition

Let  $A$  be a square matrix of order  $n$ . A matrix  $B$  is called an inverse of  $A$  if

$$AB = I_n \quad \text{and} \quad BA = I_n.$$

If such a matrix exists, it is denoted by  $A^{-1}$ , and we write

$$AA^{-1} = I_n, \quad A^{-1}A = I_n.$$

The two equations are both important. Matrix multiplication is not generally commutative, so the statements  $AB = I$  and  $BA = I$  are not automatically the same piece of information. The inverse must undo the matrix on both sides.

#### Remark

The notation  $A^{-1}$  does not mean  $\frac{1}{A}$ . Matrices are not numbers, and matrix division is not defined in the same way as numerical division. The symbol  $A^{-1}$  means: the matrix which reverses the action of  $A$ , if such a matrix exists.

#### 4.1.1 Undoing an operation

The idea of an inverse is the idea of undoing. If a matrix  $A$  transforms a vector  $x$  into a vector  $y$ , then

$$Ax = y.$$

If  $A^{-1}$  exists, we can apply it to both sides:

$$A^{-1}Ax = A^{-1}y.$$

Since  $A^{-1}A = I$ , this becomes

$$x = A^{-1}y.$$

Thus the inverse matrix recovers the original vector from the transformed vector.

This explains why singular matrices cannot have inverses. If a matrix collapses different inputs into the same output, then there is no way to recover the original input uniquely. Information has been lost.

### Example

Consider the diagonal matrix

$$D = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}.$$

Multiplication by  $D$  doubles the first coordinate and triples the second coordinate:

$$D \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x \\ 3y \end{pmatrix}.$$

To reverse this action, we should divide the first coordinate by 2 and the second coordinate by 3. Therefore

$$D^{-1} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{pmatrix}.$$

Indeed,

$$DD^{-1} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I_2.$$

This example is simple because the matrix is diagonal. Each coordinate is changed independently. More general matrices mix coordinates, so their inverses are harder to find. But the meaning stays the same: an inverse matrix reverses the action of the original matrix.

### 4.1.2 When does an inverse exist?

The determinant gives the fundamental test.

#### Theorem

A square matrix  $A$  is invertible if and only if

$$\det A \neq 0.$$

If  $\det A = 0$ , then  $A$  is singular and has no inverse.

This theorem connects the present chapter with the previous one. Determinants do not merely produce numbers. They answer the structural question of whether a matrix can be reversed.

#### Summary

The inverse matrix should be understood through three questions:

- What operation does the original matrix perform?

- Has this operation lost information?
- If no information was lost, what matrix reverses the operation?

The rest of this chapter develops methods for finding inverse matrices and for using them correctly.

## 4.2 Inverses of $2 \times 2$ matrices

The first non-trivial case is a  $2 \times 2$  matrix. It is small enough that the inverse can be written explicitly, but rich enough to show the main ideas: the determinant appears, the order of entries matters, and the inverse exists only under a clear condition.

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

We want to find a matrix

$$A^{-1} = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$$

such that

$$AA^{-1} = I_2.$$

Multiplying, we get

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} p & q \\ r & s \end{pmatrix} = \begin{pmatrix} ap + br & aq + bs \\ cp + dr & cq + ds \end{pmatrix}.$$

For this to be the identity matrix, we need

$$\begin{cases} ap + br = 1, \\ cp + dr = 0, \end{cases} \quad \begin{cases} aq + bs = 0, \\ cq + ds = 1. \end{cases}$$

Thus finding the inverse is already a problem about solving systems of linear equations. Instead of solving these systems every time, we use the known formula.

### Theorem

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

If

$$\det A = ad - bc \neq 0,$$

then

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

If  $ad - bc = 0$ , then  $A$  has no inverse.

The formula has two visible parts. First, the determinant appears in the denominator. This is not accidental: if the determinant is zero, the inverse cannot exist. Second, the matrix

$$\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

is obtained by switching the diagonal entries  $a, d$  and changing signs of the off-diagonal entries  $b, c$ .

### Example

Find the inverse of

$$A = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}.$$

First compute the determinant:

$$\det A = 2 \cdot 3 - 1 \cdot 5 = 6 - 5 = 1.$$

Since the determinant is non-zero, the inverse exists. Therefore

$$A^{-1} = \frac{1}{1} \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix} = \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}.$$

We should verify the result:

$$AA^{-1} = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix} \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix} = \begin{pmatrix} 6-5 & -2+2 \\ 15-15 & -5+6 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Verification is not a decorative step. It helps us remember the meaning of the inverse. We are not only producing a matrix; we are producing a matrix that must multiply with the original one to give the identity.

### Example

Consider

$$B = \begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix}.$$

The determinant is

$$\det B = 1 \cdot 6 - 2 \cdot 3 = 6 - 6 = 0.$$

The formula cannot be used because it would require division by zero. More importantly, the inverse does not exist. The second row is three times the first row, so the matrix has lost independent information.

This example shows why the determinant test is not just a technical condition. A zero determinant indicates structural collapse. There is no inverse because there is no unique way to undo the action of the matrix.

#### 4.2.1 A formula should still be read

It is tempting to memorize the  $2 \times 2$  inverse formula mechanically. But it should be read as a statement with meaning:

$$A^{-1} = \frac{1}{\det A} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

The determinant controls whether the inverse exists and how strongly the inverse scales the result. The second matrix rearranges the entries in the only way that makes multiplication return the identity.

### Summary

For a  $2 \times 2$  inverse:

- compute  $\det A = ad - bc$ ,
- if  $\det A = 0$ , stop: the matrix is not invertible,
- if  $\det A \neq 0$ , use the inverse formula,
- verify the result by multiplying with the original matrix.

## 4.3 The Gauss–Jordan method

The formula for a  $2 \times 2$  inverse is useful, but it does not give a practical method for larger matrices. For a  $3 \times 3$  matrix, and especially for larger matrices, we need a systematic procedure. The most important procedure is the Gauss–Jordan method.

The idea is simple and powerful. If we want to find  $A^{-1}$ , we want to transform  $A$  into the identity matrix. Every row operation we apply to  $A$  must also be applied to the identity matrix beside it. When the left side becomes  $I$ , the right side becomes  $A^{-1}$ .

We write this as an augmented matrix:

$$[A \mid I].$$

The goal is

$$[A \mid I] \longrightarrow [I \mid A^{-1}].$$

### Definition

An elementary matrix is a matrix obtained from an identity matrix by performing exactly one elementary row operation. If  $E$  represents that operation, then multiplying  $EA$  applies the same row operation to  $A$ .

### Remark

The method does not work by magic. Each row operation corresponds to multiplication by an elementary matrix. If

$$E_k \cdots E_2 E_1 A = I,$$

then

$$E_k \cdots E_2 E_1 = A^{-1}.$$

Applying the same operations to the identity block therefore constructs the matrix that reverses  $A$ .

### 4.3.1 A complete $2 \times 2$ example

Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix}.$$

We form the augmented matrix

$$\left[ \begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 7 & 0 & 1 \end{array} \right].$$

Our goal is to turn the left block into the identity matrix.

First eliminate the entry below the first pivot:

$$R_2 \leftarrow R_2 - 3R_1.$$

Then

$$\left[ \begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & -3 & 1 \end{array} \right].$$

Now eliminate the entry above the second pivot:

$$R_1 \leftarrow R_1 - 2R_2.$$

This gives

$$\left[ \begin{array}{cc|cc} 1 & 0 & 7 & -2 \\ 0 & 1 & -3 & 1 \end{array} \right].$$

The left side is now  $I_2$ , so

$$A^{-1} = \begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix}.$$

We verify:

$$\begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix} \begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} 7-6 & -2+2 \\ 21-21 & -6+7 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

#### 4.3.2 A complete $3 \times 3$ example

Now consider

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}.$$

We start with

$$\left[ \begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array} \right].$$

Use the first row to eliminate the entry below the first pivot:

$$R_3 \leftarrow R_3 - R_1.$$

Then

$$\left[ \begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & -1 & 1 & -1 & 0 & 1 \end{array} \right].$$

Use the second row to eliminate the entry below the second pivot:

$$R_3 \leftarrow R_3 + R_2.$$

This gives

$$\left[ \begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 2 & -1 & 1 & 1 \end{array} \right].$$

Normalize the third row:

$$R_3 \leftarrow \frac{1}{2}R_3.$$

Thus

$$\left[ \begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{array} \right].$$

Now eliminate the entry above the third pivot in row 2:

$$R_2 \leftarrow R_2 - R_3.$$

We get

$$\left[ \begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{array} \right].$$

Finally eliminate the entry above the second pivot in row 1:

$$R_1 \leftarrow R_1 - R_2.$$

Therefore

$$\left[ \begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{array} \right].$$

So

$$A^{-1} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

This example is longer than the  $2 \times 2$  case, but the logic is the same. We create pivots, eliminate entries below and above them, and keep track of every operation on the identity matrix.

### 4.3.3 What if the method fails?

The Gauss–Jordan method may fail to produce the identity matrix on the left. This happens precisely when the matrix is singular.

#### Example

Let

$$B = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}.$$

Start with

$$\left[ \begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & 1 \end{array} \right].$$

Apply

$$R_2 \leftarrow R_2 - 2R_1.$$

We obtain

$$\left[ \begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 0 & -2 & 1 \end{array} \right].$$

The left side cannot become the identity matrix, because the second row of the left block is zero. Thus  $B$  has no inverse.

This is not a failure of the method. It is information. The method reveals that the matrix is singular.

## Summary

The Gauss–Jordan method for finding  $A^{-1}$ :

1. Form the augmented matrix  $[A \mid I]$ .
2. Use elementary row operations to reduce the left block.
3. If the left block becomes  $I$ , the right block is  $A^{-1}$ .
4. If the left block cannot become  $I$ , then  $A$  is not invertible.

## 4.4 The adjugate method

The Gauss–Jordan method is often the most practical method for finding inverse matrices by hand. There is also another method, built from determinants and cofactors. This method is called the adjugate method.

The adjugate method is important for two reasons. First, it gives an explicit formula for the inverse of a matrix. Second, it shows a deep connection between determinants, cofactors, and invertibility.

### 4.4.1 The cofactor matrix and the adjugate matrix

Let  $A = (a_{ij})$  be an  $n \times n$  matrix. For each entry  $a_{ij}$ , we can compute its cofactor

$$C_{ij} = (-1)^{i+j} M_{ij},$$

where  $M_{ij}$  is the minor obtained by deleting row  $i$  and column  $j$ .

The matrix of cofactors is

$$C = \begin{pmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{pmatrix}.$$

The adjugate matrix is the transpose of the cofactor matrix.

## Definition

The adjugate of a square matrix  $A$ , denoted by  $\text{adj}(A)$ , is defined as

$$\text{adj}(A) = C^{\top},$$

where  $C$  is the cofactor matrix of  $A$ .

The transpose is essential. The adjugate is not simply the cofactor matrix; it is the cofactor matrix with rows and columns exchanged.

### 4.4.2 The inverse formula

The central formula is the following.

### Theorem

If  $A$  is a square matrix and  $\det A \neq 0$ , then

$$A^{-1} = \frac{1}{\det A} \operatorname{adj}(A).$$

This formula should be read slowly. It says that the inverse exists only when  $\det A \neq 0$ . It also says that the inverse can be built from cofactors. Thus the same objects used to compute determinants also describe the inverse matrix.

A compact way to express the reason behind the formula is

$$A \operatorname{adj}(A) = (\det A)I.$$

If  $\det A \neq 0$ , we divide by  $\det A$  and obtain

$$A \left( \frac{1}{\det A} \operatorname{adj}(A) \right) = I.$$

So

$$\frac{1}{\det A} \operatorname{adj}(A)$$

is the inverse matrix.

#### 4.4.3 A $3 \times 3$ example

Let

$$A = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \\ 0 & 2 & 1 \end{pmatrix}.$$

First compute the determinant. Expanding along the first row,

$$\det A = 1 \det \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix} + 2 \det \begin{pmatrix} -1 & 3 \\ 0 & 2 \end{pmatrix}.$$

The first minor gives

$$3 \cdot 1 - 1 \cdot 2 = 1,$$

and the second minor gives

$$(-1) \cdot 2 - 3 \cdot 0 = -2.$$

Therefore

$$\det A = 1 + 2(-2) = -3.$$

Since  $\det A \neq 0$ , the inverse exists.

Now compute the cofactors. We list them carefully:

$$C_{11} = \det \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix} = 1,$$

$$C_{12} = -\det \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix} = 1,$$

$$C_{13} = \det \begin{pmatrix} -1 & 3 \\ 0 & 2 \end{pmatrix} = -2,$$

$$C_{21} = -\det \begin{pmatrix} 0 & 2 \\ 2 & 1 \end{pmatrix} = 4,$$

$$\begin{aligned}
C_{22} &= \det \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = 1, \\
C_{23} &= -\det \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} = -2, \\
C_{31} &= \det \begin{pmatrix} 0 & 2 \\ 3 & 1 \end{pmatrix} = -6, \\
C_{32} &= -\det \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix} = -3, \\
C_{33} &= \det \begin{pmatrix} 1 & 0 \\ -1 & 3 \end{pmatrix} = 3.
\end{aligned}$$

Thus the cofactor matrix is

$$C = \begin{pmatrix} 1 & 1 & -2 \\ 4 & 1 & -2 \\ -6 & -3 & 3 \end{pmatrix}.$$

Therefore

$$\text{adj}(A) = C^T = \begin{pmatrix} 1 & 4 & -6 \\ 1 & 1 & -3 \\ -2 & -2 & 3 \end{pmatrix}.$$

Finally,

$$A^{-1} = \frac{1}{-3} \begin{pmatrix} 1 & 4 & -6 \\ 1 & 1 & -3 \\ -2 & -2 & 3 \end{pmatrix}.$$

So

$$A^{-1} = \begin{pmatrix} -\frac{1}{3} & -\frac{4}{3} & 2 \\ -\frac{1}{3} & -\frac{1}{3} & 1 \\ \frac{2}{3} & \frac{2}{3} & -1 \end{pmatrix}.$$

This method is longer than Gauss–Jordan for this example, but it shows exactly how determinants and cofactors build the inverse.

#### 4.4.4 When should this method be used?

For hand calculations, the adjugate method is convenient for  $2 \times 2$  matrices and sometimes manageable for  $3 \times 3$  matrices. For larger matrices it becomes computationally heavy because many cofactors must be computed.

Its conceptual value is very high: it explains why the determinant appears in inverse formulas and why zero determinant prevents invertibility.

##### Summary

The adjugate method:

1. Compute  $\det A$ .
2. If  $\det A = 0$ , the inverse does not exist.
3. Compute all cofactors  $C_{ij}$ .
4. Form the cofactor matrix  $C$ .
5. Transpose it to obtain  $\text{adj}(A)$ .

6. Use  $A^{-1} = \frac{1}{\det A} \operatorname{adj}(A)$ .

## 4.5 Invertibility and structure

It is possible to treat invertibility as a purely computational question: compute the determinant, or try Gauss–Jordan elimination, and see what happens. This works, but it is not the only way to think. Often the structure of a matrix already tells us something about whether an inverse should exist.

The central rule remains the same:

$$A \text{ is invertible} \iff \det A \neq 0.$$

But before computing the determinant, we should learn to look at the matrix.

### 4.5.1 Diagonal matrices

A diagonal matrix is one of the clearest cases. Let

$$D = \operatorname{diag}(d_1, d_2, \dots, d_n).$$

Then

$$\det D = d_1 d_2 \cdots d_n.$$

Therefore  $D$  is invertible exactly when none of the diagonal entries is zero.

If every  $d_i \neq 0$ , then

$$D^{-1} = \operatorname{diag}\left(\frac{1}{d_1}, \frac{1}{d_2}, \dots, \frac{1}{d_n}\right).$$

#### Example

Let

$$D = \begin{pmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 5 \end{pmatrix}.$$

Then

$$D^{-1} = \begin{pmatrix} \frac{1}{4} & 0 & 0 \\ 0 & -\frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{5} \end{pmatrix}.$$

This inverse is easy to understand: each coordinate was scaled independently, so the inverse simply rescales each coordinate back.

If a diagonal entry is zero, one coordinate is collapsed to zero. That information cannot be recovered.

### 4.5.2 Triangular matrices

Triangular matrices are also easy to test. If  $T$  is triangular, then

$$\det T = \text{product of diagonal entries.}$$

Thus  $T$  is invertible if and only if every diagonal entry is non-zero.

### Example

Let

$$T = \begin{pmatrix} 1 & 2 & 3 \\ 0 & -1 & 4 \\ 0 & 0 & 6 \end{pmatrix}.$$

Since

$$\det T = 1 \cdot (-1) \cdot 6 = -6 \neq 0,$$

the matrix is invertible.

On the other hand,

$$S = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 0 & 5 \\ 0 & 0 & 3 \end{pmatrix}$$

has determinant

$$\det S = 2 \cdot 0 \cdot 3 = 0.$$

So  $S$  is not invertible.

This is not merely a shortcut. A zero on the diagonal of a triangular matrix means that one pivot is missing. Without all pivots, the matrix cannot be reduced to the identity matrix.

### 4.5.3 Repeated or dependent rows

If two rows of a matrix are equal, the determinant is zero. If one row is a multiple of another, the determinant is also zero. More generally, if one row can be built from the others, the determinant is zero.

### Example

Consider

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 3 & 6 & 3 \\ 0 & 1 & 4 \end{pmatrix}.$$

The second row is three times the first row. Therefore the rows are dependent, and

$$\det A = 0.$$

So  $A$  is not invertible.

The interpretation is important. If one row repeats information already contained in another row, then the matrix does not contain enough independent information to be reversed.

### 4.5.4 Products of matrices

The determinant rule

$$\det(AB) = \det A \det B$$

has a direct consequence for invertibility.

### Theorem

If  $A$  and  $B$  are invertible square matrices of the same size, then  $AB$  is invertible, and

$$(AB)^{-1} = B^{-1}A^{-1}.$$

The order in this formula is essential. To undo the product  $AB$ , we must undo the last action first.

Suppose a vector  $x$  is transformed by  $B$  and then by  $A$ :

$$x \mapsto Bx \mapsto A(Bx) = ABx.$$

To recover  $x$ , first undo  $A$ , and then undo  $B$ :

$$ABx \mapsto Bx \mapsto x.$$

Thus the inverse is

$$B^{-1}A^{-1}.$$

We can verify algebraically:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I.$$

Similarly,

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}IB = B^{-1}B = I.$$

#### 4.5.5 Matrices satisfying special identities

Sometimes a matrix is invertible because it satisfies an identity that already reveals its inverse.

##### Example

Suppose a square matrix  $A$  satisfies

$$A^2 = I.$$

Then  $A$  is its own inverse, because

$$AA = I.$$

Therefore

$$A^{-1} = A.$$

Such a matrix is called an involution.

##### Example

Suppose a square matrix  $A$  satisfies

$$A^2 - 3A + 2I = 0.$$

Then

$$A^2 - 3A = -2I.$$

Factor the left side:

$$A(A - 3I) = -2I.$$

Multiplying by  $-\frac{1}{2}$ , we get

$$A \left( -\frac{1}{2}(A - 3I) \right) = I.$$

Therefore

$$A^{-1} = -\frac{1}{2}(A - 3I) = \frac{1}{2}(3I - A).$$

This argument finds the inverse without performing Gauss–Jordan elimination.

These examples show that invertibility is not only a computational topic. It is also a structural topic. Sometimes the matrix tells us how it can be undone.

## Summary

To recognize invertibility, look for structure:

- diagonal matrices are invertible when no diagonal entry is zero,
- triangular matrices are invertible when all diagonal entries are non-zero,
- repeated or dependent rows imply non-invertibility,
- products of invertible matrices are invertible,
- special identities may reveal the inverse directly.

## 4.6 Matrix equations

One of the main reasons for studying inverse matrices is that they allow us to solve matrix equations. The idea is similar to solving ordinary equations, but there is one crucial difference: matrix multiplication is not commutative. Therefore the side on which we multiply by an inverse matters.

### 4.6.1 Equations of the form $AX = B$

Suppose

$$AX = B,$$

where  $A$  is a known square matrix,  $B$  is a known matrix, and  $X$  is unknown. If  $A$  is invertible, we multiply on the left by  $A^{-1}$ :

$$A^{-1}AX = A^{-1}B.$$

Since  $A^{-1}A = I$ , this gives

$$X = A^{-1}B.$$

The inverse must be placed on the left because  $A$  is on the left of  $X$ . Writing  $XA^{-1}$  would not undo  $AX$ .

### Example

Solve

$$AX = B,$$

where

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix}, \quad B = \begin{pmatrix} 4 & 1 \\ 5 & 2 \end{pmatrix}.$$

From an earlier example,

$$A^{-1} = \begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix}.$$

Therefore

$$X = A^{-1}B = \begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix} \begin{pmatrix} 4 & 1 \\ 5 & 2 \end{pmatrix}.$$

Compute the product:

$$X = \begin{pmatrix} 28 - 10 & 7 - 4 \\ -12 + 5 & -3 + 2 \end{pmatrix} = \begin{pmatrix} 18 & 3 \\ -7 & -1 \end{pmatrix}.$$

Thus

$$X = \begin{pmatrix} 18 & 3 \\ -7 & -1 \end{pmatrix}.$$

#### 4.6.2 Equations of the form $XA = B$

Now suppose

$$XA = B.$$

Here  $A$  is on the right of  $X$ . To undo multiplication by  $A$ , we must multiply on the right by  $A^{-1}$ :

$$XAA^{-1} = BA^{-1}.$$

Since  $AA^{-1} = I$ , we get

$$X = BA^{-1}.$$

##### Example

Solve

$$XA = B,$$

where

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix}, \quad B = \begin{pmatrix} 4 & 1 \\ 5 & 2 \end{pmatrix}.$$

Again,

$$A^{-1} = \begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix}.$$

But now

$$X = BA^{-1} = \begin{pmatrix} 4 & 1 \\ 5 & 2 \end{pmatrix} \begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix}.$$

Thus

$$X = \begin{pmatrix} 28 - 3 & -8 + 1 \\ 35 - 6 & -10 + 2 \end{pmatrix} = \begin{pmatrix} 25 & -7 \\ 29 & -8 \end{pmatrix}.$$

This is not the same answer as in the previous example. The equation is different because the unknown matrix is multiplied by  $A$  on a different side.

#### 4.6.3 Equations with matrices on both sides

Some equations have the unknown matrix between two known matrices:

$$AXB = C.$$

If  $A$  and  $B$  are invertible, then we undo  $A$  on the left and  $B$  on the right:

$$A^{-1}(AXB)B^{-1} = A^{-1}CB^{-1}.$$

Therefore

$$X = A^{-1}CB^{-1}.$$

The order is not optional. Each inverse must be placed on the side where it can cancel the corresponding matrix.

#### 4.6.4 Solving vector equations

The equation

$$Ax = b$$

where  $x$  and  $b$  are column vectors is a special case of  $AX = B$ . If  $A$  is invertible, then

$$x = A^{-1}b.$$

This gives a conceptual method for solving linear systems. In the next chapter we will study systems of linear equations in detail, and we will compare different methods: elimination, inverse matrices, and Cramer's rule.

##### Example

Solve

$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}.$$

Let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 5 \\ 3 \end{pmatrix}.$$

The determinant is

$$\det A = 2 \cdot 1 - 1 \cdot 1 = 1,$$

so

$$A^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}.$$

Therefore

$$\begin{pmatrix} x \\ y \end{pmatrix} = A^{-1}b = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 5 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 - 3 \\ -5 + 6 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Thus

$$x = 2, \quad y = 1.$$

#### 4.6.5 Common mistakes

The most common mistake is to move matrices as if they were numbers. For example, from

$$AX = B$$

one may be tempted to write

$$X = BA^{-1}.$$

This is wrong in general. Multiplying by  $A^{-1}$  on the right does not cancel the  $A$  on the left of  $X$ . The correct step is

$$A^{-1}AX = A^{-1}B,$$

so

$$X = A^{-1}B.$$

##### Summary

When solving matrix equations:

- identify where the unknown matrix is placed,
- multiply by inverses on the side where cancellation is valid,

- preserve the order of multiplication,
- check that the matrices involved are invertible,
- verify the final answer by substitution when possible.

Inverse matrices are powerful because they allow equations to be solved by undoing operations. But the power comes with a responsibility: the order of multiplication must be respected at every step.

# Chapter 5

## Systems of Linear Equations

### 5.1 What systems of linear equations describe

A system of linear equations is not only a collection of equations to solve. It is a collection of conditions that must be satisfied at the same time. Each equation gives one condition. A solution is an assignment of values to the unknowns that satisfies all conditions simultaneously.

This point is important. When we solve a system, we are not solving the equations one after another as separate problems. We are looking for values that make the whole system true.

#### Example

Consider the system

$$\begin{cases} x + y = 5, \\ 2x - y = 1. \end{cases}$$

The pair  $(2, 3)$  satisfies the first equation because

$$2 + 3 = 5.$$

It also satisfies the second equation because

$$2 \cdot 2 - 3 = 1.$$

Therefore  $(2, 3)$  is a solution of the system.

The pair  $(1, 4)$  satisfies the first equation, since  $1 + 4 = 5$ , but it does not satisfy the second equation, since  $2 \cdot 1 - 4 = -2$ . Therefore  $(1, 4)$  is not a solution of the system.

The example should be read slowly. A solution of a system must pass every test in the system. Passing one equation is not enough.

#### Unknowns, coefficients, and constants

A linear equation in the unknowns  $x_1, x_2, \dots, x_n$  has the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b.$$

The numbers  $a_1, a_2, \dots, a_n$  are coefficients. The number  $b$  is the constant term or right-hand side.

A system of linear equations is made of several such equations:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2, \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m. \end{cases}$$

Here  $m$  is the number of equations and  $n$  is the number of unknowns.

### Remark

The coefficients describe how the unknowns enter the equations. The right-hand side describes what values the linear combinations must equal. A system compares these two pieces of information.

### Solution sets

The solution set of a system is the set of all solutions. Sometimes this set contains one element. Sometimes it is empty. Sometimes it contains infinitely many elements.

These are the three basic possibilities for a linear system:

- exactly one solution,
- no solution,
- infinitely many solutions.

The goal of solving a system is not only to find numbers. The goal is to determine which of these cases occurs and to describe the solution set accurately.

### Matrix form

Systems of linear equations are naturally connected with matrices. The coefficients form a matrix, the unknowns form a column vector, and the right-hand side forms another column vector.

For example, the system

$$\begin{cases} 2x + y = 5, \\ x - y = 1 \end{cases}$$

can be written as

$$\begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 1 \end{pmatrix}.$$

Using the notation

$$A = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}, \quad x = \begin{pmatrix} x \\ y \end{pmatrix}, \quad b = \begin{pmatrix} 5 \\ 1 \end{pmatrix},$$

we write the system compactly as

$$Ax = b.$$

This notation is not merely shorter. It reveals the structure of the problem. The matrix  $A$  transforms the unknown vector  $x$  into the right-hand side vector  $b$ . Solving the system means finding which input vector  $x$ , if any, is sent to  $b$ .

### The augmented matrix

When solving a system, it is often convenient to write the coefficients and the right-hand side together in an augmented matrix:

$$\left[ \begin{array}{cc|c} 2 & 1 & 5 \\ 1 & -1 & 1 \end{array} \right].$$

The vertical line reminds us that the last column is not another column of coefficients. It is the right-hand side.

The augmented matrix is a compact record of the system. Row operations on this matrix correspond to controlled transformations of the equations.

## Summary

When reading a system of linear equations, ask:

- What are the unknowns?
- What are the equations, and what condition does each one impose?
- What are the coefficients and right-hand sides?
- What would it mean for one assignment to satisfy all equations at once?
- Can the system be written as  $Ax = b$ ?
- What does the augmented matrix record?

A system of equations should first be read as a structured mathematical object. Only then should we choose a method for solving it.

## 5.2 Geometric meaning of solutions

Linear systems have an algebraic form, but they also have a geometric meaning. This geometric interpretation is often the easiest way to understand why a system may have one solution, no solution, or infinitely many solutions.

The central idea is simple: each equation describes a set of points. The solution set of the system is the intersection of all these sets.

### Two variables: lines in the plane

In two variables, a linear equation

$$ax + by = c$$

usually describes a line in the plane. A system of two linear equations describes two lines, and a solution is a point that lies on both lines.

#### Example

Consider

$$\begin{cases} x + y = 5, \\ 2x - y = 1. \end{cases}$$

The first equation describes one line. The second equation describes another line. The pair  $(2, 3)$  is a solution because the point  $(2, 3)$  lies on both lines.

This is the geometric meaning of solving the system: find the common point of the two lines.

### One solution

If two lines in the plane intersect in one point, the system has exactly one solution.

#### Example

The system

$$\begin{cases} x + y = 5, \\ 2x - y = 1 \end{cases}$$

has exactly one solution. Solving gives

$$x = 2, \quad y = 3.$$

Geometrically, the two lines meet at the point  $(2, 3)$ .

When a system has exactly one solution, the conditions are independent enough to determine the unknowns completely.

### No solution

If two lines are parallel and distinct, they never meet. Then the system has no solution.

#### Example

Consider

$$\begin{cases} x + y = 1, \\ x + y = 4. \end{cases}$$

The left-hand sides are the same, but the right-hand sides are different. The same expression  $x + y$  cannot equal both 1 and 4 at the same time.

Geometrically, these are two parallel distinct lines. Therefore the system has no solution.

A system with no solution is called inconsistent. The equations impose conditions that cannot all be satisfied simultaneously.

### Infinitely many solutions

If two equations describe the same line, then every point on that line is a solution. The system has infinitely many solutions.

#### Example

Consider

$$\begin{cases} x + y = 3, \\ 2x + 2y = 6. \end{cases}$$

The second equation is just twice the first equation. It gives the same condition, not a new independent condition.

Therefore the solution set is the whole line

$$x + y = 3.$$

There are infinitely many solutions, for example  $(0, 3)$ ,  $(1, 2)$ , and  $(3, 0)$ .

This case is not a failure of mathematics. It means that the equations do not contain enough independent information to determine one point.

### Three variables: planes in space

In three variables, a linear equation

$$ax + by + cz = d$$

usually describes a plane in space. A system of linear equations describes the common points of several planes.

Several things may happen:

- three planes may meet at one point,
- they may have no common point,
- they may share a line,
- they may all be the same plane.

The algebraic classification remains the same: one solution, no solution, or infinitely many solutions. The geometry becomes richer because planes can intersect in more ways than lines in the plane.

### Reading geometry from algebra

The algebraic form of a system often reveals the geometric situation. For example, proportional left-hand sides may indicate parallel or identical lines. A contradiction such as

$$0 = 5$$

means that the conditions cannot all be satisfied. A row of zeros such as

$$0 = 0$$

means that one equation became redundant.

These patterns will appear naturally during Gaussian elimination.

#### Summary

When interpreting a system geometrically, ask:

- What geometric object does each equation describe?
- Is the solution set an intersection of lines, planes, or other linear objects?
- Do the objects meet in one point?
- Are they parallel or inconsistent?
- Is one condition a repetition of another condition?
- Does the geometry suggest one solution, no solution, or infinitely many solutions?

The geometric viewpoint does not replace algebraic methods. It explains what those methods are detecting.

## 5.3 Gaussian elimination

Gaussian elimination is a method for simplifying a system of linear equations without changing its solution set. The method uses elementary row operations on the augmented matrix. These operations are not random changes of notation. They correspond to replacing equations by equivalent equations.

The aim is to transform a complicated system into a readable one.

## Equivalent systems

Two systems are equivalent if they have the same solution set. Gaussian elimination works because elementary row operations produce equivalent systems.

The allowed row operations are:

- swapping two rows,
- multiplying a row by a non-zero number,
- adding a multiple of one row to another row.

These are the same operations introduced earlier for matrices. Here their meaning is sharper: they change the written form of the equations, but not the set of solutions.

### Remark

The purpose of row operations is not to make the matrix look different. The purpose is to reveal the solution set by simplifying the conditions.

## A first example

Consider the system

$$\begin{cases} x + y = 5, \\ 2x - y = 1. \end{cases}$$

Its augmented matrix is

$$\left[ \begin{array}{cc|c} 1 & 1 & 5 \\ 2 & -1 & 1 \end{array} \right].$$

We use the first row to eliminate the  $x$ -coefficient in the second row:

$$R_2 \leftarrow R_2 - 2R_1.$$

This gives

$$\left[ \begin{array}{cc|c} 1 & 1 & 5 \\ 0 & -3 & -9 \end{array} \right].$$

The second row now means

$$-3y = -9,$$

so

$$y = 3.$$

Substitute into the first equation:

$$x + 3 = 5,$$

so

$$x = 2.$$

Therefore the solution is

$$(x, y) = (2, 3).$$

This example shows the basic strategy. We simplify the system until the unknowns can be read one by one.

## Pivots

A pivot is an entry used to eliminate other entries in its column. In the previous example, the first pivot was the 1 in the upper-left corner. It allowed us to remove the 2 below it.

Pivots are not only computational markers. They identify independent information in the system. Each pivot usually corresponds to a variable that can be solved from the system.

### Example

Consider

$$\left[ \begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 1 & 4 & 5 \\ 0 & 0 & 2 & 6 \end{array} \right].$$

The pivot positions are in the first, second, and third variable columns. The last row gives

$$2z = 6,$$

so  $z = 3$ . The second row then gives

$$y + 4z = 5,$$

so  $y + 12 = 5$ , hence  $y = -7$ . The first row gives

$$x + 2y - z = 3,$$

so  $x + 2(-7) - 3 = 3$ , and therefore  $x = 20$ .

This process is called back substitution. We solve from the bottom upward after the system has been simplified.

### Row echelon form

A matrix is in row echelon form when the leading non-zero entries move to the right as we move downward, and zero rows, if any, are placed at the bottom. Row echelon form is useful because it makes the structure of the system visible.

For example,

$$\left[ \begin{array}{ccc|c} 1 & 2 & 0 & 4 \\ 0 & 1 & -3 & 2 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

is in row echelon form. The last row represents the equation

$$0 = 0,$$

which gives no new restriction.

### Detecting no solution

During elimination, a row may become

$$\left[ \begin{array}{ccc|c} 0 & 0 & 0 & 5 \end{array} \right].$$

This row represents

$$0 = 5.$$

That is impossible. Therefore the system has no solution.

### Example

Consider

$$\begin{cases} x + y = 1, \\ 2x + 2y = 5. \end{cases}$$

The augmented matrix is

$$\left[ \begin{array}{cc|c} 1 & 1 & 1 \\ 2 & 2 & 5 \end{array} \right].$$

Apply

$$R_2 \leftarrow R_2 - 2R_1.$$

We get

$$\left[ \begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 0 & 3 \end{array} \right].$$

The second row says

$$0 = 3,$$

which is impossible. The system has no solution.

The contradiction is not a technical detail. It is the algebraic sign that the conditions are inconsistent.

### Free variables and infinitely many solutions

If a variable column has no pivot, that variable is free. A free variable can be chosen as a parameter. The remaining variables are expressed in terms of it.

#### Example

Consider the row echelon form

$$\left[ \begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 0 & 1 & 4 \end{array} \right].$$

This represents

$$\begin{cases} x + 2y - z = 3, \\ z = 4. \end{cases}$$

The variable  $y$  has no pivot, so let

$$y = t.$$

Then  $z = 4$ , and the first equation gives

$$x + 2t - 4 = 3,$$

so

$$x = 7 - 2t.$$

Thus the solution set is

$$(x, y, z) = (7 - 2t, t, 4), \quad t \in \mathbb{R}.$$

There are infinitely many solutions.

A parameter is not a decoration. It records the freedom that remains after all independent conditions have been used.

### What elimination tells us

Gaussian elimination does more than compute answers. It classifies the system.

## Summary

After row reduction, look for these patterns:

- A pivot in every variable column and no contradiction: exactly one solution.
- A row of the form  $0 = \text{non-zero}$ : no solution.
- At least one free variable and no contradiction: infinitely many solutions.

Gaussian elimination is the most flexible method in this chapter. It works for square and non-square systems, and it reveals not only the solution but also the structure of the solution set.

## 5.4 Cramer's rule

Cramer's rule is a method for solving certain square systems of linear equations using determinants. It is not the most efficient method for large systems, but it is conceptually important because it shows a direct connection between determinants and solvability.

The method applies to systems where the number of equations equals the number of unknowns and the coefficient matrix has non-zero determinant.

### The two by two case

Consider the system

$$\begin{cases} ax + by = e, \\ cx + dy = f. \end{cases}$$

The coefficient matrix is

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

and its determinant is

$$\det A = ad - bc.$$

If  $\det A \neq 0$ , the system has exactly one solution.

Cramer's rule says that the solution can be computed by replacing one column of the coefficient matrix at a time by the right-hand side column.

Define

$$A_x = \begin{pmatrix} e & b \\ f & d \end{pmatrix}, \quad A_y = \begin{pmatrix} a & e \\ c & f \end{pmatrix}.$$

Then

$$x = \frac{\det A_x}{\det A}, \quad y = \frac{\det A_y}{\det A}.$$

### Remark

The denominator  $\det A$  is not a technical detail. If  $\det A = 0$ , the coefficient matrix is singular, and Cramer's rule does not apply.

## A worked example

### Example

Solve the system

$$\begin{cases} 2x + y = 5, \\ x - y = 1. \end{cases}$$

The coefficient matrix is

$$A = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}.$$

Its determinant is

$$\det A = 2 \cdot (-1) - 1 \cdot 1 = -2 - 1 = -3.$$

Since  $\det A \neq 0$ , Cramer's rule applies.

For  $x$ , replace the first column by the right-hand side:

$$A_x = \begin{pmatrix} 5 & 1 \\ 1 & -1 \end{pmatrix}.$$

Then

$$\det A_x = 5 \cdot (-1) - 1 \cdot 1 = -5 - 1 = -6.$$

So

$$x = \frac{\det A_x}{\det A} = \frac{-6}{-3} = 2.$$

For  $y$ , replace the second column:

$$A_y = \begin{pmatrix} 2 & 5 \\ 1 & 1 \end{pmatrix}.$$

Then

$$\det A_y = 2 \cdot 1 - 5 \cdot 1 = 2 - 5 = -3.$$

So

$$y = \frac{\det A_y}{\det A} = \frac{-3}{-3} = 1.$$

Thus

$$(x, y) = (2, 1).$$

This result should be checked by substitution. We have

$$2 \cdot 2 + 1 = 5, \quad 2 - 1 = 1.$$

Both equations are satisfied.

## The general square case

For a square system

$$Ax = b,$$

where  $A$  is an  $n \times n$  matrix and  $b$  is a column vector, Cramer's rule applies if

$$\det A \neq 0.$$

For each unknown  $x_i$ , form the matrix  $A_i$  by replacing the  $i$ -th column of  $A$  with  $b$ . Then

$$x_i = \frac{\det A_i}{\det A}.$$

The rule is elegant, but the number of determinant computations grows quickly. For this reason, Gaussian elimination is usually more practical for larger systems.

### Why the determinant condition matters

The condition

$$\det A \neq 0$$

means that the coefficient matrix is invertible. Geometrically and structurally, the equations contain enough independent information to determine one solution.

If

$$\det A = 0,$$

then the matrix is singular. This does not automatically mean that the system has no solution. It means that the system cannot have a unique solution determined by Cramer's rule. The system must be analyzed further: it may have no solution or infinitely many solutions.

#### Example

Consider

$$\begin{cases} x + y = 2, \\ 2x + 2y = 4. \end{cases}$$

The coefficient matrix is

$$A = \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix},$$

and

$$\det A = 1 \cdot 2 - 1 \cdot 2 = 0.$$

Cramer's rule does not apply. But the system is not inconsistent. The second equation is twice the first, so there are infinitely many solutions.

This example is important. A zero determinant means loss of uniqueness, not automatically loss of existence.

#### Summary

When using Cramer's rule, ask:

- Is the system square?
- What is the coefficient matrix  $A$ ?
- Is  $\det A \neq 0$ ?
- Which column must be replaced to compute each unknown?
- Does the final answer satisfy the original equations?
- If  $\det A = 0$ , what further classification is needed?

Cramer's rule shows how determinants can solve systems, but it must be used only when its structural condition is satisfied.

## 5.5 The inverse matrix method

Some systems of linear equations can be solved by using the inverse of the coefficient matrix. This method is based on the matrix equation

$$Ax = b.$$

Here  $A$  is the coefficient matrix,  $x$  is the column vector of unknowns, and  $b$  is the right-hand side vector.

The method is simple in form, but its meaning should be understood carefully. If  $A$  is invertible, then multiplying by  $A^{-1}$  undoes the action of  $A$ .

### Solving $Ax = b$

Suppose

$$Ax = b$$

and  $A$  is invertible. Multiply both sides on the left by  $A^{-1}$ :

$$A^{-1}Ax = A^{-1}b.$$

Since

$$A^{-1}A = I,$$

we get

$$Ix = A^{-1}b.$$

Therefore

$$x = A^{-1}b.$$

### Theorem

If  $A$  is an invertible square matrix, then the system

$$Ax = b$$

has the unique solution

$$x = A^{-1}b.$$

The word “invertible” is essential. If  $A^{-1}$  does not exist, this method cannot be used.

### A worked example

#### Example

Solve

$$\begin{cases} 2x + y = 5, \\ x + y = 3. \end{cases}$$

Write the system as

$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}.$$

Thus

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 5 \\ 3 \end{pmatrix}.$$

The determinant is

$$\det A = 2 \cdot 1 - 1 \cdot 1 = 1,$$

so  $A$  is invertible. For a  $2 \times 2$  matrix,

$$A^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}.$$

Therefore

$$\begin{pmatrix} x \\ y \end{pmatrix} = A^{-1}b = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 5 \\ 3 \end{pmatrix}.$$

Compute the product:

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 - 3 \\ -5 + 6 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Thus

$$x = 2, \quad y = 1.$$

The inverse matrix method gives the same result as other correct methods. The difference is the language: here solving means applying the inverse transformation.

### Why the method works

The equation

$$Ax = b$$

can be read as follows: the matrix  $A$  acts on the unknown vector  $x$  and produces the vector  $b$ . If  $A$  is invertible, then this action can be reversed. Applying  $A^{-1}$  to  $b$  recovers the original vector  $x$ .

This interpretation is important. The formula

$$x = A^{-1}b$$

is not magic. It is the matrix version of undoing an operation.

### Connection with determinants

For a square matrix  $A$ , invertibility is equivalent to

$$\det A \neq 0.$$

Therefore the inverse matrix method applies exactly when the coefficient matrix has non-zero determinant.

If

$$\det A = 0,$$

then  $A^{-1}$  does not exist. The system may have no solution or infinitely many solutions, but it cannot be solved by multiplying by  $A^{-1}$ .

### Example

Consider

$$\begin{cases} x + y = 2, \\ 2x + 2y = 4. \end{cases}$$

The coefficient matrix is

$$A = \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix},$$

and

$$\det A = 0.$$

The inverse matrix method cannot be used. However, the system does have solutions: every pair  $(x, y)$  satisfying  $x + y = 2$  is a solution. There are infinitely many such pairs.

Again, the failure of invertibility means loss of uniqueness, not automatically inconsistency.

### When this method is useful

The inverse matrix method is conceptually clear, but it is not always the most practical computational method. Finding  $A^{-1}$  can be more work than solving the system directly by elimination.

The method is especially useful when:

- the inverse matrix is already known,
- several systems have the same coefficient matrix  $A$  but different right-hand sides,
- we want to understand the system as a matrix equation.

For routine computation, Gaussian elimination is usually more flexible.

### Summary

When using the inverse matrix method, ask:

- Can the system be written as  $Ax = b$ ?
- Is  $A$  square?
- Is  $A$  invertible, equivalently  $\det A \neq 0$ ?
- Have I multiplied on the correct side by  $A^{-1}$ ?
- Does the vector  $A^{-1}b$  satisfy the original system?

The inverse matrix method is powerful because it connects solving systems with the idea of reversing a linear transformation.

## 5.6 Parametric systems and classification

Some systems contain a parameter. Then the question is not only “What is the solution?” but also “For which parameter values does the structure of the system change?”

This is where determinants, row reduction, and geometric interpretation come together.

### The three possible outcomes

Every linear system has one of the following outcomes:

- exactly one solution,
- no solution,
- infinitely many solutions.

For a square system

$$Ax = b,$$

one important test is the determinant of  $A$ . If

$$\det A \neq 0,$$

then  $A$  is invertible and the system has exactly one solution. If

$$\det A = 0,$$

then the system does not have a unique solution obtained by inversion. But we must still decide whether it has no solution or infinitely many solutions.

#### Remark

The condition  $\det A = 0$  is a warning sign, not a complete answer. It says that uniqueness is lost. It does not by itself decide whether solutions exist.

### A simple parameter example

Consider the system

$$\begin{cases} x + y = 1, \\ 2x + 2y = a. \end{cases}$$

The left-hand side of the second equation is twice the left-hand side of the first equation. Therefore consistency depends on the right-hand side.

If we multiply the first equation by 2, we get

$$2x + 2y = 2.$$

So for the second equation to describe the same condition, we need

$$a = 2.$$

If  $a = 2$ , the two equations describe the same line, and there are infinitely many solutions:

$$x + y = 1.$$

If  $a \neq 2$ , the equations describe parallel distinct lines, and there is no solution.

#### Summary

For

$$\begin{cases} x + y = 1, \\ 2x + 2y = a, \end{cases}$$

we have:

- if  $a = 2$ , infinitely many solutions,
- if  $a \neq 2$ , no solution.

This example shows that a parameter may affect only the right-hand side, but that can still change the solution set completely.

## Using determinants to find exceptional values

Now consider a parameter in the coefficient matrix. Let

$$A(t) = \begin{pmatrix} t & 1 \\ 4 & t \end{pmatrix}.$$

The determinant is

$$\det A(t) = t^2 - 4.$$

The exceptional values occur when

$$t^2 - 4 = 0,$$

so

$$t = 2 \quad \text{or} \quad t = -2.$$

For all other values of  $t$ , the matrix is invertible. Therefore any system

$$A(t)x = b$$

has exactly one solution for  $t \neq 2$  and  $t \neq -2$ .

The exceptional values  $t = 2$  and  $t = -2$  require separate analysis. At those values, the determinant test tells us that the system is not uniquely solvable, but not whether it is inconsistent or has infinitely many solutions.

## A full classification example

### Example

Classify the system depending on the parameter  $a$ :

$$\begin{cases} x + y = 3, \\ 2x + ay = 6. \end{cases}$$

The coefficient matrix is

$$A(a) = \begin{pmatrix} 1 & 1 \\ 2 & a \end{pmatrix}.$$

Its determinant is

$$\det A(a) = a - 2.$$

If

$$a \neq 2,$$

then  $\det A(a) \neq 0$ , so the system has exactly one solution.

Now consider the exceptional value  $a = 2$ . The system becomes

$$\begin{cases} x + y = 3, \\ 2x + 2y = 6. \end{cases}$$

The second equation is twice the first. Therefore the system has infinitely many solutions. Thus:

- if  $a \neq 2$ , the system has exactly one solution,
- if  $a = 2$ , the system has infinitely many solutions.

The determinant found the value where the structure changes. Row interpretation then classified what happens at that value.

### Example

Classify the system depending on  $a$ :

$$\begin{cases} x + y = 3, \\ 2x + 2y = a. \end{cases}$$

The coefficient matrix is

$$A = \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix},$$

so

$$\det A = 0.$$

There is no value of  $a$  that makes this coefficient matrix invertible. We must compare the equations.

Twice the first equation gives

$$2x + 2y = 6.$$

Therefore:

- if  $a = 6$ , the two equations describe the same line, so there are infinitely many solutions,
- if  $a \neq 6$ , the two equations are inconsistent, so there is no solution.

This example shows that a singular coefficient matrix may still lead to different outcomes depending on the right-hand side.

### Parameters and row reduction

For larger systems, row reduction is often the safest method. It allows us to see contradictions, zero rows, and free variables directly.

A row such as

$$0 = 0$$

indicates redundancy. A row such as

$$0 = 5$$

indicates inconsistency. A missing pivot indicates a free variable. These patterns remain valid even when parameters are present.

The difficulty is that some row operations may require dividing by an expression containing a parameter. Before dividing by such an expression, we must consider when it might be zero.

### Remark

When a parameter appears, do not divide by an expression such as  $a - 2$  without asking whether  $a - 2 = 0$  is possible. Parameter values that make a denominator zero are usually exceptional cases and must be treated separately.

### A classification strategy

A good solution to a parameter system should not only list parameter values. It should explain why the classification changes.

### Summary

For a square parameter system  $A(t)x = b(t)$ , a useful strategy is:

1. Compute  $\det A(t)$ .
2. Solve  $\det A(t) = 0$  to find exceptional parameter values.
3. For non-exceptional values, conclude that there is exactly one solution.
4. For exceptional values, analyze the system separately using row reduction or equation comparison.
5. Classify each case as exactly one solution, no solution, or infinitely many solutions.

The determinant identifies where uniqueness can fail. Elimination and interpretation decide what happens there.

### What the chapter has built

Systems of linear equations bring together the earlier algebraic tools.

Matrices organize the coefficients. Row operations simplify the system. Determinants detect whether a square coefficient matrix is invertible. Inverse matrices give the formula  $x = A^{-1}b$  when the inverse exists. Cramer's rule gives a determinant-based expression for the solution when  $\det A \neq 0$ .

These methods are not isolated tricks. They are different ways of reading the same object: a system of linear conditions.

### Summary

At the end of this chapter, remember:

- A system is a set of conditions to be satisfied simultaneously.
- The augmented matrix records the coefficients and the right-hand side.
- Gaussian elimination is the most flexible method for solving and classifying systems.
- If  $\det A \neq 0$ , a square system  $Ax = b$  has exactly one solution.
- If  $\det A = 0$ , uniqueness is lost, but existence must still be checked.
- A linear system may have exactly one solution, no solution, or infinitely many solutions.

The goal is not only to solve systems. The goal is to understand what the equations say about structure, dependence, and compatibility.

# Chapter 6

## Vectors and Coordinates

### 6.1 Points and coordinates

Analytic geometry begins with a simple but powerful idea: geometric objects can be described by numbers. A point on a line can be described by one number. A point in a plane can be described by two numbers. A point in space can be described by three numbers. This numerical description allows us to use algebra to answer geometric questions.

The important word here is description. A point is not the same thing as the symbols used to describe it. The symbols depend on a coordinate system. Once a coordinate system is chosen, the symbols become a precise way to say where the point is.

#### Definition

In the plane, a point is usually written as an ordered pair

$$P = (x, y).$$

The first coordinate  $x$  describes horizontal position, and the second coordinate  $y$  describes vertical position. In three-dimensional space, a point is usually written as an ordered triple

$$P = (x, y, z).$$

The order of the coordinates matters. The points  $(2, 5)$  and  $(5, 2)$  are generally different points. The numbers are the same, but their roles are not the same. The first number answers one question, and the second number answers another question.

#### Example

The point

$$P = (3, -2)$$

means the following: starting from the origin, move 3 units in the positive horizontal direction and 2 units in the negative vertical direction.

The point

$$Q = (-2, 3)$$

uses the same two numbers, but in different positions. It describes a different point.

This is a small example of a general habit that will appear throughout analytic geometry. Before doing a calculation, we should read the object. A pair of numbers is not just a pair of numbers. It is an ordered description of position.

#### The origin and the coordinate axes

The origin is the point from which positions are measured. In the plane it is

$$O = (0, 0),$$

and in space it is

$$O = (0, 0, 0).$$

The coordinate axes give directions of measurement. In the plane, the  $x$ -axis records horizontal displacement from the origin, while the  $y$ -axis records vertical displacement. In space, the  $z$ -axis adds a third independent direction.

A coordinate system therefore does two things at once. It gives a point of reference, and it gives directions in which numbers are measured. Without this structure, a coordinate tuple has no geometric meaning.

#### Remark

Coordinates are not only labels. They are labels chosen relative to a system of measurement. Changing the coordinate system may change the coordinates of a point, even though the point itself has not moved.

### Coordinates as variables

Sometimes coordinates are fixed numbers. For example,

$$A = (1, 4), \quad B = (-3, 2)$$

are particular points. But often coordinates are variables. When we write

$$P = (x, y),$$

we may be describing an arbitrary point in the plane. The letters  $x$  and  $y$  are not yet known numbers. They are placeholders for possible positions.

This distinction is essential. A fixed point is one object. A point with variable coordinates can represent many possible points. Equations will later use this idea. For instance, an equation in  $x$  and  $y$  may describe all points whose coordinates satisfy a certain condition.

#### Example

The expression

$$P = (x, 2)$$

does not describe one fixed point unless  $x$  is known. It describes all points whose second coordinate is 2. Geometrically, these points lie on a horizontal line.

Similarly,

$$P = (1, y)$$

describes all points whose first coordinate is 1. These points lie on a vertical line.

This example already hints at the main idea of analytic geometry: an algebraic condition can describe a geometric object. The condition  $y = 2$  describes a horizontal line. The condition  $x = 1$  describes a vertical line. Later we will use more complicated conditions to describe more complicated objects.

### Reading before calculating

The first task in coordinate geometry is not to apply a formula. It is to understand what kind of object is being described.

#### Summary

When reading coordinates, ask:

- Am I looking at a fixed point or a point with variable coordinates?

- Which coordinate describes which direction?
- Is the order of the coordinates important in this problem?
- Is an equation describing one point or a whole set of points?
- What geometric information is encoded in the algebraic notation?

The aim is to develop a controlled language. Coordinates allow us to speak about geometry with numbers, but the numbers must be read with care.

## 6.2 Vectors as displacements

A point describes a position. A vector describes a displacement. This difference is one of the most important distinctions in analytic geometry.

If we say that a point is  $(3, 2)$ , we are saying where something is. If we say that a vector is  $(3, 2)$ , we are saying how something moves: three units in one coordinate direction and two units in another coordinate direction. The same pair of numbers may appear in both situations, but the meaning is different.

### Definition

Let

$$A = (a_1, a_2), \quad B = (b_1, b_2)$$

be points in the plane. The vector from  $A$  to  $B$  is

$$\overrightarrow{AB} = B - A = (b_1 - a_1, b_2 - a_2).$$

In space, if

$$A = (a_1, a_2, a_3), \quad B = (b_1, b_2, b_3),$$

then

$$\overrightarrow{AB} = (b_1 - a_1, b_2 - a_2, b_3 - a_3).$$

The formula is simple, but it should be read carefully. We subtract the starting coordinates from the ending coordinates. A displacement is not only about where we end. It is about how we get from one position to another.

### Example

Let

$$A = (1, 4), \quad B = (6, 2).$$

Then

$$\overrightarrow{AB} = B - A = (6 - 1, 2 - 4) = (5, -2).$$

This means that to move from  $A$  to  $B$ , we move 5 units in the positive  $x$ -direction and 2 units in the negative  $y$ -direction.

The vector  $(5, -2)$  does not remember the absolute position of the starting point. It remembers the displacement. The same displacement could begin somewhere else.

### Example

Let

$$C = (-3, 1), \quad D = (2, -1).$$

Then

$$\overrightarrow{CD} = D - C = (2 - (-3), -1 - 1) = (5, -2).$$

Thus

$$\overrightarrow{AB} = \overrightarrow{CD},$$

even though the points  $A, B, C, D$  are not the same. The arrows begin in different places, but they describe the same displacement.

This is why vectors are often called free objects in elementary geometry. A vector can be represented by an arrow, but the arrow may be moved parallel to itself without changing the vector. What matters is direction, length, and orientation, not the particular location of the drawing.

### Position vectors

There is one special way in which points and vectors are connected. If a vector begins at the origin and ends at a point  $P$ , then the coordinates of the vector and the coordinates of the point are the same.

If

$$P = (p_1, p_2),$$

then

$$\overrightarrow{OP} = (p_1, p_2),$$

where  $O = (0, 0)$ .

### Remark

The equality of coordinates does not mean that points and vectors are the same kind of object. It means that a point can be represented by the vector from the origin to that point. This representation is useful, but the distinction should not be forgotten.

This distinction will help us later. A line can be described by starting from one point and moving in a vector direction. A plane can be described by a point and two independent directions, or by a point and one normal vector. In every case, we must know whether an expression describes a position, a displacement, or a direction.

### Opposite displacements

The vector from  $B$  to  $A$  is the opposite of the vector from  $A$  to  $B$ :

$$\overrightarrow{BA} = A - B = -(B - A) = -\overrightarrow{AB}.$$

This is not a new formula to memorize. It simply says that going back reverses the displacement.

### Example

For the points

$$A = (1, 4), \quad B = (6, 2),$$

we found

$$\overrightarrow{AB} = (5, -2).$$

In the opposite direction,

$$\overrightarrow{BA} = A - B = (1 - 6, 4 - 2) = (-5, 2).$$

Thus

$$\overrightarrow{BA} = -\overrightarrow{AB}.$$

The sign of a vector therefore carries geometric information. It tells us orientation. The vectors  $(5, -2)$  and  $(-5, 2)$  are parallel and lie along the same supporting line, but they point in opposite directions.

### Summary

When working with vectors as displacements, ask:

- Which point is the starting point?
- Which point is the ending point?
- Am I describing a position or a displacement?
- Would the same displacement be obtained if the arrow were moved elsewhere?
- What changes when the direction of the arrow is reversed?

A vector is not merely a pair or triple of numbers. It is a controlled way of recording movement from one position to another.

## 6.3 Vector operations

Vectors can be added, subtracted, and multiplied by numbers. These operations look algebraic, but each of them has a geometric meaning. If we treat them only as coordinate formulas, we lose much of their value.

The guiding idea is this: vectors describe displacements, and vector operations describe how displacements are combined or changed.

### Adding vectors

Suppose a first displacement moves an object by  $u$ , and a second displacement moves it by  $v$ . The total displacement is  $u + v$ . In coordinates, addition is performed component by component.

#### Definition

For vectors

$$u = (u_1, u_2), \quad v = (v_1, v_2),$$

their sum is

$$u + v = (u_1 + v_1, u_2 + v_2).$$

In space, for

$$u = (u_1, u_2, u_3), \quad v = (v_1, v_2, v_3),$$

we define

$$u + v = (u_1 + v_1, u_2 + v_2, u_3 + v_3).$$

The formula says that horizontal changes add to horizontal changes, vertical changes add to

vertical changes, and third-coordinate changes add to third-coordinate changes. We do not mix the roles of the coordinates.

### Example

Let

$$u = (3, -1), \quad v = (-2, 5).$$

Then

$$u + v = (3 + (-2), -1 + 5) = (1, 4).$$

Geometrically, moving by  $u$  and then by  $v$  produces the same final displacement as moving once by  $(1, 4)$ .

This interpretation explains why vector addition is commutative:

$$u + v = v + u.$$

If the same two displacements are applied, the final total displacement is the same, even though the path used to draw the two arrows may look different.

## Subtracting vectors

Subtraction measures the difference between displacements. If  $u$  and  $v$  are two vectors, then

$$u - v = u + (-v).$$

Here  $-v$  is the vector with the same length and opposite orientation.

### Definition

For vectors

$$u = (u_1, u_2), \quad v = (v_1, v_2),$$

their difference is

$$u - v = (u_1 - v_1, u_2 - v_2).$$

### Example

Let

$$u = (4, 2), \quad v = (1, 5).$$

Then

$$u - v = (4 - 1, 2 - 5) = (3, -3).$$

This vector records how the displacement  $v$  must be changed to become the displacement  $u$ .

Subtraction should be read with care. The vectors  $u - v$  and  $v - u$  are opposites:

$$v - u = -(u - v).$$

Changing the order changes the orientation.

## Multiplying a vector by a number

Multiplying a vector by a number changes its scale. If the number is positive, the orientation is preserved. If the number is negative, the orientation is reversed. If the number is zero, the vector collapses to the zero vector.

### Definition

If

$$u = (u_1, u_2),$$

and  $\lambda$  is a real number, then

$$\lambda u = (\lambda u_1, \lambda u_2).$$

In space,

$$\lambda(u_1, u_2, u_3) = (\lambda u_1, \lambda u_2, \lambda u_3).$$

### Example

Let

$$u = (2, -3).$$

Then

$$3u = (6, -9), \quad -2u = (-4, 6), \quad 0u = (0, 0).$$

The vector  $3u$  points in the same direction as  $u$ , but is three times as long. The vector  $-2u$  lies on the same direction line, but points in the opposite direction.

## Parallel vectors

Scalar multiplication gives a simple way to recognize parallel vectors. Two non-zero vectors are parallel when one is a scalar multiple of the other.

### Definition

Non-zero vectors  $u$  and  $v$  are parallel if there exists a real number  $\lambda \neq 0$  such that

$$v = \lambda u.$$

### Example

The vectors

$$u = (2, 3), \quad v = (6, 9)$$

are parallel because

$$v = 3u.$$

The vector

$$w = (-4, -6)$$

is also parallel to  $u$ , because

$$w = -2u.$$

The negative scalar means that  $w$  points in the opposite orientation along the same direction line.

This idea will be central in the next chapter. A line can be described by a point and a direction vector. All vectors parallel to that direction vector describe the same direction line, although they may have different lengths or orientations.

## The zero vector

The zero vector is the displacement that changes nothing:

$$0 = (0, 0)$$

in the plane, and

$$0 = (0, 0, 0)$$

in space.

It is useful, but it must be handled carefully. The zero vector has no direction. For this reason, when we discuss directions or parallelism, we usually require direction vectors to be non-zero.

#### Remark

A zero displacement is meaningful: it says that the starting point and ending point coincide. But it should not be used as a direction vector for a line or a plane, because it does not point anywhere.

#### Summary

When using vector operations, ask:

- Am I combining displacements or comparing them?
- Does scalar multiplication preserve or reverse orientation?
- Is a vector being used as a direction?
- If so, is it non-zero?
- Are two vectors parallel because one is a scalar multiple of the other?

The purpose of vector operations is not only to produce new coordinates. The purpose is to control displacements, directions, and geometric relationships.

## 6.4 Length, distance, and midpoints

Coordinates allow us to measure geometric quantities. Once a displacement is written as a vector, its length can be computed. Once two points are written in coordinates, the distance between them can be computed. The formulas are familiar, but they should not be treated as isolated rules. They come from the Pythagorean theorem applied to coordinate differences.

### Length of a vector

A vector tells us how far and in which direction to move. Its length measures only the size of the displacement, not its orientation.

#### Definition

For a vector

$$v = (v_1, v_2)$$

in the plane, its length is

$$\|v\| = \sqrt{v_1^2 + v_2^2}.$$

For a vector

$$v = (v_1, v_2, v_3)$$

in space, its length is

$$\|v\| = \sqrt{v_1^2 + v_2^2 + v_3^2}.$$

The squares appear because coordinate directions are perpendicular. A displacement in the horizontal direction and a displacement in the vertical direction form the legs of a right triangle. The vector is the diagonal.

### Example

Let

$$v = (3, 4).$$

Then

$$\|v\| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5.$$

The vector has length 5. The coordinates  $(3, 4)$  describe how the displacement is decomposed into coordinate directions; the length 5 describes the total size of the displacement.

A vector and its opposite have the same length:

$$\|-v\| = \|v\|.$$

This is natural. Reversing a displacement changes its orientation, but not how long it is.

### Distance between two points

The distance between two points is the length of the vector joining them. If

$$A = (a_1, a_2), \quad B = (b_1, b_2),$$

then

$$\overrightarrow{AB} = (b_1 - a_1, b_2 - a_2).$$

Therefore the distance between  $A$  and  $B$  is

$$d(A, B) = \|\overrightarrow{AB}\|.$$

### Definition

For points

$$A = (a_1, a_2), \quad B = (b_1, b_2)$$

in the plane,

$$d(A, B) = \sqrt{(b_1 - a_1)^2 + (b_2 - a_2)^2}.$$

For points in space,

$$A = (a_1, a_2, a_3), \quad B = (b_1, b_2, b_3),$$

we have

$$d(A, B) = \sqrt{(b_1 - a_1)^2 + (b_2 - a_2)^2 + (b_3 - a_3)^2}.$$

The formula is not new information. It is the length formula applied to the displacement from  $A$  to  $B$ .

### Example

Let

$$A = (1, -2), \quad B = (5, 1).$$

Then

$$\overrightarrow{AB} = (5 - 1, 1 - (-2)) = (4, 3).$$

Hence

$$d(A, B) = \|\vec{AB}\| = \sqrt{4^2 + 3^2} = 5.$$

This calculation should be read in two layers. Algebraically, we subtract coordinates and take a square root. Geometrically, we first form the displacement from one point to the other, and then measure its length.

## Midpoints

A midpoint is the point halfway between two given points. In coordinates, it is obtained by averaging corresponding coordinates.

### Definition

For points

$$A = (a_1, a_2), \quad B = (b_1, b_2),$$

the midpoint of the segment  $AB$  is

$$M = \left( \frac{a_1 + b_1}{2}, \frac{a_2 + b_2}{2} \right).$$

In space, for

$$A = (a_1, a_2, a_3), \quad B = (b_1, b_2, b_3),$$

the midpoint is

$$M = \left( \frac{a_1 + b_1}{2}, \frac{a_2 + b_2}{2}, \frac{a_3 + b_3}{2} \right).$$

The formula is easy to remember, but again we should understand it. Each coordinate is placed halfway between the corresponding coordinates of  $A$  and  $B$ .

### Example

Let

$$A = (-1, 4), \quad B = (5, 2).$$

The midpoint is

$$M = \left( \frac{-1 + 5}{2}, \frac{4 + 2}{2} \right) = (2, 3).$$

The first coordinate 2 is halfway between  $-1$  and  $5$ . The second coordinate 3 is halfway between 4 and 2.

There is also a vector way to read the midpoint. Starting from  $A$ , the full displacement to  $B$  is  $B - A$ . Half of that displacement is  $\frac{1}{2}(B - A)$ . Therefore

$$M = A + \frac{1}{2}(B - A).$$

After simplifying coordinates, this gives the same midpoint formula. This form is often more meaningful: the midpoint is reached by starting at  $A$  and moving halfway toward  $B$ .

## What is being measured?

Length, distance, and midpoint calculations are common, but they answer different questions.

A length belongs to a vector. It asks how large a displacement is. A distance belongs to a pair of points. It asks how far apart two positions are. A midpoint is itself a point. It asks

where the halfway position lies.

### Summary

When measuring with coordinates, ask:

- Am I measuring a vector or comparing two points?
- Have I first formed the correct displacement?
- Does the result represent a number, such as a distance, or a point, such as a mid-point?
- Can the calculation be interpreted geometrically after it is done?

Analytic geometry becomes clearer when every formula is connected to the object it measures.

## 6.5 Dot product and angles

Some geometric questions are not only about length. We often want to know whether two directions are similar, opposite, or perpendicular. We may want to measure the angle between two vectors. For this purpose, one of the most important operations is the dot product.

The dot product takes two vectors and produces a number. This number is not a new vector. It is a measurement of how the two vectors interact directionally.

### Definition

For vectors

$$u = (u_1, u_2), \quad v = (v_1, v_2),$$

the dot product is

$$u \cdot v = u_1 v_1 + u_2 v_2.$$

In space, for

$$u = (u_1, u_2, u_3), \quad v = (v_1, v_2, v_3),$$

we define

$$u \cdot v = u_1 v_1 + u_2 v_2 + u_3 v_3.$$

The formula multiplies corresponding coordinates and then adds the results. But the meaning is not merely arithmetic. The dot product measures how much two vectors point in compatible directions.

### Example

Let

$$u = (2, 3), \quad v = (4, -1).$$

Then

$$u \cdot v = 2 \cdot 4 + 3 \cdot (-1) = 8 - 3 = 5.$$

The result is a number. It is not a vector, and it should not be interpreted as a point or a displacement.

## The dot product and length

The dot product of a vector with itself gives the square of its length:

$$v \cdot v = \|v\|^2.$$

Indeed, if  $v = (v_1, v_2)$ , then

$$v \cdot v = v_1^2 + v_2^2,$$

while

$$\|v\| = \sqrt{v_1^2 + v_2^2}.$$

Therefore

$$\|v\| = \sqrt{v \cdot v}.$$

This connection is important. It shows that the dot product contains metric information: it knows something about lengths.

### Example

For

$$v = (-2, 5),$$

we have

$$v \cdot v = (-2)^2 + 5^2 = 4 + 25 = 29.$$

Therefore

$$\|v\| = \sqrt{29}.$$

## The dot product and angles

The deeper geometric meaning of the dot product is expressed by the formula

$$u \cdot v = \|u\| \|v\| \cos \theta,$$

where  $\theta$  is the angle between the non-zero vectors  $u$  and  $v$ .

This formula should be read as an interpretation. The dot product is large and positive when the vectors point in similar directions. It is negative when they point more against each other. It is zero when the vectors are perpendicular.

### Theorem

For non-zero vectors  $u$  and  $v$ ,

$$\cos \theta = \frac{u \cdot v}{\|u\| \|v\|},$$

where  $\theta$  is the angle between the vectors.

The denominator  $\|u\| \|v\|$  removes the effect of the lengths. What remains is information about direction.

### Example

Let

$$u = (1, 0), \quad v = (0, 1).$$

Then

$$u \cdot v = 1 \cdot 0 + 0 \cdot 1 = 0.$$

Both vectors have length 1. Hence

$$\cos \theta = 0,$$

so

$$\theta = 90^\circ.$$

The vectors are perpendicular.

## Perpendicular vectors

The most important special case is orthogonality. Two non-zero vectors are perpendicular exactly when their dot product is zero.

### Theorem

Non-zero vectors  $u$  and  $v$  are perpendicular if and only if

$$u \cdot v = 0.$$

This criterion is one of the main reasons the dot product is so useful in analytic geometry. It turns a geometric condition, perpendicularity, into an algebraic equation.

### Example

Let

$$u = (3, 2), \quad v = (4, -6).$$

Then

$$u \cdot v = 3 \cdot 4 + 2 \cdot (-6) = 12 - 12 = 0.$$

Therefore the two vectors are perpendicular.

This is not only a convenient test. It is a model for much of analytic geometry: translate a geometric relation into an algebraic condition, solve or simplify the condition, and then interpret the result geometrically.

## The sign of the dot product

The sign of the dot product gives qualitative information about the angle.

- If  $u \cdot v > 0$ , then the angle between  $u$  and  $v$  is acute.
- If  $u \cdot v = 0$ , then the angle is right.
- If  $u \cdot v < 0$ , then the angle is obtuse.

This follows from the sign of  $\cos \theta$ . Positive cosine corresponds to an acute angle, zero cosine to a right angle, and negative cosine to an obtuse angle.

### Remark

The dot product does not only help us compute angles. It helps us read the relationship between directions. Sometimes the sign of the dot product is already enough to understand the geometry of a problem.

## Why this matters for lines and planes

The dot product will appear repeatedly in the next chapters. A line may be described by a direction vector. A plane may be described by a normal vector. Perpendicularity between a vector and a direction, or between a vector and a plane, can be expressed using the dot product.

For example, if  $n$  is a normal vector to a plane, then every vector lying inside that plane is perpendicular to  $n$ . Algebraically, this becomes an equation of the form

$$n \cdot v = 0.$$

This is the beginning of the standard equation of a plane.

### Summary

When using the dot product, ask:

- Am I trying to measure length, angle, or perpendicularity?
- Is the result supposed to be a number rather than a vector?
- Does a zero dot product indicate a right angle?
- Does the sign of the dot product already tell me something about the angle?
- Can a geometric condition be written as a dot product equation?

The dot product is a bridge between algebra and geometry. It allows angles and perpendicularity to be handled with equations.

## 6.6 Cross product and normal vectors

The dot product produces a number and measures directional agreement. In three-dimensional space there is another important operation: the cross product. It takes two vectors and produces a new vector perpendicular to both of them.

This operation is specific to three-dimensional vector geometry in the form used here. It will provide the missing link between vectors and equations of planes.

### Definition in coordinates

Let

$$u = (u_1, u_2, u_3), \quad v = (v_1, v_2, v_3).$$

Their cross product is defined by

$$u \times v = (u_2v_3 - u_3v_2, u_3v_1 - u_1v_3, u_1v_2 - u_2v_1).$$

### Definition

For vectors  $u, v \in \mathbb{R}^3$ , the cross product  $u \times v$  is the vector

$$u \times v = \begin{pmatrix} u_2v_3 - u_3v_2 \\ u_3v_1 - u_1v_3 \\ u_1v_2 - u_2v_1 \end{pmatrix}.$$

It is perpendicular to both  $u$  and  $v$ .

A common mnemonic writes this computation as

$$u \times v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix},$$

with the usual alternating signs in the expansion. The coordinate formula above is the definition; the determinant notation is only a compact memory aid.

### Example

Let

$$u = (1, 2, 3), \quad v = (2, -1, 1).$$

Then

$$u \times v = (2 \cdot 1 - 3(-1), 3 \cdot 2 - 1 \cdot 1, 1(-1) - 2 \cdot 2),$$

so

$$u \times v = (5, 5, -5).$$

We can check perpendicularity:

$$(5, 5, -5) \cdot (1, 2, 3) = 5 + 10 - 15 = 0,$$

and

$$(5, 5, -5) \cdot (2, -1, 1) = 10 - 5 - 5 = 0.$$

Thus the resulting vector is perpendicular to both original vectors.

## Order and orientation

The order of the vectors matters:

$$v \times u = -(u \times v).$$

Reversing the order reverses the direction of the resulting normal vector, but it does not change the plane perpendicular to that normal direction.

### Remark

The vectors  $u \times v$  and  $v \times u$  have opposite orientations. In plane equations, either one may be used as a normal vector because multiplying a normal vector by a non-zero scalar does not change the plane.

## Parallel vectors and the zero cross product

If  $u$  and  $v$  are parallel, they do not determine two independent directions. In that case,

$$u \times v = 0.$$

Conversely, for non-zero vectors in  $\mathbb{R}^3$ ,

$$u \times v = 0$$

implies that the vectors are parallel.

### Theorem

For  $u, v \in \mathbb{R}^3$ ,

$$u \times v = 0$$

if and only if  $u$  and  $v$  are linearly dependent.

This criterion will be useful when deciding whether three points are collinear and whether two direction vectors really span a plane.

### Geometric size

If  $\theta$  is the angle between non-zero vectors  $u$  and  $v$ , then

$$\|u \times v\| = \|u\|\|v\|\sin\theta.$$

Therefore  $\|u \times v\|$  is the area of the parallelogram spanned by  $u$  and  $v$ .

This formula also explains why the cross product is zero for parallel vectors: their angle is 0 or  $\pi$ , so  $\sin\theta = 0$ .

### Finding a normal vector to a plane

Suppose a plane contains two non-parallel direction vectors  $u$  and  $v$ . A normal vector must be perpendicular to both directions. Therefore

$$n = u \times v$$

is a natural choice.

### Example

Consider the three points

$$A = (1, 0, 0), \quad B = (0, 1, 0), \quad C = (0, 0, 1).$$

Two direction vectors in their plane are

$$u = B - A = (-1, 1, 0),$$

and

$$v = C - A = (-1, 0, 1).$$

Their cross product is

$$u \times v = (1 \cdot 1 - 0 \cdot 0, 0(-1) - (-1) \cdot 1, (-1) \cdot 0 - 1(-1)) = (1, 1, 1).$$

Thus a normal vector is

$$n = (1, 1, 1).$$

Using the point  $A$ , the plane equation is

$$n \cdot (P - A) = 0,$$

which becomes

$$(x - 1) + y + z = 0,$$

or

$$x + y + z = 1.$$

The chain of ideas is now complete:

three points  $\longrightarrow$  two direction vectors  $\longrightarrow$  cross product  $\longrightarrow$  normal vector  $\longrightarrow$  plane equation.

### Summary

When using the cross product, ask:

- Are the vectors in  $\mathbb{R}^3$ ?
- Do I need a vector perpendicular to two given directions?
- Have I preserved the alternating signs in the coordinate formula?
- Does reversing the order only reverse orientation?
- Is the cross product zero, indicating dependent directions?
- Can the result be used as a normal vector to a plane?

The cross product turns two independent directions inside a plane into one direction perpendicular to the plane.

## 6.7 Coordinates as a language

The purpose of this chapter is not only to introduce several formulas. Its purpose is to build a language. Coordinates, points, vectors, lengths, distances, and dot products are the words of that language. Once we can read them correctly, we can use them to describe more complicated geometric objects.

The next chapters will study lines and planes. These objects will not appear from nowhere. They will be built from the ideas already introduced here.

### Points give position

A point tells us where we are. In the plane, a point has the form

$$P = (x, y).$$

In space, it has the form

$$P = (x, y, z).$$

A point may be fixed, such as

$$A = (2, -1),$$

or variable, such as

$$P = (x, y).$$

When the coordinates are variable, the point is allowed to move through a set of possible positions.

This distinction is essential for equations. An equation in coordinates usually does not describe one number. It describes all points whose coordinates satisfy a condition.

### Example

The condition

$$y = 3$$

describes all points  $(x, y)$  in the plane whose second coordinate is 3. The first coordinate

is free. Thus the equation describes a horizontal line.  
The point  $(2, 3)$  is one point on that line, but it is not the whole line.

A common mistake is to confuse one solution with the entire geometric object. A single point may satisfy an equation, but the equation may describe a whole set of points.

## Vectors give displacement and direction

A vector tells us how to move. If

$$v = (v_1, v_2),$$

then  $v_1$  and  $v_2$  describe coordinate changes. The vector may be used as an actual displacement from one point to another, or it may be used only as a direction.

For example, the vector

$$v = (2, 1)$$

can mean: move two units in the  $x$ -direction and one unit in the  $y$ -direction. But it can also mean: consider the direction determined by this displacement.

This distinction will matter for lines. A line can be described by one point on the line and one non-zero direction vector. The point anchors the line. The vector gives the direction in which the line extends.

### Remark

A point and a vector may have the same coordinates, but they play different roles. A point answers the question “where?” A vector answers the question “how far and in which direction?”

## Equations describe conditions

An equation in analytic geometry is usually a condition on points. It says which points are allowed and which points are not.

For example, the equation

$$x^2 + y^2 = 1$$

describes all points whose distance from the origin is 1. This is a circle of radius 1 centered at the origin.

The formula is algebraic, but the object is geometric. The equation does not merely ask us to solve for  $x$  or  $y$ . It describes a set of points.

### Example

Consider the condition

$$(x - 2)^2 + (y + 1)^2 = 9.$$

A point  $(x, y)$  satisfies this condition exactly when its distance from  $(2, -1)$  is 3. Therefore the equation describes a circle with center  $(2, -1)$  and radius 3.

This example shows the main power of analytic geometry. A geometric statement about distance becomes an algebraic equation. Conversely, an algebraic equation can be read as a geometric statement.

## Fixed and variable quantities

Many mistakes in analytic geometry come from not distinguishing fixed objects from variable ones. A point may be fixed. A direction vector may be fixed. But the point  $(x, y)$  or  $(x, y, z)$

often represents a variable point moving through a set.

When we write a line in the form

$$P = A + tv,$$

the point  $A$  is fixed, the vector  $v$  is fixed and non-zero, and the parameter  $t$  is variable. As  $t$  changes, the point  $P$  moves along the line.

This formula belongs to the next chapter, but the language is already available. A fixed point gives a starting position. A direction vector gives a possible movement. A parameter controls how far we move.

#### Remark

Whenever a formula contains letters, ask which letters represent fixed data and which letters are allowed to vary. This question often reveals the meaning of the formula.

### From this chapter to lines and planes

The chapter has prepared three ideas that will be used repeatedly.

First, coordinates describe position. Second, vectors describe displacements and directions. Third, equations describe conditions on variable points.

Lines will be described by combining points and direction vectors. Planes will be described either by points and directions or by a point and a normal vector. The dot product will allow us to express perpendicularity, and therefore it will help us write equations of planes.

#### Summary

When working with coordinates and vectors, ask:

- Am I describing a point, a vector, or a set of points?
- Which quantities are fixed, and which are variable?
- Is a vector being used as a displacement, a direction, or a normal vector?
- Can a geometric condition be translated into an algebraic equation?
- After computing, can I interpret the result geometrically?

Analytic geometry is not a collection of unrelated formulas. It is a way of translating between geometric objects and algebraic statements. The translation is useful only when we understand both languages.

# Chapter 7

## Lines

### 7.1 What information defines a line

A line is one of the simplest geometric objects, but it is also one of the easiest objects to misunderstand. In elementary algebra a line is often introduced through an equation such as  $y = mx + b$ . That equation is useful, but it should not be confused with the object itself. A line is not a formula. A line is a set of points arranged in one direction.

Analytic geometry studies this set of points by translating geometric information into algebraic language. The first question is therefore not “Which formula should I use?” The first question is: what information is enough to determine the line?

#### A point and a direction

The most natural description of a line uses two pieces of information:

- one point on the line,
- one non-zero vector that gives the direction of the line.

If  $A$  is a point and  $v \neq 0$  is a direction vector, then the line consists of all points obtained by starting at  $A$  and moving by arbitrary multiples of  $v$ .

#### Definition

Let  $A$  be a point and let  $v \neq 0$  be a vector. The line through  $A$  in direction  $v$  is the set of points

$$P = A + tv,$$

where  $t$  runs through all real numbers.

The parameter  $t$  controls the movement along the line. When  $t = 0$ , we are at the point  $A$ . When  $t = 1$ , we move from  $A$  by exactly one copy of the vector  $v$ . When  $t = -1$ , we move by the same vector in the opposite direction.

#### Example

Let

$$A = (1, 2), \quad v = (3, -1).$$

The line through  $A$  in direction  $v$  is

$$P = (1, 2) + t(3, -1).$$

For  $t = 0$ , we get

$$P = (1, 2).$$

For  $t = 1$ , we get

$$P = (4, 1).$$

For  $t = -1$ , we get

$$P = (-2, 3).$$

These are different points, but they all lie on the same line because they are obtained from the same starting point by moving in the same direction.

This description matches the language developed in the previous chapter. A point gives position. A vector gives direction. A parameter tells us how much of that direction to use.

### Two distinct points

A second common way to determine a line is to give two distinct points. If the points are different, then the vector from one point to the other gives a direction.

Let

$$A = (a_1, a_2), \quad B = (b_1, b_2), \quad A \neq B.$$

Then

$$\overrightarrow{AB} = B - A = (b_1 - a_1, b_2 - a_2)$$

is a non-zero direction vector for the line through  $A$  and  $B$ . Therefore the line can be written as

$$P = A + t(B - A).$$

The condition  $A \neq B$  is not a technical decoration. If  $A = B$ , then the vector  $B - A$  is the zero vector, and the zero vector has no direction. One point alone does not determine a unique line.

#### Example

Let

$$A = (2, -1), \quad B = (5, 3).$$

Then

$$B - A = (3, 4).$$

The line through  $A$  and  $B$  can be written as

$$P = (2, -1) + t(3, 4).$$

For  $t = 1$ , this gives  $(5, 3)$ , so the point  $B$  is indeed on the line.

### A point and a normal vector in the plane

In the plane there is another important way to determine a line. Instead of giving a direction along the line, we may give a direction perpendicular to the line. Such a vector is called a normal vector.

If  $n \neq 0$  is perpendicular to a line and  $A$  is a point on the line, then a point  $P$  lies on the line exactly when the displacement  $P - A$  is perpendicular to  $n$ . Using the dot product, this becomes

$$n \cdot (P - A) = 0.$$

This form is the beginning of the normal description of a line. It is extremely useful because perpendicularity, distance from a point to a line, and the general equation of a line all become natural from it.

#### Remark

A direction vector points along the line. A normal vector points across the line. Both can describe the same line, but they reveal different information.

## The same object, different data

A line may therefore be determined from different kinds of information. We may know a point and a direction. We may know two points. In the plane, we may know a point and a normal vector. We may also know an equation and need to read from it what the point, direction, or normal vector is.

This is why this chapter will not present only one formula for a line. A line has many algebraic descriptions. Each description is a way of reading the same geometric object.

### Summary

When first meeting a line problem, ask:

- Am I given a point on the line?
- Am I given a direction vector?
- Am I given two points?
- Am I given a normal vector or an implicit equation?
- Is the line in the plane or in space?
- Which description makes the given information visible?

The goal is not to memorize all possible formulas at once. The goal is to understand what information each formula is carrying.

## 7.2 Many descriptions of the same line

A line is a single geometric object, but it can be described in many algebraic languages. This is not a defect of geometry. It is a strength. Different descriptions make different features visible.

A good student should not ask only: “What is the formula for a line?” A better question is: “Which description of the line is most useful for the question I am trying to answer?”

In this section we collect the most common descriptions of a line in the plane. Later sections will explain them more carefully and show how to move between them.

### Vector form

The vector form is

$$P = A + tv,$$

where  $A$  is a point on the line,  $v \neq 0$  is a direction vector, and  $t \in \mathbb{R}$ .

This form makes motion along the line visible. It says: start at a point, then move in a fixed direction by an arbitrary amount.

### Coordinate parametric form

If

$$A = (x_0, y_0), \quad v = (a, b),$$

then the vector form becomes

$$(x, y) = (x_0, y_0) + t(a, b).$$

Equivalently,

$$x = x_0 + ta, \quad y = y_0 + tb.$$

This is the coordinate parametric form. It is useful when we want to generate points on the line, check membership, or describe a line in space later.

### Two-point form

If a line passes through two distinct points  $A$  and  $B$ , then the vector  $B - A$  gives a direction. The line can be written as

$$P = A + t(B - A).$$

This form is useful because two points often occur naturally in geometric problems.

### Point-slope form

For a non-vertical line in the plane, if the line passes through  $(x_0, y_0)$  and has slope  $m$ , then

$$y - y_0 = m(x - x_0).$$

This form makes the slope and one point visible. It is useful in elementary coordinate geometry, but it does not describe vertical lines.

### Slope-intercept form

The form

$$y = mx + b$$

shows the slope  $m$  and the intersection point  $(0, b)$  with the  $y$ -axis. It is familiar and convenient, but it is not the most general way to describe a line. A vertical line cannot be written in this form.

### Implicit or general form

The general form of a line in the plane is

$$Ax + By + C = 0,$$

where  $A$  and  $B$  are not both zero.

This form is very important. It describes vertical, horizontal, and oblique lines in one language. It is also useful for intersections, side-of-line tests, and distance from a point to a line.

### Normal form

The normal form uses a point  $P_0$  on the line and a normal vector  $n \neq 0$ , perpendicular to the line:

$$n \cdot (P - P_0) = 0.$$

If  $n = (A, B)$ ,  $P = (x, y)$ , and  $P_0 = (x_0, y_0)$ , this becomes

$$A(x - x_0) + B(y - y_0) = 0.$$

After expanding, it becomes an equation of the form

$$Ax + By + C = 0.$$

Thus the general form hides a normal vector.

## Why so many forms?

The forms are not separate topics to memorize independently. They are different ways of reading the same object.

| Description             | Visible information                  | Useful for                  |
|-------------------------|--------------------------------------|-----------------------------|
| $P = A + tv$            | point and direction                  | movement, parametrization   |
| $P = A + t(B - A)$      | two points                           | building a line from points |
| $y - y_0 = m(x - x_0)$  | point and slope                      | non-vertical plane lines    |
| $y = mx + b$            | slope and intercept                  | quick graphing              |
| $Ax + By + C = 0$       | normal vector hidden in coefficients | intersections, distance     |
| $n \cdot (P - P_0) = 0$ | point and normal vector              | perpendicularity, distance  |

This table should not be read as a list of unrelated formulas. It is a map. When a problem gives certain information, one description may be natural. When a problem asks for a certain property, another description may be more useful.

### Remark

Changing the description of a line is like translating a sentence from one language into another. The meaning should remain the same, but different words may make different parts of the meaning easier to see.

### Summary

When choosing a line description, ask:

- Do I need to see a direction vector?
- Do I need to use a normal vector?
- Do I need to check whether a point lies on the line?
- Do I need to compare this line with another line?
- Do I need to compute a distance?
- Is a special case, such as a vertical line, possible?

The rest of the chapter develops this map into a working method.

## 7.3 Parametric description of lines

The parametric description is one of the most natural ways to describe a line. It uses the idea that a line can be traced by a moving point.

We start with a fixed point and a fixed direction. Then we allow a parameter to vary. As the parameter changes, the point moves along the line.

### Vector parametric form

Let  $A$  be a point and let  $v \neq 0$  be a direction vector. The line through  $A$  in direction  $v$  is

$$P = A + tv, \quad t \in \mathbb{R}.$$

This is a compact formula, but it contains three different roles:

- $A$  is fixed and anchors the line,
- $v$  is fixed and gives the direction,
- $t$  varies and moves the point along the line.

If we do not distinguish these roles, the formula becomes only a string of symbols. If we do distinguish them, the formula becomes a description of motion.

### Example

Consider the line

$$P = (2, 1) + t(3, 4).$$

For  $t = 0$ , we get

$$P = (2, 1).$$

For  $t = 1$ , we get

$$P = (5, 5).$$

For  $t = -1$ , we get

$$P = (-1, -3).$$

The same formula produces different points on the same line. The parameter controls how far we move in the direction  $(3, 4)$ .

The parameter is not something mysterious. It is a coordinate along the line, measured in copies of the direction vector.

## Coordinate parametric form in the plane

If

$$A = (x_0, y_0), \quad v = (a, b),$$

then

$$P = A + tv$$

means

$$(x, y) = (x_0, y_0) + t(a, b).$$

Therefore

$$x = x_0 + ta, \quad y = y_0 + tb.$$

These are the parametric equations of the line in the plane.

### Example

Let

$$A = (-1, 3), \quad v = (2, -5).$$

Then the line is

$$(x, y) = (-1, 3) + t(2, -5),$$

or equivalently

$$x = -1 + 2t, \quad y = 3 - 5t.$$

For  $t = 2$ , the point is

$$x = -1 + 4 = 3, \quad y = 3 - 10 = -7.$$

Thus  $(3, -7)$  lies on the line.

The coordinate equations should be read together. The same value of  $t$  must be used in both equations. We are not choosing one parameter for  $x$  and another parameter for  $y$ . One moving point has one parameter.

### Checking whether a point lies on a parametric line

To check whether a point belongs to a parametric line, we ask whether there exists a value of the parameter that produces the point.

#### Example

Consider the line

$$x = 1 + 3t, \quad y = 2 - t.$$

Does the point  $(7, 0)$  lie on the line?

From the first equation,

$$7 = 1 + 3t,$$

so

$$t = 2.$$

Using the same value in the second equation gives

$$y = 2 - 2 = 0.$$

Therefore  $(7, 0)$  lies on the line.

Now consider the point  $(7, 1)$ . The first coordinate again forces  $t = 2$ , but then the second coordinate would be 0, not 1. Therefore  $(7, 1)$  does not lie on the line.

This example shows why the parameter must be shared. A point lies on the line only if all coordinate equations agree on the same parameter value.

### Different parametrizations of the same line

A line does not have a unique parametric equation. We may choose a different point on the same line, or a different non-zero vector parallel to the same direction.

For example,

$$P = (2, 1) + t(3, 4)$$

and

$$P = (5, 5) + s(6, 8)$$

describe the same line. The point  $(5, 5)$  lies on the first line, and the vector  $(6, 8)$  is twice the direction vector  $(3, 4)$ . The second description moves along the same line, but uses a different starting point and a different scale for the parameter.

#### Remark

A parametric equation describes a line, but the parametrization itself is not unique. The geometric line remains the same even if we choose another point on it or replace the direction vector by a non-zero scalar multiple.

### Why parametric form is useful

Parametric form is especially useful when the natural data are a point and a direction, or two points. It is also the most natural form for lines in space, where slope-intercept equations are no longer available.

## Summary

When reading a parametric line, ask:

- What point is fixed on the line?
- What vector gives the direction?
- What does the parameter do geometrically?
- Does a given point arise from one common parameter value?
- Could a different point or parallel direction vector describe the same line?

The parametric form is not merely a formula. It is a way to watch a point move.

## 7.4 Two points and direction vectors

Many line problems begin with two points. This is natural: in Euclidean geometry, two distinct points determine exactly one line. In analytic geometry we translate this statement into coordinates and vectors.

The translation is simple. Two points give a displacement. That displacement gives a direction.

### Building the direction vector

Let

$$A = (a_1, a_2), \quad B = (b_1, b_2),$$

and suppose  $A \neq B$ . The vector from  $A$  to  $B$  is

$$\overrightarrow{AB} = B - A = (b_1 - a_1, b_2 - a_2).$$

This vector is non-zero because the points are distinct. It is therefore a valid direction vector for the line through  $A$  and  $B$ .

The line can then be written as

$$P = A + t(B - A), \quad t \in \mathbb{R}.$$

### Example

Find a parametric equation of the line through

$$A = (1, 2), \quad B = (4, -1).$$

First compute the direction vector:

$$B - A = (4 - 1, -1 - 2) = (3, -3).$$

Therefore the line is

$$P = (1, 2) + t(3, -3).$$

In coordinate form,

$$x = 1 + 3t, \quad y = 2 - 3t.$$

This procedure should be read as a chain of meanings. The two points determine a displacement. The displacement gives a direction. A point and a direction determine the line.

## The order of the points

If we reverse the order of the two points, the direction vector changes sign:

$$A - B = -(B - A).$$

This does not change the line. It only reverses the orientation of the parametrization.

Using the previous example, we found

$$B - A = (3, -3).$$

If we start from  $B$  instead, then

$$A - B = (-3, 3).$$

The line may also be written as

$$P = (4, -1) + s(-3, 3).$$

This is the same line, but the parameter moves along it in the opposite orientation.

### Remark

Changing the orientation of the direction vector does not change the geometric line. It changes how the line is traced by the parameter.

## Non-unique direction vectors

A direction vector for a line is never unique. If  $v$  is a direction vector, then every non-zero scalar multiple  $\lambda v$  is also a direction vector for the same line.

For instance, the vectors

$$(3, -3), \quad (1, -1), \quad (-2, 2)$$

all describe the same direction. They have different lengths and possibly different orientations, but they are parallel.

### Example

The line

$$P = (1, 2) + t(3, -3)$$

can also be written as

$$P = (1, 2) + s(1, -1).$$

The second version uses a shorter direction vector. Both equations describe the same set of points, because  $(3, -3) = 3(1, -1)$ .

This is often useful. We may simplify a direction vector by dividing out a common factor. The line remains the same, but the notation becomes easier to read.

## Finding a point from a parametrization

A parametric line already contains a point. In

$$P = A + tv,$$

the point  $A$  appears when  $t = 0$ . But every value of  $t$  gives another point on the same line. We may choose any of them as a new anchor point.

### Example

Consider

$$P = (2, -1) + t(4, 2).$$

For  $t = \frac{1}{2}$ , we get

$$P = (2, -1) + \frac{1}{2}(4, 2) = (4, 0).$$

Thus  $(4, 0)$  is also a point on the line. We could write the same line as

$$P = (4, 0) + s(4, 2).$$

This flexibility is not a problem. It is part of the geometry. A line has infinitely many points and infinitely many possible non-zero direction vectors.

### When the two points are the same

If the two given points are equal, then they do not determine a unique line. The vector between them is the zero vector:

$$B - A = 0.$$

The zero vector cannot be a direction vector, because it gives no direction.

This is a common hidden issue in problems with parameters. If two points depend on a parameter, there may be special parameter values for which the points coincide. Those values must be checked separately.

### Summary

When a line is given by two points, ask:

- Are the two points distinct?
- What is the vector from one point to the other?
- Can the direction vector be simplified?
- Does reversing the order only reverse orientation?
- Are there parameter values for which the two points might coincide?

The formula  $P = A + t(B - A)$  is not just a shortcut. It expresses the idea that a line is obtained by moving from one point in the direction of another.

## 7.5 Normal and implicit forms in the plane

The parametric form describes a line by moving along it. The normal form describes a line in a different way: by using a vector perpendicular to it. This change of viewpoint is important. A direction vector tells us how to travel along the line. A normal vector tells us how to stand across the line.

Both viewpoints describe the same geometric object, but they make different properties visible.

### Normal vector to a line

A non-zero vector  $n$  is called normal to a line if it is perpendicular to every direction vector of the line.

If a line has direction vector  $v$ , then a normal vector  $n$  satisfies

$$n \cdot v = 0.$$

In the plane, if

$$v = (a, b),$$

then one possible normal vector is

$$n = (-b, a),$$

because

$$(a, b) \cdot (-b, a) = -ab + ab = 0.$$

Another possible normal vector is  $(b, -a)$ . The normal vector is not unique, just as the direction vector is not unique.

### Example

Let a line have direction vector

$$v = (3, 2).$$

Then

$$n = (-2, 3)$$

is a normal vector, because

$$(3, 2) \cdot (-2, 3) = -6 + 6 = 0.$$

Thus  $(-2, 3)$  points perpendicular to the line.

### Normal form from a point and a normal vector

Suppose a line passes through a fixed point  $P_0 = (x_0, y_0)$ , and suppose  $n = (A, B) \neq (0, 0)$  is normal to the line. A variable point  $P = (x, y)$  lies on the line exactly when the displacement from  $P_0$  to  $P$  is perpendicular to  $n$ .

The displacement is

$$P - P_0 = (x - x_0, y - y_0).$$

The perpendicularity condition is

$$n \cdot (P - P_0) = 0.$$

Therefore

$$(A, B) \cdot (x - x_0, y - y_0) = 0,$$

so

$$A(x - x_0) + B(y - y_0) = 0.$$

### Definition

A line in the plane with point  $P_0 = (x_0, y_0)$  and normal vector  $n = (A, B) \neq (0, 0)$  can be written in normal form as

$$A(x - x_0) + B(y - y_0) = 0.$$

Equivalently,

$$n \cdot (P - P_0) = 0.$$

This equation is not arbitrary. It says: the vector from the fixed point to the variable point must be perpendicular to the normal vector.

### Example

Find an equation of the line through

$$P_0 = (1, -2)$$

with normal vector

$$n = (3, 4).$$

The normal form is

$$3(x - 1) + 4(y - (-2)) = 0.$$

Thus

$$3(x - 1) + 4(y + 2) = 0.$$

Expanding,

$$3x - 3 + 4y + 8 = 0,$$

so

$$3x + 4y + 5 = 0.$$

### Implicit or general form

After expanding the normal form, every line in the plane can be written as

$$Ax + By + C = 0,$$

where  $A$  and  $B$  are not both zero.

This is called the implicit form or general form of a line. The word “implicit” means that the line is described as a condition. A point  $(x, y)$  lies on the line if its coordinates satisfy the equation.

### Definition

The general form of a line in the plane is

$$Ax + By + C = 0,$$

where  $(A, B) \neq (0, 0)$ . The vector

$$n = (A, B)$$

is normal to the line.

The last sentence is essential. The coefficients of  $x$  and  $y$  are not just coefficients. Together they form a normal vector.

### Example

Consider the line

$$2x - 5y + 7 = 0.$$

A normal vector is

$$n = (2, -5).$$

A direction vector must be perpendicular to  $n$ . One possible choice is

$$v = (5, 2),$$

because

$$(2, -5) \cdot (5, 2) = 10 - 10 = 0.$$

This example shows how information can be read from an equation. The equation does not display a direction vector directly, but it displays a normal vector.

### Checking membership

The implicit form is often the easiest way to check whether a point lies on a line. We substitute the coordinates into the left-hand side.

#### Example

Let

$$\ell : 2x - 5y + 7 = 0.$$

Check whether  $P = (4, 3)$  lies on  $\ell$ . We compute

$$2 \cdot 4 - 5 \cdot 3 + 7 = 8 - 15 + 7 = 0.$$

Therefore  $(4, 3)$  lies on the line.

For  $Q = (1, 1)$ , we get

$$2 \cdot 1 - 5 \cdot 1 + 7 = 4.$$

Since the result is not zero,  $(1, 1)$  does not lie on the line.

The number obtained after substitution also has meaning. It measures, up to a scale factor, how far the point is from satisfying the line condition. This idea will later become the distance formula.

### Why implicit form is useful

The general form is especially useful because it treats all lines in the plane uniformly. It includes vertical lines, horizontal lines, and oblique lines. It also makes normal vectors visible, which helps with perpendicularity and distance.

#### Summary

When reading an implicit equation  $Ax + By + C = 0$ , ask:

- Are  $A$  and  $B$  not both zero?
- What normal vector is visible in the equation?
- Can I find a direction vector perpendicular to that normal vector?
- Does a given point satisfy the equation?
- Would this form help with distance or intersection questions?

The implicit form is not merely another formula. It is the language of conditions, normal vectors, and perpendicularity.

## 7.6 Slope forms and special cases

Many students first meet lines through equations of the form  $y = mx + b$ . This form is useful, but it is only one description among several. It works well for non-vertical lines in the plane. It does not describe every line.

The purpose of this section is therefore not to reject slope forms, but to place them correctly.

They are convenient when slope is the natural information, but they should not replace the more general geometric understanding of a line.

### Slope as a ratio of changes

For a non-vertical line, the slope measures the ratio of vertical change to horizontal change. If a line has direction vector

$$v = (a, b)$$

with  $a \neq 0$ , then its slope is

$$m = \frac{b}{a}.$$

This says that when the horizontal coordinate changes by  $a$ , the vertical coordinate changes by  $b$ .

#### Example

If a line has direction vector

$$v = (4, 6),$$

then its slope is

$$m = \frac{6}{4} = \frac{3}{2}.$$

The vector  $(4, 6)$  and the vector  $(2, 3)$  describe the same direction, and both give the same slope.

Slope is therefore not a new kind of direction. It is a way of encoding direction for non-vertical lines.

### Point-slope form

Suppose a non-vertical line passes through  $(x_0, y_0)$  and has slope  $m$ . The point-slope form is

$$y - y_0 = m(x - x_0).$$

This equation says that every point  $(x, y)$  on the line has the same ratio of vertical change to horizontal change relative to  $(x_0, y_0)$ .

#### Example

Find an equation of the line through  $(2, -1)$  with slope 3.

Using point-slope form,

$$y - (-1) = 3(x - 2),$$

so

$$y + 1 = 3x - 6.$$

Thus

$$y = 3x - 7.$$

The computation is short, but the meaning is important. The equation does not simply contain the number 3. It says that vertical change is always three times horizontal change.

### Slope-intercept form

If a non-vertical line crosses the  $y$ -axis at  $(0, b)$  and has slope  $m$ , then its equation is

$$y = mx + b.$$

The number  $b$  is the value of  $y$  when  $x = 0$ . The number  $m$  describes the slope.

### Example

The line

$$y = -2x + 5$$

has slope  $-2$  and intersects the  $y$ -axis at

$$(0, 5).$$

A possible direction vector is

$$v = (1, -2),$$

because when  $x$  increases by 1,  $y$  changes by  $-2$ .

This form is good for quick graphing and for reading slope. It is not always the best form for distance, intersections, or vertical lines.

### Horizontal lines

A horizontal line has slope 0. Its equation has the form

$$y = c.$$

Every point on the line has the same second coordinate.

For example,

$$y = 4$$

describes all points  $(x, y)$  with  $y = 4$ . The first coordinate is free.

In general form, this line is

$$y - 4 = 0.$$

A normal vector is  $(0, 1)$ , and a direction vector is  $(1, 0)$ .

### Vertical lines

A vertical line has the form

$$x = c.$$

Every point on the line has the same first coordinate. Such a line does not have a finite slope and cannot be written as  $y = mx + b$ .

For example,

$$x = -2$$

describes all points  $(x, y)$  with first coordinate  $-2$ . The second coordinate is free.

In general form, this line is

$$x + 2 = 0.$$

A normal vector is  $(1, 0)$ , and a direction vector is  $(0, 1)$ .

### Remark

The equation  $y = mx + b$  is familiar, but it is not universal. Whenever vertical lines may appear, use a description that can include them, such as parametric form or general form.

## From slope form to general form

Every non-vertical line written as

$$y = mx + b$$

can be rewritten in general form:

$$mx - y + b = 0.$$

If we prefer integer coefficients, we may multiply by a common denominator.

### Example

Rewrite

$$y = \frac{2}{3}x - 5$$

in general form. Move all terms to one side:

$$\frac{2}{3}x - y - 5 = 0.$$

Multiplying by 3, we get

$$2x - 3y - 15 = 0.$$

A normal vector is therefore

$$n = (2, -3).$$

This example shows again that changing form changes what is visible. The slope form made the slope visible. The general form made the normal vector visible.

### Summary

When using slope forms, ask:

- Is the line definitely non-vertical?
- Is slope the most useful information in this problem?
- Would a general form make a normal vector visible?
- Is the line horizontal or vertical?
- Which form avoids unnecessary special cases?

Slope forms are useful tools, but they should be used with awareness of their limits.

## 7.7 Converting between line descriptions

The same line can be written in many forms. Therefore one of the basic skills in analytic geometry is translation. We translate from the form that is given to the form that makes the next question easier.

This translation should not be mechanical. Each conversion has a meaning. When we convert a parametric equation into an implicit equation, we are changing from a direction-based description to a condition-based description. When we convert an implicit equation into a parametric equation, we are finding a point and a direction hidden inside a condition.

## From parametric form to implicit form

Suppose a line is given in parametric form:

$$P = P_0 + tv.$$

In the plane, if

$$P_0 = (x_0, y_0), \quad v = (a, b),$$

then a normal vector may be chosen as

$$n = (-b, a).$$

The implicit equation is then obtained from

$$n \cdot (P - P_0) = 0.$$

### Example

Convert the line

$$(x, y) = (1, 2) + t(3, -1)$$

to implicit form.

The direction vector is

$$v = (3, -1).$$

A normal vector is

$$n = (1, 3),$$

because

$$(3, -1) \cdot (1, 3) = 3 - 3 = 0.$$

Using the point  $(1, 2)$ , the normal form is

$$1(x - 1) + 3(y - 2) = 0.$$

Thus

$$x - 1 + 3y - 6 = 0,$$

so

$$x + 3y - 7 = 0.$$

The direction vector was visible in the parametric form. The normal vector became visible after conversion to implicit form.

## From implicit form to parametric form

Now suppose the line is given by

$$Ax + By + C = 0.$$

A normal vector is immediately visible:

$$n = (A, B).$$

A direction vector must be perpendicular to  $n$ . One convenient choice is

$$v = (-B, A),$$

because

$$(A, B) \cdot (-B, A) = -AB + AB = 0.$$

We also need one point on the line. This can be found by choosing one coordinate and solving for the other, whenever possible.

### Example

Convert

$$2x - 5y + 10 = 0$$

to parametric form.

A normal vector is

$$n = (2, -5).$$

A direction vector is

$$v = (5, 2),$$

because

$$(2, -5) \cdot (5, 2) = 10 - 10 = 0.$$

To find a point, choose  $x = 0$ . Then

$$-5y + 10 = 0,$$

so

$$y = 2.$$

Thus  $(0, 2)$  lies on the line. A parametric equation is

$$(x, y) = (0, 2) + t(5, 2).$$

There are many possible choices of the point. If we choose another point on the same line, we get another parametrization of the same geometric object.

### From two points to implicit form

If a line is given by two distinct points, first build a direction vector. Then choose a normal vector perpendicular to that direction vector.

### Example

Find an implicit equation of the line through

$$A = (1, -1), \quad B = (4, 5).$$

The direction vector is

$$B - A = (3, 6).$$

We may simplify the direction to

$$v = (1, 2).$$

A normal vector is

$$n = (-2, 1),$$

because

$$(1, 2) \cdot (-2, 1) = -2 + 2 = 0.$$

Using the point  $A = (1, -1)$ , the normal form is

$$-2(x - 1) + 1(y - (-1)) = 0.$$

Thus

$$-2(x - 1) + y + 1 = 0.$$

Expanding,

$$-2x + 2 + y + 1 = 0,$$

so

$$-2x + y + 3 = 0.$$

Equivalently,

$$2x - y - 3 = 0.$$

Both final equations describe the same line. Multiplying every coefficient by a non-zero number does not change the set of solutions.

### From slope form to direction and normal vectors

If a non-vertical line has equation

$$y = mx + b,$$

then a direction vector is

$$v = (1, m).$$

Indeed, when  $x$  increases by 1, the value of  $y$  changes by  $m$ .

After rewriting as

$$mx - y + b = 0,$$

we also see a normal vector:

$$n = (m, -1).$$

These two vectors are perpendicular, because

$$(1, m) \cdot (m, -1) = m - m = 0.$$

#### Example

For the line

$$y = -3x + 4,$$

a direction vector is

$$v = (1, -3).$$

Rewriting as

$$3x + y - 4 = 0,$$

we see a normal vector

$$n = (3, 1).$$

Indeed,

$$(1, -3) \cdot (3, 1) = 3 - 3 = 0.$$

### Checking the translation

After converting one form into another, it is good practice to check that the descriptions agree. We can check that the chosen point satisfies the new equation. We can check that the direction vector is perpendicular to the normal vector. We can also test a second point generated by the parametrization.

## Summary

When converting between descriptions, ask:

- What information is visible in the given form?
- Do I need a direction vector or a normal vector?
- Have I found at least one point on the line?
- Is the direction vector perpendicular to the normal vector?
- Does the converted equation describe the same set of points?

Conversion is not a ritual of rearranging symbols. It is a way of making hidden geometric information visible.

## 7.8 Parallel, perpendicular, and intersecting lines

Once we can describe one line, we can compare two lines. Direction vectors make parallelism natural. Dot products make perpendicularity natural. Implicit equations make intersections natural because they become systems of linear equations.

### Parallel lines from direction vectors

Two lines are parallel when their non-zero direction vectors are scalar multiples of one another.

#### Definition

Let  $\ell_1$  and  $\ell_2$  have direction vectors  $v_1$  and  $v_2$ . They are parallel if there exists a non-zero scalar  $\lambda$  such that

$$v_2 = \lambda v_1.$$

#### Example

For

$$\ell_1 : P = (1, 2) + t(2, -3)$$

and

$$\ell_2 : P = (-4, 0) + s(-4, 6),$$

we have  $(-4, 6) = -2(2, -3)$ , so the lines are parallel.

Parallel lines may be distinct or may describe the same line. If they share a point, they are the same line; otherwise they are distinct parallel lines.

#### Example

The lines

$$\ell_1 : P = (1, 2) + t(2, -3)$$

and

$$\ell_2 : P = (3, -1) + s(4, -6)$$

have parallel directions. Moreover,  $(3, -1)$  is obtained from  $\ell_1$  when  $t = 1$ . Therefore the two parametrizations describe the same line.

## Perpendicular lines

In the plane, two non-parallel lines automatically intersect. In space, position must also be checked: skew lines can have perpendicular direction vectors without meeting.

### Definition

Let  $\ell_1$  and  $\ell_2$  be intersecting lines with non-zero direction vectors  $v_1$  and  $v_2$ . The lines are perpendicular if

$$v_1 \cdot v_2 = 0.$$

For lines in the plane, perpendicular direction vectors already guarantee that the lines intersect.

### Example

Let two lines in the plane have directions

$$v_1 = (3, 1), \quad v_2 = (2, -6).$$

Then

$$v_1 \cdot v_2 = 3 \cdot 2 + 1 \cdot (-6) = 0.$$

The directions are perpendicular, and the plane lines therefore meet at a right angle.

### Remark

For lines in  $\mathbb{R}^3$ , the equation  $v_1 \cdot v_2 = 0$  describes perpendicular directions. To call the lines themselves perpendicular, one must additionally verify that they intersect.

## Using normal vectors in the plane

For lines

$$\ell_1 : A_1x + B_1y + C_1 = 0, \quad \ell_2 : A_2x + B_2y + C_2 = 0,$$

the normal vectors are

$$n_1 = (A_1, B_1), \quad n_2 = (A_2, B_2).$$

The lines are parallel when their normal vectors are parallel. They are perpendicular when their normal vectors are perpendicular.

### Example

For

$$\ell_1 : 2x - 3y + 1 = 0, \quad \ell_2 : 4x - 6y - 5 = 0,$$

we have

$$(4, -6) = 2(2, -3),$$

so the lines are parallel.

### Example

For

$$\ell_1 : x + 2y - 4 = 0, \quad \ell_2 : 2x - y + 3 = 0,$$

the normals satisfy

$$(1, 2) \cdot (2, -1) = 0.$$

Therefore the lines are perpendicular.

### Intersecting lines as a system of equations

Two non-parallel lines in the plane intersect in exactly one point. Their intersection is found by solving their equations simultaneously.

#### Example

For

$$\ell_1 : x + y - 3 = 0, \quad \ell_2 : 2x - y = 0,$$

we solve

$$\begin{cases} x + y = 3, \\ 2x - y = 0. \end{cases}$$

Adding the equations gives  $3x = 3$ , so  $x = 1$ , and then  $y = 2$ . The intersection point is

$$(1, 2).$$

#### Summary

When comparing two lines, ask:

- Are the lines in the plane or in space?
- Are their direction vectors parallel or perpendicular?
- In space, do lines with perpendicular directions actually intersect?
- If the lines are parallel, are they identical or distinct?
- If plane lines are not parallel, what is their intersection point?

Direction, position, and dimension must all be read before the geometric relationship is named.

## 7.9 Distance from a point to a line

The distance from a point to a line is not the distance to an arbitrary point on the line. It is the shortest possible distance. Geometrically, the shortest path from a point to a line is perpendicular to the line.

This is why the normal form of a line is so useful. A normal vector already points in the direction in which this shortest distance is measured.

### The geometric idea

Let a line be given in general form:

$$\ell : Ax + By + C = 0.$$

A normal vector to the line is

$$n = (A, B).$$

If a point  $P_0 = (x_0, y_0)$  does not lie on the line, then the shortest segment from  $P_0$  to the line is parallel to  $n$ . The distance should therefore be measured in the normal direction.

The expression

$$Ax_0 + By_0 + C$$

tells us how far the point is from satisfying the equation of the line, but it still depends on the length of the normal vector. To obtain the actual distance, we divide by the length of the normal vector:

$$\sqrt{A^2 + B^2}.$$

### Theorem

Let

$$\ell : Ax + By + C = 0$$

be a line in the plane, with  $(A, B) \neq (0, 0)$ . The distance from the point  $P_0 = (x_0, y_0)$  to  $\ell$  is

$$d(P_0, \ell) = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}.$$

The absolute value appears because distance is non-negative. Without the absolute value, the sign tells us on which side of the oriented line the point lies.

### Example of the distance formula

#### Example

Find the distance from the point

$$P_0 = (3, 1)$$

to the line

$$\ell : 2x - y - 4 = 0.$$

Here

$$A = 2, \quad B = -1, \quad C = -4.$$

Substitute  $(x_0, y_0) = (3, 1)$ :

$$Ax_0 + By_0 + C = 2 \cdot 3 + (-1) \cdot 1 - 4 = 6 - 1 - 4 = 1.$$

The length of the normal vector is

$$\sqrt{A^2 + B^2} = \sqrt{2^2 + (-1)^2} = \sqrt{5}.$$

Therefore

$$d(P_0, \ell) = \frac{|1|}{\sqrt{5}} = \frac{1}{\sqrt{5}}.$$

The computation is short, but it should be interpreted. The point does not satisfy the line equation because substitution gives 1, not 0. Dividing by  $\sqrt{5}$  corrects for the length of the normal vector.

### Why the denominator is needed

The equation of a line is not unique. For example,

$$2x - y - 4 = 0$$

and

$$4x - 2y - 8 = 0$$

describe the same line. The second equation is obtained by multiplying the first by 2.

If we used only  $|Ax_0 + By_0 + C|$ , the value would also be multiplied by 2. But the geometric distance should not change. The denominator  $\sqrt{A^2 + B^2}$  is multiplied by the same factor, so the ratio remains the same.

#### Remark

The distance formula uses the equation of the line, but the result is geometric. It cannot depend on which equivalent equation we choose.

### Signed distance and sides of a line

The expression

$$Ax_0 + By_0 + C$$

without absolute value has a sign. Points on one side of the line give one sign, and points on the other side give the opposite sign. Points on the line give zero.

A normalized signed quantity is

$$\frac{Ax_0 + By_0 + C}{\sqrt{A^2 + B^2}}.$$

Its absolute value is the distance to the line.

#### Example

For the line

$$\ell : x + y - 1 = 0,$$

consider the points

$$P = (2, 0), \quad Q = (0, -2).$$

For  $P$ ,

$$2 + 0 - 1 = 1.$$

For  $Q$ ,

$$0 + (-2) - 1 = -3.$$

The signs are different, so the points lie on opposite sides of the oriented line.

This interpretation is useful in geometry and in applications. The equation of a line does more than describe the points on the line. It also separates the plane into two sides.

### Distance and projection idea

There is another way to understand the formula. Choose any point  $A$  on the line. The vector  $P_0 - A$  goes from the line to the point. The distance from  $P_0$  to the line is the length of the projection of this vector onto the normal direction.

If  $n = (A, B)$ , then a unit normal vector is

$$\frac{n}{\|n\|}.$$

The signed length of the projection is

$$(P_0 - A) \cdot \frac{n}{\|n\|}.$$

This is the same idea hidden inside the distance formula.

## Summary

When computing distance from a point to a line, ask:

- Is the line written in general form  $Ax + By + C = 0$ ?
- What is the normal vector  $(A, B)$ ?
- What value do I get after substituting the point?
- Did I divide by the length of the normal vector?
- Does the sign, before taking absolute value, carry side-of-line information?

The distance formula is not an isolated trick. It is the normal-vector interpretation of a line turned into a measurement.

## 7.10 Lines in space

Lines in space are best understood through points and direction vectors. The parametric form works in the plane and in space, while slope-intercept form belongs only to the plane. This is one reason why the vector description is so important.

In three-dimensional space, a line still needs the same kind of information: one point and one non-zero direction vector.

### Parametric form in space

Let

$$A = (x_0, y_0, z_0)$$

be a point in space, and let

$$v = (a, b, c) \neq (0, 0, 0)$$

be a direction vector. The line through  $A$  in direction  $v$  is

$$(x, y, z) = (x_0, y_0, z_0) + t(a, b, c), \quad t \in \mathbb{R}.$$

Equivalently,

$$x = x_0 + ta, \quad y = y_0 + tb, \quad z = z_0 + tc.$$

### Example

The line

$$(x, y, z) = (1, -2, 3) + t(4, 0, -1)$$

has point  $(1, -2, 3)$  and direction vector  $(4, 0, -1)$ . Its coordinate equations are

$$x = 1 + 4t, \quad y = -2, \quad z = 3 - t.$$

The second coordinate is constant because the second component of the direction vector is zero.

This example is important. A zero component of a direction vector does not mean that the vector is invalid. It means that the line does not move in that coordinate direction.

## A line through two points in space

Two distinct points in space determine a line in the same way as in the plane. If

$$A = (a_1, a_2, a_3), \quad B = (b_1, b_2, b_3), \quad A \neq B,$$

then

$$B - A = (b_1 - a_1, b_2 - a_2, b_3 - a_3)$$

is a direction vector. The line is

$$P = A + t(B - A).$$

### Example

Find a parametric equation of the line through

$$A = (2, 1, -1), \quad B = (5, 3, 4).$$

The direction vector is

$$B - A = (3, 2, 5).$$

Therefore the line is

$$(x, y, z) = (2, 1, -1) + t(3, 2, 5).$$

In coordinates,

$$x = 2 + 3t, \quad y = 1 + 2t, \quad z = -1 + 5t.$$

The idea is unchanged: two points give a displacement, and the displacement gives a direction.

## Symmetric form

If all components of the direction vector are non-zero, then the parametric equations

$$x = x_0 + ta, \quad y = y_0 + tb, \quad z = z_0 + tc$$

can be rewritten as

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}.$$

This is called the symmetric form of a line in space.

### Example

For the line

$$(x, y, z) = (1, 2, -3) + t(2, -1, 4),$$

all components of the direction vector are non-zero. Therefore

$$\frac{x - 1}{2} = \frac{y - 2}{-1} = \frac{z + 3}{4}.$$

The symmetric form is compact, but it must be used carefully. It comes from solving each parametric equation for the same parameter  $t$ . If a component of the direction vector is zero, we cannot divide by it.

## Zero components in symmetric form

Suppose

$$(x, y, z) = (1, -2, 3) + t(4, 0, -1).$$

Then

$$x = 1 + 4t, \quad y = -2, \quad z = 3 - t.$$

The second component of the direction vector is zero, so the line has constant  $y$ -coordinate. We may write

$$\frac{x-1}{4} = \frac{z-3}{-1}, \quad y = -2.$$

We should not write a fraction with denominator 0. The equation  $y = -2$  must be kept separately.

#### Remark

Symmetric form is convenient only when used honestly. A zero component of the direction vector becomes a constant-coordinate condition, not a division by zero.

### Why one equation is not enough in space

In the plane, a single linear equation

$$Ax + By + C = 0$$

describes a line. In space, a single linear equation

$$Ax + By + Cz + D = 0$$

usually describes a plane, not a line.

This is a fundamental difference. In the plane, one linear condition leaves one degree of freedom, which gives a line. In space, one linear condition leaves two degrees of freedom, which gives a plane.

A line in space can often be described as the intersection of two planes, that is, by a system

$$\begin{cases} A_1x + B_1y + C_1z + D_1 = 0, \\ A_2x + B_2y + C_2z + D_2 = 0. \end{cases}$$

However, the parametric form is usually the clearest first description of a line in space.

#### Example

The equation

$$z = 2$$

in space does not describe a line. It describes all points whose third coordinate is 2. This is a plane parallel to the  $xy$ -plane.

To describe a line in that plane, we need another independent condition or a parametric description.

### Comparing lines in space

Lines in space may be parallel, intersecting, or skew. Skew lines are lines that do not intersect and are not parallel. This possibility does not occur for lines in the plane.

For now, the most important point is that direction vectors still detect parallelism. If two direction vectors are scalar multiples, the lines are parallel or the same line. If they are not scalar multiples, the lines may intersect or may be skew. To decide, one must solve the parametric equations together.

## Summary

When working with lines in space, ask:

- What point lies on the line?
- What non-zero vector gives the direction?
- Are any direction-vector components zero?
- Is symmetric form safe to use?
- Am I mistakenly treating one plane equation as a line?
- Are two lines parallel, intersecting, or skew?

The vector-parametric description is the most reliable language for lines in space. It keeps the point, the direction, and the movement visible.

## 7.11 Choosing the right description

This chapter introduced many descriptions of a line. At first this may seem like too much notation. But the purpose is not to collect formulas. The purpose is to learn how to choose a description that fits the problem.

The same line can be written in several ways. The best form depends on what is known and what is being asked.

### When a point and direction are given

If a problem gives a point and a direction vector, the parametric form is usually the most natural:

$$P = A + tv.$$

It keeps the given information visible. It also makes it easy to generate points and to understand movement along the line.

This form is also the safest first choice in space.

### When two points are given

If a problem gives two distinct points  $A$  and  $B$ , first form the direction vector

$$B - A.$$

Then write

$$P = A + t(B - A).$$

From there, if needed, convert to another form.

The important point is that the two given points are not just coordinates to insert into a memorized formula. They define a displacement, and that displacement defines a direction.

### When a normal vector is visible

If a problem gives an equation

$$Ax + By + C = 0,$$

then the normal vector

$$n = (A, B)$$

is immediately visible. This form is useful for distance questions, side-of-line questions, and comparisons using normal vectors.

If a point  $P_0$  and a normal vector  $n$  are given, use

$$n \cdot (P - P_0) = 0.$$

This is often more meaningful than jumping directly to expanded coordinates.

### When slope is useful

If the line is non-vertical and slope is central to the question, then slope forms may be convenient:

$$y - y_0 = m(x - x_0)$$

or

$$y = mx + b.$$

But these forms should be used with awareness. Vertical lines require a different description, such as

$$x = c,$$

or a parametric or general equation.

### Which form helps which question?

The following guide is often useful.

| Question                               | Useful description                   |
|----------------------------------------|--------------------------------------|
| Generate points on the line            | parametric form                      |
| Line through two points                | two-point form, then parametric form |
| Check whether a point lies on the line | implicit or parametric form          |
| Find an intersection in the plane      | general form as a system             |
| Check parallelism                      | direction vectors or normal vectors  |
| Check perpendicularity                 | dot product of directions or normals |
| Compute distance from a point          | general form $Ax + By + C = 0$       |
| Work in space                          | parametric form                      |

This table should not be used mechanically. It is a guide for choosing what to read from the line.

### A complete reading example

Consider the line

$$\ell : 2x - 3y + 6 = 0.$$

From this form we immediately read a normal vector:

$$n = (2, -3).$$

A direction vector can be chosen perpendicular to  $n$ :

$$v = (3, 2),$$

because

$$(2, -3) \cdot (3, 2) = 6 - 6 = 0.$$

To find a point on the line, set  $x = 0$ . Then

$$-3y + 6 = 0,$$

so

$$y = 2.$$

Thus  $(0, 2)$  lies on the line. A parametric form is

$$P = (0, 2) + t(3, 2).$$

The same line can therefore be read in two ways:

$$2x - 3y + 6 = 0$$

shows the normal vector and is good for distance or side tests, while

$$P = (0, 2) + t(3, 2)$$

shows a point and direction and is good for motion along the line.

#### Remark

A good solution often changes the description of a line. This is not extra work. It is how we choose the language in which the problem becomes clear.

### What to remember

A line is a set of points. Equations and parametrizations are descriptions of that set. The main descriptions are not competitors; they are tools.

#### Summary

When working with lines, ask:

- Is the line in the plane or in space?
- Am I given a point, two points, a direction vector, or a normal vector?
- Which description makes the given information visible?
- Which description makes the required property easiest to study?
- Am I studying membership, intersection, parallelism, perpendicularity, or distance?
- After computing, can I explain what the result means geometrically?

The most important lesson is not that lines have many formulas. The most important lesson is that each formula reveals a different aspect of the same geometric object.

# Chapter 8

## Planes

### 8.1 What information defines a plane

A plane is the three-dimensional analogue of a line in one important sense: it is a flat object. But a plane is not one-dimensional. A line has one direction. A plane has two independent directions.

This distinction changes how we describe planes. A line can be determined by a point and one non-zero direction vector. A plane can be determined by a point and two non-parallel direction vectors, or by a point and one normal vector.

#### A point and two directions

Imagine standing at a point on a flat surface. To move around the surface, one direction is not enough. You need two independent directions. If those directions are not parallel, then every point of the plane can be reached by combining them.

Let  $P_0$  be a point in space and let  $u$  and  $v$  be two non-zero, non-parallel vectors. Then the plane through  $P_0$  spanned by  $u$  and  $v$  is described by

$$P = P_0 + su + tv, \quad s, t \in \mathbb{R}.$$

The two parameters  $s$  and  $t$  move independently in the two directions.

#### Example

Let

$$P_0 = (1, 2, 0), \quad u = (1, 0, 1), \quad v = (0, 1, 2).$$

Then a plane is given by

$$P = (1, 2, 0) + s(1, 0, 1) + t(0, 1, 2).$$

In coordinates,

$$x = 1 + s, \quad y = 2 + t, \quad z = s + 2t.$$

Changing  $s$  and  $t$  moves the point across the whole plane.

The condition that  $u$  and  $v$  are not parallel is essential. If they were parallel, they would describe only one direction, and the formula would trace a line, not a plane.

#### A point and a normal vector

There is another very powerful way to describe a plane. Instead of giving two directions inside the plane, we may give one direction perpendicular to the plane. Such a vector is called a normal vector.

If  $n \neq 0$  is perpendicular to a plane and  $P_0$  is a point on the plane, then a point  $P$  lies on the plane exactly when the displacement  $P - P_0$  is perpendicular to  $n$ . Using the dot product, this condition is

$$n \cdot (P - P_0) = 0.$$

This is the normal description of a plane. It is especially useful for equations, distances, angles, and parallelism.

#### Remark

Two direction vectors describe movement inside the plane. A normal vector describes the direction across the plane. Both viewpoints describe the same geometric object.

### Three non-collinear points

A plane can also be determined by three points, provided they are not collinear. If the points are  $A$ ,  $B$ , and  $C$ , then the vectors

$$\overrightarrow{AB} = B - A, \quad \overrightarrow{AC} = C - A$$

lie in the plane. If these vectors are not parallel, then they give the two independent directions needed to describe the plane.

#### Example

The points

$$A = (1, 0, 0), \quad B = (0, 1, 0), \quad C = (0, 0, 1)$$

are not collinear. The vectors

$$B - A = (-1, 1, 0), \quad C - A = (-1, 0, 1)$$

are not parallel, so they determine a plane.

### The same plane, different descriptions

As with lines, a plane is not the same thing as one formula. The same plane may be described parametrically, by a normal equation, by a general equation, or by three points.

#### Summary

A plane may be determined by:

- one point and two non-parallel direction vectors,
- one point and one non-zero normal vector,
- three non-collinear points,
- an equation of the form  $Ax + By + Cz + D = 0$ , where  $(A, B, C) \neq (0, 0, 0)$ .

The goal of this chapter is to learn how to move between these descriptions and how to use each one for the questions it answers best.

## 8.2 Normal and general equation of a plane

The normal form of a plane is one of the most useful descriptions in analytic geometry. It uses a point on the plane and a vector perpendicular to the plane.

Let

$$P_0 = (x_0, y_0, z_0)$$

be a point on the plane, and let

$$n = (A, B, C) \neq (0, 0, 0)$$

be a normal vector. A point

$$P = (x, y, z)$$

lies on the plane exactly when  $P - P_0$  is perpendicular to  $n$ . Therefore

$$n \cdot (P - P_0) = 0.$$

#### Definition

A plane through  $P_0 = (x_0, y_0, z_0)$  with normal vector  $n = (A, B, C) \neq (0, 0, 0)$  can be written as

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0.$$

This is the normal equation of the plane.

The equation is not a memorized pattern. It says that the displacement from the fixed point to any point of the plane must be perpendicular to the normal vector.

#### From normal form to general form

Expanding the normal equation gives

$$Ax + By + Cz + D = 0,$$

where

$$D = -Ax_0 - By_0 - Cz_0.$$

This is called the general equation of a plane.

#### Definition

The general equation of a plane in space is

$$Ax + By + Cz + D = 0,$$

where  $(A, B, C) \neq (0, 0, 0)$ . The vector

$$n = (A, B, C)$$

is normal to the plane.

The coefficients of  $x$ ,  $y$ , and  $z$  are not only numbers in an equation. Together they form a normal vector.

#### Example from a point and normal vector

##### Example

Find the equation of the plane passing through

$$A = (1, 0, 2)$$

with normal vector

$$n = (2, -1, 1).$$

Use the normal form:

$$2(x - 1) - 1(y - 0) + 1(z - 2) = 0.$$

Thus

$$2(x - 1) - y + z - 2 = 0.$$

Expand:

$$2x - 2 - y + z - 2 = 0,$$

so

$$2x - y + z - 4 = 0.$$

We can check the result. The point  $(1, 0, 2)$  satisfies the equation:

$$2 \cdot 1 - 0 + 2 - 4 = 0.$$

The normal vector visible in the final equation is  $(2, -1, 1)$ , as required.

### Reading information from an equation

If a plane is given by

$$3x - 2y + 5z - 7 = 0,$$

then a normal vector is immediately visible:

$$n = (3, -2, 5).$$

To find a point on the plane, choose convenient values for two variables and solve for the third.

#### Example

Find one point on

$$3x - 2y + 5z - 7 = 0.$$

Choose  $y = 0$  and  $z = 0$ . Then

$$3x - 7 = 0,$$

so

$$x = \frac{7}{3}.$$

Thus

$$\left(\frac{7}{3}, 0, 0\right)$$

lies on the plane.

There are infinitely many possible points on a plane. We choose one that is easy to compute.

#### Summary

When reading a plane equation  $Ax + By + Cz + D = 0$ , ask:

- Are  $A$ ,  $B$ , and  $C$  not all zero?
- What normal vector is visible?
- Can I find a convenient point on the plane?
- Does a given point satisfy the equation?
- Would the normal vector help with distance, angle, or parallelism questions?

The general equation is powerful because it stores the normal direction directly in its coefficients.

## 8.3 Planes through three points

Three points can determine a plane, but only if they are not collinear. If the three points lie on one line, then infinitely many planes pass through that line, and the plane is not uniquely determined.

This condition is important. In analytic geometry, we must check whether the data really define the object we want.

### From three points to two directions

Let

$$A, \quad B, \quad C$$

be three points in space. If they are not collinear, then the vectors

$$\overrightarrow{AB} = B - A, \quad \overrightarrow{AC} = C - A$$

are two non-parallel direction vectors in the plane.

Then the plane can be written parametrically as

$$P = A + s(B - A) + t(C - A), \quad s, t \in \mathbb{R}.$$

#### Example

Find a parametric equation of the plane through

$$A = (1, 0, 0), \quad B = (0, 1, 0), \quad C = (0, 0, 1).$$

First compute two direction vectors:

$$B - A = (-1, 1, 0),$$

$$C - A = (-1, 0, 1).$$

Therefore the plane is

$$P = (1, 0, 0) + s(-1, 1, 0) + t(-1, 0, 1).$$

In coordinates,

$$x = 1 - s - t, \quad y = s, \quad z = t.$$

This equation says that starting from  $A$ , we can move independently toward  $B$  and toward  $C$ .

### Finding the normal vector

To obtain the general equation of the plane, we need a normal vector. A normal vector must be perpendicular to both direction vectors in the plane.

If

$$u = B - A, \quad v = C - A,$$

then a normal vector is given by the cross product

$$n = u \times v.$$

This vector is perpendicular to both  $u$  and  $v$ .

### Example

Continue with

$$A = (1, 0, 0), \quad B = (0, 1, 0), \quad C = (0, 0, 1).$$

We found

$$u = (-1, 1, 0), \quad v = (-1, 0, 1).$$

Compute

$$u \times v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix}.$$

Thus

$$u \times v = (1, 1, 1).$$

So a normal vector is

$$n = (1, 1, 1).$$

Using the point  $A = (1, 0, 0)$ , the normal equation is

$$1(x - 1) + 1(y - 0) + 1(z - 0) = 0.$$

Therefore

$$x + y + z - 1 = 0.$$

Equivalently,

$$x + y + z = 1.$$

This example is a useful model. Three points give two directions. Two directions give a normal vector. A point and a normal vector give the equation.

### Checking non-collinearity

The points  $A$ ,  $B$ , and  $C$  are collinear exactly when the vectors  $B - A$  and  $C - A$  are parallel. Equivalently, their cross product is the zero vector:

$$(B - A) \times (C - A) = 0.$$

If the cross product is non-zero, the points determine a unique plane.

### Remark

A zero cross product means that the two direction vectors are not independent. In that case, the three points do not determine a unique plane.

### Summary

To find a plane through three points:

1. Choose one point as an anchor point, for example  $A$ .
2. Compute  $B - A$  and  $C - A$ .
3. Check that these vectors are not parallel.
4. Use them as direction vectors for parametric form, or compute their cross product to find a normal vector.

5. Use the normal vector and one point to write the general equation.

The method should be understood as a chain of meanings, not as a memorized procedure.

## 8.4 Parametric description of planes

The parametric description of a plane is based on movement. A point moves in two independent directions. This is the main difference between the parametrization of a line and the parametrization of a plane.

A line needs one parameter. A plane needs two parameters.

### Vector parametric form

Let  $P_0$  be a point in space and let  $u$  and  $v$  be two non-parallel vectors. The plane through  $P_0$  in the directions  $u$  and  $v$  is

$$P = P_0 + su + tv, \quad s, t \in \mathbb{R}.$$

The parameters  $s$  and  $t$  move independently.

#### Example

Consider the plane

$$P = (2, -1, 3) + s(1, 0, 2) + t(0, 3, -1).$$

For  $s = 0, t = 0$ , we get

$$P = (2, -1, 3).$$

For  $s = 1, t = 0$ , we move by  $(1, 0, 2)$ :

$$P = (3, -1, 5).$$

For  $s = 0, t = 1$ , we move by  $(0, 3, -1)$ :

$$P = (2, 2, 2).$$

All these points lie on the same plane.

The two direction vectors should not be parallel. If they are parallel, the two-parameter formula does not sweep out a plane; it only moves along one line.

### Coordinate parametric form

If

$$P_0 = (x_0, y_0, z_0), \quad u = (a_1, a_2, a_3), \quad v = (b_1, b_2, b_3),$$

then

$$(x, y, z) = (x_0, y_0, z_0) + s(a_1, a_2, a_3) + t(b_1, b_2, b_3).$$

In coordinate form,

$$x = x_0 + sa_1 + tb_1,$$

$$y = y_0 + sa_2 + tb_2,$$

$$z = z_0 + sa_3 + tb_3.$$

The same values of  $s$  and  $t$  must be used in all three coordinates.

## Checking whether a point lies on a parametric plane

To check whether a point lies on a parametric plane, we ask whether there exist values of the two parameters that produce the point.

### Example

Consider

$$(x, y, z) = (1, 0, 2) + s(1, 1, 0) + t(0, 2, 1).$$

Does the point  $(3, 4, 3)$  lie on the plane?

The coordinate equations are

$$x = 1 + s, \quad y = s + 2t, \quad z = 2 + t.$$

From  $x = 3$ , we get

$$s = 2.$$

From  $z = 3$ , we get

$$t = 1.$$

Then the middle coordinate gives

$$y = s + 2t = 2 + 2 = 4.$$

Therefore  $(3, 4, 3)$  lies on the plane.

A point belongs to the plane only if all coordinate equations are satisfied by the same pair  $(s, t)$ .

## From parametric form to normal form

If a plane is given parametrically by

$$P = P_0 + su + tv,$$

then a normal vector can be found by

$$n = u \times v.$$

Then the normal equation is

$$n \cdot (P - P_0) = 0.$$

This conversion is useful because distances and angles are usually easier to compute from normal vectors.

### Summary

When reading a parametric plane, ask:

- What point anchors the plane?
- What are the two direction vectors?
- Are the direction vectors non-parallel?
- What do the two parameters control?
- Can I find a normal vector using a cross product?
- Does a given point arise from one common pair of parameter values?

The parametric form describes a plane by motion. It is the language of two independent directions.

## 8.5 Parallelism, perpendicularity, and angles

The normal vector is the most efficient tool for comparing planes. A plane may be visually hard to compare with another plane, but its normal vector turns many geometric questions into vector questions.

Let

$$\pi_1 : A_1x + B_1y + C_1z + D_1 = 0,$$

and

$$\pi_2 : A_2x + B_2y + C_2z + D_2 = 0.$$

Their normal vectors are

$$n_1 = (A_1, B_1, C_1), \quad n_2 = (A_2, B_2, C_2).$$

### Parallel planes

Two planes are parallel when their normal vectors are parallel. This means that one normal vector is a scalar multiple of the other.

#### Definition

The planes  $\pi_1$  and  $\pi_2$  are parallel if there exists a non-zero scalar  $\lambda$  such that

$$n_2 = \lambda n_1.$$

#### Example

Consider

$$\pi_1 : 2x - y + 3z - 1 = 0$$

and

$$\pi_2 : 4x - 2y + 6z + 5 = 0.$$

Their normal vectors are

$$n_1 = (2, -1, 3), \quad n_2 = (4, -2, 6).$$

Since

$$n_2 = 2n_1,$$

the planes are parallel.

Parallel planes may be distinct, or they may be the same plane. To check whether they are the same, test whether a point from one plane satisfies the equation of the other.

### Perpendicular planes

Two planes are perpendicular when their normal vectors are perpendicular.

#### Definition

The planes  $\pi_1$  and  $\pi_2$  are perpendicular if

$$n_1 \cdot n_2 = 0.$$

### Example

Consider

$$\pi_1 : x + 2y - z = 0$$

and

$$\pi_2 : 2x - y + 0z + 3 = 0.$$

The normal vectors are

$$n_1 = (1, 2, -1), \quad n_2 = (2, -1, 0).$$

Their dot product is

$$n_1 \cdot n_2 = 1 \cdot 2 + 2 \cdot (-1) + (-1) \cdot 0 = 0.$$

Therefore the planes are perpendicular.

The test is simple because the normal vectors carry the orientation of the planes.

### Angle between planes

The angle between two planes is measured by the angle between their normal vectors. Usually we take the acute angle, so we use an absolute value in the formula.

### Theorem

Let  $\theta$  be the acute angle between two planes with normal vectors  $n_1$  and  $n_2$ . Then

$$\cos \theta = \frac{|n_1 \cdot n_2|}{\|n_1\| \|n_2\|}.$$

### Example

Determine the angle between

$$\pi_1 : x + 2y - 2z + 1 = 0$$

and

$$\pi_2 : 2x - y + z - 3 = 0.$$

The normal vectors are

$$n_1 = (1, 2, -2), \quad n_2 = (2, -1, 1).$$

Their dot product is

$$n_1 \cdot n_2 = 1 \cdot 2 + 2 \cdot (-1) + (-2) \cdot 1 = 2 - 2 - 2 = -2.$$

Their lengths are

$$\|n_1\| = \sqrt{1^2 + 2^2 + (-2)^2} = 3,$$

and

$$\|n_2\| = \sqrt{2^2 + (-1)^2 + 1^2} = \sqrt{6}.$$

Therefore

$$\cos \theta = \frac{|-2|}{3\sqrt{6}} = \frac{2}{3\sqrt{6}}.$$

So

$$\theta = \arccos \frac{2}{3\sqrt{6}}.$$

The calculation is a direct use of the dot product, but the interpretation is geometric: we compare the orientations of the planes through their normal directions.

### Summary

When comparing two planes, ask:

- What are their normal vectors?
- Are the normal vectors parallel?
- Are the normal vectors perpendicular?
- If an angle is needed, should it be the acute angle?
- Do proportional normal vectors mean distinct parallel planes or the same plane?

Plane comparison becomes much simpler when the normal vectors are read first.

## 8.6 Distance from a point to a plane

The distance from a point to a plane is the shortest distance from the point to any point of the plane. Geometrically, the shortest segment is perpendicular to the plane. This is why the normal vector appears in the distance formula.

Let the plane be given by

$$\pi : Ax + By + Cz + D = 0,$$

where  $(A, B, C) \neq (0, 0, 0)$ . A normal vector is

$$n = (A, B, C).$$

### Theorem

The distance from a point  $P_0 = (x_0, y_0, z_0)$  to the plane

$$Ax + By + Cz + D = 0$$

is

$$d(P_0, \pi) = \frac{|Ax_0 + By_0 + Cz_0 + D|}{\sqrt{A^2 + B^2 + C^2}}.$$

The numerator tells us how far the point is from satisfying the plane equation, scaled by the chosen equation. The denominator corrects for the length of the normal vector.

### Example

#### Example

For the plane

$$\pi : x - 2y + 2z - 4 = 0,$$

find the distance from

$$P = (3, 0, 1)$$

to  $\pi$ .

Here

$$A = 1, \quad B = -2, \quad C = 2, \quad D = -4.$$

Substitute the point into the numerator:

$$Ax_0 + By_0 + Cz_0 + D = 1 \cdot 3 + (-2) \cdot 0 + 2 \cdot 1 - 4 = 1.$$

The length of the normal vector is

$$\sqrt{1^2 + (-2)^2 + 2^2} = \sqrt{9} = 3.$$

Therefore

$$d(P, \pi) = \frac{|1|}{3} = \frac{1}{3}.$$

This calculation should be interpreted. The point is not on the plane because substitution does not give zero. The distance is the perpendicular distance, measured in the normal direction.

### Why the denominator appears

The same plane can be represented by many equivalent equations. For example,

$$x - 2y + 2z - 4 = 0$$

and

$$2x - 4y + 4z - 8 = 0$$

describe the same plane. If we used only the absolute value in the numerator, the second equation would give twice the value. But the geometric distance cannot depend on the chosen equation.

The denominator  $\sqrt{A^2 + B^2 + C^2}$  scales in the same way as the numerator, so the quotient remains unchanged.

#### Remark

The distance formula is geometric, even though it uses an equation. Equivalent equations of the same plane give the same distance.

### Signed distance

Without the absolute value, the quantity

$$\frac{Ax_0 + By_0 + Cz_0 + D}{\sqrt{A^2 + B^2 + C^2}}$$

is a signed distance. Its sign depends on the chosen orientation of the normal vector. Points on opposite sides of the plane have opposite signs.

This idea is useful when one wants to know not only how far a point is from the plane, but also on which side of the plane it lies.

## Summary

When computing distance from a point to a plane, ask:

- Is the plane in general form?
- What is the normal vector?
- What value is obtained by substituting the point?
- Did I divide by the length of the normal vector?
- Does the sign before the absolute value carry side information?

The formula is the three-dimensional version of the point-to-line distance formula in the plane.

## 8.7 A line and a plane in space

A line and a plane in space may intersect in one point, be parallel and disjoint, or the line may lie entirely in the plane. These cases can be detected by combining the parametric equation of the line with the equation of the plane.

Let the line be

$$\ell : P = A + tv, \quad t \in \mathbb{R},$$

and let the plane be

$$\pi : n \cdot (P - P_0) = 0.$$

The direction vector of the line is  $v$ , and the normal vector of the plane is  $n$ .

### Direction test

The dot product

$$n \cdot v$$

tells us how the line direction relates to the plane.

If

$$n \cdot v \neq 0,$$

then the line is not parallel to the plane and intersects it in exactly one point.

If

$$n \cdot v = 0,$$

then the line direction is perpendicular to the normal vector, so the line direction is parallel to the plane. In this case the line is either contained in the plane or parallel and disjoint from it.

### Remark

The condition  $n \cdot v = 0$  does not by itself say that the line lies in the plane. It says that the direction of the line is parallel to the plane. We must still check a point of the line.

### Intersection by substitution

The most direct method is to substitute the parametric equations of the line into the plane equation.

### Example

Check whether the line

$$\ell : x = 1 + 2t, \quad y = -1 + t, \quad z = 3 - t$$

intersects the plane

$$\pi : 2x - y + z - 4 = 0.$$

Substitute the line into the plane:

$$2(1 + 2t) - (-1 + t) + (3 - t) - 4 = 0.$$

Simplify:

$$2 + 4t + 1 - t + 3 - t - 4 = 0,$$

so

$$2 + 2t = 0.$$

Thus

$$t = -1.$$

Now substitute  $t = -1$  into the line:

$$x = 1 + 2(-1) = -1,$$

$$y = -1 + (-1) = -2,$$

$$z = 3 - (-1) = 4.$$

Therefore the intersection point is

$$(-1, -2, 4).$$

The parameter value is not the final answer. It is the value that gives the point of intersection.

### Contained or parallel disjoint

If substitution gives an identity such as

$$0 = 0,$$

then every point of the line satisfies the plane equation, so the line lies in the plane.

If substitution gives a contradiction such as

$$0 = 5,$$

then no point of the line satisfies the plane equation, so the line is parallel to the plane and disjoint from it.

### Example

Let

$$\ell : P = (1, 0, 0) + t(1, 1, 0)$$

and

$$\pi : z = 0.$$

Every point of the line has  $z = 0$ , so the line lies in the plane.

Now let

$$\ell : P = (1, 0, 1) + t(1, 1, 0)$$

with the same plane  $z = 0$ . Every point of this line has  $z = 1$ , so the line is parallel to the plane and disjoint from it.

The two lines have the same direction. Their position relative to the plane is different.

### Parameter-dependent cases

Sometimes a line or plane depends on a parameter. The same tests still apply, but the answer may depend on the parameter value.

#### Example

Consider

$$\ell(\lambda) : x = \lambda + t, \quad y = 1 + 2t, \quad z = 2 - t$$

and

$$\pi : x - (\lambda - 1)y + z - 3 = 0.$$

The direction vector of the line is

$$v = (1, 2, -1).$$

A normal vector of the plane is

$$n = (1, -(\lambda - 1), 1) = (1, 1 - \lambda, 1).$$

Compute

$$n \cdot v = 1 + 2(1 - \lambda) - 1 = 2 - 2\lambda.$$

If

$$\lambda \neq 1,$$

then  $n \cdot v \neq 0$ , so the line intersects the plane in one point.

If  $\lambda = 1$ , then the line direction is parallel to the plane. Substitute  $\lambda = 1$  into the line and plane. The plane becomes

$$x + z - 3 = 0.$$

The line becomes

$$x = 1 + t, \quad y = 1 + 2t, \quad z = 2 - t.$$

Then

$$x + z - 3 = (1 + t) + (2 - t) - 3 = 0.$$

This is true for every  $t$ . Therefore, for  $\lambda = 1$ , the line is contained in the plane.

Parameter problems are not a new method. They are the same method with cases separated carefully.

#### Summary

For a line and a plane, ask:

- What is the direction vector of the line?
- What is the normal vector of the plane?
- Is  $n \cdot v$  zero or non-zero?

- If non-zero, what point is obtained by substitution?
- If zero, does one point of the line satisfy the plane equation?
- Are there parameter values where the relationship changes?

The line-plane relationship is a direct application of the languages developed in the chapters on lines and planes.

## 8.8 Choosing the right plane description

A plane can be described in several ways. The best description depends on what information is given and what question is being asked.

This is the same lesson that appeared for lines. A formula is not the object itself. A formula is a language for reading the object.

### When a normal vector is given

If a point and a normal vector are given, use the normal form:

$$n \cdot (P - P_0) = 0.$$

In coordinates, if  $n = (A, B, C)$ , this becomes

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0.$$

This form is usually the fastest route to the general equation.

### When three points are given

If three non-collinear points are given, first build two direction vectors:

$$B - A, \quad C - A.$$

Then choose whether to write the plane parametrically or compute a normal vector using the cross product:

$$n = (B - A) \times (C - A).$$

The normal vector then gives the general equation.

### When an equation is given

If a plane is given by

$$Ax + By + Cz + D = 0,$$

then the normal vector is immediately visible:

$$n = (A, B, C).$$

This is useful for distances, angles, parallelism, and perpendicularity.

## Which form helps which question?

| Question                                | Useful description                      |
|-----------------------------------------|-----------------------------------------|
| Plane through point and normal vector   | normal form                             |
| Plane through three points              | two direction vectors and cross product |
| Generate points on the plane            | parametric form                         |
| Check whether a point lies on the plane | general equation or parametric form     |
| Find distance from point to plane       | general equation                        |
| Angle between two planes                | normal vectors                          |
| Parallel or perpendicular planes        | normal vectors                          |
| Line-plane intersection                 | substitute line into plane equation     |

The table should be read as a guide. A good solution often converts from one description to another.

## A complete reading example

Consider the plane

$$2x - y + z - 4 = 0.$$

From the equation we immediately read a normal vector:

$$n = (2, -1, 1).$$

A point on the plane can be found by choosing  $y = 0$  and  $z = 0$ . Then

$$2x - 4 = 0,$$

so

$$x = 2.$$

Thus  $(2, 0, 0)$  lies on the plane.

To find two direction vectors, choose any two vectors perpendicular to  $n$ . For example,

$$u = (1, 2, 0)$$

works because

$$(2, -1, 1) \cdot (1, 2, 0) = 2 - 2 + 0 = 0.$$

Also,

$$v = (0, 1, 1)$$

works because

$$(2, -1, 1) \cdot (0, 1, 1) = 0 - 1 + 1 = 0.$$

Therefore one parametric description is

$$P = (2, 0, 0) + s(1, 2, 0) + t(0, 1, 1).$$

The same plane has been read in two languages: the general equation shows the normal vector, while the parametric equation shows directions inside the plane.

### Summary

At the end of this chapter, remember:

- A plane needs either two independent directions or one normal direction.

- The equation  $Ax + By + Cz + D = 0$  contains the normal vector  $(A, B, C)$ .
- Three points determine a plane only when they are not collinear.
- Distances and angles are easiest with normal vectors.
- A line-plane relationship is studied by combining the line parametrization with the plane equation.

The chapter completes the analytic geometry sequence: points, vectors, lines, and planes are now all described through coordinates and vector language.

# Chapter 9

## Sequences and Functions

### 9.1 What a function is

The word “function” appears very often in mathematics, but it is sometimes introduced too quickly. A student may first meet functions as formulas such as  $f(x) = 2x + 1$ , and then begin to think that a function is the same thing as a formula. This is not quite right.

A formula can describe a function, but a function is more basic than a formula. A function is a rule of assignment. It tells us how to assign one output to each allowed input.

#### Definition

Let  $X$  and  $Y$  be sets. A function from  $X$  to  $Y$  is a rule that assigns to every element  $x \in X$  exactly one element of  $Y$ .

We write

$$f : X \rightarrow Y.$$

The set  $X$  is called the domain, and the set  $Y$  is called the codomain.

The phrase “exactly one” is essential. For each allowed input, the function must give one output, not two different outputs and not no output at all. This is the basic discipline of functions.

#### Inputs and outputs

If  $f : X \rightarrow Y$ , then an element  $x \in X$  is an input. The value assigned to this input is written as  $f(x)$ . This value belongs to  $Y$ .

We may write this assignment as

$$x \mapsto f(x).$$

This notation should be read as: the input  $x$  is sent to the output  $f(x)$ .

#### Example

Let

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x^2 + 1.$$

The input 3 is sent to

$$f(3) = 3^2 + 1 = 10.$$

The input  $-2$  is sent to

$$f(-2) = (-2)^2 + 1 = 5.$$

The rule assigns one real number to each real input.

The notation  $f(3) = 10$  describes one value of the function. It does not describe the whole function. The whole function is the entire rule that works for every allowed input.

#### A function is not only a formula

Functions can be described in several ways. A formula is common, but it is not the only possibility. A function may be described by a table, by a graph, by a verbal rule, or by a

recursive rule.

### Example

Suppose a small machine accepts the inputs 1, 2, 3 and returns the following outputs:

|        |   |   |   |
|--------|---|---|---|
| $x$    | 1 | 2 | 3 |
| $f(x)$ | 4 | 4 | 7 |

This table describes a function with domain  $\{1, 2, 3\}$ . Each input has exactly one output. There is no need for a formula such as  $f(x) = \dots$  in order for this assignment to be a function.

A function is therefore not a decorative name for an algebraic expression. It is a structured assignment from one set to another.

### What can go wrong

Not every relation between two sets is a function. The problem occurs when one input is assigned more than one output, or when an allowed input is assigned no output.

### Example

Suppose the input 2 is assigned both 5 and 8. Then the assignment is not a function, because one input has two outputs.

Suppose instead that the domain is  $\{1, 2, 3\}$ , but no output is assigned to 3. Then the assignment is also not a function, because an allowed input has no output.

The definition of a function is strict because it allows us to ask clear questions. If we ask for  $f(2)$ , there must be one definite answer.

### Summary

When reading a function, ask:

- What is the domain?
- What is the codomain?
- What are the allowed inputs?
- What output is assigned to each input?
- Is each allowed input assigned exactly one output?
- Is the function described by a formula, a table, a graph, or another rule?

The main idea is simple but important: a function is a controlled way of assigning outputs to inputs.

## 9.2 Domain, codomain, and values

The notation

$$f : X \rightarrow Y$$

contains more information than it may first seem. It tells us where inputs are taken from and where outputs are expected to live. Before computing with a function, we should learn to read

this notation.

## Domain and codomain

In the notation

$$f : X \rightarrow Y,$$

the set  $X$  is the domain. It contains the inputs for which the function is defined. The set  $Y$  is the codomain. It is the set in which the outputs are declared to live.

If  $x \in X$ , then the value of the function at  $x$  is written

$$f(x).$$

This value belongs to  $Y$ :

$$f(x) \in Y.$$

### Example

Let

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = 2x - 1.$$

The domain is  $\mathbb{R}$ . This means every real number is an allowed input. The codomain is also  $\mathbb{R}$ . For example,

$$f(4) = 2 \cdot 4 - 1 = 7.$$

The input is 4, and the output is 7.

The domain is part of the function. Two rules may look the same as formulas but describe different functions if their domains are different.

### Example

Consider the rule

$$f(x) = x^2.$$

If the domain is  $\mathbb{R}$ , then negative numbers, zero, and positive numbers are all allowed inputs. If the domain is  $[0, \infty)$ , then only non-negative inputs are allowed. The formula is the same, but the function is not the same because the domain has changed.

## Range or image

The codomain is the declared target set. The range, also called the image, is the set of outputs that are actually obtained.

### Definition

For a function  $f : X \rightarrow Y$ , the range or image of  $f$  is the set

$$f(X) = \{f(x) : x \in X\}.$$

It consists of all values that the function actually takes.

The range may be smaller than the codomain.

### Example

Let

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x^2.$$

The codomain is  $\mathbb{R}$ , but the range is

$$[0, \infty),$$

because a square is never negative and every non-negative real number is the square of some real number.

This distinction matters. The codomain is part of how the function is declared. The range is discovered by studying what values the function actually produces.

### The function and its values

A common source of confusion is the difference between  $f$  and  $f(x)$ . The symbol  $f$  denotes the whole function. The expression  $f(x)$  denotes the value of the function at the input  $x$ .

For the function

$$f(x) = x^2 + 1,$$

the value at 3 is

$$f(3) = 10.$$

The value 10 is a number. The function  $f$  is not a number; it is the entire assignment rule.

### Remark

The expression  $f(x)$  should be read as “the value of  $f$  at  $x$ ,” not as another name for the whole function. The function is  $f$ . The value at an input is  $f(x)$ .

### Arguments and values

The input of a function is often called an argument. If we write  $f(2)$ , then 2 is the argument. If we write  $f(a)$ , then  $a$  is the argument. If we write  $f(x + h)$ , then the whole expression  $x + h$  is the argument.

This point will become important later. When we substitute into a function, we substitute the entire argument, not only one letter.

### Example

Let

$$f(x) = x^2.$$

Then

$$f(3) = 3^2 = 9,$$

$$f(a) = a^2,$$

and

$$f(x + h) = (x + h)^2.$$

The last expression is not  $x^2 + h$ . The whole argument  $x + h$  is placed into the rule.

## Summary

When reading function notation, ask:

- What is the domain?
- What is the codomain?
- What values are actually obtained?
- Am I talking about the whole function  $f$  or one value  $f(x)$ ?
- What is the entire argument being substituted into the function?

Careful reading of notation prevents many later mistakes. Functions are not difficult because the symbols are complicated; they become difficult when the roles of the symbols are not read.

## 9.3 Function notation and evaluation

Function notation is compact. This is useful, but it also means that we must read it carefully. Many mistakes with functions are not caused by difficult calculations. They are caused by not reading what is being substituted into the function.

If a function is given by a rule such as

$$f(x) = 2x - 1,$$

then the letter  $x$  is a placeholder for an input. When we evaluate the function, we replace that placeholder by the chosen argument.

### Evaluating at numbers

To evaluate a function at a number, substitute that number into the rule.

#### Example

Let

$$f(x) = 2x - 1.$$

Then

$$f(4) = 2 \cdot 4 - 1 = 7.$$

Also,

$$f(-3) = 2 \cdot (-3) - 1 = -6 - 1 = -7.$$

The notation  $f(4)$  means the value of the function at the input 4.

The parentheses are not decoration. They indicate the argument of the function.

### Evaluating at expressions

The argument of a function does not have to be a number. It may be a letter or a more complicated expression. The rule is the same: replace the input placeholder by the entire argument.

### Example

Let

$$f(x) = 2x - 1.$$

Then

$$f(a) = 2a - 1.$$

Also,

$$f(x + h) = 2(x + h) - 1 = 2x + 2h - 1.$$

The whole expression  $x + h$  is substituted in place of  $x$ .

This habit is especially important in calculus, where expressions such as  $f(x + h)$  appear frequently. If the whole argument is not substituted correctly, later formulas become meaningless.

### Example

Let

$$g(x) = x^2.$$

Then

$$g(x + h) = (x + h)^2.$$

Expanding,

$$g(x + h) = x^2 + 2xh + h^2.$$

It would be wrong to write  $g(x + h) = x^2 + h$ , because the function squares the whole input.

A function acts on its entire argument. This is one of the simplest and most important rules of function notation.

### Different letters, same role

The letter used in a formula is not sacred. The function

$$f(x) = x^2 + 1$$

can also be described by writing

$$f(t) = t^2 + 1.$$

The input letter changed, but the function did not. What matters is the rule: take the input, square it, and add 1.

### Example

If

$$f(x) = x^2 + 1,$$

then

$$f(t) = t^2 + 1, \quad f(u) = u^2 + 1, \quad f(3) = 3^2 + 1 = 10.$$

The symbols  $x, t, u$  are placeholders. They indicate where the input goes.

This does not mean that letters never matter. In a larger expression, each letter may already have a role. The point is only that the name of the input variable is not the function itself.

## Functions with named rules

Sometimes functions are defined without writing  $f(x) = \dots$  first. For example, we may say: let  $f$  be the function that assigns to each real number its square plus one. This means exactly the same rule:

$$f(x) = x^2 + 1.$$

A verbal description may be just as precise as a formula if it clearly assigns one output to each input.

### Example

Let  $f$  assign to each positive integer the number of its positive divisors. Then

$$f(1) = 1, \quad f(2) = 2, \quad f(6) = 4,$$

because 6 has the positive divisors 1, 2, 3, 6.

This function is not introduced by a simple algebraic formula, but it is still a function.

## Reading equations involving functions

An equation such as

$$f(x) = 0$$

asks for inputs whose output is zero. It is not asking for the function itself to disappear. It asks for the arguments  $x$  at which the value of the function equals zero.

### Example

Let

$$f(x) = x^2 - 4.$$

The equation

$$f(x) = 0$$

means

$$x^2 - 4 = 0.$$

Thus

$$x = -2 \quad \text{or} \quad x = 2.$$

These are the inputs where the function has value zero.

This kind of question will reappear often. Zeros of a function, intersections with axes, solutions of equations, and sign changes are all connected to evaluating functions carefully.

### Summary

When evaluating a function, ask:

- What is the rule of the function?
- What is the entire argument being substituted?
- Am I substituting a number, a letter, or an expression?
- Does the function act on the whole argument?
- Am I looking for a value  $f(a)$ , or solving an equation such as  $f(x) = 0$ ?

Function notation becomes manageable when we treat it as careful substitution into a rule.

## 9.4 Graphs as pictures of functions

A graph is a picture of a function. More precisely, it is a picture of input-output pairs. If the input is  $x$  and the output is  $f(x)$ , then the corresponding point on the graph is

$$(x, f(x)).$$

This simple idea is the key to reading graphs. A graph is not a drawing to guess from. It is a geometric representation of values of a function.

### Definition

Let  $f : D \rightarrow \mathbb{R}$ , where  $D \subseteq \mathbb{R}$ . The graph of  $f$  is the set

$$\{(x, f(x)) : x \in D\}.$$

Each point on the graph has first coordinate equal to an input and second coordinate equal to the corresponding output.

If the point  $(2, 5)$  lies on the graph of  $f$ , then

$$f(2) = 5.$$

If  $f(2) = 5$ , then  $(2, 5)$  lies on the graph. These are two ways of saying the same thing.

### Reading values from a graph

To read  $f(a)$  from a graph, we look at the input  $a$  on the horizontal axis and find the height of the graph above that input. That height is the value  $f(a)$ .

### Example

Suppose a graph contains the points

$$(-1, 3), \quad (0, 1), \quad (2, 5).$$

Then the graph tells us that

$$f(-1) = 3, \quad f(0) = 1, \quad f(2) = 5.$$

The first coordinate is the input. The second coordinate is the output.

This reading is basic, but it is not trivial. Many later questions about functions depend on reading coordinates of points on a graph correctly.

### Intercepts

The point where a graph meets the vertical axis has input 0. Therefore, if the point exists, it has the form

$$(0, f(0)).$$

This is called the  $y$ -intercept.

A point where the graph meets the horizontal axis has output 0. Such points have the form

$$(x, 0).$$

They correspond to solutions of the equation

$$f(x) = 0.$$

These are called zeros or roots of the function.

### Example

Let

$$f(x) = x^2 - 4.$$

The  $y$ -intercept is found by computing

$$f(0) = 0^2 - 4 = -4.$$

So the graph meets the  $y$ -axis at

$$(0, -4).$$

The zeros are found from

$$x^2 - 4 = 0,$$

so

$$x = -2 \quad \text{or} \quad x = 2.$$

Thus the graph meets the  $x$ -axis at

$$(-2, 0) \quad \text{and} \quad (2, 0).$$

The graph and the equation give the same information in different forms. The graph shows the intercepts visually. The equation allows us to compute them exactly.

### The vertical line test

A graph represents a function of  $x$  only if each input  $x$  has at most one output. Visually, this means that no vertical line should intersect the graph more than once.

### Theorem

A curve in the plane represents the graph of a function of  $x$  if every vertical line intersects the curve at most once.

The reason is the definition of a function. One input cannot have two different outputs.

### Example

A circle is not the graph of a function of  $x$  on its full domain, because many vertical lines meet the circle in two points. For the same value of  $x$ , there may be two possible values of  $y$ .

This does not mean that circles are not important. It means that a full circle is not the graph of one function  $y = f(x)$ .

### Graphs and domains

The domain of a function is visible in the graph as the set of  $x$ -values for which the graph exists. If there is no point above a certain value of  $x$ , then that value is not in the domain.

The range is visible as the set of  $y$ -values reached by the graph.

### Example

For the function

$$f(x) = \sqrt{x},$$

real outputs exist only for  $x \geq 0$ . Therefore the graph begins at  $x = 0$  and continues to the right. The domain is

$$[0, \infty),$$

and the range is also

$$[0, \infty).$$

### Summary

When reading a graph, ask:

- What input does the horizontal coordinate represent?
- What output does the vertical coordinate represent?
- What does the point  $(a, b)$  say about the value of the function?
- Where does the graph meet the axes?
- What  $x$ -values belong to the domain?
- What  $y$ -values belong to the range?
- Does the graph pass the vertical line test?

A graph should be read as information, not only as a shape.

## 9.5 Real functions and basic examples

In this course we will often study functions whose inputs and outputs are real numbers. Such functions are called real functions of a real variable. They are central in calculus because they allow us to describe quantities that depend on one real parameter.

A typical notation is

$$f : D \rightarrow \mathbb{R}, \quad D \subseteq \mathbb{R}.$$

This means that the inputs are taken from a set  $D$  of real numbers, and the outputs are real numbers.

### Natural domains

Sometimes the domain is given explicitly. Sometimes it is understood from the formula. When a formula is given and no domain is stated, we often look for the largest set of real inputs for which the formula makes sense. This is called the natural domain.

### Example

For

$$f(x) = 2x + 1,$$

the natural domain is  $\mathbb{R}$ . For

$$g(x) = \frac{1}{x},$$

the natural domain is  $\mathbb{R} \setminus \{0\}$ . For

$$h(x) = \sqrt{x},$$

the natural domain over the real numbers is  $[0, \infty)$ .

These examples show that the formula is not the whole story. We must also ask where the formula is meaningful.

### Affine and linear functions

A function of the form

$$f(x) = mx + b$$

is called an affine function. Its graph is a line. The number  $m$  is the slope, and  $b = f(0)$  is the vertical intercept.

In elementary school terminology, functions of the form  $mx + b$  are often called linear. In the strict language of linear algebra, however, a real function of one variable is linear only when

$$f(x) = mx,$$

so that  $f(0) = 0$ .

#### Remark

This distinction matters because this book also studies matrices and linear algebra. We use *affine* for  $f(x) = mx + b$ , and reserve *linear* in the strict sense for  $f(x) = mx$ .

#### Example

The function

$$f(x) = 3x - 2$$

is affine but not linear in the strict sense, because

$$f(0) = -2 \neq 0.$$

Its slope is 3, so the output increases by 3 whenever the input increases by 1.

### Quadratic, reciprocal, and square root functions

A quadratic function has the form

$$f(x) = ax^2 + bx + c, \quad a \neq 0.$$

Its graph is a parabola. The reciprocal function  $1/x$  illustrates a domain exclusion and a vertical asymptote, while  $\sqrt{x}$  illustrates a function whose real domain begins at an endpoint.

#### Example

For  $f(x) = x^2$ ,

$$f(-2) = f(2) = 4.$$

For  $g(x) = 1/x$ , the value  $g(0)$  is not defined. For  $h(x) = \sqrt{x}$ , negative real inputs are excluded.

## Angles and trigonometric functions

Calculus formulas use angles measured in radians. For a circle of radius  $r$ , an angle  $\theta$  measured in radians satisfies

$$s = r\theta,$$

where  $s$  is the corresponding arc length. In particular,

$$2\pi \text{ radians} = 360^\circ, \quad \pi \text{ radians} = 180^\circ.$$

On the unit circle, the point determined by an angle  $x$  has coordinates

$$(\cos x, \sin x).$$

Both sine and cosine are defined on  $\mathbb{R}$ , take values in  $[-1, 1]$ , and satisfy

$$\sin(x + 2\pi) = \sin x, \quad \cos(x + 2\pi) = \cos x,$$

and

$$\sin^2 x + \cos^2 x = 1.$$

The tangent function is

$$\tan x = \frac{\sin x}{\cos x}.$$

It is defined only when  $\cos x \neq 0$ , so its natural domain is

$$\mathbb{R} \setminus \left\{ \frac{\pi}{2} + k\pi : k \in \mathbb{Z} \right\}.$$

### Remark

The standard limit

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

and the usual derivative formulas for trigonometric functions have their standard form only when angles are measured in radians.

## Exponential functions

For  $a > 0$ ,  $a \neq 1$ , the exponential function with base  $a$  is

$$f(x) = a^x.$$

Its domain is  $\mathbb{R}$ , and its values are positive. The exponent laws include

$$a^{x+y} = a^x a^y, \quad a^{x-y} = \frac{a^x}{a^y}, \quad (a^x)^y = a^{xy}.$$

If  $a > 1$ , the function is increasing. If  $0 < a < 1$ , it is decreasing.

A distinguished base is

$$e \approx 2.71828.$$

The natural exponential function  $e^x$  has a central role in calculus because its derivative is itself. It also appears in the standard limit

$$\lim_{x \rightarrow 0} (1 + x)^{1/x} = e.$$

### Example

The model

$$N(t) = N_0 e^{kt}$$

describes exponential growth when  $k > 0$  and exponential decay when  $k < 0$ . Its initial value is  $N(0) = N_0$ .

## Logarithmic functions

The logarithm with base  $a > 0$ ,  $a \neq 1$ , is the inverse of the exponential function  $a^x$ . The statement

$$y = \log_a x$$

means

$$a^y = x.$$

Because exponential values are positive, logarithms are defined only for  $x > 0$ .

The logarithm with base  $e$  is the natural logarithm  $\ln x$ . The inverse relationships are

$$\ln(e^x) = x$$

for every real  $x$ , and

$$e^{\ln x} = x$$

for  $x > 0$ . The basic laws include

$$\ln(xy) = \ln x + \ln y, \quad \ln \frac{x}{y} = \ln x - \ln y,$$

and

$$\ln(x^p) = p \ln x,$$

whenever all expressions are defined.

### Remark

The condition  $x > 0$  is essential for  $\ln x$ . In antiderivatives,  $\ln |x|$  appears because its derivative is  $1/x$  on both intervals  $(-\infty, 0)$  and  $(0, \infty)$ .

## Inverse trigonometric functions

Trigonometric functions are periodic, so restricted domains are needed before inverse functions can be defined. The standard choices are

$$\arcsin : [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right],$$

$$\arccos : [-1, 1] \rightarrow [0, \pi],$$

and

$$\arctan : \mathbb{R} \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$$

For example,

$$\arcsin \frac{1}{2} = \frac{\pi}{6}, \quad \arctan 1 = \frac{\pi}{4}.$$

## Piecewise functions

A function may use different formulas on different parts of its domain. Such functions are called piecewise-defined functions.

### Example

Define

$$f(x) = \begin{cases} x + 1, & x < 0, \\ x^2, & x \geq 0. \end{cases}$$

Then  $f(-2) = -1$ , while  $f(3) = 9$ . The input determines which rule applies.

### Summary

When studying a real function, ask:

- What is the domain?
- Is the function algebraic, trigonometric, exponential, logarithmic, or piecewise?
- If its graph is a line, is the function affine or linear in the strict sense?
- Are angles measured in radians?
- Are logarithmic arguments positive?
- What restricted range defines an inverse trigonometric function?
- Which rule applies to a piecewise-defined input?

Real functions are the main objects of calculus. Their study begins with reading domains, formulas, inverse relationships, and graph structure carefully.

## 9.6 Reading information from graphs

A graph can contain a large amount of information, even when we do not know the formula of the function. Reading a graph means extracting that information carefully.

The aim is not to guess. The aim is to connect visible geometric features with statements about the function.

### Reading values

If a point  $(a, b)$  lies on the graph of  $f$ , then

$$f(a) = b.$$

The first coordinate is the input. The second coordinate is the output.

### Example

Suppose the graph of  $f$  contains the point  $(3, -2)$ . Then

$$f(3) = -2.$$

If the graph also contains  $(-1, 4)$ , then

$$f(-1) = 4.$$

This is the most basic skill in reading graphs. It should become automatic: point on graph means input-output pair.

## Zeros and signs

A zero of a function is an input where the output is zero. On the graph, zeros are the points where the graph meets the horizontal axis.

If

$$f(a) = 0,$$

then  $(a, 0)$  is on the graph.

The sign of  $f(x)$  can also be read from the graph. Where the graph lies above the horizontal axis,  $f(x) > 0$ . Where it lies below the horizontal axis,  $f(x) < 0$ .

### Example

Suppose a graph crosses the horizontal axis at  $x = -2$  and  $x = 5$ . Then

$$f(-2) = 0, \quad f(5) = 0.$$

If the graph is above the axis between these points, then

$$f(x) > 0 \quad \text{for } -2 < x < 5.$$

If it is below the axis to the right of 5, then

$$f(x) < 0 \quad \text{for } x > 5$$

on the visible part of the graph.

A graph therefore answers not only numerical questions, but also qualitative questions about positivity and negativity.

## Increasing and decreasing behavior

A function is increasing on an interval if its values go upward as the input moves to the right. It is decreasing if its values go downward as the input moves to the right.

At this stage, we do not need a formal definition with quantifiers. The basic reading is visual: moving from left to right, does the graph rise or fall?

### Example

If a graph rises from left to right on the interval  $[1, 4]$ , then the function is increasing on that interval. This means that larger inputs in that interval give larger function values. If the graph falls from left to right on  $[4, 7]$ , then the function is decreasing on that interval.

Later, derivatives will give a powerful method for studying increasing and decreasing behavior. But the geometric idea should be understood first.

## Maximum and minimum values

A graph can also show where a function has high or low values. A local maximum is a point where the graph is higher than nearby points. A local minimum is a point where the graph is lower than nearby points.

### Example

Suppose a graph reaches a visible peak at the point  $(2, 5)$ . Then near  $x = 2$ , the function has a local maximum value 5.

If the graph reaches a visible valley at  $(-1, -3)$ , then near  $x = -1$ , the function has a local minimum value  $-3$ .

The input tells us where the maximum or minimum occurs. The output tells us what the maximum or minimum value is.

### Graph reading is not formula guessing

Sometimes a graph is given without a formula. This is not a weakness. A graph can still answer many questions.

From a graph we may read:

- values such as  $f(2)$ ,
- zeros of the function,
- intervals where the function is positive or negative,
- intervals where the function is increasing or decreasing,
- approximate maximum and minimum values,
- the domain and range visible in the picture.

But we should be honest about precision. A graph may show approximate values, while a formula may give exact values. Both forms of information are useful, but they are not the same.

#### Remark

Reading a graph means translating geometric features into mathematical statements. It does not always mean finding a formula.

#### Summary

When reading a graph, ask:

- What is the input coordinate?
- What is the output coordinate?
- Where is the graph equal to zero?
- Where is the graph above or below the horizontal axis?
- Where does the graph rise or fall as we move from left to right?
- Where are visible peaks or valleys?
- Am I reading exact information or approximate information?

A graph is a language. Learning to read it is part of learning functions.

## 9.7 Sequences as functions

Sequences are often introduced as lists of numbers:

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$$

or

$$2, 4, 6, 8, \dots$$

This way of writing sequences is useful, but it can hide the most important idea. A sequence is not a completely different kind of object from a function. A sequence is a special kind of function.

The difference is in the domain.

### Definition

A real sequence is a function

$$a : \mathbb{N} \rightarrow \mathbb{R}.$$

For each natural number  $n$ , the function assigns a real number  $a(n)$ . Instead of writing  $a(n)$ , we usually write

$$a_n.$$

Thus

$$a_n = a(n).$$

The notation changes, but the idea is the same: one input gives one output. For sequences, the inputs are natural numbers. The values are called terms of the sequence.

### Terms of a sequence

If  $a : \mathbb{N} \rightarrow \mathbb{R}$ , then

$$a_1, \quad a_2, \quad a_3, \quad \dots$$

are the values of the function at the inputs  $1, 2, 3, \dots$ . Depending on convention, some sequences start at  $n = 0$ . What matters is that the starting index must be clear.

### Example

Let

$$a_n = 2n + 1, \quad n \in \mathbb{N}.$$

Then

$$a_1 = 2 \cdot 1 + 1 = 3,$$

$$a_2 = 2 \cdot 2 + 1 = 5,$$

and

$$a_3 = 2 \cdot 3 + 1 = 7.$$

The sequence begins

$$3, 5, 7, 9, \dots$$

The formula  $a_n = 2n + 1$  tells us the value of the function  $a$  at the natural number  $n$ . It is function notation written in sequence language.

### Index and value

In a sequence, the input is called the index. The output is called the term. These two roles should not be confused.

In the sequence

$$a_n = 2n + 1,$$

the number  $n$  is the index. The number  $a_n$  is the term at index  $n$ . For example, when  $n = 4$ ,

$$a_4 = 9.$$

The index is 4, and the term is 9.

### Remark

The subscript in  $a_n$  is not decoration. It records the input of the sequence-function. The expression  $a_5$  means the value of the sequence at index 5.

## Graphs of sequences

Since a sequence is a function, it can have a graph. But its graph looks different from the graph of a function defined on an interval. For a sequence  $a : \mathbb{N} \rightarrow \mathbb{R}$ , the graph consists of points

$$(n, a_n), \quad n \in \mathbb{N}.$$

These points are separated because the inputs are separated.

### Example

For the sequence

$$a_n = \frac{1}{n}, \quad n \geq 1,$$

the graph consists of the points

$$(1, 1), \quad \left(2, \frac{1}{2}\right), \quad \left(3, \frac{1}{3}\right), \quad \left(4, \frac{1}{4}\right), \quad \dots$$

We do not automatically draw a continuous curve through these points, because the sequence is not defined for every real number between 1 and 2. It is defined only at natural-number inputs.

This is one of the main differences between ordinary real functions on intervals and sequences. A sequence has a discrete domain.

## Sequences as ordered data

Because the domain is  $\mathbb{N}$ , the values of a sequence come with a natural order:

$$a_1, \quad a_2, \quad a_3, \quad \dots$$

We can ask how the terms change as  $n$  increases. Do they grow? Do they decrease? Do they approach a number? Do they repeat a pattern?

These questions are function questions, but adapted to the special domain  $\mathbb{N}$ .

### Summary

When reading a sequence, ask:

- What is the starting index?
- What is the domain of the sequence?
- What is the formula or rule for  $a_n$ ?
- What are the first few terms?
- What is the difference between the index  $n$  and the value  $a_n$ ?
- Should the graph be read as separated points rather than a continuous curve?

A sequence is a function with discrete input. This simple fact explains both why sequences belong in a chapter on functions and why they are studied with some different questions.

## 9.8 Explicit and recursive sequences

A sequence is a function on natural-number inputs. To define such a function, we must say how its terms are obtained. There are two common ways to do this: an explicit formula and a recursive definition.

Both are useful, but they answer different practical questions.

### Explicit formulas

An explicit formula gives  $a_n$  directly in terms of  $n$ . If we want the tenth term, we substitute  $n = 10$ . If we want the hundredth term, we substitute  $n = 100$ .

#### Definition

An explicit formula for a sequence gives the term  $a_n$  directly as an expression depending on the index  $n$ .

#### Example

Let

$$a_n = 3n - 2, \quad n \geq 1.$$

Then

$$a_1 = 3 \cdot 1 - 2 = 1,$$

$$a_2 = 3 \cdot 2 - 2 = 4,$$

$$a_3 = 3 \cdot 3 - 2 = 7.$$

The sequence begins

$$1, 4, 7, 10, \dots$$

If we want the tenth term, we compute

$$a_{10} = 3 \cdot 10 - 2 = 28.$$

The advantage of an explicit formula is direct access. We can compute a distant term without computing all previous terms.

### Recursive definitions

A recursive definition describes how to obtain the next term from earlier terms. Such a definition needs a starting value. Without a starting value, the sequence is not determined.

#### Definition

A recursive definition of a sequence gives one or more initial terms and a rule for computing later terms from earlier terms.

#### Example

Define a sequence by

$$a_1 = 5, \quad a_{n+1} = a_n + 2.$$

Then

$$\begin{aligned}a_1 &= 5, \\a_2 &= a_1 + 2 = 7, \\a_3 &= a_2 + 2 = 9, \\a_4 &= a_3 + 2 = 11.\end{aligned}$$

The sequence begins

$$5, 7, 9, 11, \dots$$

The recursive rule says how to move from one term to the next. It does not immediately give  $a_{100}$  unless we either compute many previous terms or find an explicit formula.

### Why the initial term matters

The recurrence relation alone may not define one sequence. It may define a whole family of possible sequences, depending on the starting value.

#### Example

The rule

$$a_{n+1} = a_n + 2$$

does not determine a unique sequence by itself.

If  $a_1 = 1$ , the sequence begins

$$1, 3, 5, 7, \dots$$

If  $a_1 = 10$ , the sequence begins

$$10, 12, 14, 16, \dots$$

The recursive rule is the same, but the sequences are different because the starting value is different.

This is similar to describing motion. A rule for how to move is not enough if we do not know where we start.

### Geometric-type recursion

Some recursions multiply instead of add.

#### Example

Define

$$a_1 = 3, \quad a_{n+1} = 2a_n.$$

Then

$$\begin{aligned}a_1 &= 3, \\a_2 &= 2a_1 = 6, \\a_3 &= 2a_2 = 12, \\a_4 &= 2a_3 = 24.\end{aligned}$$

The sequence begins

$$3, 6, 12, 24, \dots$$

Each term is twice the previous one.

The rule is simple, but the meaning is important. The next value depends on the current value. The sequence is built step by step.

### Reading recursive notation

The expression  $a_{n+1}$  means the term after  $a_n$ . If  $n = 1$ , then  $a_{n+1} = a_2$ . If  $n = 2$ , then  $a_{n+1} = a_3$ .

A recurrence relation such as

$$a_{n+1} = a_n + 2$$

should be read as a rule that works repeatedly:

$$a_2 = a_1 + 2, \quad a_3 = a_2 + 2, \quad a_4 = a_3 + 2,$$

and so on.

#### Remark

Recursive notation is compact because one line contains infinitely many instructions. To understand it, expand the first few cases and read what the rule is doing.

#### Summary

When reading a sequence definition, ask:

- Is the sequence given explicitly or recursively?
- If it is explicit, can I compute  $a_n$  directly from  $n$ ?
- If it is recursive, what is the starting value?
- How is the next term obtained from earlier terms?
- What are the first few terms?
- Does the notation  $a_{n+1}$  refer to the next term after  $a_n$ ?

Explicit and recursive descriptions are two languages for sequences. Each one reveals a different aspect of the same kind of object: a function defined on natural-number inputs.

## 9.9 Discrete and interval domains

Sequences and real functions belong to the same general idea: they are functions. But they often have very different domains. This difference changes the kinds of questions we naturally ask.

A function such as

$$f : [0, 1] \rightarrow \mathbb{R}$$

has inputs from an interval. Between two different inputs in  $[0, 1]$ , there are many other inputs. A sequence

$$a : \mathbb{N} \rightarrow \mathbb{R}$$

has inputs  $1, 2, 3, \dots$ . Between 1 and 2, there is no natural-number input.

## Interval domains

Intervals of real numbers are sometimes informally called continuous domains in elementary calculus. This phrase describes the structure of the input set: its points are not isolated. It does not imply that every function defined on the interval is continuous.

### Remark

A domain may be an interval while the function itself has discontinuities. Continuity of a function is a separate property defined using limits.

For example, the interval

$$[0, 1]$$

contains 0, 1, and also

$$\frac{1}{2}, \quad \frac{1}{3}, \quad 0.72, \quad \sqrt{2} - 1,$$

and infinitely many other numbers between 0 and 1.

A function defined on an interval can be studied by asking what happens as  $x$  moves through nearby real inputs. A connected curve is appropriate only when the behavior of the function justifies that picture.

## Discrete domains

The set of natural numbers is different:

$$\mathbb{N} = \{1, 2, 3, \dots\}$$

or, in some conventions,

$$\mathbb{N} = \{0, 1, 2, 3, \dots\}.$$

The exact convention should be stated when it matters. In either case, the natural numbers are separated. There is a next natural number after 1, and there is no natural number halfway between 1 and 2.

This is why a sequence graph consists of separated points:

$$(n, a_n), \quad n \in \mathbb{N}.$$

The sequence is not automatically defined at  $n = 1.5$ , because  $1.5 \notin \mathbb{N}$ .

### Example

Let

$$a_n = n^2, \quad n \in \mathbb{N}.$$

Then

$$a_1 = 1, \quad a_2 = 4, \quad a_3 = 9.$$

The expression  $a_{1.5}$  is not part of this sequence, because 1.5 is not a natural-number index.

This does not mean sequences are less important. It means that their input structure is different.

## Why this changes the questions

For functions defined on intervals, we can ask how the function behaves near a point from both sides. We can ask whether a graph has a break, whether values approach a limit as  $x \rightarrow a$ , or whether the function is continuous at  $a$ .

For sequences, the natural local picture is different. The inputs jump from one natural number to the next. Therefore, in an introductory course, we do not usually study continuity of a sequence at a natural-number input in the same way we study continuity of a real function on an interval.

Instead, for sequences we often ask questions such as:

- What happens to  $a_n$  as  $n \rightarrow \infty$ ?
- Is the sequence increasing or decreasing?
- Is the sequence bounded?
- Does the sequence approach a number?
- Is the sequence defined recursively or explicitly?

The object is still a function. The domain changes the natural questions.

### Remark

A sequence is not outside the world of functions. It is a function with a discrete domain. Because the domain is discrete, we study it with questions adapted to discrete input.

## Drawing responsibly

When drawing a function defined on an interval, it may be appropriate to draw a connected curve if the function behaves continuously. When drawing a sequence, we should usually draw separated points.

Connecting sequence points with line segments may be useful as a visual guide, but it can also be misleading. The line segment between  $(1, a_1)$  and  $(2, a_2)$  suggests values at non-natural inputs, but the sequence itself does not provide those values.

### Example

For

$$a_n = \frac{1}{n},$$

the points

$$(1, 1), \quad \left(2, \frac{1}{2}\right), \quad \left(3, \frac{1}{3}\right)$$

belong to the graph of the sequence. A smooth curve through these points may help the eye see a trend, but the sequence itself is defined only at natural-number inputs.

### Summary

When comparing ordinary real functions and sequences, ask:

- What is the domain?
- Are the inputs separated or do they fill an interval?
- Am I describing the domain, or continuity of the function?

- Should the graph be drawn as points or as a curve?
- Is the natural question about behavior near a real point or behavior as the index goes to infinity?
- Does the notation hide the fact that a sequence is still a function?

The difference between interval and discrete domains is not cosmetic. It shapes the way we read, draw, and investigate mathematical objects.

## 9.10 What questions can we ask?

Once we know that functions and sequences are related, we should also understand why they are not studied in exactly the same way. A sequence is a function, but its domain is usually  $\mathbb{N}$ . A real function in calculus often has a domain that is an interval or a union of intervals. This changes the natural questions.

The purpose of this section is to organize those questions.

### Questions about real functions

For a real function

$$f : D \rightarrow \mathbb{R}, \quad D \subseteq \mathbb{R},$$

we often ask questions connected with its graph, values, and behavior as the input varies through real numbers.

Typical questions include:

- What is the domain?
- What is the range?
- What is  $f(a)$  for a given input  $a$ ?
- For which  $x$  do we have  $f(x) = 0$ ?
- Where is the function positive or negative?
- Is the function increasing or decreasing on an interval?
- What happens to  $f(x)$  as  $x$  approaches a point?
- Is the function continuous?

Not all of these questions are answered in this chapter. Some of them will appear later, especially limits and continuity. But the language needed to ask them begins here.

### Questions about sequences

For a sequence

$$a : \mathbb{N} \rightarrow \mathbb{R},$$

we ask questions adapted to natural-number inputs.

Typical questions include:

- What is the starting index?
- What are the first few terms?

- Is the sequence given by an explicit formula or recursively?
- Is the sequence increasing or decreasing as  $n$  grows?
- Is the sequence bounded above or below?
- What happens to  $a_n$  as  $n \rightarrow \infty$ ?
- Does the sequence approach a limiting value?

The question about  $n \rightarrow \infty$  is especially important. Since the inputs of a sequence continue without end, we often want to know the long-term behavior of the terms.

## A comparison

| Object        | Typical domain                     | Natural questions                                                             |
|---------------|------------------------------------|-------------------------------------------------------------------------------|
| Real function | interval or subset of $\mathbb{R}$ | graph, zeros, signs, continuity, limits as $x \rightarrow a$                  |
| Sequence      | $\mathbb{N}$                       | terms, recursion, monotonicity, boundedness, limits as $n \rightarrow \infty$ |

This table is not meant to separate functions and sequences completely. It is meant to show that different domains lead to different habits of investigation.

## The common idea

Despite these differences, the common idea remains the same. A function assigns outputs to inputs. A sequence also assigns outputs to inputs. The input set is special, but the structure is still functional.

For a real function, we may write

$$f(x)$$

for the value at input  $x$ . For a sequence, we usually write

$$a_n$$

for the value at input  $n$ . These notations look different, but they express the same idea: value at an input.

### Remark

The notation changes because the situation changes. But the underlying idea remains: one input gives one output.

## Preparing for limits

The next chapter will study limits and continuity. The present chapter prepares for that in two ways.

First, it gives the language of functions, values, domains, and graphs. Without that language, the expression

$$\lim_{x \rightarrow a} f(x)$$

has no clear meaning.

Second, it introduces sequences as functions on  $\mathbb{N}$ . This prepares the expression

$$\lim_{n \rightarrow \infty} a_n.$$

Both limits involve the idea of approaching. But the input variable approaches in different ways:  $x$  may approach a real point through real values, while  $n$  goes through natural numbers toward infinity.

## Summary

At the end of this chapter, remember:

- A function is an assignment from inputs to outputs.
- The domain is part of the function.
- The notation  $f(x)$  means the value of  $f$  at  $x$ .
- A graph is a picture of input-output pairs.
- A sequence is a function whose domain is usually  $\mathbb{N}$ .
- Because sequences have discrete domains, we ask different questions about them.
- The type of domain shapes the type of mathematics we do.

Functions and sequences are not two unrelated topics placed side by side. Sequences are a special case of functions, and this viewpoint will help us understand limits, continuity, and later calculus.

# Chapter 10

## Limits and Continuity

### 10.1 What a limit tries to describe

Limits are one of the central ideas of calculus. They are also one of the first ideas that require a new way of reading mathematical notation. A limit does not ask only what happens at a point. It asks what values are approached near a point, or far along a sequence.

This distinction is essential. The value of a function at a point and the limit of a function at that point are related, but they are not the same question.

#### Approaching is not the same as arriving

Consider a function  $f$  and a point  $a$ . The expression

$$f(a)$$

asks for the value of the function at the input  $a$ . The expression

$$\lim_{x \rightarrow a} f(x)$$

asks what value  $f(x)$  approaches when  $x$  approaches  $a$ .

The variable  $x \rightarrow a$  means that  $x$  gets close to  $a$ . It does not mean that  $x = a$ . This is why a limit can exist even when the function is not defined at the point.

#### Example

Consider

$$f(x) = \frac{x^2 - 1}{x - 1}, \quad x \neq 1.$$

The function is not defined at  $x = 1$ , because the denominator becomes zero. But for  $x \neq 1$ , we can factor:

$$\frac{x^2 - 1}{x - 1} = \frac{(x - 1)(x + 1)}{x - 1} = x + 1.$$

When  $x$  approaches 1, the expression  $x + 1$  approaches 2. Therefore

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = 2.$$

The limit exists even though  $f(1)$  is not defined.

This example should be read as a lesson about the question being asked. The limit does not ask what happens after substituting  $x = 1$ . It asks what values are approached when  $x$  gets close to 1.

#### Limits of sequences and limits of functions

There are two main limit situations in this chapter.

For a sequence  $(a_n)$ , we ask what happens to the terms when the index grows:

$$\lim_{n \rightarrow \infty} a_n.$$

Here the input is discrete. The index  $n$  runs through natural numbers.

For a function  $f(x)$ , we may ask what happens when  $x$  approaches a real number  $a$ :

$$\lim_{x \rightarrow a} f(x),$$

or when  $x$  grows without bound:

$$\lim_{x \rightarrow \infty} f(x).$$

The notation is different because the input situation is different. But the common idea is the same: we study what values are approached.

### What a limit can tell us

Limits help us describe behavior that may not be visible from one single value. They allow us to discuss:

- long-term behavior of sequences,
- behavior of functions near a point,
- holes in graphs,
- vertical and horizontal asymptotes,
- continuity,
- the idea of a derivative in the next chapter.

A limit is therefore not just another calculation. It is a way of reading behavior.

#### Summary

When first reading a limit, ask:

- What object is being studied: a sequence or a function?
- What input is approaching something?
- Is the input approaching a finite point or infinity?
- Am I being asked about the value at the point or behavior near the point?
- Does direct substitution answer the question, or does it reveal a difficulty?

The main habit is this: before calculating a limit, read what kind of approaching is being described.

## 10.2 Limits of sequences

A sequence is a function whose input is usually a natural number. Therefore the most natural limiting question for a sequence is not what happens near a finite real input. The natural question is what happens when the index becomes larger and larger.

If  $(a_n)$  is a sequence, we write

$$\lim_{n \rightarrow \infty} a_n = L$$

when the terms of the sequence approach the number  $L$  as  $n$  grows.

### Definition

We say that a sequence  $(a_n)$  converges to  $L$  if the terms  $a_n$  become arbitrarily close to  $L$  for sufficiently large  $n$ . We write

$$\lim_{n \rightarrow \infty} a_n = L.$$

If no finite limit exists, we say that the sequence diverges.

The phrase “for sufficiently large  $n$ ” is important. A sequence may behave irregularly at the beginning and still have a limit. Limits describe long-term behavior, not the first few terms.

### A first convergent sequence

Consider

$$a_n = \frac{1}{n}.$$

The first terms are

$$1, \quad \frac{1}{2}, \quad \frac{1}{3}, \quad \frac{1}{4}, \quad \dots$$

These terms get closer and closer to 0. Therefore

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

The sequence never actually reaches 0, because  $\frac{1}{n} > 0$  for every natural number  $n$ . But reaching the limit is not required. The question is whether the terms approach the limit.

### Remark

A limit is about approaching, not necessarily about becoming equal. The terms  $1/n$  approach 0, even though no term of the sequence is equal to 0.

### Divergence to infinity

Some sequences do not approach a finite number. For example,

$$a_n = n$$

grows without bound. We write informally

$$\lim_{n \rightarrow \infty} n = \infty,$$

meaning that the terms become larger than any fixed bound. This is not convergence to a real number. It is divergence to infinity.

### Example

The sequence

$$a_n = n^2$$

has terms

$$1, 4, 9, 16, \dots$$

The terms grow without bound. The sequence does not converge to a finite real number.

### Oscillation

A sequence may also fail to converge because it oscillates.

### Example

Consider

$$a_n = (-1)^n.$$

The terms are

$$-1, 1, -1, 1, \dots$$

They do not approach one number. The sequence alternates forever between two values. Therefore the sequence diverges.

Oscillation does not automatically prevent convergence. If the oscillation becomes smaller, a limit may still exist.

### Example

Consider

$$a_n = \frac{(-1)^n}{n}.$$

The sign alternates, but the size of the terms becomes smaller:

$$\left| \frac{(-1)^n}{n} \right| = \frac{1}{n} \rightarrow 0.$$

Therefore

$$\lim_{n \rightarrow \infty} \frac{(-1)^n}{n} = 0.$$

The difference between these two examples is the size of the oscillation. In  $(-1)^n$ , the oscillation never becomes small. In  $(-1)^n/n$ , the oscillation is squeezed toward 0.

## Reading sequence limits

When reading a sequence limit, we should identify the long-term structure of the formula. Does the numerator and denominator grow at comparable rates? Does a power dominate? Does a term go to zero? Does an alternating sign matter?

### Example

Consider

$$a_n = \frac{n}{n+1}.$$

For large  $n$ , the numerator and denominator have almost the same size. Divide numerator and denominator by  $n$ :

$$\frac{n}{n+1} = \frac{1}{1 + \frac{1}{n}}.$$

Since  $\frac{1}{n} \rightarrow 0$ , we get

$$\lim_{n \rightarrow \infty} \frac{n}{n+1} = 1.$$

This calculation should be understood conceptually: the extra 1 in the denominator becomes insignificant compared with  $n$  when  $n$  is large.

## Summary

When studying a sequence limit, ask:

- What happens when  $n$  becomes large?
- Are the terms approaching a finite number?
- Are the terms growing without bound?
- Is there oscillation?
- If there is oscillation, does its size shrink?
- Which part of the formula dominates for large  $n$ ?

A sequence limit describes long-term behavior. The first few terms may help us guess, but the limit is determined by what happens as  $n$  grows without bound.

## 10.3 Basic techniques for sequence limits

Many sequence limits can be found by reading which terms dominate when  $n$  becomes large. The goal is not to memorize many separate tricks. The goal is to learn how to simplify a formula so that its long-term behavior becomes visible.

### Dividing by the highest power

For rational expressions in  $n$ , a common method is to divide numerator and denominator by the highest power of  $n$  that appears in the denominator or numerator.

#### Example

Compute

$$\lim_{n \rightarrow \infty} \frac{4n - 3}{6 - 5n}.$$

Divide numerator and denominator by  $n$ :

$$\frac{4n - 3}{6 - 5n} = \frac{4 - \frac{3}{n}}{\frac{6}{n} - 5}.$$

Since

$$\frac{3}{n} \rightarrow 0, \quad \frac{6}{n} \rightarrow 0,$$

we get

$$\lim_{n \rightarrow \infty} \frac{4n - 3}{6 - 5n} = \frac{4}{-5} = -\frac{4}{5}.$$

The method works because lower-order terms become small compared with the highest power of  $n$ .

#### Example

Compute

$$\lim_{n \rightarrow \infty} \frac{n^2 - 1}{3 - n^3}.$$

The denominator has degree 3, while the numerator has degree 2. Divide by  $n^3$ :

$$\frac{n^2 - 1}{3 - n^3} = \frac{\frac{1}{n} - \frac{1}{n^3}}{\frac{3}{n^3} - 1}.$$

The numerator tends to 0, and the denominator tends to  $-1$ . Hence

$$\lim_{n \rightarrow \infty} \frac{n^2 - 1}{3 - n^3} = 0.$$

This example shows that when the denominator grows faster than the numerator, the quotient may tend to zero.

### Using dominant terms

Sometimes we can read the answer by comparing leading terms. For example,

$$\frac{2n^3 - 4n - 1}{6n + 3n^2 - n^3}$$

has leading term  $2n^3$  in the numerator and  $-n^3$  in the denominator. Therefore the limit should be

$$\frac{2}{-1} = -2.$$

A careful calculation confirms this:

$$\frac{2n^3 - 4n - 1}{6n + 3n^2 - n^3} = \frac{2 - \frac{4}{n^2} - \frac{1}{n^3}}{\frac{6}{n^2} + \frac{3}{n} - 1} \rightarrow -2.$$

The phrase “dominant term” means the term that grows fastest as  $n \rightarrow \infty$ .

### Roots and rationalization

Expressions with square roots often need to be rewritten before the limit is visible. A standard method is to multiply by the conjugate.

#### Example

Compute

$$\lim_{n \rightarrow \infty} (\sqrt{n^2 + n} - n).$$

Direct reading is not enough because both terms grow without bound. Multiply by the conjugate:

$$\sqrt{n^2 + n} - n = \frac{(\sqrt{n^2 + n} - n)(\sqrt{n^2 + n} + n)}{\sqrt{n^2 + n} + n}.$$

The numerator becomes

$$(n^2 + n) - n^2 = n.$$

Thus

$$\sqrt{n^2 + n} - n = \frac{n}{\sqrt{n^2 + n} + n}.$$

Factor  $n$  from the square root:

$$\frac{n}{n\sqrt{1 + \frac{1}{n}} + n} = \frac{1}{\sqrt{1 + \frac{1}{n}} + 1}.$$

Therefore

$$\lim_{n \rightarrow \infty} (\sqrt{n^2 + n} - n) = \frac{1}{2}.$$

The conjugate method changes a difference of large quantities into a quotient whose behavior is easier to read.

### Alternating signs

Terms such as  $(-1)^n$  should be read carefully. They may cause oscillation, but if they are multiplied by something tending to zero, the whole sequence may still converge.

#### Example

Compute

$$\lim_{n \rightarrow \infty} \frac{2n + (-1)^n}{n}.$$

Rewrite as

$$\frac{2n + (-1)^n}{n} = 2 + \frac{(-1)^n}{n}.$$

The term  $\frac{(-1)^n}{n}$  tends to 0, because its absolute value is  $1/n$ . Therefore

$$\lim_{n \rightarrow \infty} \frac{2n + (-1)^n}{n} = 2.$$

The alternating part does not disappear because signs are unimportant. It disappears because its size tends to zero.

### Powers and roots

For expressions involving powers, we compare growth rates. If  $|q| < 1$ , then

$$q^n \rightarrow 0.$$

If  $q > 1$ , then  $q^n$  grows without bound.

#### Example

Compute

$$\lim_{n \rightarrow \infty} \frac{4^n - 1}{2^{2n} - 7}.$$

Since

$$2^{2n} = 4^n,$$

we have

$$\frac{4^n - 1}{2^{2n} - 7} = \frac{4^n - 1}{4^n - 7}.$$

Divide by  $4^n$ :

$$\frac{1 - \frac{1}{4^n}}{1 - \frac{7}{4^n}} \rightarrow \frac{1}{1} = 1.$$

## Summary

Useful habits for sequence limits:

- For rational expressions, divide by the dominant power of  $n$ .
- For square-root differences, consider multiplying by the conjugate.
- For alternating signs, check whether the size of the alternating term tends to zero.
- For powers, compare the dominant exponential terms.
- After transforming the expression, interpret which terms vanish and which remain.

Techniques are useful only when connected with meaning. Each transformation should make the long-term behavior easier to see.

## 10.4 Limits of functions

A function limit describes what happens to the values  $f(x)$  when the input  $x$  approaches a certain point. The point may or may not belong to the domain of the function. What matters for the limit is the behavior near the point.

We write

$$\lim_{x \rightarrow a} f(x) = L$$

when the values  $f(x)$  approach  $L$  as  $x$  approaches  $a$ .

### Definition

The notation

$$\lim_{x \rightarrow a} f(x) = L$$

means that the values of  $f(x)$  can be made arbitrarily close to  $L$  by taking  $x$  sufficiently close to  $a$ , with  $x \neq a$ .

The condition  $x \neq a$  is not a technicality. A limit studies nearby values. The value at the point itself is a separate question.

### Direct substitution

Sometimes the limit can be found by direct substitution. This often happens for polynomials and many simple functions.

### Example

Compute

$$\lim_{x \rightarrow 3} (x^2 + 2x - 1).$$

The expression is defined and behaves regularly near  $x = 3$ , so we substitute:

$$3^2 + 2 \cdot 3 - 1 = 9 + 6 - 1 = 14.$$

Therefore

$$\lim_{x \rightarrow 3} (x^2 + 2x - 1) = 14.$$

Direct substitution is not a rule to use blindly. It is a first test. If it gives a meaningful value and the function behaves regularly, it may solve the problem. If it produces a form such as  $0/0$ , more work is needed.

## A removable difficulty

Consider

$$\lim_{x \rightarrow 3} \frac{x^2 - 4x + 3}{x - 3}.$$

Direct substitution gives

$$\frac{9 - 12 + 3}{0} = \frac{0}{0},$$

which is not a number. This form tells us that direct substitution has failed. It does not tell us that the limit does not exist.

Factor the numerator:

$$x^2 - 4x + 3 = (x - 1)(x - 3).$$

For  $x \neq 3$ ,

$$\frac{x^2 - 4x + 3}{x - 3} = x - 1.$$

Therefore

$$\lim_{x \rightarrow 3} \frac{x^2 - 4x + 3}{x - 3} = \lim_{x \rightarrow 3} (x - 1) = 2.$$

The original expression is not defined at  $x = 3$ , but near 3 it behaves like  $x - 1$ . The limit reads the nearby behavior.

## The meaning of $0/0$

The expression  $0/0$  is called an indeterminate form. It means that the information from direct substitution is insufficient. It is not an answer.

The limit may exist, may fail to exist, or may be infinite. We must transform the expression or analyze it in another way.

### Example

The limit

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$$

gives  $0/0$  by direct substitution. After factoring,

$$\frac{x^2 - 1}{x - 1} = x + 1$$

for  $x \neq 1$ , so the limit is 2.

But

$$\lim_{x \rightarrow 0} \frac{|x|}{x}$$

also cannot be found by direct substitution, and the two-sided limit does not exist because the left and right behaviors are different.

This is why we must read the situation before choosing a technique.

## Limits and graphs

On a graph, the limit  $\lim_{x \rightarrow a} f(x)$  asks what height the graph approaches near  $x = a$ . A hole in the graph may not prevent a limit from existing. A jump may prevent a two-sided limit from existing.

### Remark

The value  $f(a)$  is the height of the graph at  $x = a$ , if that point exists. The limit  $\lim_{x \rightarrow a} f(x)$  is the height approached by nearby points of the graph.

### Summary

When computing a function limit, ask:

- What point is  $x$  approaching?
- Is the function defined at nearby points?
- Does direct substitution give a meaningful value?
- If direct substitution gives  $0/0$ , can the expression be simplified?
- Is the left-hand behavior the same as the right-hand behavior?
- Does the graph suggest a hole, a jump, or an asymptote?

A function limit is a statement about local behavior. The point itself matters for continuity, but the limit first asks what happens nearby.

## 10.5 One-sided limits

A two-sided limit asks what happens as  $x$  approaches a point from both sides. Sometimes the behavior from the left and from the right is different. To describe this, we use one-sided limits.

The left-hand limit is written as

$$\lim_{x \rightarrow a^-} f(x).$$

It describes what happens when  $x$  approaches  $a$  through values smaller than  $a$ .

The right-hand limit is written as

$$\lim_{x \rightarrow a^+} f(x).$$

It describes what happens when  $x$  approaches  $a$  through values greater than  $a$ .

### When the two-sided limit exists

A two-sided limit exists exactly when the left-hand and right-hand limits both exist and are equal.

### Theorem

The limit

$$\lim_{x \rightarrow a} f(x) = L$$

exists if and only if

$$\lim_{x \rightarrow a^-} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow a^+} f(x) = L.$$

If the two one-sided limits are different, then the two-sided limit does not exist.

### Example

Consider

$$f(x) = \frac{|x|}{x}, \quad x \neq 0.$$

If  $x > 0$ , then  $|x| = x$ , so

$$f(x) = 1.$$

If  $x < 0$ , then  $|x| = -x$ , so

$$f(x) = -1.$$

Therefore

$$\lim_{x \rightarrow 0^-} \frac{|x|}{x} = -1, \quad \lim_{x \rightarrow 0^+} \frac{|x|}{x} = 1.$$

The one-sided limits are different, so

$$\lim_{x \rightarrow 0} \frac{|x|}{x}$$

does not exist.

This example shows why looking from both sides matters. A formula may look simple, but its behavior can depend on the direction of approach.

## Piecewise functions

One-sided limits are especially important for piecewise-defined functions. Each side of the point may use a different formula.

### Example

Let

$$f(x) = \begin{cases} x + 2, & x < 1, \\ 3x, & x \geq 1. \end{cases}$$

The left-hand limit at 1 uses the formula  $x + 2$ :

$$\lim_{x \rightarrow 1^-} f(x) = 1 + 2 = 3.$$

The right-hand limit at 1 uses the formula  $3x$ :

$$\lim_{x \rightarrow 1^+} f(x) = 3 \cdot 1 = 3.$$

The two one-sided limits are equal, so

$$\lim_{x \rightarrow 1} f(x) = 3.$$

Also, because  $f(1) = 3 \cdot 1 = 3$ , the function is continuous at 1. We will define continuity later in this chapter.

### Example

Let

$$g(x) = \begin{cases} x + 2, & x < 1, \\ 3x + 1, & x \geq 1. \end{cases}$$

Then

$$\lim_{x \rightarrow 1^-} g(x) = 3,$$

but

$$\lim_{x \rightarrow 1^+} g(x) = 4.$$

Since the one-sided limits are different,

$$\lim_{x \rightarrow 1} g(x)$$

does not exist.

The point of the calculation is not only to obtain numbers. The calculation tells us whether the two sides of the graph meet at the same height.

### Reading from a graph

On a graph, the left-hand limit is the height approached as we move toward the point from the left. The right-hand limit is the height approached as we move toward the point from the right.

If the graph approaches the same height from both sides, the two-sided limit exists. If the graph approaches different heights, there is a jump.

### Summary

When studying one-sided limits, ask:

- What happens when  $x$  approaches from the left?
- What happens when  $x$  approaches from the right?
- Are the two one-sided limits equal?
- Does the formula change on different sides of the point?
- Does the graph show a jump or a meeting point?

One-sided limits are the language we need when behavior depends on direction.

## 10.6 Limits at infinity

Some limits describe behavior near a finite point. Other limits describe behavior far away. When we write

$$\lim_{x \rightarrow \infty} f(x),$$

we ask what happens to  $f(x)$  as  $x$  becomes larger and larger. When we write

$$\lim_{x \rightarrow -\infty} f(x),$$

we ask what happens as  $x$  becomes very large in the negative direction.

This is closely related to sequence limits, where we ask what happens as  $n \rightarrow \infty$ . The difference is that  $n$  runs through natural numbers, while  $x$  runs through real values.

## A first example

Consider

$$f(x) = \frac{1}{x}.$$

As  $x$  becomes very large and positive,  $1/x$  becomes close to 0. Therefore

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0.$$

As  $x$  becomes very large and negative,  $1/x$  also becomes close to 0. Therefore

$$\lim_{x \rightarrow -\infty} \frac{1}{x} = 0.$$

The sign is different on the two sides, but the values still approach the same number 0.

## Rational functions

For rational functions, limits at infinity are often found by comparing the highest powers of  $x$ .

### Example

Compute

$$\lim_{x \rightarrow \infty} \frac{3x^2 - 5x + 1}{2x^2 + 7}.$$

Divide numerator and denominator by  $x^2$ :

$$\frac{3x^2 - 5x + 1}{2x^2 + 7} = \frac{3 - \frac{5}{x} + \frac{1}{x^2}}{2 + \frac{7}{x^2}}.$$

As  $x \rightarrow \infty$ , the terms  $1/x$  and  $1/x^2$  tend to 0. Hence

$$\lim_{x \rightarrow \infty} \frac{3x^2 - 5x + 1}{2x^2 + 7} = \frac{3}{2}.$$

This is the same dominant-term idea used for sequence limits. The largest powers determine the long-term behavior.

### Example

Compute

$$\lim_{x \rightarrow \infty} \frac{2x + 1}{x^2 + 3}.$$

The denominator grows faster than the numerator. Divide by  $x^2$ :

$$\frac{2x + 1}{x^2 + 3} = \frac{\frac{2}{x} + \frac{1}{x^2}}{1 + \frac{3}{x^2}}.$$

The numerator tends to 0, and the denominator tends to 1. Therefore

$$\lim_{x \rightarrow \infty} \frac{2x + 1}{x^2 + 3} = 0.$$

## Horizontal asymptotes

If

$$\lim_{x \rightarrow \infty} f(x) = L$$

or

$$\lim_{x \rightarrow -\infty} f(x) = L,$$

then the graph approaches the horizontal line

$$y = L$$

in that direction. This line is called a horizontal asymptote.

### Example

For

$$f(x) = \frac{3x^2 - 5x + 1}{2x^2 + 7},$$

we found

$$\lim_{x \rightarrow \infty} f(x) = \frac{3}{2}.$$

Also,

$$\lim_{x \rightarrow -\infty} f(x) = \frac{3}{2}.$$

Thus the graph has horizontal asymptote

$$y = \frac{3}{2}.$$

The asymptote is not necessarily a line the graph never touches. It is a line the graph approaches far away.

### Infinite limits

Sometimes a function grows without bound as  $x \rightarrow \infty$ . For example,

$$\lim_{x \rightarrow \infty} x^2 = \infty.$$

This means the values become larger than any fixed number. It is not convergence to a real number.

Similarly,

$$\lim_{x \rightarrow -\infty} x^3 = -\infty.$$

The values become negative with arbitrarily large magnitude.

### Summary

When studying limits at infinity, ask:

- Is the input going to  $\infty$  or  $-\infty$ ?
- Which terms dominate for large  $|x|$ ?
- Does the function approach a finite number?
- Does it grow without bound?
- Is there a horizontal asymptote?
- Is the behavior different for  $x \rightarrow \infty$  and  $x \rightarrow -\infty$ ?

Limits at infinity describe long-term behavior. They tell us what the function does far from the origin.

## 10.7 Algebra of limits and indeterminate forms

Limit computations often use algebra. But algebra should be used with understanding. Some transformations preserve the behavior near a point; others may accidentally change the problem. The guiding question should always be: what happens to the expression as the input approaches the limiting value?

### Basic algebra of limits

If two limits exist,

$$\lim_{x \rightarrow a} f(x) = L, \quad \lim_{x \rightarrow a} g(x) = M,$$

then we can combine them in predictable ways:

$$\lim_{x \rightarrow a} (f(x) + g(x)) = L + M,$$

$$\lim_{x \rightarrow a} (f(x)g(x)) = LM,$$

and, if  $M \neq 0$ ,

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{L}{M}.$$

These rules are useful when the individual limits are known and the resulting expression is meaningful.

#### Example

Compute

$$\lim_{x \rightarrow 2} (x^2 + 3x - 1).$$

Using direct substitution and algebra of limits,

$$\lim_{x \rightarrow 2} (x^2 + 3x - 1) = 2^2 + 3 \cdot 2 - 1 = 9.$$

For polynomials, this direct substitution works because polynomial functions are continuous everywhere. Rational functions behave similarly at points where the denominator is not zero.

### When direct substitution fails

Direct substitution may produce expressions such as

$$\frac{0}{0}, \quad \frac{\infty}{\infty}, \quad 0 \cdot \infty, \quad \infty - \infty.$$

These are called indeterminate forms. They are not answers. They indicate that the expression needs to be transformed or analyzed more carefully.

#### Remark

An indeterminate form means that competing behaviors are present. The expression may have a finite limit, an infinite limit, or no limit. The form itself does not decide the answer.

### Factoring

Factoring is often used when direct substitution gives  $0/0$ .

### Example

Compute

$$\lim_{x \rightarrow 3} \frac{x^2 - 4x + 3}{2x - 6}.$$

Substitution gives  $0/0$ . Factor:

$$x^2 - 4x + 3 = (x - 1)(x - 3), \quad 2x - 6 = 2(x - 3).$$

For  $x \neq 3$ ,

$$\frac{x^2 - 4x + 3}{2x - 6} = \frac{(x - 1)(x - 3)}{2(x - 3)} = \frac{x - 1}{2}.$$

Therefore

$$\lim_{x \rightarrow 3} \frac{x^2 - 4x + 3}{2x - 6} = \frac{3 - 1}{2} = 1.$$

The cancellation is valid for  $x \neq 3$ . This is enough for the limit, because the limit studies values near 3, not necessarily at 3.

### Rationalizing

Expressions with radicals may require multiplication by the conjugate.

### Example

Compute

$$\lim_{x \rightarrow 25} \frac{\sqrt{x} - 5}{x - 25}.$$

Substitution gives  $0/0$ . Multiply by the conjugate:

$$\frac{\sqrt{x} - 5}{x - 25} \cdot \frac{\sqrt{x} + 5}{\sqrt{x} + 5} = \frac{x - 25}{(x - 25)(\sqrt{x} + 5)}.$$

For  $x \neq 25$ , this becomes

$$\frac{1}{\sqrt{x} + 5}.$$

Therefore

$$\lim_{x \rightarrow 25} \frac{\sqrt{x} - 5}{x - 25} = \frac{1}{5 + 5} = \frac{1}{10}.$$

Rationalizing is not a trick for its own sake. It removes a difference of square roots that hides the limiting behavior.

### Rewriting before deciding

A common mistake is to see  $0/0$  and conclude that the limit is zero or undefined. This is incorrect. The form  $0/0$  tells us that direct substitution has not answered the question.

### Summary

When direct substitution gives an indeterminate form, ask:

- Can I factor the numerator or denominator?
- Is there a common factor that cancels for nearby inputs?

- Is there a square root expression that can be rationalized?
- Can I divide by a dominant power for limits at infinity?
- After rewriting, what expression describes the same nearby behavior?

The algebra of limits is useful because it helps us reveal behavior. The algebra should serve the limit, not replace the meaning of the limit.

## 10.8 Standard limits

Some limits occur so often that they become standard tools. They should not be treated as isolated formulas to memorize without meaning. Each standard limit captures a basic local behavior of an important function.

### The basic trigonometric limit

One of the most important limits is

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1,$$

where  $x$  is measured in radians.

This limit says that, near zero,  $\sin x$  behaves like  $x$ . The functions are not equal, but their ratio approaches 1.

#### Example

Compute

$$\lim_{x \rightarrow 0} \frac{\sin 3x}{4x}.$$

Rewrite the expression to use the standard limit:

$$\frac{\sin 3x}{4x} = \frac{3}{4} \cdot \frac{\sin 3x}{3x}.$$

As  $x \rightarrow 0$ , also  $3x \rightarrow 0$ , so

$$\frac{\sin 3x}{3x} \rightarrow 1.$$

Therefore

$$\lim_{x \rightarrow 0} \frac{\sin 3x}{4x} = \frac{3}{4}.$$

The rewriting is the main step. We reshape the expression until the standard limit becomes visible.

### Tangent and arcsine type limits

Since

$$\tan x = \frac{\sin x}{\cos x},$$

and  $\cos x \rightarrow 1$  as  $x \rightarrow 0$ , we also have

$$\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1.$$

Similarly, in elementary calculus we use

$$\lim_{x \rightarrow 0} \frac{\arctan x}{x} = 1.$$

These limits express the fact that near zero, these functions are locally close to the identity function.

### Example

Compute

$$\lim_{x \rightarrow 0} \frac{\tan 2x}{\tan x}.$$

Rewrite as

$$\frac{\tan 2x}{\tan x} = 2 \cdot \frac{\tan 2x}{2x} \cdot \frac{x}{\tan x}.$$

As  $x \rightarrow 0$ , both fractions tend to 1. Therefore

$$\lim_{x \rightarrow 0} \frac{\tan 2x}{\tan x} = 2.$$

## The exponential standard limit

Another standard limit is

$$\lim_{x \rightarrow 0} (1 + x)^{1/x} = e.$$

More generally, if  $u \rightarrow 0$ , then

$$(1 + u)^{1/u} \rightarrow e.$$

This limit is the basis for many expressions involving variable exponents.

### Example

Compute

$$\lim_{x \rightarrow 0} (1 - 3x)^{1/x}.$$

Let

$$u = -3x.$$

Then  $u \rightarrow 0$  as  $x \rightarrow 0$ , and

$$\frac{1}{x} = \frac{-3}{u}.$$

Therefore

$$(1 - 3x)^{1/x} = (1 + u)^{-3/u} = \left( (1 + u)^{1/u} \right)^{-3}.$$

Taking the limit gives

$$e^{-3}.$$

This example shows why substitution is useful: it changes the expression into a recognizable standard form.

## Standard limits as local models

The standard limits tell us how certain functions behave near a point. Near zero,

$$\sin x \sim x, \quad \tan x \sim x, \quad \arctan x \sim x.$$

The symbol  $\sim$  means that the ratio of the two expressions tends to 1. This notation should be used carefully, but the idea is helpful: near zero, these functions can often be compared to simpler expressions.

## Summary

Useful standard limits include:

- $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1,$
- $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1,$
- $\lim_{x \rightarrow 0} \frac{\arctan x}{x} = 1,$
- $\lim_{x \rightarrow 0} (1 + x)^{1/x} = e.$

Standard limits are not a substitute for understanding. They are reliable local models that help us simplify common limiting situations.

## 10.9 Continuity at a point

Continuity connects the value of a function at a point with the behavior of the function near that point. A function is continuous at a point when there is no break between the nearby behavior and the value at the point.

The definition uses limits.

### Definition

A function  $f$  is continuous at a point  $a$  if all three conditions hold:

1.  $f(a)$  is defined,
2.  $\lim_{x \rightarrow a} f(x)$  exists,
3.  $\lim_{x \rightarrow a} f(x) = f(a).$

Each condition matters. If the value is missing, the function cannot be continuous at the point. If the limit does not exist, there is no single nearby behavior. If the limit exists but differs from the value, the graph has a mismatch at the point.

### A continuous example

#### Example

Let

$$f(x) = x^2 + 1.$$

Check continuity at  $a = 2$ .

First,

$$f(2) = 2^2 + 1 = 5.$$

Second,

$$\lim_{x \rightarrow 2} (x^2 + 1) = 5.$$

The limit exists and equals the value of the function. Therefore  $f$  is continuous at 2.

Polynomials are continuous at every real point. This is why direct substitution works for

polynomial limits.

### A removable discontinuity

A function may fail to be continuous because the value at one point is missing or incorrect, even though the nearby behavior has a clear limit.

#### Example

Let

$$f(x) = \frac{x^2 - 25}{x + 5}$$

for  $x \neq -5$ . The function is not defined at  $-5$ . For  $x \neq -5$ , factor:

$$\frac{x^2 - 25}{x + 5} = \frac{(x - 5)(x + 5)}{x + 5} = x - 5.$$

Therefore

$$\lim_{x \rightarrow -5} f(x) = -10.$$

If we define

$$f(-5) = -10,$$

then the function becomes continuous at  $-5$ . The discontinuity is removable.

The word “removable” means that a suitable value at the point can repair continuity.

### When the value is wrong

A function may be defined at a point but still fail to be continuous there.

#### Example

Define

$$f(x) = \begin{cases} \frac{x^2 - 1}{x - 1}, & x \neq 1, \\ 5, & x = 1. \end{cases}$$

For  $x \neq 1$ ,

$$\frac{x^2 - 1}{x - 1} = x + 1.$$

Thus

$$\lim_{x \rightarrow 1} f(x) = 2.$$

But

$$f(1) = 5.$$

Since the limit and the value are different,  $f$  is not continuous at 1. If we changed the value to  $f(1) = 2$ , continuity would be restored.

This example shows why continuity is not only about having a value. The value must agree with the nearby behavior.

### Continuity on intervals

A function is continuous on an interval if it is continuous at every point of that interval. Many familiar functions have this property on their natural domains.

For example:

- polynomial functions are continuous on  $\mathbb{R}$ ,
- rational functions are continuous wherever the denominator is not zero,
- $\sqrt{x}$  is continuous on  $[0, \infty)$ ,
- trigonometric functions such as  $\sin x$  and  $\cos x$  are continuous on  $\mathbb{R}$ .

These facts allow many limits to be computed by substitution, but only at points where the function is continuous.

### Summary

To check continuity at  $a$ , ask:

- Is  $f(a)$  defined?
- Does  $\lim_{x \rightarrow a} f(x)$  exist?
- Is the limit equal to  $f(a)$ ?
- If continuity fails, is it because of a missing value, a wrong value, or different one-sided behavior?

Continuity means agreement between value and nearby behavior. It is the point where limits become a statement about the function itself.

## 10.10 Types of discontinuities

When a function is not continuous at a point, the failure can happen in different ways. Classifying the discontinuity helps us understand what kind of break appears in the graph and whether it can be repaired.

### Removable discontinuity

A removable discontinuity occurs when the limit exists, but the value at the point is missing or does not agree with the limit.

#### Example

Let

$$f(x) = \frac{x^2 - 4}{x - 2}, \quad x \neq 2.$$

For  $x \neq 2$ ,

$$f(x) = x + 2.$$

Therefore

$$\lim_{x \rightarrow 2} f(x) = 4.$$

The function is not defined at 2, but if we define  $f(2) = 4$ , the function becomes continuous there. The discontinuity is removable.

Graphically, a removable discontinuity is often seen as a hole.

## Jump discontinuity

A jump discontinuity occurs when the left-hand and right-hand limits exist but are different.

### Example

Let

$$f(x) = \begin{cases} 0, & x < 1, \\ 2, & x \geq 1. \end{cases}$$

Then

$$\lim_{x \rightarrow 1^-} f(x) = 0, \quad \lim_{x \rightarrow 1^+} f(x) = 2.$$

The one-sided limits are different, so the two-sided limit does not exist. The function has a jump discontinuity at 1.

A jump cannot be repaired by changing only the value at the point. The two sides approach different heights.

## Infinite discontinuity

An infinite discontinuity occurs when the function grows without bound near a point. This often happens near vertical asymptotes.

### Example

Consider

$$f(x) = \frac{1}{x-2}.$$

As  $x \rightarrow 2^+$ , the denominator is small and positive, so

$$f(x) \rightarrow \infty.$$

As  $x \rightarrow 2^-$ , the denominator is small and negative, so

$$f(x) \rightarrow -\infty.$$

The function has an infinite discontinuity at 2, and the line  $x = 2$  is a vertical asymptote.

This is different from a removable discontinuity. There is no finite value that can be assigned at  $x = 2$  to make the function continuous.

## Oscillatory behavior

Some functions fail to have a limit because they oscillate more and more rapidly near a point.

### Example

The function

$$f(x) = \sin \frac{1}{x}, \quad x \neq 0,$$

oscillates infinitely many times as  $x \rightarrow 0$ . It does not approach one number, so

$$\lim_{x \rightarrow 0} \sin \frac{1}{x}$$

does not exist.

This kind of example is more advanced than a simple jump, but it teaches an important lesson: a limit requires approaching one value.

### Summary

Common types of discontinuities:

- Removable: the limit exists, but the value is missing or incorrect.
- Jump: the left-hand and right-hand limits are different.
- Infinite: the function grows without bound near the point.
- Oscillatory: the function fails to approach one value because of persistent oscillation.

A discontinuity is not merely a place where a formula looks difficult. It is a specific kind of failure of continuity.

## 10.11 Reading limits from graphs

Graphs are one of the best ways to understand limits, provided we read them carefully. A graph can show whether values approach one height, whether there is a hole, whether there is a jump, or whether the function grows without bound.

The graph should be read as information about nearby behavior.

### Value versus limit on a graph

The value  $f(a)$  is represented by the point of the graph at  $x = a$ , if such a point is included. The limit  $\lim_{x \rightarrow a} f(x)$  is represented by the height approached by the graph near  $x = a$ .

These may be the same, but they do not have to be the same.

### Example

Suppose a graph has a hole at  $(2, 5)$ , but near  $x = 2$  the curve approaches height 5 from both sides. Suppose also that the function has a filled point at  $(2, 1)$ . Then

$$f(2) = 1,$$

but

$$\lim_{x \rightarrow 2} f(x) = 5.$$

The function is not continuous at 2, because the value and the limit are different.

This example is one of the most important pictures in the study of continuity. A graph can have a limit at a point even if the value at that point is different.

### Reading one-sided limits

To read

$$\lim_{x \rightarrow a^-} f(x),$$

follow the graph toward  $x = a$  from the left. To read

$$\lim_{x \rightarrow a^+} f(x),$$

follow the graph toward  $x = a$  from the right.

If the two approached heights agree, the two-sided limit exists. If they do not agree, the two-sided limit does not exist.

### Example

Suppose that as  $x$  approaches 3 from the left, the graph approaches height 2. As  $x$  approaches 3 from the right, the graph approaches height 7. Then

$$\lim_{x \rightarrow 3^-} f(x) = 2, \quad \lim_{x \rightarrow 3^+} f(x) = 7.$$

Since these are different,

$$\lim_{x \rightarrow 3} f(x)$$

does not exist.

The issue is not the value at  $x = 3$ . The issue is that the two sides do not approach the same height.

### Vertical and horizontal asymptotes

If the graph grows without bound as  $x$  approaches  $a$ , then there may be a vertical asymptote:

$$x = a.$$

For example, if

$$\lim_{x \rightarrow a^+} f(x) = \infty,$$

then the graph rises without bound on the right side of  $a$ .

If the graph approaches a fixed height as  $x \rightarrow \infty$  or  $x \rightarrow -\infty$ , there may be a horizontal asymptote:

$$y = L.$$

This means that far away, the graph gets close to the line  $y = L$ .

### What graphs can and cannot prove

A graph can strongly suggest a limit, and it is an excellent tool for understanding. But a drawn graph may be approximate. If exactness is required, algebraic or theoretical reasoning may be needed.

### Remark

Use graphs to understand the question and predict the answer. Use algebra or definitions when an exact justification is required.

### Summary

When reading limits from a graph, ask:

- What height is approached from the left?
- What height is approached from the right?
- Is the value at the point included, missing, or different from the limit?
- Is there a hole, a jump, or a vertical asymptote?
- What happens far to the left or far to the right?

- Is the graph exact, or is it only a guide for reasoning?

Reading limits from graphs is not a separate skill from computing limits. It is another language for the same idea: describing approached behavior.

## 10.12 Choosing the right limit method

This chapter introduced several kinds of limits and several methods for working with them. The purpose is not to create a list of tricks. The purpose is to learn how to recognize what kind of limiting question is being asked and which method makes the behavior visible.

A good limit solution begins by reading the expression.

### First identify the type of limit

Before calculating, ask what kind of input is approaching what.

- If the expression is  $\lim_{n \rightarrow \infty} a_n$ , it is a sequence limit.
- If the expression is  $\lim_{x \rightarrow a} f(x)$ , it is a function limit near a finite point.
- If the expression is  $\lim_{x \rightarrow a^-} f(x)$  or  $\lim_{x \rightarrow a^+} f(x)$ , it is a one-sided limit.
- If the expression is  $\lim_{x \rightarrow \infty} f(x)$ , it is a limit at infinity.

Different kinds of limits naturally suggest different tools.

### Then try the simplest reading

For a function limit at a finite point, direct substitution is often the first test. If it gives a meaningful value and the function is continuous there, the limit is found.

If direct substitution gives  $0/0$ , do not stop. Try to factor, cancel, rationalize, or use a standard limit.

For a sequence or a limit at infinity, direct substitution is not the right idea. Instead, look for dominant terms and long-term behavior.

### A practical guide

| Situation                                                               | Useful method                                     |
|-------------------------------------------------------------------------|---------------------------------------------------|
| Polynomial at a finite point                                            | direct substitution                               |
| Rational function with non-zero denominator                             | direct substitution                               |
| Form $0/0$ with polynomials                                             | factor and cancel                                 |
| Form $0/0$ with square roots                                            | multiply by the conjugate                         |
| Trigonometric expression near 0                                         | use standard trigonometric limits                 |
| Rational expression as $x \rightarrow \infty$ or $n \rightarrow \infty$ | divide by dominant power                          |
| Different formulas on two sides                                         | compute one-sided limits                          |
| Question about continuity                                               | compare $f(a)$ with $\lim_{x \rightarrow a} f(x)$ |

The table is a guide, not a replacement for thinking. A problem may require more than one step.

## A complete reading example

Consider

$$\lim_{x \rightarrow 4} \frac{x^2 - 2x - 8}{x^2 - 9x + 20}.$$

First read the type of problem: this is a function limit at a finite point. Try substitution:

$$\frac{16 - 8 - 8}{16 - 36 + 20} = \frac{0}{0}.$$

This is indeterminate, so we factor:

$$x^2 - 2x - 8 = (x - 4)(x + 2),$$

$$x^2 - 9x + 20 = (x - 4)(x - 5).$$

For  $x \neq 4$ ,

$$\frac{x^2 - 2x - 8}{x^2 - 9x + 20} = \frac{x + 2}{x - 5}.$$

Now substitute:

$$\frac{4 + 2}{4 - 5} = -6.$$

Therefore

$$\lim_{x \rightarrow 4} \frac{x^2 - 2x - 8}{x^2 - 9x + 20} = -6.$$

The important point is the order of thought: identify the type of limit, test direct substitution, recognize  $0/0$ , transform the expression, then compute.

## Preparing for derivatives

Limits are not the end of the story. They are the language needed to define derivatives. In the next chapter, we will study expressions such as

$$\lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}.$$

This is a limit of a quotient. It measures the limiting rate of change of a function.

The work in this chapter prepares that idea. We learned how to read  $f(x + h)$ , how to interpret  $h \rightarrow 0$ , and how to handle expressions that initially look like  $0/0$ .

### Summary

At the end of this chapter, remember:

- A limit describes approached behavior.
- A sequence limit describes what happens as  $n \rightarrow \infty$ .
- A function limit near a point describes what happens as  $x \rightarrow a$ , with  $x \neq a$ .
- One-sided limits must agree for a two-sided limit to exist.
- Continuity means  $\lim_{x \rightarrow a} f(x) = f(a)$ .
- Indeterminate forms are not answers; they are signals to transform or analyze further.
- The method should be chosen from the structure of the expression.

A limit calculation should always end with interpretation. The final number is important, but the reasoning tells us what behavior the number describes.

# Chapter 11

## Derivatives and Function Analysis

### 11.1 What a derivative measures

The derivative is one of the main ideas of calculus. It measures how a function changes when its input changes. More precisely, it measures the limiting rate of change of the function at a point.

This idea should not be introduced only as a list of rules. Rules are useful later, but first we need to understand what the derivative is trying to describe.

#### Change in input and change in output

Let  $f$  be a function. If the input changes from  $a$  to  $a + h$ , then the change in input is

$$h.$$

The corresponding change in output is

$$f(a + h) - f(a).$$

The quotient

$$\frac{f(a + h) - f(a)}{h}$$

measures the average rate of change of the function between  $a$  and  $a + h$ .

#### Remark

The numerator measures change in the function value. The denominator measures change in the input. The quotient compares these two changes.

#### From average change to instantaneous change

The average rate of change depends on the size of the interval. To measure the rate of change at one point, we let the interval shrink. This leads to a limit.

#### Definition

The derivative of  $f$  at a point  $a$  is

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h},$$

provided this limit exists.

The derivative is therefore not only a formula. It is a limit of average rates of change.

#### Example

Let

$$f(x) = x^2.$$

At the point  $a$ , the difference quotient is

$$\frac{f(a+h) - f(a)}{h} = \frac{(a+h)^2 - a^2}{h}.$$

Expand the numerator:

$$(a+h)^2 - a^2 = a^2 + 2ah + h^2 - a^2 = 2ah + h^2.$$

Thus, for  $h \neq 0$ ,

$$\frac{f(a+h) - f(a)}{h} = \frac{2ah + h^2}{h} = 2a + h.$$

Taking the limit as  $h \rightarrow 0$ , we get

$$f'(a) = \lim_{h \rightarrow 0} (2a + h) = 2a.$$

So the derivative of  $x^2$  at  $a$  is  $2a$ .

This computation shows why the previous chapter on limits matters. The derivative is born from a limit that initially has the form  $0/0$ . We simplify the expression and then take the limit.

### Geometric meaning

On the graph of a function, the average rate of change between two points is the slope of the secant line through those points. As the second point moves closer to the first, the secant line approaches the tangent line.

The derivative  $f'(a)$  is the slope of the tangent line to the graph at  $x = a$ , if this tangent line exists.

#### Summary

When reading the derivative definition, ask:

- What is the change in input?
- What is the change in output?
- What average rate of change is being formed?
- What happens as the input change tends to zero?
- What does the result mean on the graph?

A derivative measures local change. It turns the visual idea of a tangent slope into a precise limit.

## 11.2 Difference quotients and tangent lines

The derivative definition contains the expression

$$\frac{f(a+h) - f(a)}{h}.$$

This expression is called a difference quotient. It is the bridge between average change and local change.

Before using derivative rules, we should be able to read this quotient. It will appear repeatedly in calculus.

### Average rate of change

For two inputs  $x_1$  and  $x_2$ , with  $x_1 \neq x_2$ , the average rate of change of  $f$  from  $x_1$  to  $x_2$  is

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

This quotient compares the change in output to the change in input.

If we set  $x_1 = a$  and  $x_2 = a + h$ , then the formula becomes

$$\frac{f(a + h) - f(a)}{h}.$$

#### Example

Let

$$f(x) = x^2.$$

The average rate of change from  $x = 1$  to  $x = 3$  is

$$\frac{f(3) - f(1)}{3 - 1} = \frac{9 - 1}{2} = 4.$$

This means that on this interval the output changes, on average, by 4 units for each unit of input.

The average rate of change is a number. Geometrically, it is the slope of the secant line through the two points on the graph.

### Secant lines

The graph of  $y = f(x)$  contains the points

$$(a, f(a))$$

and

$$(a + h, f(a + h)).$$

The slope of the line through these two points is

$$\frac{f(a + h) - f(a)}{(a + h) - a} = \frac{f(a + h) - f(a)}{h}.$$

This is exactly the difference quotient.

#### Remark

A difference quotient is not an arbitrary algebraic expression. It is the slope of a secant line on the graph of the function.

### Tangent lines

A tangent line is the limiting position of secant lines as the second point approaches the first. Algebraically, this means letting  $h \rightarrow 0$  in the difference quotient.

If the derivative  $f'(a)$  exists, then the tangent line to the graph at  $x = a$  has slope  $f'(a)$  and passes through the point  $(a, f(a))$ . Its equation is

$$y - f(a) = f'(a)(x - a).$$

### Example

Find the tangent line to

$$f(x) = x^2$$

at  $x = 2$ .

From the previous section, we know that

$$f'(a) = 2a.$$

Thus

$$f'(2) = 4.$$

Also,

$$f(2) = 4.$$

The tangent line passes through  $(2, 4)$  with slope 4, so

$$y - 4 = 4(x - 2).$$

Equivalently,

$$y = 4x - 4.$$

The tangent line is a local linear approximation to the graph. Near the point of tangency, the graph behaves approximately like this line.

### Why the limit may fail

The derivative at a point exists only if the difference quotient approaches one finite number as  $h \rightarrow 0$ . If the left and right behaviors are different, or if the quotient grows without bound, the derivative does not exist.

For example, the function  $f(x) = |x|$  has a corner at  $x = 0$ . From the right, the slope is 1. From the left, the slope is  $-1$ . Since these do not agree,  $f'(0)$  does not exist.

### Summary

When working with a difference quotient, ask:

- Which two inputs are being compared?
- What are the corresponding function values?
- What secant slope does the quotient represent?
- What happens when the two inputs move together?
- Does the limiting slope exist from both sides?

The derivative is the limit of secant slopes. This geometric picture should remain visible even when computations become more algebraic.

## 11.3 The derivative as a function

The derivative can be computed at a single point. But often we want to know the derivative at many points. When we assign to each input  $x$  the derivative of  $f$  at  $x$ , we obtain a new function.

This new function is called the derivative function and is written as

$$f'(x).$$

### From $f'(a)$ to $f'(x)$

The number  $f'(a)$  is the derivative at one point. It measures the local rate of change at  $x = a$ . The expression  $f'(x)$  describes the derivative at a variable point  $x$ . It tells us how the local rate of change depends on the input.

#### Example

For

$$f(x) = x^2,$$

we found from the definition that

$$f'(a) = 2a.$$

Since the point  $a$  was arbitrary, we can write

$$f'(x) = 2x.$$

This means that the slope of the tangent line to  $y = x^2$  at the point with input  $x$  is  $2x$ .

The derivative function is therefore a function about another function. It describes how the original function changes.

### Different notations

Several notations are commonly used for derivatives. If  $y = f(x)$ , then we may write

$$f'(x), \quad y', \quad \frac{dy}{dx}, \quad \frac{d}{dx}f(x).$$

These notations emphasize different aspects of the same idea.

The notation  $f'(x)$  emphasizes that the derivative is a function. The notation  $dy/dx$  emphasizes a ratio of changes in output and input. The notation  $\frac{d}{dx}$  emphasizes the operation of differentiating with respect to  $x$ .

#### Remark

The notation changes, but the mathematical object is the same: the derivative describes local rate of change.

### Where the derivative is defined

The derivative function may not be defined everywhere that the original function is defined. A function may have corners, cusps, vertical tangents, or discontinuities where the derivative does not exist.

#### Example

The function

$$f(x) = |x|$$

is defined for all real numbers. But its derivative does not exist at  $x = 0$ , because the left-hand slope is  $-1$  and the right-hand slope is  $1$ .

For  $x > 0$ , the function is  $f(x) = x$ , so  $f'(x) = 1$ . For  $x < 0$ , the function is  $f(x) = -x$ , so  $f'(x) = -1$ . At  $x = 0$ , there is no derivative.

Thus the domain of  $f'$  can be smaller than the domain of  $f$ .

## Differentiability and continuity

If a function is differentiable at a point, then it is continuous at that point. This means that having a derivative is a stronger condition than being continuous.

### Theorem

If  $f$  is differentiable at  $a$ , then  $f$  is continuous at  $a$ .

The converse is not true. A function may be continuous but not differentiable. The function  $f(x) = |x|$  is continuous at 0, but it is not differentiable there.

### Summary

When reading a derivative function, ask:

- Is the derivative being evaluated at one point or treated as a function?
- What notation is being used for the derivative?
- On which inputs does the derivative exist?
- Does the graph have corners or breaks where differentiability may fail?
- What does the derivative say about local change?

The derivative function is a tool for reading how the original function changes from point to point.

## 11.4 Basic differentiation rules

Differentiation rules are shortcuts justified by the limit definition.

### Constants and powers

A constant function has derivative zero. For every real exponent  $p$ , the power rule is valid on  $x > 0$ :

$$\frac{d}{dx}x^p = px^{p-1}.$$

For integer exponents, the formula extends to all points where the power is defined. Fractional powers require their real domains to be checked separately.

### Example

We have

$$\begin{aligned}\frac{d}{dx}x^5 &= 5x^4, \\ \frac{d}{dx}x^{-3} &= -3x^{-4}, \quad x \neq 0,\end{aligned}$$

and

$$\frac{d}{dx}\sqrt{x} = \frac{1}{2\sqrt{x}}, \quad x > 0.$$

Although  $\sqrt{x}$  is defined at zero, its derivative is not finite there.

### Constant multiples, sums, and differences

For a constant  $c$ ,

$$(cf)' = cf'.$$

For differentiable functions  $f$  and  $g$ ,

$$(f + g)' = f' + g', \quad (f - g)' = f' - g'.$$

#### Example

For

$$y = 5x^{15} - x^2 + \frac{1}{3}x - 2,$$

we obtain

$$y' = 75x^{14} - 2x + \frac{1}{3}.$$

### Rewrite before differentiating

Roots and fractions often become powers, but the original domain must remain visible. On  $x > 0$ ,

$$y = \frac{2}{\sqrt{x}} - \frac{3}{\sqrt{x^3}} = 2x^{-1/2} - 3x^{-3/2},$$

so

$$y' = -x^{-3/2} + \frac{9}{2}x^{-5/2}.$$

#### Summary

Before differentiating, identify the domain, rewrite roots and fractions when useful, and check that every formula is valid on the interval being studied.

## 11.5 Derivatives of elementary functions

Many functions appear so often that their derivatives should become familiar. The goal is not only to memorize a table, but to recognize the local behavior of standard functions.

### Exponential and logarithmic functions

The exponential function  $e^x$  has a special role: it is its own derivative.

#### Theorem

$$\frac{d}{dx}e^x = e^x.$$

For  $x > 0$ ,

$$\frac{d}{dx}\ln x = \frac{1}{x}.$$

### Example

Differentiate

$$f(x) = 3e^x - 2 \ln x.$$

Using the rules,

$$f'(x) = 3e^x - \frac{2}{x},$$

for  $x > 0$ .

The logarithm already reminds us to watch the domain. The formula  $(\ln x)' = 1/x$  is used where  $x > 0$ .

## Trigonometric functions

The basic trigonometric derivatives are:

$$\frac{d}{dx} \sin x = \cos x,$$

$$\frac{d}{dx} \cos x = -\sin x,$$

$$\frac{d}{dx} \tan x = \frac{1}{\cos^2 x}.$$

The last expression is also written as  $\sec^2 x$ , but this course will mostly use  $1/\cos^2 x$ .

### Example

Differentiate

$$f(x) = 2 \sin x - 5 \cos x.$$

Then

$$f'(x) = 2 \cos x + 5 \sin x.$$

The sign changes because the derivative of  $\cos x$  is  $-\sin x$ .

## Inverse trigonometric functions

Some later examples require inverse trigonometric functions. The basic derivatives are:

$$\frac{d}{dx} \arctan x = \frac{1}{1+x^2},$$

$$\frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1-x^2}}, \quad -1 < x < 1,$$

$$\frac{d}{dx} \arccos x = -\frac{1}{\sqrt{1-x^2}}, \quad -1 < x < 1.$$

### Example

Differentiate

$$f(x) = x \arctan x.$$

This expression is a product, so the product rule will be needed. Once that rule is introduced, we obtain

$$f'(x) = \arctan x + \frac{x}{1+x^2}.$$

This example previews an important point: knowing elementary derivatives is not enough. We must also know how functions are combined.

### A compact table

| Function    | Derivative        |
|-------------|-------------------|
| $c$         | $0$               |
| $x^p$       | $px^{p-1}$        |
| $e^x$       | $e^x$             |
| $\ln x$     | $1/x$             |
| $\sin x$    | $\cos x$          |
| $\cos x$    | $-\sin x$         |
| $\tan x$    | $1/\cos^2 x$      |
| $\arctan x$ | $1/(1+x^2)$       |
| $\arcsin x$ | $1/\sqrt{1-x^2}$  |
| $\arccos x$ | $-1/\sqrt{1-x^2}$ |

A derivative table is useful only if it is combined with careful reading of the expression. Before applying a formula, identify the function and its domain.

#### Summary

When using elementary derivatives, ask:

- Which elementary function appears?
- Is the expression exactly elementary, or is it composed with another function?
- What domain restrictions are present?
- Are product, quotient, or chain rules also needed?

Elementary derivatives are building blocks. Most real examples are built by combining them.

## 11.6 Chain, product, and quotient rules

Most functions are not single elementary building blocks. They are combinations of simpler functions. To differentiate them, we need rules that describe how derivatives behave under products, quotients, and composition.

The first task is always to read the structure of the expression.

### The product rule

The derivative of a product is not the product of the derivatives. Both factors may change, and the rule must account for both changes.

#### Theorem

If  $f$  and  $g$  are differentiable, then

$$(fg)' = f'g + fg'.$$

Equivalently,

$$\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x).$$

### Example

Differentiate

$$y = x^2 e^x.$$

Let

$$f(x) = x^2, \quad g(x) = e^x.$$

Then

$$f'(x) = 2x, \quad g'(x) = e^x.$$

By the product rule,

$$y' = 2xe^x + x^2 e^x.$$

We may factor:

$$y' = e^x(2x + x^2).$$

## The quotient rule

For a quotient, the denominator also changes. The quotient rule is:

### Theorem

If  $f$  and  $g$  are differentiable and  $g(x) \neq 0$ , then

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}.$$

### Example

Differentiate

$$y = \frac{x^2 + 1}{x - 1}.$$

Let

$$f(x) = x^2 + 1, \quad g(x) = x - 1.$$

Then

$$f'(x) = 2x, \quad g'(x) = 1.$$

By the quotient rule,

$$y' = \frac{2x(x - 1) - (x^2 + 1) \cdot 1}{(x - 1)^2}.$$

Simplify the numerator:

$$2x(x - 1) - (x^2 + 1) = 2x^2 - 2x - x^2 - 1 = x^2 - 2x - 1.$$

Thus

$$y' = \frac{x^2 - 2x - 1}{(x - 1)^2}.$$

The quotient rule should be used with attention to the domain. The original function is

defined only where the denominator is not zero.

### The chain rule

The chain rule is used for compositions of functions. If an expression has an inside function and an outside function, the derivative must account for both.

#### Theorem

If

$$y = f(g(x)),$$

then

$$y' = f'(g(x)) g'(x).$$

A useful way to say this is: differentiate the outside function, keep the inside function, and multiply by the derivative of the inside function.

#### Example

Differentiate

$$y = (3x + 1)^7.$$

The outside function is the seventh power, and the inside function is

$$3x + 1.$$

By the chain rule,

$$y' = 7(3x + 1)^6 \cdot 3.$$

Therefore

$$y' = 21(3x + 1)^6.$$

#### Example

Differentiate

$$y = \sin(x^2).$$

The outside function is  $\sin u$ , and the inside function is  $u = x^2$ . Thus

$$y' = \cos(x^2) \cdot 2x.$$

So

$$y' = 2x \cos(x^2).$$

### Reading nested expressions

The chain rule is often the most important rule because many expressions are nested. Before differentiating, identify the layers.

#### Example

Differentiate

$$y = e^{\cos^2 x}.$$

This expression has several layers:

$$x \mapsto \cos x \mapsto (\cos x)^2 \mapsto e^{\cos^2 x}.$$

Let

$$u = \cos^2 x.$$

Then

$$y = e^u, \quad y' = e^u u'.$$

Now

$$u = (\cos x)^2,$$

so by the chain rule again,

$$u' = 2 \cos x \cdot (-\sin x).$$

Therefore

$$y' = e^{\cos^2 x} \cdot 2 \cos x (-\sin x).$$

Thus

$$y' = -2e^{\cos^2 x} \sin x \cos x.$$

### Summary

Before differentiating a combined expression, ask:

- Is the expression a sum, product, quotient, or composition?
- Are there nested functions?
- What is the inside function?
- What is the outside function?
- Do I need more than one rule?
- Can the result be simplified without hiding its structure?

Most differentiation errors come from misreading the structure of the expression. Read first, then apply the rule.

## 11.7 Monotonicity and the first derivative

The derivative measures local change. This makes it a powerful tool for understanding whether a function is increasing or decreasing. The sign of the derivative tells us the local direction of change.

This is the first major use of derivatives in function analysis.

### Positive and negative derivative

If  $f'(x) > 0$  on an interval, then the function is increasing on that interval. If  $f'(x) < 0$ , then the function is decreasing.

### Theorem

Let  $f$  be differentiable on an interval.

- If  $f'(x) > 0$  for all  $x$  in the interval, then  $f$  is increasing there.
- If  $f'(x) < 0$  for all  $x$  in the interval, then  $f$  is decreasing there.

This theorem should be read geometrically. A positive derivative means that tangent lines have positive slope. The graph rises as we move from left to right. A negative derivative means that tangent lines have negative slope. The graph falls.

### A basic example

#### Example

Study the monotonicity of

$$f(x) = x^2.$$

The derivative is

$$f'(x) = 2x.$$

The sign of  $2x$  depends on  $x$ :

$$2x < 0 \quad \text{for } x < 0,$$

$$2x = 0 \quad \text{for } x = 0,$$

$$2x > 0 \quad \text{for } x > 0.$$

Therefore  $f$  is decreasing on  $(-\infty, 0)$  and increasing on  $(0, \infty)$ .

This conclusion matches the graph of the parabola. The derivative gives an algebraic way to detect what the graph shows visually.

### Sign charts

A sign chart is a simple way to organize where the derivative is positive, negative, or zero. The important points are usually where the derivative is zero or undefined. These points divide the domain into intervals. On each interval, we check the sign of the derivative.

#### Example

Study the monotonicity of

$$f(x) = x^3 - 3x.$$

First compute the derivative:

$$f'(x) = 3x^2 - 3 = 3(x^2 - 1) = 3(x - 1)(x + 1).$$

The derivative is zero at

$$x = -1 \quad \text{and} \quad x = 1.$$

These points divide the real line into three intervals:

$$(-\infty, -1), \quad (-1, 1), \quad (1, \infty).$$

Check the sign of  $f'(x) = 3(x - 1)(x + 1)$ :

- on  $(-\infty, -1)$ , both factors are negative, so  $f'(x) > 0$ ,

- on  $(-1, 1)$ , one factor is negative and one is positive, so  $f'(x) < 0$ ,
- on  $(1, \infty)$ , both factors are positive, so  $f'(x) > 0$ .

Therefore  $f$  is increasing on  $(-\infty, -1)$ , decreasing on  $(-1, 1)$ , and increasing on  $(1, \infty)$ .

The derivative does not only compute slopes at isolated points. It organizes the whole graph into intervals of increasing and decreasing behavior.

### Where the derivative is zero

A derivative equal to zero means that the tangent line is horizontal. This may indicate a local maximum or local minimum, but it does not always do so.

#### Example

For

$$f(x) = x^3,$$

we have

$$f'(x) = 3x^2.$$

Thus

$$f'(0) = 0.$$

However,  $x^3$  is increasing on both sides of 0. The point  $x = 0$  is not a local maximum or a local minimum.

This example is important. A zero derivative is a signal to investigate, not an automatic conclusion.

#### Summary

When using the first derivative for monotonicity, ask:

- What is the domain of the function?
- Where is  $f'(x) = 0$ ?
- Where is  $f'(x)$  undefined?
- What intervals are created by these points?
- What is the sign of  $f'(x)$  on each interval?
- What does that sign say about the graph?

The first derivative is a language for reading growth and decline.

## 11.8 Local extrema and critical points

A local maximum or local minimum is a place where the function is higher or lower than nearby values. Derivatives help us find candidates for such points, but they do not replace interpretation.

The main idea is this: if a smooth graph changes from increasing to decreasing, it has a local maximum. If it changes from decreasing to increasing, it has a local minimum.

## Local maxima and minima

### Definition

A function  $f$  has a local maximum at  $a$  if  $f(a)$  is greater than or equal to nearby values of the function.

A function  $f$  has a local minimum at  $a$  if  $f(a)$  is less than or equal to nearby values of the function.

The word local means nearby. A local maximum does not have to be the largest value of the function on its whole domain. It only has to be largest in a neighborhood of the point.

## Critical points

Local extrema often occur at points where the derivative is zero or undefined.

### Definition

A critical point of  $f$  is a point  $a$  in the domain of  $f$  such that

$$f'(a) = 0$$

or  $f'(a)$  does not exist.

Critical points are candidates for local extrema. They must be checked. Not every critical point is an extremum.

### Example

For

$$f(x) = x^2,$$

we have

$$f'(x) = 2x.$$

The derivative is zero at  $x = 0$ . The function decreases before 0 and increases after 0. Therefore  $x = 0$  is a local minimum.

### Example

For

$$f(x) = x^3,$$

we have

$$f'(x) = 3x^2.$$

The derivative is zero at  $x = 0$ . But the function is increasing both before and after 0. Therefore  $x = 0$  is a critical point, but not a local extremum.

These examples show why finding  $f'(x) = 0$  is only the beginning of the analysis.

## The first derivative test

The first derivative test uses the sign of  $f'$  around a critical point.

### Theorem

Let  $a$  be a critical point of  $f$ .

- If  $f'$  changes from positive to negative at  $a$ , then  $f$  has a local maximum at  $a$ .
- If  $f'$  changes from negative to positive at  $a$ , then  $f$  has a local minimum at  $a$ .
- If  $f'$  does not change sign at  $a$ , then  $a$  is not a local extremum by this test.

### Example

Find the local extrema of

$$f(x) = x^3 - 3x.$$

We computed earlier that

$$f'(x) = 3(x - 1)(x + 1).$$

The critical points are

$$x = -1 \quad \text{and} \quad x = 1.$$

The sign of  $f'$  is positive on  $(-\infty, -1)$ , negative on  $(-1, 1)$ , and positive on  $(1, \infty)$ .

At  $x = -1$ , the derivative changes from positive to negative, so there is a local maximum. Its value is

$$f(-1) = (-1)^3 - 3(-1) = 2.$$

At  $x = 1$ , the derivative changes from negative to positive, so there is a local minimum. Its value is

$$f(1) = 1^3 - 3 = -2.$$

The derivative sign tells us the shape of the graph: increasing, then decreasing, then increasing.

### Endpoints

On a closed interval, endpoints must be considered separately. A derivative test describes behavior inside an interval, but an endpoint can still be the largest or smallest value on the interval.

For example, on the interval  $[0, 2]$ , the function  $f(x) = x^2$  has its smallest value at  $x = 0$  and largest value at  $x = 2$ . The endpoint  $x = 2$  is not found by solving  $f'(x) = 0$ , but it matters for the interval.

### Summary

When looking for extrema, ask:

- What is the domain or interval being studied?
- Where is  $f'(x) = 0$ ?
- Where does  $f'(x)$  fail to exist while  $f$  is defined?
- How does the sign of  $f'$  change around each critical point?
- Are endpoints part of the problem?
- What are the actual function values at the candidate points?

Extrema are not only derivative facts. They are statements about function values near a point or on an interval.

## 11.9 Second derivative and convexity

The first derivative tells us whether a function is increasing or decreasing. The second derivative tells us how the first derivative changes. This gives information about the bending of the graph.

If the first derivative is increasing, slopes are becoming larger. If the first derivative is decreasing, slopes are becoming smaller. This is the idea behind convexity and concavity.

### The second derivative

If  $f'$  is differentiable, then the derivative of  $f'$  is called the second derivative of  $f$ . It is written

$$f''(x).$$

Thus

$$f''(x) = (f'(x))'.$$

#### Example

Let

$$f(x) = x^3 - 3x.$$

Then

$$f'(x) = 3x^2 - 3.$$

Differentiate again:

$$f''(x) = 6x.$$

The second derivative tells us how the slope  $f'(x)$  changes.

### Convexity and concavity

If  $f''(x) > 0$  on an interval, the function is convex there. The graph bends upward. If  $f''(x) < 0$ , the function is concave there. The graph bends downward.

#### Theorem

Let  $f$  be twice differentiable on an interval.

- If  $f''(x) > 0$  on the interval, then  $f$  is convex on that interval.
- If  $f''(x) < 0$  on the interval, then  $f$  is concave on that interval.

#### Remark

Terminology varies between textbooks. The phrases *concave up* and *concave down* are also common. In these notes,  $f'' > 0$  is called *convex*, and  $f'' < 0$  is called *concave*.

This should be connected to slopes. When  $f'' > 0$ , the slopes increase. When  $f'' < 0$ , the slopes decrease.

### Example

Study the convexity of

$$f(x) = x^3 - 3x.$$

We have

$$f''(x) = 6x.$$

Thus

$$f''(x) < 0 \quad \text{for } x < 0,$$

and

$$f''(x) > 0 \quad \text{for } x > 0.$$

Therefore the function is concave on  $(-\infty, 0)$  and convex on  $(0, \infty)$ .

### Inflection points

An inflection point is a point where the graph changes convexity. Usually, this occurs where the second derivative changes sign.

#### Definition

A point  $a$  is an inflection point of  $f$  if the graph of  $f$  changes convexity at  $a$ .

### Example

For

$$f(x) = x^3 - 3x,$$

we found

$$f''(x) = 6x.$$

The second derivative changes sign at  $x = 0$ . Therefore the graph has an inflection point at  $x = 0$ . The corresponding point on the graph is

$$(0, f(0)) = (0, 0).$$

The condition  $f''(a) = 0$  by itself is not enough. The second derivative must change sign for convexity to change.

### Second derivative test for extrema

The second derivative can sometimes classify critical points.

#### Theorem

Suppose  $f'(a) = 0$ .

- If  $f''(a) > 0$ , then  $f$  has a local minimum at  $a$ .
- If  $f''(a) < 0$ , then  $f$  has a local maximum at  $a$ .
- If  $f''(a) = 0$ , the test gives no conclusion.

The test matches the geometric picture. If the graph is convex at a horizontal tangent, it bends upward and forms a local minimum. If it is concave at a horizontal tangent, it bends downward and forms a local maximum.

### Example

Let

$$f(x) = x^2.$$

Then

$$f'(x) = 2x, \quad f''(x) = 2.$$

The critical point is  $x = 0$ . Since

$$f''(0) = 2 > 0,$$

there is a local minimum at  $x = 0$ .

### Summary

When using the second derivative, ask:

- What is  $f''(x)$ ?
- Where is  $f''(x) > 0$  or  $f''(x) < 0$ ?
- Which convexity terminology is being used?
- Does the graph change convexity?
- Are there inflection points?
- Can the second derivative test classify a critical point?
- If  $f''(a) = 0$ , is another test needed?

The second derivative helps us read the bending of a graph, not only its direction of movement.

## 11.10 Asymptotes and global function analysis

Derivatives describe local change, but function analysis asks for a global picture. We want to understand the domain, intercepts, signs, monotonicity, extrema, convexity, asymptotes, and overall shape of the graph.

No single tool answers all of these questions. A complete analysis combines algebra, limits, and derivatives.

### Domain first

The domain should be found before differentiating. If a function is not defined at certain points, those points may become vertical asymptotes, gaps, or boundaries of intervals.

### Example

For

$$f(x) = \frac{1}{x-2},$$

the domain is

$$\mathbb{R} \setminus \{2\}.$$

The point  $x = 2$  must be treated separately. It is not allowed as an input, and it may create a vertical asymptote.

Differentiating before reading the domain can hide important structure.

### Vertical asymptotes

A vertical asymptote occurs when the function grows without bound near a finite point. If

$$\lim_{x \rightarrow a^-} f(x) = \pm\infty$$

or

$$\lim_{x \rightarrow a^+} f(x) = \pm\infty,$$

then the line

$$x = a$$

is a vertical asymptote.

#### Example

For

$$f(x) = \frac{1}{x-2},$$

we have

$$\lim_{x \rightarrow 2^+} \frac{1}{x-2} = \infty,$$

and

$$\lim_{x \rightarrow 2^-} \frac{1}{x-2} = -\infty.$$

Thus  $x = 2$  is a vertical asymptote.

Vertical asymptotes are detected by limits, not by derivatives.

### Horizontal asymptotes

Horizontal asymptotes describe long-term behavior. If

$$\lim_{x \rightarrow \infty} f(x) = L$$

or

$$\lim_{x \rightarrow -\infty} f(x) = L,$$

then

$$y = L$$

is a horizontal asymptote in that direction.

#### Example

Let

$$f(x) = \frac{2x+1}{x-3}.$$

As  $x \rightarrow \infty$ , the leading terms dominate:

$$\frac{2x+1}{x-3} \rightarrow 2.$$

Also, as  $x \rightarrow -\infty$ , the same limit is 2. Therefore

$$y = 2$$

is a horizontal asymptote.

Horizontal asymptotes tell us what the graph approaches far away, not necessarily what values it never reaches.

### A general analysis plan

A complete function analysis should be organized. The following order is often useful:

1. Find the domain.
2. Check intercepts and simple values if useful.
3. Study limits at excluded points and at infinity.
4. Find vertical and horizontal asymptotes.
5. Compute the first derivative.
6. Use the first derivative to determine monotonicity and local extrema.
7. Compute the second derivative if convexity is required.
8. Use the second derivative to determine convexity and inflection points.
9. Combine all information into a coherent graph description.

This plan is not a rigid algorithm. Some functions require special attention. But it prevents us from looking only at derivatives while ignoring the domain and limits.

#### Summary

For global function analysis, ask:

- What is the domain?
- What happens near excluded points?
- What happens as  $x \rightarrow \infty$  and  $x \rightarrow -\infty$ ?
- Where is the function increasing or decreasing?
- Where are local extrema?
- Where is the function convex or concave?
- What graph is consistent with all this information?

Function analysis is the art of combining many local and global facts into one picture.

## 11.11 Complete function analysis examples

A complete function analysis combines the tools developed in this chapter. We do not want isolated derivative calculations. We want a coherent description of the function.

The following example shows how the pieces fit together.

## A polynomial example

### Example

Analyze the function

$$f(x) = x^3 - 3x.$$

**Domain.** The function is a polynomial, so its domain is

$$\mathbb{R}.$$

**Intercepts.** We factor:

$$f(x) = x(x^2 - 3).$$

Thus the zeros are

$$x = 0, \quad x = \sqrt{3}, \quad x = -\sqrt{3}.$$

**Limits at infinity.** The leading term is  $x^3$ . Therefore

$$\lim_{x \rightarrow \infty} f(x) = \infty, \quad \lim_{x \rightarrow -\infty} f(x) = -\infty.$$

There are no horizontal asymptotes.

**First derivative.** We compute

$$f'(x) = 3x^2 - 3 = 3(x - 1)(x + 1).$$

The critical points are

$$x = -1 \quad \text{and} \quad x = 1.$$

The sign of  $f'$  is positive on  $(-\infty, -1)$ , negative on  $(-1, 1)$ , and positive on  $(1, \infty)$ . Therefore  $f$  is increasing, then decreasing, then increasing.

At  $x = -1$ , the derivative changes from positive to negative, so there is a local maximum:

$$f(-1) = 2.$$

At  $x = 1$ , the derivative changes from negative to positive, so there is a local minimum:

$$f(1) = -2.$$

**Second derivative.** We compute

$$f''(x) = 6x.$$

Thus  $f''(x) < 0$  for  $x < 0$ , and  $f''(x) > 0$  for  $x > 0$ . The graph is concave on  $(-\infty, 0)$  and convex on  $(0, \infty)$ . Since the second derivative changes sign at 0, there is an inflection point at

$$(0, 0).$$

**Conclusion.** The graph starts down on the left, increases to a local maximum at  $(-1, 2)$ , decreases to a local minimum at  $(1, -2)$ , and then increases to infinity. Its convexity changes at the origin.

The example is not important because the function is difficult. It is important because it shows the order of analysis. Each step answers a different question.

## A rational example

### Example

Analyze the basic behavior of

$$f(x) = \frac{x+1}{x-1}.$$

**Domain.** The denominator cannot be zero, so

$$D_f = \mathbb{R} \setminus \{1\}.$$

**Vertical asymptote.** Near  $x = 1$ , the denominator tends to zero. Since the numerator tends to 2, the function grows without bound. Thus

$$x = 1$$

is a vertical asymptote.

**Horizontal asymptote.** For large  $|x|$ , the leading terms dominate:

$$\frac{x+1}{x-1} \rightarrow 1.$$

So

$$y = 1$$

is a horizontal asymptote.

**Derivative.** Using the quotient rule,

$$f'(x) = \frac{1 \cdot (x-1) - (x+1) \cdot 1}{(x-1)^2} = \frac{x-1-x-1}{(x-1)^2} = \frac{-2}{(x-1)^2}.$$

Since

$$(x-1)^2 > 0$$

for every  $x \neq 1$ , we have

$$f'(x) < 0$$

on both intervals of the domain. Therefore the function is decreasing on  $(-\infty, 1)$  and on  $(1, \infty)$ .

**Conclusion.** The graph has a vertical asymptote at  $x = 1$ , a horizontal asymptote at  $y = 1$ , and decreases on each interval of its domain.

This example shows why domain and limits must be studied before or together with derivatives. The derivative alone does not reveal the vertical asymptote unless we already know where the function is not defined.

## How to write a final analysis

A final solution should not be only a list of calculations. It should connect calculations to conclusions. For example, after finding  $f'(x) > 0$ , state that the function is increasing. After finding a limit at infinity, state the horizontal asymptote. After finding a sign change in  $f''$ , state the inflection point.

## Summary

A complete function analysis should usually include:

- domain,
- intercepts or zeros, when useful,
- limits at boundary points and infinity,
- asymptotes,
- first derivative and monotonicity,
- local extrema,
- second derivative and convexity,
- inflection points,
- a final coherent description of the graph.

The purpose of function analysis is understanding. Derivatives are tools for that understanding, not the whole story by themselves.

# Chapter 12

## Integrals

### 12.1 What an integral measures

Integration begins with a question that is complementary to differentiation. Differentiation asks how a function changes locally. Integration asks how quantities accumulate.

There are two closely related ways to meet integrals. An indefinite integral asks for a function whose derivative is known. A definite integral asks for accumulated signed area or total change over an interval. Both ideas are important, and both will appear in this chapter.

#### Undoing differentiation

Suppose we know that

$$F'(x) = f(x).$$

Then  $F$  is called an antiderivative of  $f$ . Finding such a function is one meaning of integration.

For example, since

$$\frac{d}{dx}x^2 = 2x,$$

we say that  $x^2$  is an antiderivative of  $2x$ .

But it is not the only one. For every constant  $C$ ,

$$\frac{d}{dx}(x^2 + C) = 2x.$$

The derivative of a constant is zero, so differentiation loses constant information.

#### Remark

Integration as antidifferentiation recovers a function from its derivative, but only up to an additive constant.

#### Accumulation

The second meaning of integration is accumulation. If a quantity changes continuously, an integral can measure the total accumulated amount over an interval.

For a non-negative function, the definite integral

$$\int_a^b f(x) dx$$

can be read geometrically as the area under the graph of  $f$  from  $x = a$  to  $x = b$ . If the function takes negative values, the integral measures signed area: regions below the horizontal axis contribute negatively.

This interpretation will become more precise later. For now, the important point is that integration adds up infinitely many small contributions.

#### The notation

The symbol

$$\int f(x) dx$$

denotes an indefinite integral. It asks for all antiderivatives of  $f$ .

The symbol

$$\int_a^b f(x) dx$$

denotes a definite integral. It has limits of integration  $a$  and  $b$ , and its result is a number.

The expression  $dx$  reminds us of the variable of integration. It also hints at the idea of summing small pieces along the  $x$ -axis.

### Example

The expression

$$\int 2x dx$$

asks for functions whose derivative is  $2x$ . The answer is

$$x^2 + C.$$

The expression

$$\int_0^3 2x dx$$

asks for a number: the accumulated value of  $2x$  from 0 to 3. Later we will compute it as

$$\left[ x^2 \right]_0^3 = 9.$$

### Summary

When reading an integral, ask:

- Is the integral indefinite or definite?
- Am I looking for a family of antiderivatives or a number?
- What is the variable of integration?
- Is the integral being used to undo differentiation or to measure accumulation?
- What role does the constant of integration play?

The integral is not only a symbol for a calculation. It is a language for recovering functions and measuring accumulated change.

## 12.2 Antiderivatives and indefinite integrals

An antiderivative reverses differentiation. If differentiating  $F$  gives  $f$ , then integrating  $f$  should recover  $F$ , up to a constant.

### Definition

A function  $F$  is an antiderivative of  $f$  on an interval if

$$F'(x) = f(x)$$

for every  $x$  in that interval.

The phrase “on an interval” matters. Antiderivatives are usually discussed where the functions involved are defined and behave consistently.

### Why the constant appears

If  $F'(x) = f(x)$ , then for every constant  $C$ ,

$$(F(x) + C)' = F'(x) + 0 = f(x).$$

Thus one antiderivative gives infinitely many antiderivatives. They differ by constants.

This is why the indefinite integral is written as

$$\int f(x) dx = F(x) + C.$$

The constant  $C$  is not decoration. It records the information lost when differentiating.

#### Example

Since

$$\frac{d}{dx} x^3 = 3x^2,$$

we have

$$\int 3x^2 dx = x^3 + C.$$

The functions  $x^3$ ,  $x^3 + 5$ , and  $x^3 - 17$  all have derivative  $3x^2$ .

### Checking an integral

The best way to check an indefinite integral is to differentiate the answer. If the derivative returns the integrand, the antiderivative is correct.

#### Example

Claim:

$$\int (4x^3 - 2) dx = x^4 - 2x + C.$$

Check by differentiating:

$$\frac{d}{dx} (x^4 - 2x + C) = 4x^3 - 2.$$

This matches the integrand, so the integral is correct.

This habit is very useful. Integration techniques can be subtle, but differentiation of the final answer is often straightforward.

### Indefinite integrals are families

The expression

$$\int f(x) dx$$

does not represent one function. It represents the family of all antiderivatives of  $f$ . The constant  $C$  describes this family.

### Remark

When the answer to an indefinite integral is written without  $+C$ , it is incomplete unless a particular antiderivative has been explicitly requested.

### Summary

When working with indefinite integrals, ask:

- What function has the given integrand as its derivative?
- Did I include the constant of integration?
- Can I check the answer by differentiating?
- Is the integral asking for one function or a family of functions?

Indefinite integration is the systematic search for antiderivatives.

## 12.3 Basic integration rules

Basic integration rules come from basic differentiation rules read backward. The derivative of a constant is zero, the derivative of a power follows the power rule, and the derivative of a sum is the sum of derivatives. Integration reverses these facts.

### The power rule for integration

For powers of  $x$ , the main rule is:

#### Theorem

If  $p \neq -1$ , then

$$\int x^p dx = \frac{x^{p+1}}{p+1} + C$$

on every interval on which the real power  $x^p$  and the displayed antiderivative are defined.

For an arbitrary real exponent, the safest common domain is  $x > 0$ . Integer powers extend to their usual real domains, while rational powers require their domains to be checked separately.

The condition  $p \neq -1$  is essential. When  $p = -1$ , the integrand is  $1/x$ , and its integral is logarithmic on any interval not crossing zero:

$$\int \frac{1}{x} dx = \ln |x| + C.$$

#### Example

Compute

$$\int x^5 dx.$$

Here  $p = 5$ , so

$$\int x^5 dx = \frac{x^6}{6} + C.$$

Check:

$$\frac{d}{dx} \left( \frac{x^6}{6} \right) = x^5.$$

## Constants and sums

Constants can be pulled out of integrals:

$$\int cf(x) dx = c \int f(x) dx.$$

Sums and differences can be integrated term by term:

$$\int (f(x) + g(x)) dx = \int f(x) dx + \int g(x) dx.$$

### Example

Compute

$$\int (5x^4 - 3x^2 + 7) dx.$$

Integrate term by term:

$$\int 5x^4 dx = x^5,$$

$$\int (-3x^2) dx = -x^3,$$

$$\int 7 dx = 7x.$$

Therefore

$$\int (5x^4 - 3x^2 + 7) dx = x^5 - x^3 + 7x + C.$$

## Rewriting before integrating

Many integrals become simple only after rewriting roots and fractions as powers. The domain of the original expression must remain visible.

### Example

For  $x > 0$ , compute

$$\int \frac{x\sqrt{x}}{\sqrt[3]{x}} dx.$$

Rewrite using exponents:

$$\frac{x\sqrt{x}}{\sqrt[3]{x}} = \frac{x \cdot x^{1/2}}{x^{1/3}} = x^{1+1/2-1/3} = x^{7/6}.$$

Thus

$$\int \frac{x\sqrt{x}}{\sqrt[3]{x}} dx = \int x^{7/6} dx = \frac{x^{13/6}}{13/6} + C = \frac{6}{13}x^{13/6} + C.$$

The integral was not difficult after the expression was read correctly. The main work was rewriting the integrand into a power without forgetting its domain.

## Elementary integrals

The following basic antiderivatives should become familiar:

| Integrand        | Antiderivative                                |
|------------------|-----------------------------------------------|
| $x^p, p \neq -1$ | $\frac{x^{p+1}}{p+1} + C$ on a valid interval |
| $1/x$            | $\ln x  + C$                                  |
| $e^x$            | $e^x + C$                                     |
| $\sin x$         | $-\cos x + C$                                 |
| $\cos x$         | $\sin x + C$                                  |
| $1/\cos^2 x$     | $\tan x + C$                                  |
| $1/(1+x^2)$      | $\arctan x + C$                               |

A table is useful only after the integrand and its domain have been read. Sometimes the integrand must first be rewritten or transformed.

### Summary

When applying basic integration rules, ask:

- What is the domain of the integrand?
- Can the integrand be rewritten as powers of  $x$ ?
- Is there a term  $1/x$ , which gives a logarithm?
- Can I integrate term by term?
- Did I include the constant  $C$ ?
- Can I check the result by differentiating on the stated interval?

Basic rules are the foundation. More advanced methods often aim to transform an integral into one of these basic forms.

## 12.4 The substitution method

The substitution method is the integration counterpart of the chain rule. It is used when the integrand contains a composite expression together with something close to its derivative.

The idea is to rename the inner expression so that the integral becomes simpler.

### The basic idea

Suppose we see an expression such as

$$\int 2x \cos(x^2) dx.$$

The inside expression is  $x^2$ , and its derivative is  $2x$ . This suggests the substitution

$$u = x^2.$$

Then

$$du = 2x dx.$$

The integral becomes

$$\int \cos u du = \sin u + C.$$

Returning to  $x$ , we obtain

$$\int 2x \cos(x^2) dx = \sin(x^2) + C.$$

### Remark

Substitution works by recognizing an inside function and its derivative. It reverses the chain rule.

## A step-by-step example

### Example

Compute

$$\int x\sqrt{1+x^2} \, dx.$$

The expression under the square root is

$$1+x^2.$$

Its derivative is  $2x$ , and the integrand contains  $x \, dx$ . Let

$$u = 1+x^2.$$

Then

$$du = 2x \, dx, \quad x \, dx = \frac{1}{2} du.$$

The integral becomes

$$\int x\sqrt{1+x^2} \, dx = \frac{1}{2} \int u^{1/2} \, du.$$

Using the power rule,

$$\frac{1}{2} \int u^{1/2} \, du = \frac{1}{2} \cdot \frac{u^{3/2}}{3/2} + C = \frac{1}{3} u^{3/2} + C.$$

Return to  $x$ :

$$\int x\sqrt{1+x^2} \, dx = \frac{1}{3} (1+x^2)^{3/2} + C.$$

The factor  $1/2$  appears because the integrand contains  $x \, dx$ , not exactly  $2x \, dx$ . Constants can be adjusted.

## Substitution with logarithms and exponentials

Substitution often appears when a logarithm or exponential contains an inner expression.

### Example

Compute

$$\int \frac{e^x}{2e^x + 1} \, dx.$$

Let

$$u = 2e^x + 1.$$

Then

$$du = 2e^x \, dx, \quad e^x \, dx = \frac{1}{2} du.$$

Therefore

$$\int \frac{e^x}{2e^x + 1} \, dx = \frac{1}{2} \int \frac{1}{u} \, du.$$

Thus

$$\int \frac{e^x}{2e^x + 1} dx = \frac{1}{2} \ln |u| + C = \frac{1}{2} \ln |2e^x + 1| + C.$$

In this example, the denominator became the new variable. Its derivative was present in the numerator up to a constant factor.

### Substitution with trigonometric functions

#### Example

Compute

$$\int \cos x e^{\sin x} dx.$$

The inside expression is  $\sin x$ , and its derivative is  $\cos x$ . Let

$$u = \sin x.$$

Then

$$du = \cos x dx.$$

The integral becomes

$$\int e^u du = e^u + C.$$

Returning to  $x$ ,

$$\int \cos x e^{\sin x} dx = e^{\sin x} + C.$$

The method is the same: identify the inside function, check whether its derivative appears, and rewrite the integral in the new variable.

### Substitution is a change of language

A substitution is not just a symbol trick. It changes the variable so that the structure becomes simpler. The final answer must be returned to the original variable unless the problem asks otherwise.

#### Summary

When trying substitution, ask:

- Is there an inside expression repeated in the integrand?
- Is its derivative present, possibly up to a constant factor?
- What should be called  $u$ ?
- What is  $du$ ?
- After integrating in  $u$ , did I return to the original variable?
- Can I check by differentiating the final answer?

Substitution is one of the most important integration methods because many complicated-looking integrals are simple chain rules read backward.

## 12.5 Integration by parts

Integration by parts is the method that reverses the product rule. It is useful when an integrand is a product of two types of functions and substitution does not directly simplify the expression.

The product rule says

$$(uv)' = u'v + uv'.$$

Rearranging and integrating gives the integration by parts formula.

### Theorem

If  $u = u(x)$  and  $v = v(x)$  are differentiable, then

$$\int u \, dv = uv - \int v \, du.$$

Equivalently,

$$\int u(x)v'(x) \, dx = u(x)v(x) - \int v(x)u'(x) \, dx.$$

The method replaces one integral by another. It is useful only if the new integral is simpler.

### A first example

#### Example

Compute

$$\int xe^x \, dx.$$

Choose

$$u = x, \quad dv = e^x \, dx.$$

Then

$$du = dx, \quad v = e^x.$$

By integration by parts,

$$\int xe^x \, dx = xe^x - \int e^x \, dx.$$

Therefore

$$\int xe^x \, dx = xe^x - e^x + C.$$

The choice of  $u$  matters. We chose  $u = x$  because differentiating it makes it simpler.

### Logarithmic functions

Integration by parts is often useful when a logarithm appears.

#### Example

Compute

$$\int \ln x \, dx, \quad x > 0.$$

Although this does not visibly look like a product, we can write

$$\ln x = 1 \cdot \ln x.$$

Choose

$$u = \ln x, \quad dv = dx.$$

Then

$$du = \frac{1}{x} dx, \quad v = x.$$

Thus

$$\int \ln x \, dx = x \ln x - \int x \cdot \frac{1}{x} \, dx.$$

So

$$\int \ln x \, dx = x \ln x - \int 1 \, dx = x \ln x - x + C.$$

This example shows that a factor of 1 may be introduced when it helps us apply the method.

### Repeated integration by parts

Sometimes integration by parts must be applied more than once.

#### Example

Compute

$$\int x^2 e^x \, dx.$$

Choose

$$u = x^2, \quad dv = e^x \, dx.$$

Then

$$du = 2x \, dx, \quad v = e^x.$$

Therefore

$$\int x^2 e^x \, dx = x^2 e^x - \int 2x e^x \, dx.$$

Now apply integration by parts to  $\int x e^x \, dx$ , which was computed earlier:

$$\int x e^x \, dx = x e^x - e^x + C.$$

Thus

$$\int x^2 e^x \, dx = x^2 e^x - 2(x e^x - e^x) + C.$$

So

$$\int x^2 e^x \, dx = e^x (x^2 - 2x + 2) + C.$$

The pattern is clear: differentiating the polynomial lowers its degree until the remaining integral is elementary.

### Choosing $u$

There is no perfect rule for choosing  $u$ , but a useful principle is to choose  $u$  so that differentiating it simplifies the expression. Logarithms, inverse trigonometric functions, and polynomial factors are often good candidates for  $u$ .

## Summary

When using integration by parts, ask:

- Does the integrand look like a product?
- Which factor becomes simpler when differentiated?
- Which factor can be integrated to find  $v$ ?
- Is the new integral simpler than the original one?
- Do I need to apply the method repeatedly?
- Can I check the final answer by differentiating?

Integration by parts is not a formula to apply mechanically. It is a way to move differentiation from one factor to another.

## 12.6 Integrating rational functions

A rational function is a quotient of polynomials. Integrals of rational functions can range from simple to quite technical. In this course, the goal is to learn the basic reading habits: simplify first, look for a derivative of the denominator, and recognize standard logarithmic and arctangent patterns.

### When the numerator is the derivative of the denominator

If the integrand has the form

$$\frac{g'(x)}{g(x)},$$

then the integral is logarithmic:

$$\int \frac{g'(x)}{g(x)} dx = \ln |g(x)| + C.$$

### Example

Compute

$$\int \frac{2x - 1}{x^2 - x + 1} dx.$$

The denominator is

$$g(x) = x^2 - x + 1,$$

and its derivative is

$$g'(x) = 2x - 1.$$

Therefore

$$\int \frac{2x - 1}{x^2 - x + 1} dx = \ln |x^2 - x + 1| + C.$$

The important step is recognizing the structure. The numerator is not random; it is the derivative of the denominator.

## Adjusting the numerator

Sometimes the numerator is close to the derivative of the denominator but not exactly equal. We can split it into a derivative part and a remainder.

### Example

Compute

$$\int \frac{x+1}{x^2-x+1} dx.$$

The derivative of the denominator is  $2x-1$ . Write

$$x+1 = \frac{1}{2}(2x-1) + \frac{3}{2}.$$

Thus

$$\int \frac{x+1}{x^2-x+1} dx = \frac{1}{2} \int \frac{2x-1}{x^2-x+1} dx + \frac{3}{2} \int \frac{dx}{x^2-x+1}.$$

The first integral is

$$\frac{1}{2} \ln(x^2-x+1),$$

because

$$x^2-x+1 = \left(x-\frac{1}{2}\right)^2 + \frac{3}{4} > 0.$$

For the second integral, write

$$x^2-x+1 = \left(x-\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2.$$

Using

$$\int \frac{du}{u^2+a^2} = \frac{1}{a} \arctan \frac{u}{a} + C,$$

with  $u = x - 1/2$  and  $a = \sqrt{3}/2$ , we obtain

$$\int \frac{dx}{x^2-x+1} = \frac{2}{\sqrt{3}} \arctan \left( \frac{2x-1}{\sqrt{3}} \right) + C.$$

Therefore

$$\boxed{\int \frac{x+1}{x^2-x+1} dx = \frac{1}{2} \ln(x^2-x+1) + \sqrt{3} \arctan \left( \frac{2x-1}{\sqrt{3}} \right) + C.}$$

This example shows that one rational integral may contain both logarithmic and arctangent pieces.

## Completing the square

The standard arctangent pattern is

$$\int \frac{dx}{x^2+a^2} = \frac{1}{a} \arctan \frac{x}{a} + C, \quad a > 0.$$

When a quadratic denominator has no real roots, completing the square may reveal this pattern.

### Example

Compute

$$\int \frac{dx}{x^2 + 4x + 8}.$$

Complete the square:

$$x^2 + 4x + 8 = (x + 2)^2 + 4.$$

Thus

$$\int \frac{dx}{x^2 + 4x + 8} = \int \frac{dx}{(x + 2)^2 + 2^2} = \frac{1}{2} \arctan \frac{x + 2}{2} + C.$$

## Polynomial division

If the degree of the numerator is greater than or equal to the degree of the denominator, divide first. Polynomial division separates the rational function into a polynomial part and a proper rational part.

### Example

Consider

$$\int \frac{x^2 + 1}{x - 1} dx.$$

Divide:

$$\frac{x^2 + 1}{x - 1} = x + 1 + \frac{2}{x - 1}.$$

Therefore

$$\int \frac{x^2 + 1}{x - 1} dx = \int (x + 1) dx + 2 \int \frac{dx}{x - 1},$$

so

$$\int \frac{x^2 + 1}{x - 1} dx = \frac{x^2}{2} + x + 2 \ln |x - 1| + C.$$

Simplifying before integrating is often the main step.

## Summary

When integrating a rational function, ask:

- Is the fraction proper, or should I divide polynomials first?
- Is the numerator the derivative of the denominator?
- Can the numerator be split into a derivative part plus a remainder?
- Does completing the square reveal an arctangent form?
- Are logarithms expected from factors in the denominator?

Rational integration is a large topic. The first skill is to read the denominator and numerator as related structures, not as isolated polynomials.

## 12.7 Trigonometric integrals

Trigonometric integrals require two kinds of reading. First, we should recognize direct derivatives, such as  $(\sin x)' = \cos x$ . Second, we should know a few identities that rewrite products or powers into simpler forms.

The goal in this section is not to list every possible trigonometric integral. The goal is to learn how to choose a useful transformation.

### Direct substitution

Some trigonometric integrals are simple substitutions.

#### Example

Compute

$$\int \sin^3 x \cos x \, dx.$$

The inside expression is  $\sin x$ , and its derivative is  $\cos x$ . Let

$$u = \sin x.$$

Then

$$du = \cos x \, dx.$$

The integral becomes

$$\int u^3 \, du = \frac{u^4}{4} + C.$$

Therefore

$$\int \sin^3 x \cos x \, dx = \frac{\sin^4 x}{4} + C.$$

The integrand was not difficult after the inside function was recognized.

### Using identities

Products of sines and cosines may require identities. Two useful identities are

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}.$$

These are useful for even powers of sine or cosine.

#### Example

Compute

$$\int \sin^2 x \, dx.$$

Use the identity

$$\sin^2 x = \frac{1 - \cos 2x}{2}.$$

Then

$$\int \sin^2 x \, dx = \frac{1}{2} \int 1 \, dx - \frac{1}{2} \int \cos 2x \, dx.$$

Since

$$\int \cos 2x \, dx = \frac{1}{2} \sin 2x,$$

we get

$$\int \sin^2 x \, dx = \frac{x}{2} - \frac{\sin 2x}{4} + C.$$

The identity changed a power into a sum of simpler terms.

### Products with different angles

Products such as  $\sin ax \cos bx$  or  $\cos ax \cos bx$  can be simplified using product-to-sum identities:

$$\sin A \cos B = \frac{1}{2}(\sin(A + B) + \sin(A - B)),$$

$$\cos A \cos B = \frac{1}{2}(\cos(A + B) + \cos(A - B)),$$

$$\sin A \sin B = \frac{1}{2}(\cos(A - B) - \cos(A + B)).$$

#### Example

Compute

$$\int \sin 3x \cos 2x \, dx.$$

Use

$$\sin A \cos B = \frac{1}{2}(\sin(A + B) + \sin(A - B)).$$

With  $A = 3x$ ,  $B = 2x$ ,

$$\sin 3x \cos 2x = \frac{1}{2}(\sin 5x + \sin x).$$

Therefore

$$\int \sin 3x \cos 2x \, dx = \frac{1}{2} \int \sin 5x \, dx + \frac{1}{2} \int \sin x \, dx.$$

Thus

$$\int \sin 3x \cos 2x \, dx = -\frac{1}{10} \cos 5x - \frac{1}{2} \cos x + C.$$

### Tangent and secant patterns

The derivative of  $\tan x$  is

$$\frac{1}{\cos^2 x}.$$

Therefore expressions containing  $1/\cos^2 x$  often lead to tangent.

#### Example

Compute

$$\int \frac{dx}{2 \cos^2 3x}.$$

Rewrite:

$$\int \frac{dx}{2 \cos^2 3x} = \frac{1}{2} \int \frac{1}{\cos^2 3x} \, dx.$$

Let

$$u = 3x, \quad du = 3 \, dx.$$

Then

$$\frac{1}{2} \int \frac{1}{\cos^2 3x} \, dx = \frac{1}{6} \int \frac{1}{\cos^2 u} \, du.$$

Thus

$$\int \frac{dx}{2 \cos^2 3x} = \frac{1}{6} \tan(3x) + C.$$

### Summary

When integrating trigonometric expressions, ask:

- Is there an inside function whose derivative is present?
- Can substitution use  $\sin x$ ,  $\cos x$ , or a multiple of  $x$ ?
- Are even powers present, suggesting half-angle identities?
- Is there a product of different angles, suggesting product-to-sum identities?
- Does the integrand contain  $1/\cos^2 x$ , suggesting tangent?

Trigonometric integration is largely the art of rewriting the expression until a familiar derivative appears backward.

## 12.8 Definite integrals and accumulation

An indefinite integral gives a family of antiderivatives. A definite integral gives a number. It measures accumulated signed area over an interval.

The notation is

$$\int_a^b f(x) dx.$$

The numbers  $a$  and  $b$  are the limits of integration. The function  $f$  is the integrand. The variable  $x$  is the variable of integration.

### Area under a non-negative function

If  $f(x) \geq 0$  on  $[a, b]$ , then

$$\int_a^b f(x) dx$$

can be interpreted as the area under the graph of  $f$ , above the  $x$ -axis, from  $x = a$  to  $x = b$ .

### Example

Let

$$f(x) = 2$$

on  $[1, 4]$ . The graph is a horizontal line. The region under the graph is a rectangle of height 2 and width 3. Therefore

$$\int_1^4 2 dx = 2 \cdot 3 = 6.$$

This example is simple, but it gives the right idea: a definite integral accumulates contributions over an interval.

## Signed area

If the function is below the  $x$ -axis, then the definite integral contributes negatively. Therefore a definite integral is not always ordinary geometric area. It is signed area.

### Example

Let

$$f(x) = -1$$

on  $[0, 3]$ . The graph lies below the  $x$ -axis. The signed area is

$$\int_0^3 (-1) dx = -3.$$

The ordinary geometric area of the region is 3, but the integral is  $-3$ .

This distinction matters. If a function is sometimes positive and sometimes negative, the integral can involve cancellation.

## Basic properties

Definite integrals satisfy natural rules:

$$\int_a^b (f(x) + g(x)) dx = \int_a^b f(x) dx + \int_a^b g(x) dx,$$

$$\int_a^b cf(x) dx = c \int_a^b f(x) dx.$$

They are also additive over intervals:

$$\int_a^c f(x) dx = \int_a^b f(x) dx + \int_b^c f(x) dx.$$

Changing the order of the limits changes the sign:

$$\int_b^a f(x) dx = - \int_a^b f(x) dx.$$

Also,

$$\int_a^a f(x) dx = 0.$$

There is no interval over which to accumulate.

## Definite integrals are numbers

The result of a definite integral is a number, not a function of  $x$ . The variable of integration is internal to the integral.

For example,

$$\int_0^2 x^2 dx$$

is a number. After integration, the variable  $x$  no longer remains.

### Remark

In an indefinite integral, the answer is a family of functions and includes  $+C$ . In a definite integral, the answer is a number and no  $+C$  remains.

## Summary

When reading a definite integral, ask:

- What interval is being used?
- Is the function positive, negative, or changing sign?
- Is the integral measuring signed area or ordinary area?
- Is the answer expected to be a number or a function?
- Can the integral be split into simpler intervals?

A definite integral measures accumulation across an interval. It is one of the main ways calculus turns local quantities into total quantities.

## 12.9 The fundamental theorem of calculus

The fundamental theorem of calculus connects the two meanings of integration. It says that accumulation and antidifferentiation are not separate ideas. They are linked.

If  $F$  is an antiderivative of  $f$ , then a definite integral of  $f$  over  $[a, b]$  can be computed by evaluating  $F$  at the endpoints.

### Theorem

If  $f$  is continuous on  $[a, b]$  and  $F'(x) = f(x)$ , then

$$\int_a^b f(x) dx = F(b) - F(a).$$

This formula is often written as

$$\int_a^b f(x) dx = [F(x)]_a^b.$$

The notation means

$$[F(x)]_a^b = F(b) - F(a).$$

### A first example

#### Example

Compute

$$\int_0^3 2x dx.$$

An antiderivative of  $2x$  is

$$F(x) = x^2.$$

Therefore

$$\int_0^3 2x dx = [x^2]_0^3 = 3^2 - 0^2 = 9.$$

The definite integral is a number. The constant of integration is not needed because it cancels:

$$(F(b) + C) - (F(a) + C) = F(b) - F(a).$$

## Another example

### Example

Compute

$$\int_1^4 \frac{1}{x} dx.$$

An antiderivative of  $1/x$  is

$$\ln |x|.$$

On the interval  $[1, 4]$ , this is simply  $\ln x$ . Hence

$$\int_1^4 \frac{1}{x} dx = [\ln x]_1^4 = \ln 4 - \ln 1 = \ln 4.$$

This example also shows why the interval matters. Since  $[1, 4]$  is positive, no sign issue appears inside the logarithm.

## Accumulation functions

The theorem also has another interpretation. If we define

$$A(x) = \int_a^x f(t) dt,$$

then, under appropriate continuity assumptions,

$$A'(x) = f(x).$$

The function  $A(x)$  accumulates the values of  $f$  from  $a$  to  $x$ . Its derivative is the rate at which accumulation changes, which is the original function  $f$ .

This explains why integration and differentiation are inverse processes in a precise sense.

### Remark

The dummy variable  $t$  in  $\int_a^x f(t) dt$  is used to avoid confusing the variable of integration with the endpoint  $x$ . The integral accumulates with respect to  $t$ , and the result depends on  $x$ .

## Using the theorem responsibly

The fundamental theorem requires that an antiderivative be valid on the interval being used. If the integrand is not defined somewhere inside the interval, the situation needs more care and may involve improper integrals, which are not treated in this introductory chapter.

### Summary

When using the fundamental theorem, ask:

- Is the integral definite?
- What antiderivative  $F$  satisfies  $F' = f$ ?
- Are the endpoints substituted correctly?
- Did I compute  $F(b) - F(a)$ , not  $F(a) - F(b)$ ?
- Is the integrand defined on the interval?

The fundamental theorem is the bridge between local change and total accumulation.

## 12.10 Choosing the right integration method

Integration is often harder than differentiation because there is no single rule that always works. The first step is to read the structure of the integrand. The method should be chosen from that structure.

This section summarizes the main choices from the chapter.

### Start with rewriting

Before choosing an advanced method, simplify the integrand if possible. Roots and fractions may become powers. A rational expression may need polynomial division. A trigonometric expression may need an identity.

#### Example

The integral

$$\int \frac{\sqrt{x}}{x^2} dx$$

becomes easier after rewriting:

$$\frac{\sqrt{x}}{x^2} = x^{1/2-2} = x^{-3/2}.$$

Then

$$\int x^{-3/2} dx = -2x^{-1/2} + C.$$

The method was not hidden. It appeared after the integrand was written in a readable form.

### A practical guide

| Structure of the integrand                        | Possible method                                 |
|---------------------------------------------------|-------------------------------------------------|
| Powers, sums, simple constants                    | basic rules                                     |
| Roots and fractions of powers                     | rewrite as powers                               |
| Inside function with its derivative               | substitution                                    |
| Product with polynomial and exponential/logarithm | integration by parts                            |
| Rational function                                 | simplify, divide, logarithm/arctangent patterns |
| Quadratic denominator without real roots          | complete the square                             |
| Trigonometric powers or products                  | identities or substitution                      |
| Definite integral                                 | find antiderivative and use endpoints           |

The table is not a mechanical algorithm. It is a guide for what to look for first.

### Checking the result

For an indefinite integral, differentiate the answer. If the derivative returns the integrand, the answer is correct.

For a definite integral, check whether the answer has the right sign and approximate size. If the integrand is positive on the interval, the integral should not be negative. If the interval has length zero, the integral should be zero.

### Example

Suppose we compute

$$\int_0^1 x^2 dx = \frac{1}{3}.$$

This is reasonable: the function is non-negative on  $[0, 1]$ , and its values lie between 0 and 1. The accumulated area should be between 0 and 1, and  $1/3$  fits that expectation.

Reasonableness checks do not replace exact work, but they help detect mistakes.

### Connecting back to derivatives

Every integration method in this chapter comes from a differentiation rule read backward:

- basic integrals reverse basic derivatives,
- substitution reverses the chain rule,
- integration by parts reverses the product rule,
- logarithmic rational integrals come from  $(\ln |g(x)|)' = g'(x)/g(x)$ .

This perspective prevents integration from becoming a bag of tricks. Each method has a reason.

### Summary

When choosing an integration method, ask:

- Can the integrand be simplified first?
- Is this a basic antiderivative after rewriting?
- Is there an inside function whose derivative appears?
- Is the integrand a product suggesting integration by parts?
- Is the integrand rational, suggesting algebraic decomposition?
- Is a trigonometric identity useful?
- For a definite integral, what interval and sign information should I keep in mind?
- Can I verify the result by differentiating or by a reasonableness check?

The goal is not to guess the method. The goal is to read the integrand until the method becomes natural.

# Chapter 13

## Applications of Calculus

### 13.1 Calculus as a modelling tool

The previous chapters developed limits, derivatives, and integrals as mathematical objects. In applications, these ideas become tools. We use them to describe change, accumulation, optimisation, motion, and many quantitative problems from geometry and physics.

The most important skill is not only to compute a derivative or an integral. The important skill is to translate a situation into a mathematical model, choose the right calculus tool, and interpret the result in the original context.

#### From a problem to a function

Many applications begin by identifying a quantity that depends on another quantity. For example:

- the area of a rectangle may depend on its width,
- the position of a particle may depend on time,
- the concentration of a drug may depend on time after administration,
- the cost of a cable may depend on where the cable crosses a river.

In each case we try to write a function. Once a function is available, calculus can be used to ask meaningful questions about it.

#### Remark

Calculus does not solve the real-world problem directly. It solves a mathematical model. The modelling step decides what the variables mean and what assumptions are being made.

#### Derivatives as local change

A derivative measures local rate of change. In applications this often means speed, acceleration, growth rate, marginal change, or sensitivity.

If  $s(t)$  is position, then

$$v(t) = s'(t)$$

is velocity, and

$$a(t) = v'(t) = s''(t)$$

is acceleration.

If  $C(t)$  is a concentration, then  $C'(t)$  tells us whether the concentration is increasing or decreasing at time  $t$ , and how fast.

If  $A(x)$  is an area depending on a design variable  $x$ , then  $A'(x) = 0$  may indicate an optimal design.

## Integrals as accumulation

An integral measures accumulated quantity. In applications this may mean total distance, total work, accumulated flow, area, volume, or average value.

If  $v(t)$  is velocity, then

$$\int_a^b v(t) dt$$

measures signed displacement over the time interval  $[a, b]$ . If  $v(t) \geq 0$ , it also measures distance travelled.

If  $F(x)$  is force as a function of displacement, then

$$\int_a^b F(x) dx$$

measures work done by the force over the displacement interval.

## A general workflow

Application problems should be solved in stages:

1. Identify the quantity to be studied.
2. Choose variables and state what they represent.
3. Write the mathematical relationship between the variables.
4. Use derivatives or integrals according to the question.
5. Interpret the mathematical answer in the original context.
6. Check whether the answer is reasonable, including units and restrictions.

Skipping the first two steps often leads to correct calculations applied to the wrong function.

### Summary

When calculus is used as a tool, ask:

- What quantity is being modelled?
- What are the variables and what do they mean?
- What assumptions are being made?
- Is the question about local change, optimisation, accumulation, or long-term behavior?
- Should I use a derivative, an integral, or both?
- Does the final answer make sense in the original situation?

Applications of calculus are applications of meaning. The calculation is only one part of the solution.

## 13.2 Optimisation problems

Optimisation problems ask for the best possible value of a quantity. We may want to maximise area, minimise cost, maximise volume, or minimise the amount of material used. Derivatives are useful because extrema often occur where the derivative is zero or at the boundary of the allowed interval.

The difficulty in optimisation is often not the derivative. The difficulty is building the function to optimise.

### The structure of an optimisation problem

A typical optimisation problem has three layers:

- an objective quantity, which we want to maximise or minimise,
- constraints, which limit the possible choices,
- a domain, which tells us which values are physically or geometrically meaningful.

The goal is to express the objective quantity as a function of one variable whenever possible. Then calculus can be applied.

#### Summary

A useful optimisation strategy is:

1. Draw or describe the situation clearly.
2. Choose variables and define them.
3. Write the objective function.
4. Use the constraint to reduce the number of variables.
5. Determine the allowed domain.
6. Differentiate and find critical points.
7. Check endpoints and interpret the result.

The domain is not optional. A critical point outside the allowed domain is not a solution to the original problem.

### Fencing next to a river

#### Example

A farmer wants to fence a rectangular pasture next to a river. No fence is needed along the river. The farmer has  $L$  meters of fence. What dimensions maximise the area?

Let  $x$  be the width perpendicular to the river, and let  $y$  be the length parallel to the river. Only three sides need fencing, so

$$2x + y = L.$$

The area is

$$A = xy.$$

Use the constraint to write

$$y = L - 2x.$$

Therefore

$$A(x) = x(L - 2x) = Lx - 2x^2.$$

The meaningful domain is

$$0 < x < \frac{L}{2}.$$

Differentiate:

$$A'(x) = L - 4x.$$

Set  $A'(x) = 0$ :

$$L - 4x = 0, \quad x = \frac{L}{4}.$$

Then

$$y = L - 2 \cdot \frac{L}{4} = \frac{L}{2}.$$

Thus the maximum area occurs when the side perpendicular to the river is  $L/4$  and the side parallel to the river is  $L/2$ .

The result should be interpreted geometrically: the side along the river is twice the perpendicular side.

**Open box from a cardboard sheet**

### Example

Squares of side length  $x$  are cut from each corner of a  $30\text{ cm} \times 48\text{ cm}$  sheet of cardboard. The sides are folded up to form an open box. Find the value of  $x$  that maximises the volume.

After cutting, the base has dimensions

$$30 - 2x \quad \text{and} \quad 48 - 2x,$$

and the height is  $x$ . Hence

$$V(x) = x(30 - 2x)(48 - 2x).$$

The domain is

$$0 < x < 15,$$

because the shorter side must remain positive.

Expand:

$$V(x) = x(1440 - 156x + 4x^2) = 1440x - 156x^2 + 4x^3.$$

Then

$$V'(x) = 1440 - 312x + 12x^2.$$

Set  $V'(x) = 0$ :

$$12x^2 - 312x + 1440 = 0.$$

Divide by 12:

$$x^2 - 26x + 120 = 0.$$

Factor:

$$(x - 6)(x - 20) = 0.$$

Only  $x = 6$  belongs to the domain  $(0, 15)$ . Therefore the cut-out squares should have side length

$$6\text{ cm}.$$

The other critical value, 20, is mathematically produced by the derivative but physically impossible for the cardboard sheet.

### Remark

Optimisation always belongs to a context. The derivative gives candidates; the original problem decides which candidates are meaningful.

### Checking maximum or minimum

After finding a critical point, we should check whether it gives a maximum or minimum. We may use the sign of the derivative, the second derivative, or endpoint comparison on a closed interval.

In the fencing problem, the objective function was a downward-opening quadratic, so the critical point gives a maximum. In the box problem, the endpoints give zero volume, while the interior critical point gives positive volume; the relevant interior point gives the maximum.

### Summary

When solving optimisation problems, ask:

- What is being optimised?

- What constraints are given?
- What is the domain of meaningful values?
- Does the derivative give critical points inside the domain?
- Are endpoints relevant?
- Does the final answer satisfy the original constraints?

Optimisation is calculus combined with modelling. The derivative helps only after the model is built correctly.

### 13.3 Geometric applications of integrals

Integrals can measure geometric quantities by accumulating small pieces. Area, volume, and length can often be expressed as integrals when the object is described by a function.

The main idea is to cut a complicated object into many thin pieces, approximate each piece by something simple, and then pass to a limit. The integral represents this limiting accumulation.

#### Area under a graph

If  $f(x) \geq 0$  on  $[a, b]$ , then

$$\int_a^b f(x) dx$$

gives the area under the graph of  $f$  above the  $x$ -axis.

#### Example

A tunnel entrance has the shape of the parabola

$$y = 4 - x^2.$$

Find the area of the entrance above the  $x$ -axis.

The parabola meets the  $x$ -axis when

$$4 - x^2 = 0,$$

so  $x = -2$  or  $x = 2$ . The area is

$$A = \int_{-2}^2 (4 - x^2) dx = \left[ 4x - \frac{x^3}{3} \right]_{-2}^2 = \frac{32}{3}.$$

The limits of integration are not arbitrary. They come from where the curve meets the axis.

#### Area of a circle

The upper semicircle of radius  $R$  centered at the origin is described by

$$y = \sqrt{R^2 - x^2}, \quad -R \leq x \leq R.$$

The area of the full circle is twice the area under the upper semicircle:

$$A = 2 \int_{-R}^R \sqrt{R^2 - x^2} dx.$$

Evaluating this integral exactly gives

$$A = \pi R^2.$$

#### Remark

The modelling step is elementary: vertical slices produce the integral. An exact evaluation of

$$\int \sqrt{R^2 - x^2} dx$$

usually uses a trigonometric substitution or another method beyond the integration techniques developed so far. The formula is stated here to show how the familiar area can be represented by accumulation, not as an exercise in an unintroduced technique.

### Volumes of solids of revolution

If a non-negative function  $f$  is rotated around the  $x$ -axis, cross-sections perpendicular to the  $x$ -axis are disks of radius  $f(x)$ . Their area is

$$\pi[f(x)]^2,$$

so the volume is

$$V = \int_a^b \pi[f(x)]^2 dx.$$

#### Example

A vessel is formed by rotating

$$y = \sqrt{x}$$

around the  $x$ -axis for  $0 \leq x \leq 4$ . Its volume is

$$V = \int_0^4 \pi(\sqrt{x})^2 dx = \pi \int_0^4 x dx = 8\pi.$$

The formula is simple because each thin slice is approximately a disk.

### Arc length

The length of a curve  $y = f(x)$  on  $[a, b]$  is given by

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx.$$

This formula combines derivatives and integrals. The derivative measures local slope; the integral accumulates small pieces of length.

#### Example

Suppose a hanging rope is approximated by

$$f(x) = \frac{1}{10}x^2, \quad -10 \leq x \leq 10.$$

Then

$$f'(x) = \frac{x}{5},$$

and the rope length is

$$L = \int_{-10}^{10} \sqrt{1 + \frac{x^2}{25}} dx.$$

This integral may require a more advanced method to evaluate exactly, but the modelling step is already clear: slope determines local stretching, and the integral accumulates length.

### Summary

For geometric integral problems, ask:

- What small pieces are being accumulated?
- What are the correct bounds of integration?
- Is the integral measuring area, volume, or length?
- Is the quantity signed or ordinary geometric size?
- Does exact evaluation require a method already introduced?

Geometric applications show that integration is a method for building whole quantities from infinitesimal pieces.

## 13.4 Motion, velocity, and acceleration

One of the clearest applications of calculus is motion along a line. If position depends on time, derivatives describe velocity and acceleration. Conversely, integrals can recover velocity and position from acceleration.

Let  $s(t)$  be the position of a particle at time  $t$ . Then

$$v(t) = s'(t)$$

is velocity, and

$$a(t) = v'(t) = s''(t)$$

is acceleration.

### Velocity and acceleration from position

#### Example

A particle moves along a line with position

$$s(t) = t^3 - 6t^2 + 9t + 2, \quad t \geq 0.$$

Find velocity and acceleration.

Velocity is the derivative of position:

$$v(t) = s'(t) = 3t^2 - 12t + 9.$$

Acceleration is the derivative of velocity:

$$a(t) = v'(t) = 6t - 12.$$

The formulas are not only calculations. Velocity tells us how position changes. Acceleration tells us how velocity changes.

## When does the particle stop?

A particle stops when its velocity is zero. In the example above,

$$v(t) = 3t^2 - 12t + 9 = 3(t^2 - 4t + 3) = 3(t - 1)(t - 3).$$

Thus

$$v(t) = 0$$

when

$$t = 1 \quad \text{or} \quad t = 3.$$

These are the times when the particle is instantaneously at rest.

## Speeding up and slowing down

A particle speeds up when velocity and acceleration have the same sign. It slows down when they have opposite signs. This is because acceleration then changes velocity in the direction opposite to the motion.

For the example,

$$v(t) = 3(t - 1)(t - 3), \quad a(t) = 6(t - 2).$$

We inspect the intervals determined by  $t = 1, 2, 3$ , with  $t \geq 0$ :

$$[0, 1), \quad (1, 2), \quad (2, 3), \quad (3, \infty).$$

On each interval we compare the signs of  $v(t)$  and  $a(t)$ .

| Interval      | sign of $v$ | sign of $a$ | behavior     |
|---------------|-------------|-------------|--------------|
| $[0, 1)$      | positive    | negative    | slowing down |
| $(1, 2)$      | negative    | negative    | speeding up  |
| $(2, 3)$      | negative    | positive    | slowing down |
| $(3, \infty)$ | positive    | positive    | speeding up  |

This is a good example of interpretation. The signs matter because velocity carries direction, while speed is the magnitude of velocity.

## Recovering motion by integration

If acceleration is known, then velocity can be recovered by integration. If velocity is known, position can be recovered by integration.

For vertical motion with constant acceleration

$$a(t) = -g,$$

we integrate:

$$v(t) = \int -g \, dt = -gt + C.$$

If the initial velocity is  $v(0) = v_0$ , then  $C = v_0$ , so

$$v(t) = v_0 - gt.$$

Now integrate again:

$$h(t) = \int (v_0 - gt) \, dt = v_0 t - \frac{1}{2}gt^2 + C_2.$$

If  $h(0) = h_0$ , then  $C_2 = h_0$ , so

$$h(t) = h_0 + v_0 t - \frac{1}{2}gt^2.$$

The maximum height occurs when velocity becomes zero:

$$v_0 - gt = 0, \quad t = \frac{v_0}{g}.$$

Substituting into  $h(t)$  gives

$$h_{\max} = h_0 + \frac{v_0^2}{2g}.$$

### Summary

For motion problems, ask:

- Is position, velocity, or acceleration given?
- Do I need to differentiate or integrate?
- What initial conditions determine constants of integration?
- Does velocity equal zero at stopping or turning points?
- Are velocity and acceleration signs the same or opposite?
- What does the answer mean physically?

Motion problems show both directions of calculus: derivatives move from position to velocity and acceleration, while integrals reconstruct motion from rates of change.

## 13.5 Work, force, and accumulation

Physics gives many situations where an integral measures accumulated effect. One of the most important examples is work. If a force acts along a displacement, then work measures the accumulated force over distance.

When the force is constant, work is simply

$$W = F \cdot d.$$

When the force changes with position, we replace this product by an integral.

### Definition

If a force  $F(x)$  acts along a line from  $x = a$  to  $x = b$ , then the work done by the force is

$$W = \int_a^b F(x) dx.$$

This formula says that work is accumulated force over displacement.

### Hooke's law

For a spring, Hooke's law says that the force needed to stretch the spring is proportional to its elongation:

$$F(x) = kx,$$

where  $k > 0$  is the spring constant and  $x$  is the elongation from the natural length.

### Example

Compute the work required to stretch a spring from  $x = 0$  to  $x = L$ , assuming

$$F(x) = kx.$$

The work is

$$W = \int_0^L kx \, dx = k \left[ \frac{x^2}{2} \right]_0^L = \frac{1}{2}kL^2.$$

The force is small near the natural length and larger when the spring is stretched more. The integral accumulates these changing force values.

### Flow as accumulated rate

If water flows out of a tank at a non-negative rate  $r(t)$ , then the amount of water leaving between times  $a$  and  $b$  is

$$\int_a^b r(t) \, dt.$$

### Example

Suppose water flows out of a tank at rate

$$r(t) = 10 - 2t$$

liters per minute for  $0 \leq t \leq 5$ . This rate is non-negative on the stated interval. The amount leaving during the first three minutes is

$$\int_0^3 (10 - 2t) \, dt = \left[ 10t - t^2 \right]_0^3 = 30 - 9 = 21.$$

Therefore 21 liters leave the tank during the first three minutes.

If instead  $r(t)$  is defined as a signed net flow rate, its sign must be tied explicitly to a chosen direction. A negative integral then represents flow opposite to that positive direction, not a negative physical amount of water.

### Average value

The average value of a continuous function on  $[a, b]$  is

$$f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) \, dx.$$

This is the continuous analogue of averaging finitely many numbers.

For example, the root mean square value of a voltage  $U(t)$  over one period  $T$  is

$$U_{\text{RMS}} = \sqrt{\frac{1}{T} \int_0^T [U(t)]^2 \, dt}.$$

The square makes positive and negative oscillations contribute to energy-like effect rather than cancel.

## Summary

For physical accumulation problems, ask:

- What rate or force is being accumulated?
- What is the variable of accumulation: time, position, or another quantity?
- What are the correct bounds?
- Is the rate an unsigned amount or a signed directional quantity?
- What units should the final answer have?

Integrals are natural in physics because many physical quantities are totals built from continuously changing contributions.

## 13.6 Growth, decay, and simple differential models

Some applications do not give the function directly. Instead, they describe how the function changes. This leads to differential equations: equations involving an unknown function and its derivative.

In this introductory chapter, we only need the simplest idea. If the rate of change is known, integration can recover the function. If the rate of change is proportional to the current value, exponential functions appear naturally.

### Rate proportional to the quantity

A common model says that a quantity grows at a rate proportional to its current size:

$$N'(t) = kN(t).$$

Here  $N(t)$  might represent a population, and  $k$  is a constant. If  $k > 0$ , the population grows. If  $k < 0$ , it decays.

A function satisfying this equation is

$$N(t) = N_0 e^{kt},$$

where  $N_0 = N(0)$  is the initial value.

### Example

Suppose a bacterial colony grows according to

$$N'(t) = kN(t), \quad N(0) = N_0.$$

Then

$$N(t) = N_0 e^{kt}.$$

To find the doubling time, solve

$$N(t) = 2N_0.$$

Thus

$$N_0 e^{kt} = 2N_0.$$

Assuming  $N_0 > 0$ , divide by  $N_0$ :

$$e^{kt} = 2.$$

Take logarithms:

$$kt = \ln 2.$$

Therefore the doubling time is

$$t = \frac{\ln 2}{k}.$$

The result depends only on  $k$ , not on the initial population size. This is a feature of pure exponential growth.

### Recovering a function from its derivative

If a rate of change is given directly, integration recovers the original quantity up to a constant. An initial condition determines that constant.

#### Example

Suppose the velocity of a particle is

$$v(t) = 3t^2 - 4t.$$

Find the position  $s(t)$ , assuming  $s(0) = 5$ .

Since velocity is the derivative of position,

$$s'(t) = v(t) = 3t^2 - 4t.$$

Integrate:

$$s(t) = \int (3t^2 - 4t) dt = t^3 - 2t^2 + C.$$

Use  $s(0) = 5$ :

$$5 = 0 - 0 + C,$$

so  $C = 5$ . Therefore

$$s(t) = t^3 - 2t^2 + 5.$$

The constant of integration is not optional. In applications it is determined by initial data.

### Reading the model

A differential equation must be read as a relationship. For example,

$$N'(t) = kN(t)$$

does not give  $N(t)$  immediately. It says that the rate of change of  $N$  at time  $t$  is proportional to the current value of  $N(t)$ .

This kind of statement is common in modelling. The derivative expresses a law of change.

#### Summary

When reading a simple differential model, ask:

- What quantity is unknown?
- What does its derivative represent?
- Is the rate given directly or in terms of the quantity itself?
- What initial condition determines the constant?

- Does the solution grow, decay, or remain constant?
- What does the solution mean in the original context?

Differential models show why derivatives and integrals belong together: derivatives describe laws of change, while integrals and antiderivatives recover quantities from those laws.

## 13.7 Mixed optimisation models

Some optimisation problems combine geometry, physics, and cost. The calculus step may be short, but the modelling step can be substantial. These problems are important because they show calculus acting as a reasoning tool rather than as a mechanical procedure.

### Minimising material for a cylindrical can

#### Example

A cylindrical can must have fixed volume  $V$ . Find the relation between radius and height that minimises the surface area of the closed can.

Let  $r$  be the radius and  $h$  the height. The volume constraint is

$$\pi r^2 h = V.$$

The surface area of a closed cylinder is

$$S = 2\pi r^2 + 2\pi r h.$$

Use the constraint to express  $h$  in terms of  $r$ :

$$h = \frac{V}{\pi r^2}.$$

Then

$$S(r) = 2\pi r^2 + 2\pi r \cdot \frac{V}{\pi r^2} = 2\pi r^2 + \frac{2V}{r}.$$

The domain is  $r > 0$ . Differentiate:

$$S'(r) = 4\pi r - \frac{2V}{r^2}.$$

Set  $S'(r) = 0$ :

$$4\pi r = \frac{2V}{r^2}.$$

Thus

$$4\pi r^3 = 2V, \quad 2\pi r^3 = V.$$

Using  $V = \pi r^2 h$ , this becomes

$$2\pi r^3 = \pi r^2 h,$$

so

$$h = 2r.$$

Therefore the material is minimised when the height equals the diameter.

The final answer is more meaningful as a design relation than as an isolated formula.

## A cable cost problem

A power station is on one side of a river, and a factory is downstream on the opposite side. Cable laid underwater is more expensive than cable laid on land. The question is where the cable should leave the water.

The modelling idea is to choose a variable  $x$  for the horizontal distance from the point directly opposite the station to the landing point on the far bank. Then the underwater length is found by the Pythagorean theorem, while the land length is the remaining downstream distance. A cost function is built from these lengths.

If the river has width 100 m and the factory is 500 m downstream, and if underwater cable costs twice as much as land cable, then a simplified cost model is

$$C(x) = 2\sqrt{x^2 + 100^2} + (500 - x), \quad 0 \leq x \leq 500.$$

The factor 2 represents the higher underwater cost, while  $500 - x$  is the remaining land distance.

To minimise cost, differentiate:

$$C'(x) = 2 \cdot \frac{x}{\sqrt{x^2 + 100^2}} - 1.$$

Set  $C'(x) = 0$ :

$$\frac{2x}{\sqrt{x^2 + 10000}} = 1.$$

Thus

$$2x = \sqrt{x^2 + 10000}.$$

Squaring gives

$$4x^2 = x^2 + 10000,$$

so

$$3x^2 = 10000, \quad x = \frac{100}{\sqrt{3}}.$$

This value lies in the allowed interval, so it is the optimal landing distance from the point directly opposite the station.

### Remark

The derivative calculation is short. The main work is choosing the variable and expressing the physical cost as a function.

## What mixed models teach

Mixed optimisation problems require careful translation. Geometry may provide distances or areas. Physics may provide rates or costs. Calculus then identifies the optimum.

### Summary

For mixed optimisation models, ask:

- What is the decision variable?
- What quantity is being minimised or maximised?
- Which geometric formulas are needed?
- What constraints reduce the model to one variable?
- What interval of values is meaningful?

- Does the critical point satisfy the original physical restrictions?

The value of calculus in applications is that it connects a model to a decision.

## 13.8 Choosing calculus tools

The final skill of this chapter is choosing the right tool. In real problems, no one announces in advance whether the solution requires a derivative, an integral, optimisation, or a differential equation. The wording of the problem must be read carefully.

### Questions that suggest derivatives

Use derivatives when the problem asks about local change, slope, rate, growth, decrease, velocity, acceleration, or sensitivity.

Typical phrases include:

- instantaneous velocity,
- rate of change,
- increasing or decreasing,
- maximum or minimum,
- fastest increase or decrease,
- tangent slope.

For example, if a concentration is given by  $C(t)$ , then the time of maximum concentration is found by studying  $C'(t)$ . If the question asks when the concentration decreases most rapidly, then the derivative and second derivative may both be relevant.

### Questions that suggest integrals

Use integrals when the problem asks for accumulation, total amount, area, volume, work, distance from velocity, or average value.

Typical phrases include:

- total distance,
- accumulated amount,
- area under a graph,
- work done by a variable force,
- volume of a solid of revolution,
- average value over an interval.

For example, if a force depends on position, work is computed by integrating force over displacement. If a flow rate depends on time, total flow is computed by integrating the rate over time.

## Questions that suggest optimisation

Optimisation problems ask for a best possible value under constraints. They often use derivatives, but the model-building step comes first.

Typical phrases include:

- maximise area,
- minimise cost,
- use the least material,
- find the best dimensions,
- largest volume,
- shortest time or distance.

The important step is to build an objective function and identify the allowed domain.

## Questions that suggest differential models

A differential model appears when the problem describes a law of change rather than the quantity itself.

Typical phrases include:

- the rate is proportional to the current amount,
- acceleration is constant,
- velocity is the derivative of position,
- the quantity changes according to its present value.

Such problems often require integration or solving a simple differential equation with an initial condition.

## A final comparison

| Problem type | Mathematical object              | Common tool                 |
|--------------|----------------------------------|-----------------------------|
| Local change | slope or rate                    | derivative                  |
| Optimisation | best value of a function         | derivative and domain check |
| Accumulation | total from small contributions   | definite integral           |
| Motion       | position, velocity, acceleration | derivatives and integrals   |
| Work         | force over distance              | definite integral           |
| Growth model | rate law                         | differential equation       |
| Geometry     | area, volume, length             | definite integral           |

This table should be read as a guide. Some problems use more than one tool. For example, an optimisation problem may require geometry to build a function, derivatives to find a critical point, and endpoint checks to confirm the result.

### Summary

When facing an application problem, ask:

- What is being asked: rate, total, best value, or model?

- What quantity should become a function?
- What is the independent variable?
- What restrictions or units are present?
- Which calculus tool matches the question?
- How should the mathematical result be interpreted in context?

This chapter closes the calculus part of the course by showing that calculus is a language for reasoning. Derivatives and integrals are not only techniques. They are tools for turning change and accumulation into mathematical understanding.

# Chapter 14

## Weekly Problem Sets

This chapter contains one problem set for each substantive chapter of the course. Each set is intended to support one week of study and contains twenty problems arranged approximately from direct practice to interpretation and synthesis.

Unless a problem explicitly asks only for a numerical answer, show the essential reasoning. When a problem asks for an interpretation, explain the meaning of the result in words rather than giving only a formula.

### 14.1 Problem Set 1: Matrices

#### 1. Problem 1

For

$$A = \begin{pmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \end{pmatrix},$$

state the dimensions of  $A$  and identify  $a_{12}$ ,  $a_{21}$ , and  $a_{23}$ .

#### 2. Problem 2

Construct the  $3 \times 2$  matrix  $B = (b_{ij})$  defined by

$$b_{ij} = 2i - j.$$

#### 3. Problem 3

Let

$$A = \begin{pmatrix} 1 & 2 \\ -3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 5 & -1 \\ 2 & 0 \end{pmatrix}.$$

Compute  $A + B$ ,  $A - B$ ,  $3A$ , and  $2A - B$ .

#### 4. Problem 4

Compute the transpose of

$$C = \begin{pmatrix} 1 & 0 & -2 \\ 4 & 3 & 5 \end{pmatrix}.$$

Then state the dimensions of  $C^T$ .

#### 5. Problem 5

Determine which of the following matrices are symmetric:

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

#### 6. Problem 6

Let  $A$  be  $2 \times 3$ ,  $B$  be  $3 \times 4$ , and  $C$  be  $4 \times 2$ . Determine which of the products  $AB$ ,  $BA$ ,  $BC$ ,  $CB$ ,  $AC$ , and  $CA$  are defined. State the dimensions of every defined product.

**7. Problem 7**

Compute

$$AB \quad \text{for} \quad A = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 \\ -1 & 0 \\ 3 & -2 \end{pmatrix}.$$

**8. Problem 8**

For

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix},$$

compute both  $AB$  and  $BA$ . Explain what the result shows about matrix multiplication.

**9. Problem 9**

Let

$$A = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}, \quad x = \begin{pmatrix} 5 \\ -2 \end{pmatrix}.$$

Compute  $Ax$  and interpret the result as a new column vector.

**10. Problem 10**

Verify directly that

$$AI_2 = I_2A = A$$

for

$$A = \begin{pmatrix} 3 & -2 \\ 1 & 5 \end{pmatrix}.$$

**11. Problem 11**

Let

$$D = \text{diag}(2, -1, 3).$$

Compute  $D^2$  and  $D^3$ . Describe the pattern.

**12. Problem 12**

Starting with

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 3 & 0 & 4 \\ -2 & 1 & 5 \end{pmatrix},$$

perform the row operation  $R_2 \leftarrow R_2 - 3R_1$ .

**13. Problem 13**

Starting with

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix},$$

first interchange  $R_1$  and  $R_2$ , and then perform  $R_2 \leftarrow R_2 - \frac{1}{3}R_1$ . Write the matrix after each step.

**14. Problem 14**

Find  $x, y, z, w$  if

$$\begin{pmatrix} x & 2y \\ z-1 & w+3 \end{pmatrix} = \begin{pmatrix} 4 & 6 \\ 2 & 1 \end{pmatrix}.$$

**15. Problem 15**

Find the matrix  $X$  satisfying

$$2X + \begin{pmatrix} 1 & -1 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 5 & 3 \\ 4 & -1 \end{pmatrix}.$$

**16. Problem 16**

Verify the distributive law  $A(B + C) = AB + AC$  for

$$A = \begin{pmatrix} 1 & 2 \\ -1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}, \quad C = \begin{pmatrix} -1 & 4 \\ 2 & 0 \end{pmatrix}.$$

**17. Problem 17**

The columns of

$$A = \begin{pmatrix} 1 & 0 \\ 2 & 1 \\ -1 & 3 \end{pmatrix}$$

are vectors  $u$  and  $v$ . Write  $u$  and  $v$ , and express

$$A \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

as a linear combination of these columns.

**18. Problem 18**

The matrix

$$T = \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}$$

acts on points in the plane by matrix multiplication. Find the images of

$$(1, 2)^T, \quad (-3, 1)^T, \quad (0, 4)^T.$$

Describe what  $T$  does geometrically.

**19. Problem 19**

A shop records sales of two products on three days in the matrix

$$S = \begin{pmatrix} 12 & 8 \\ 15 & 11 \\ 9 & 14 \end{pmatrix}.$$

Explain what the rows and columns could represent. Compute the total sales of each product using matrix or vector notation.

**20. Problem 20**

For each expression below, decide whether it is defined. If it is defined, compute it; if it is not, explain the dimension mismatch:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ 1 & 4 \\ -2 & 3 \end{pmatrix}, \quad u = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}.$$

Consider  $AB$ ,  $BA$ ,  $Au$ ,  $Bu$ , and  $u^T B$ .

## 14.2 Problem Set 2: Determinants

### 1. Problem 1

Compute

$$\det \begin{pmatrix} 4 & -3 \\ 2 & 5 \end{pmatrix}.$$

### 2. Problem 2

Find the determinant of

$$A = \begin{pmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \\ 2 & 0 & 5 \end{pmatrix}$$

by expanding along the first row.

### 3. Problem 3

Compute the determinant of

$$B = \begin{pmatrix} 2 & 0 & 0 \\ -1 & 3 & 0 \\ 4 & 5 & -2 \end{pmatrix}$$

without a Laplace expansion. Explain why your method works.

### 4. Problem 4

Expand

$$\det \begin{pmatrix} 1 & 0 & 2 \\ 3 & -1 & 4 \\ 0 & 5 & 2 \end{pmatrix}$$

along the row or column that gives the shortest calculation.

### 5. Problem 5

A matrix  $B$  is obtained from a  $3 \times 3$  matrix  $A$  by interchanging two rows. If  $\det A = -7$ , find  $\det B$ .

### 6. Problem 6

A matrix  $C$  is obtained from a  $4 \times 4$  matrix  $A$  by multiplying one row by 5. If  $\det A = 3$ , find  $\det C$ .

### 7. Problem 7

A matrix  $D$  is obtained from  $A$  by adding 4 times the first row to the third row. If  $\det A = -2$ , determine  $\det D$ .

### 8. Problem 8

Suppose  $\det A = 3$  and  $\det B = -4$  for two  $3 \times 3$  matrices. Find  $\det(AB)$ ,  $\det(A^T)$ , and  $\det(2A)$ .

### 9. Problem 9

Determine whether

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{pmatrix}$$

is singular without carrying out a full determinant expansion.

10. **Problem 10**

Find all values of  $k$  for which

$$A(k) = \begin{pmatrix} k & 2 \\ 3 & k-1 \end{pmatrix}$$

is singular.

11. **Problem 11**

Find  $a$  if

$$\det \begin{pmatrix} a & 1 \\ 2 & 3 \end{pmatrix} = 10.$$

12. **Problem 12**

Compute

$$\det \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix}$$

and explain the general rule suggested by the result.

13. **Problem 13**

Let

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 3 & 1 & 4 \\ 2 & -1 & 5 \end{pmatrix}.$$

Use row operations to transform  $A$  into an upper triangular matrix, keeping track of every change to the determinant. Then compute  $\det A$ .

14. **Problem 14**

A  $3 \times 3$  matrix has determinant 6. What happens to its determinant if:

- (a) the first and second rows are interchanged;
- (b) the third row is multiplied by  $-2$ ;
- (c) twice the first row is added to the second row?

15. **Problem 15**

The columns of a  $3 \times 3$  matrix satisfy  $c_3 = 2c_1 - c_2$ . What is the determinant? Explain the structural reason.

16. **Problem 16**

The linear transformation

$$T(x) = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix} x$$

acts on the plane. By what factor does  $T$  scale areas? Does it preserve orientation?

17. **Problem 17**

Find the area of the parallelogram spanned by

$$u = (3, 1), \quad v = (-2, 4)$$

using a determinant.

**18. Problem 18**

Compute the Vandermonde determinant

$$\det \begin{pmatrix} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \\ x_1^2 & x_2^2 & x_3^2 \end{pmatrix}$$

for  $x_1 = 1$ ,  $x_2 = 2$ , and  $x_3 = 4$ , first directly and then using the product formula.

**19. Problem 19**

Consider

$$A(t) = \begin{pmatrix} 1 & t & 0 \\ 0 & 1 & t \\ t & 0 & 1 \end{pmatrix}.$$

Compute  $\det A(t)$  and determine all real values of  $t$  for which the matrix is singular.

**20. Problem 20**

For

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 0 & 1 & 4 \end{pmatrix},$$

compute the determinant in two different ways: by a Laplace expansion and by row reduction. Compare the amount of work and explain which method is more efficient.

## 14.3 Problem Set 3: Matrix Inversion

**1. Problem 1**

Decide whether each matrix is invertible:

$$A = \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}.$$

**2. Problem 2**

Find the inverse of

$$A = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}$$

using the  $2 \times 2$  inverse formula.

**3. Problem 3**

Verify your answer to the previous problem by computing both  $AA^{-1}$  and  $A^{-1}A$ .

**4. Problem 4**

Find the inverse of

$$D = \text{diag}(2, -3, 5).$$

**5. Problem 5**

Find all values of  $k$  for which

$$A(k) = \begin{pmatrix} k & 1 \\ 2 & k \end{pmatrix}$$

has an inverse.

**6. Problem 6**

Use Gauss–Jordan elimination to find the inverse of

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix}.$$

**7. Problem 7**

Use Gauss–Jordan elimination to find the inverse of

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}.$$

**8. Problem 8**

Find the adjugate of

$$A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$$

and use it to compute  $A^{-1}$ .

**9. Problem 9**

Let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 5 \\ 3 \end{pmatrix}.$$

Solve  $Ax = b$  using  $A^{-1}$ .

**10. Problem 10**

Solve

$$XA = B$$

for  $X$ , where

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}.$$

Pay attention to the side on which  $A^{-1}$  must be multiplied.

**11. Problem 11**

Solve

$$2X - A = B$$

for  $X$ , where

$$A = \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & 4 \\ 0 & 5 \end{pmatrix}.$$

**12. Problem 12**

If

$$A^{-1} = \begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix} \quad \text{and} \quad B^{-1} = \begin{pmatrix} 1 & 0 \\ 4 & 2 \end{pmatrix},$$

find  $(AB)^{-1}$ .

**13. Problem 13**

Prove by direct multiplication that

$$(A^T)^{-1} = (A^{-1})^T$$

for

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix}.$$

**14. Problem 14**

Find the inverse of the elementary matrix

$$E = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{pmatrix}$$

and explain which row operation  $E$  represents.

**15. Problem 15**

A proposed inverse of

$$A = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix}$$

is

$$C = \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix}.$$

Check whether the proposal is correct.

**16. Problem 16**

Explain why a matrix with two identical rows cannot have an inverse. Use both a determinant argument and an information-loss interpretation.

**17. Problem 17**

The transformation

$$y = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} x$$

maps an unknown vector  $x$  to

$$y = \begin{pmatrix} 8 \\ 5 \end{pmatrix}.$$

Recover  $x$ .

**18. Problem 18**

Choose the most efficient method for finding the inverse of each matrix and justify the choice:

$$A = \begin{pmatrix} 5 & 0 \\ 0 & -2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$

Then compute the inverses.

**19. Problem 19**

Let

$$A = \begin{pmatrix} 1 & a \\ 2 & 2a \end{pmatrix}.$$

Explain, without attempting Gauss–Jordan elimination, why  $A^{-1}$  does not exist for any real  $a$ .

**20. Problem 20**

For

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix},$$

find  $A^{-1}$ , solve  $Ax = b$  for

$$b = \begin{pmatrix} 5 \\ 11 \end{pmatrix},$$

and verify the solution in the original equation. Finally, explain what the inverse accomplishes conceptually.

## 14.4 Problem Set 4: Systems of Linear Equations

**1. Problem 1**

Solve by substitution:

$$\begin{cases} x + y = 7, \\ 2x - y = 2. \end{cases}$$

**2. Problem 2**

Write the augmented matrix for

$$\begin{cases} 2x - y + 3z = 5, \\ x + 4y - z = 2, \\ 3x + 2z = 7. \end{cases}$$

**3. Problem 3**

Solve by Gaussian elimination:

$$\begin{cases} x + 2y = 5, \\ 3x - y = 4. \end{cases}$$

**4. Problem 4**

Solve by Gaussian elimination:

$$\begin{cases} x + y + z = 6, \\ 2x - y + z = 3, \\ x + 3y - 2z = 1. \end{cases}$$

**5. Problem 5**

Reduce the augmented matrix

$$\left( \begin{array}{cc|c} 1 & 2 & 3 \\ 2 & 4 & 8 \end{array} \right)$$

to echelon form and classify the system.

**6. Problem 6**

Reduce

$$\left( \begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 2 & 4 & -2 & 6 \\ 0 & 1 & 1 & 2 \end{array} \right)$$

and express all solutions using a parameter.

**7. Problem 7**

Determine whether

$$\begin{cases} x - 2y = 1, \\ 2x - 4y = 5 \end{cases}$$

has no solution, one solution, or infinitely many solutions. Interpret the result geometrically.

**8. Problem 8**

Solve

$$\begin{cases} 2x + y = 4, \\ x - 3y = -5 \end{cases}$$

using Cramer's rule.

**9. Problem 9**

Solve the system in the previous problem using the inverse matrix method. Compare the two methods.

**10. Problem 10**

For which values of  $k$  does

$$\begin{cases} kx + y = 2, \\ 2x + ky = 3 \end{cases}$$

have a unique solution?

**11. Problem 11**

Classify the system according to the value of  $a$ :

$$\begin{cases} x + y = 2, \\ 2x + 2y = a. \end{cases}$$

**12. Problem 12**

Find all values of  $k$  for which

$$\begin{cases} x + y + z = 1, \\ 2x + 2y + 2z = 2, \\ x + y + kz = 3 \end{cases}$$

is consistent. Describe the solution set in each consistent case.

**13. Problem 13**

The row-reduced augmented matrix of a system is

$$\left( \begin{array}{cccc|c} 1 & 0 & 2 & 0 & 3 \\ 0 & 1 & -1 & 0 & 4 \\ 0 & 0 & 0 & 1 & 2 \end{array} \right).$$

Identify the pivot and free variables and write the complete solution parametrically.

**14. Problem 14**

Determine whether the point  $(2, -1)$  is the solution of

$$\begin{cases} 3x + y = 5, \\ x - 2y = 4. \end{cases}$$

If it is not, find the correct solution.

**15. Problem 15**

Find the intersection point of the lines

$$2x - y - 1 = 0, \quad x + 3y - 9 = 0.$$

**16. Problem 16**

Find the intersection of the three planes

$$\begin{cases} x + y + z = 3, \\ x - y + z = 1, \\ 2x + z = 2. \end{cases}$$

**17. Problem 17**

A cinema sells adult tickets for 12 units and student tickets for 8 units. In one evening it sells 180 tickets for a total of 1840 units. Form and solve a linear system to find the number of each type of ticket.

**18. Problem 18**

A system reduces to a row

$$(0 \quad 0 \quad 0 \mid 6).$$

What does this row mean, and what can be concluded about the original system?

**19. Problem 19**

Construct a system of two linear equations in two variables that has:

- (a) exactly one solution;
- (b) no solution;
- (c) infinitely many solutions.

Explain how the coefficients produce each case.

**20. Problem 20**

Consider

$$\begin{cases} x + ay = 1, \\ ax + y = 1. \end{cases}$$

Classify the system for every real value of  $a$ . For each case, determine whether the solution is unique, nonexistent, or non-unique, and find the solution set where possible.

## 14.5 Problem Set 5: Vectors and Coordinates

### 1. Problem 1

Explain the difference between the point  $P = (3, -2)$  and the vector  $v = (3, -2)$ . What does each object describe?

### 2. Problem 2

For

$$A = (1, 4, -2), \quad B = (5, -1, 3),$$

find  $\overrightarrow{AB}$  and  $\overrightarrow{BA}$ .

### 3. Problem 3

Let

$$u = (2, -1, 3), \quad v = (-4, 5, 1).$$

Compute  $u + v$ ,  $u - v$ ,  $3u$ , and  $2u - v$ .

### 4. Problem 4

Find the position vector of the point  $P = (-2, 5, 1)$ . Then find the endpoint of this vector if it begins at  $A = (3, -1, 4)$  instead of the origin.

### 5. Problem 5

Find the length of

$$v = (3, -4, 12).$$

### 6. Problem 6

Find the distance between

$$A = (1, -2, 3) \quad \text{and} \quad B = (5, 1, -1).$$

### 7. Problem 7

Find the midpoint of the segment joining

$$A = (-3, 4, 2) \quad \text{and} \quad B = (5, -2, 8).$$

### 8. Problem 8

Find a unit vector in the direction of

$$v = (2, -2, 1).$$

### 9. Problem 9

For

$$u = (1, 2, -1), \quad v = (3, 0, 4),$$

compute  $u \cdot v$ .

### 10. Problem 10

Find the angle between

$$u = (1, 0, 1), \quad v = (1, 1, 0).$$

**11. Problem 11**

Find the value of  $k$  for which

$$u = (2, k, -1) \quad \text{and} \quad v = (1, 3, 5)$$

are perpendicular.

**12. Problem 12**

Determine whether the triangle with vertices

$$A = (0, 0), \quad B = (4, 0), \quad C = (4, 3)$$

is right-angled. Use vectors and the dot product.

**13. Problem 13**

Compute

$$u \times v \quad \text{for} \quad u = (1, 2, 0), \quad v = (3, -1, 4).$$

**14. Problem 14**

Verify that the vector found in the previous problem is perpendicular to both  $u$  and  $v$ .

**15. Problem 15**

Find the area of the parallelogram spanned by

$$u = (2, 1, 0), \quad v = (1, -1, 3).$$

Then find the area of the triangle spanned by the same vectors.

**16. Problem 16**

Find a normal vector to the plane containing the direction vectors

$$u = (1, 2, -1), \quad v = (2, 0, 3).$$

**17. Problem 17**

Determine whether the points

$$A = (1, 2, 3), \quad B = (3, 6, 5), \quad C = (5, 10, 7)$$

are collinear.

**18. Problem 18**

A particle moves from  $A = (2, -1)$  by the displacement  $v = (5, 3)$ , and then by  $w = (-2, 4)$ . Find its final position and its total displacement from  $A$ .

**19. Problem 19**

The midpoint of  $AB$  is  $M = (2, 1, -3)$ , and  $A = (-1, 4, 2)$ . Find  $B$ .

**20. Problem 20**

For the points

$$A = (1, 0, 2), \quad B = (4, 2, 1), \quad C = (2, 5, 3),$$

find  $\overrightarrow{AB}$  and  $\overrightarrow{AC}$ , compute their dot product and cross product, determine the angle at  $A$ , and find the area of triangle  $ABC$ .

## 14.6 Problem Set 6: Lines

### 1. Problem 1

Write a vector-parametric equation of the line through

$$A = (2, -1)$$

with direction vector

$$v = (3, 4).$$

### 2. Problem 2

Find a parametric equation of the line through

$$A = (-1, 2) \quad \text{and} \quad B = (5, -4).$$

### 3. Problem 3

Determine whether the point  $P = (7, 3)$  lies on the line

$$(x, y) = (1, -1) + t(2, 1).$$

### 4. Problem 4

Write the coordinate parametric equations of

$$P = (2, 3) + t(-1, 4).$$

Find the points corresponding to  $t = -2$ ,  $t = 0$ , and  $t = 3$ .

### 5. Problem 5

Find the point-slope equation of the line through  $(3, -2)$  with slope 5.

### 6. Problem 6

Convert

$$y - 4 = -2(x + 1)$$

to slope-intercept form and general form.

### 7. Problem 7

Write equations for:

- (a) the vertical line through  $(3, -1)$ ;
- (b) the horizontal line through  $(-2, 5)$ .

### 8. Problem 8

Convert the line

$$(x, y) = (1, 2) + t(3, -1)$$

to general form.

### 9. Problem 9

Convert

$$2x - 5y + 10 = 0$$

to a parametric equation.

**10. Problem 10**

Determine whether the lines

$$\ell_1 : (x, y) = (1, 0) + t(2, 3), \quad \ell_2 : (x, y) = (-1, 4) + s(4, 6)$$

are distinct parallel lines or the same line.

**11. Problem 11**

Determine whether

$$\ell_1 : 2x - y + 3 = 0 \quad \text{and} \quad \ell_2 : x + 2y - 4 = 0$$

are perpendicular.

**12. Problem 12**

Find the intersection point of

$$3x - y - 5 = 0, \quad x + 2y - 8 = 0.$$

**13. Problem 13**

Find the distance from  $P = (4, -1)$  to the line

$$2x + 3y - 6 = 0.$$

**14. Problem 14**

For the oriented line

$$x - 2y + 1 = 0,$$

determine on which side of the line the points  $P = (0, 0)$  and  $Q = (4, 1)$  lie. Are they on the same side?

**15. Problem 15**

Write a parametric equation of the line in space through

$$A = (1, -2, 3)$$

with direction vector

$$v = (2, 0, -1).$$

**16. Problem 16**

Convert

$$(x, y, z) = (2, 1, -3) + t(4, -2, 5)$$

to symmetric form.

**17. Problem 17**

Write an appropriate symmetric description of

$$(x, y, z) = (1, -2, 3) + t(4, 0, -1),$$

taking proper account of the zero component.

**18. Problem 18**

Classify the following pair of lines in space as parallel, intersecting, or skew:

$$\ell_1 : (x, y, z) = (0, 0, 0) + t(1, 1, 0),$$

$$\ell_2 : (x, y, z) = (1, 0, 1) + s(0, 1, 1).$$

**19. Problem 19**

A line in the plane is given by a point  $A = (2, 1)$  and a normal vector  $n = (3, -4)$ . Write its normal equation, general equation, and one direction vector.

**20. Problem 20**

The line passes through  $A = (-1, 3)$  and  $B = (5, -1)$ . Find:

- (a) a direction vector;
- (b) a parametric equation;
- (c) a normal vector;
- (d) a general equation;
- (e) its slope, if defined;
- (f) the distance from the origin to the line.

## 14.7 Problem Set 7: Planes

**1. Problem 1**

Write the normal equation and general equation of the plane through

$$P_0 = (1, -2, 3)$$

with normal vector

$$n = (2, 1, -1).$$

**2. Problem 2**

For the plane

$$3x - 2y + 4z - 7 = 0,$$

state a normal vector and find one convenient point on the plane.

**3. Problem 3**

Determine whether  $P = (2, 1, -1)$  lies on

$$x + 2y - z - 5 = 0.$$

**4. Problem 4**

Write a parametric equation of the plane through

$$P_0 = (1, 0, 2)$$

with direction vectors

$$u = (1, 2, 0), \quad v = (0, 1, 3).$$

**5. Problem 5**

Find a parametric equation of the plane through

$$A = (1, 0, 0), \quad B = (0, 2, 0), \quad C = (0, 0, 3).$$

**6. Problem 6**

Find a normal vector to the plane through

$$A = (1, 0, 1), \quad B = (3, 1, 0), \quad C = (0, 2, 4).$$

**7. Problem 7**

Find the general equation of the plane through the three points in the previous problem.

**8. Problem 8**

Convert the parametric plane

$$P = (1, 2, 0) + s(1, 0, 1) + t(0, 2, -1)$$

to general form.

**9. Problem 9**

Convert

$$2x - y + z - 4 = 0$$

to one possible parametric form.

**10. Problem 10**

Determine whether the planes

$$\pi_1 : x - 2y + 3z - 1 = 0, \quad \pi_2 : 2x - 4y + 6z + 5 = 0$$

are distinct parallel planes or the same plane.

**11. Problem 11**

Determine whether

$$\pi_1 : x + y - z = 0, \quad \pi_2 : 2x - y + z - 3 = 0$$

are perpendicular.

**12. Problem 12**

Find the acute angle between

$$\pi_1 : x + 2y + 2z = 0 \quad \text{and} \quad \pi_2 : 2x - y + 2z - 4 = 0.$$

**13. Problem 13**

Find the distance from

$$P = (3, -1, 2)$$

to the plane

$$2x - y + 2z - 7 = 0.$$

**14. Problem 14**

Find the intersection point of the line

$$(x, y, z) = (1, 0, 2) + t(2, -1, 1)$$

with the plane

$$x + y + z - 6 = 0.$$

**15. Problem 15**

Determine whether the line

$$(x, y, z) = (1, 2, 0) + t(1, -1, 2)$$

lies in, is parallel to, or intersects the plane

$$2x + 2y - z - 6 = 0.$$

**16. Problem 16**

Find the equation of the plane through  $P = (2, -1, 4)$  parallel to

$$3x - y + 2z + 5 = 0.$$

**17. Problem 17**

Find a plane through  $P = (1, 2, 3)$  that is perpendicular to the plane

$$x - y + z = 0.$$

Give one valid answer and explain why it works.

**18. Problem 18**

Find a parametric equation of the line of intersection of

$$x + y + z = 3, \quad x - y + z = 1.$$

**19. Problem 19**

A plane contains the line

$$\ell : (x, y, z) = (1, 0, 2) + t(2, 1, -1)$$

and the point  $Q = (0, 3, 1)$ . Find a general equation of the plane.

**20. Problem 20**

For

$$A = (1, 0, 0), \quad B = (0, 1, 0), \quad C = (0, 0, 1),$$

find a parametric and a general equation of the plane through the points, verify that all three points satisfy the equation, find the distance from the origin to the plane, and determine where the line

$$(x, y, z) = t(1, 1, 1)$$

intersects it.

## 14.8 Problem Set 8: Sequences and Functions

### 1. Problem 1

Decide which of the following assignments define functions:

- (a) each real number  $x$  is assigned  $x^2$ ;
- (b) each positive number  $x$  is assigned both  $\sqrt{x}$  and  $-\sqrt{x}$ ;
- (c) each student in a class is assigned the student's date of birth.

Explain each decision.

### 2. Problem 2

For

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x^2 - 4,$$

state the domain and codomain, and determine the range.

### 3. Problem 3

Let

$$f(x) = 2x^2 - 3x + 1.$$

Compute  $f(0)$ ,  $f(-2)$ ,  $f(a)$ , and  $f(x+h)$ .

### 4. Problem 4

If

$$g(x) = \frac{1}{x-3},$$

determine the natural real domain of  $g$  and compute  $g(1)$ .

### 5. Problem 5

The graph of a function contains the points

$$(-2, 3), \quad (0, -1), \quad (4, 5).$$

State the corresponding function values.

### 6. Problem 6

For

$$f(x) = x^2 - 5x + 6,$$

find the  $y$ -intercept and all  $x$ -intercepts.

### 7. Problem 7

Determine the natural real domain of each function:

$$f(x) = \sqrt{x+2}, \quad g(x) = \frac{x+1}{x^2-9}, \quad h(x) = \ln(4-x).$$

### 8. Problem 8

Find the range of each function:

$$f(x) = x^2 + 2, \quad g(x) = -3x + 1, \quad h(x) = \sqrt{x}.$$

**9. Problem 9**

Classify each function as constant, affine, quadratic, polynomial, rational, exponential, logarithmic, or trigonometric:

$$3, \quad 2x - 5, \quad x^2 + 1, \quad \frac{x+1}{x-2}, \quad 4^x, \quad \ln x, \quad \sin x.$$

**10. Problem 10**

Determine which of the functions

$$f(x) = 3x, \quad g(x) = 3x + 2, \quad h(x) = -x$$

are linear in the sense  $f(x+y) = f(x) + f(y)$ . Explain why an affine function need not be linear.

**11. Problem 11**

For

$$f(x) = 2^x \quad \text{and} \quad g(x) = \log_2 x,$$

compute  $f(-2)$ ,  $f(3)$ ,  $g(1)$ ,  $g(8)$ , and explain the relationship between the two functions.

**12. Problem 12**

Let

$$a_n = 3n - 1, \quad n \geq 1.$$

Write the first five terms and compute  $a_{20}$ .

**13. Problem 13**

A sequence is defined by

$$a_1 = 4, \quad a_{n+1} = a_n - 3.$$

Write the first six terms.

**14. Problem 14**

Find an explicit formula for the arithmetic sequence

$$5, 9, 13, 17, \dots$$

**15. Problem 15**

Find an explicit formula for the geometric sequence

$$3, 6, 12, 24, \dots$$

**16. Problem 16**

Define recursively the sequence

$$2, 6, 18, 54, \dots$$

and state the starting index.

**17. Problem 17**

For

$$a_n = \frac{1}{n},$$

list the first five graph points  $(n, a_n)$ . Explain why the graph should normally be drawn as separated points.

**18. Problem 18**

Compare the domains of

$$a_n = n^2, \quad n \in \mathbb{N},$$

and

$$f(x) = x^2, \quad x \in \mathbb{R}.$$

Which expressions  $a_{3/2}$  and  $f(3/2)$  are meaningful?

**19. Problem 19**

A graph rises on  $(-3, 1)$ , falls on  $(1, 4)$ , crosses the  $x$ -axis at  $-2$  and  $3$ , and has a local maximum at  $(1, 5)$ . Translate this information into statements about the function.

**20. Problem 20**

A taxi fare consists of a fixed fee of 6 units and 2.5 units for every kilometre travelled. Define a function  $C(d)$ , state a meaningful domain, compute  $C(8)$ , and explain whether the function is linear or affine.

## 14.9 Problem Set 9: Limits and Continuity

**1. Problem 1**

Explain in words the difference between

$$f(a) \quad \text{and} \quad \lim_{x \rightarrow a} f(x).$$

**2. Problem 2**

Compute

$$\lim_{n \rightarrow \infty} \frac{1}{n}, \quad \lim_{n \rightarrow \infty} \frac{5n + 2}{2n - 1}.$$

**3. Problem 3**

Compute

$$\lim_{n \rightarrow \infty} \frac{3n^2 - n + 1}{2 - n^2}.$$

**4. Problem 4**

Determine whether each sequence converges:

$$a_n = (-1)^n, \quad b_n = \frac{(-1)^n}{n}, \quad c_n = 2 + \frac{3}{n}.$$

**5. Problem 5**

Compute

$$\lim_{n \rightarrow \infty} \left( \sqrt{n^2 + 4n} - n \right)$$

by rationalizing.

**6. Problem 6**

Compute by direct substitution:

$$\lim_{x \rightarrow 2} (3x^2 - x + 4).$$

**7. Problem 7**

Compute

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}.$$

**8. Problem 8**

Compute

$$\lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x - 9}.$$

**9. Problem 9**

Let

$$f(x) = \begin{cases} 2x + 1, & x < 2, \\ x^2 - 1, & x \geq 2. \end{cases}$$

Find the left-hand limit, right-hand limit, two-sided limit, and value  $f(2)$ .

**10. Problem 10**

Compute

$$\lim_{x \rightarrow 0^-} \frac{|x|}{x}, \quad \lim_{x \rightarrow 0^+} \frac{|x|}{x},$$

and decide whether the two-sided limit exists.

**11. Problem 11**

Compute

$$\lim_{x \rightarrow \infty} \frac{4x^2 - 3x + 1}{2x^2 + 5}.$$

**12. Problem 12**

Find all horizontal asymptotes of

$$f(x) = \frac{3x - 1}{x + 4}.$$

**13. Problem 13**

Determine the one-sided infinite limits of

$$f(x) = \frac{1}{x - 2}$$

as  $x \rightarrow 2^-$  and  $x \rightarrow 2^+$ . State the vertical asymptote.

**14. Problem 14**

Check whether

$$f(x) = \frac{x^2 - 4}{x - 2}, \quad x \neq 2,$$

is continuous at  $x = 2$ . If not, give a value of  $f(2)$  that makes it continuous.

**15. Problem 15**

Find  $k$  so that

$$f(x) = \begin{cases} kx + 1, & x < 3, \\ x^2 - 2, & x \geq 3 \end{cases}$$

is continuous at  $x = 3$ .

**16. Problem 16**

Classify the discontinuity at the indicated point:

(a)  $\frac{x^2 - 1}{x - 1}$  at  $x = 1$ ;

(b)  $\frac{1}{x + 2}$  at  $x = -2$ ;

(c)  $\frac{|x|}{x}$  at  $x = 0$ ;

(d)  $\sin(1/x)$  at  $x = 0$ .

**17. Problem 17**

Compute

$$\lim_{x \rightarrow 0} \frac{\sin 5x}{2x}.$$

**18. Problem 18**

Compute

$$\lim_{x \rightarrow 0} \frac{\tan 3x}{\sin 2x}.$$

**19. Problem 19**

Compute

$$\lim_{x \rightarrow 0} (1 + 4x)^{1/x}.$$

**20. Problem 20**

For each limit, identify the most appropriate first method and then compute:

$$\lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1}, \quad \lim_{x \rightarrow \infty} \frac{2x + 5}{x^2 + 1}, \quad \lim_{x \rightarrow 0} \frac{\sin x}{x}, \quad \lim_{x \rightarrow 4} \frac{x - 4}{\sqrt{x} - 2}.$$

**14.10 Problem Set 10: Derivatives and Function Analysis****1. Problem 1**

For

$$f(x) = x^2 + 1,$$

form and simplify the difference quotient

$$\frac{f(a + h) - f(a)}{h}.$$

**2. Problem 2**

Use the limit definition to find the derivative of

$$f(x) = 3x + 2.$$

**3. Problem 3**

Find the tangent line to

$$f(x) = x^2 - 1$$

at  $x = 2$ .

4. **Problem 4**

Differentiate, stating the relevant domain:

$$f(x) = x^7, \quad g(x) = x^{-3}, \quad h(x) = \sqrt{x}.$$

5. **Problem 5**

Differentiate:

$$f(x) = 4e^x - 3 \ln x + 2 \sin x - \cos x.$$

6. **Problem 6**

Differentiate using the product rule:

$$f(x) = x^3 e^x.$$

7. **Problem 7**

Differentiate using the quotient rule:

$$f(x) = \frac{x^2 + 1}{x + 2}.$$

8. **Problem 8**

Differentiate using the chain rule:

$$f(x) = (2x - 3)^6.$$

9. **Problem 9**

Differentiate:

$$f(x) = e^{\sin(x^2)}.$$

Identify all layers of composition.

10. **Problem 10**

Find all critical points of

$$f(x) = x^3 - 6x^2 + 9x.$$

11. **Problem 11**

Study the monotonicity of

$$f(x) = x^2 - 4x + 1$$

using a sign chart for  $f'$ .

12. **Problem 12**

Find and classify the local extrema of

$$f(x) = x^3 - 3x.$$

13. **Problem 13**

Find the absolute maximum and minimum of

$$f(x) = x^2 - 2x$$

on  $[-1, 3]$ .

**14. Problem 14**

Determine the intervals of convexity and concavity of

$$f(x) = x^3 - 6x.$$

**15. Problem 15**

Find all inflection points of

$$f(x) = x^4 - 4x^3.$$

Remember that  $f''(a) = 0$  gives only a candidate.

**16. Problem 16**

Use the second derivative test to classify the critical points of

$$f(x) = x^4 - 4x^2.$$

For any inconclusive point, use another method.

**17. Problem 17**

Find all vertical and horizontal asymptotes of

$$f(x) = \frac{2x + 1}{x - 3}.$$

**18. Problem 18**

Perform a complete analysis of

$$f(x) = x^3 - 3x^2.$$

Include domain, zeros, behavior at infinity, monotonicity, extrema, convexity, and inflection points.

**19. Problem 19**

Perform a complete analysis of

$$f(x) = \frac{x}{x - 1}.$$

Include domain, asymptotes, monotonicity, and a coherent description of the graph.

**20. Problem 20**

A quantity is modelled by

$$Q(t) = t^3 - 6t^2 + 9t + 4, \quad t \geq 0.$$

Find  $Q'(t)$ , determine when the quantity is increasing or decreasing, identify local extrema, and explain what the sign of  $Q'(t)$  means in a modelling context.

## 14.11 Problem Set 11: Integrals

**1. Problem 1**

Find all antiderivatives of

$$f(x) = 6x^2 - 4.$$

**2. Problem 2**

Compute, on appropriate intervals,

$$\int x^5 dx, \quad \int x^{-4} dx, \quad \int \sqrt{x} dx.$$

**3. Problem 3**

Compute

$$\int \left( 3e^x - \frac{2}{x} + 4 \cos x \right) dx.$$

**4. Problem 4**

Rewrite and integrate:

$$\int \frac{\sqrt{x}}{x^3} dx.$$

**5. Problem 5**

Use substitution to compute

$$\int 2x(1+x^2)^5 dx.$$

**6. Problem 6**

Use substitution to compute

$$\int \frac{e^x}{1+e^x} dx.$$

**7. Problem 7**

Use integration by parts to compute

$$\int x e^x dx.$$

**8. Problem 8**

Compute

$$\int x^2 \cos x dx$$

using integration by parts more than once.

**9. Problem 9**

Compute

$$\int \frac{2x+3}{x^2+3x+5} dx.$$

**10. Problem 10**

Compute

$$\int \frac{x}{x^2-2x+5} dx$$

by splitting the numerator into a derivative part and a remainder.

**11. Problem 11**

Compute

$$\int \frac{x^2+2}{x-1} dx$$

by polynomial division.

**12. Problem 12**

Compute

$$\int \sin^4 x \cos x dx.$$

**13. Problem 13**

Compute

$$\int \cos^2 x \, dx$$

using a trigonometric identity.

**14. Problem 14**

Evaluate

$$\int_0^2 (3x + 1) \, dx.$$

**15. Problem 15**

Find the signed area represented by

$$\int_{-2}^2 x \, dx$$

and compare it with the ordinary geometric area between  $y = x$  and the  $x$ -axis on  $[-2, 2]$ .

**16. Problem 16**

Use properties of definite integrals to simplify:

$$\int_0^5 f(x) \, dx - \int_2^5 f(x) \, dx,$$

and

$$\int_3^1 g(x) \, dx.$$

**17. Problem 17**

Use the fundamental theorem of calculus to evaluate

$$\int_1^4 \frac{1}{x} \, dx.$$

**18. Problem 18**

Let

$$A(x) = \int_0^x (t^2 + 1) \, dt.$$

Find  $A'(x)$  and compute  $A(2)$ .

**19. Problem 19**

Choose an appropriate integration method for each integral and justify the choice before computing:

$$\int x \sin(x^2) \, dx, \quad \int x \ln x \, dx, \quad \int \frac{3x^2}{x^3 + 1} \, dx.$$

**20. Problem 20**

Compute and verify by differentiation:

$$\int \left( \frac{2x}{1+x^2} + xe^{x^2} + \sin x \cos x \right) dx.$$

## 14.12 Problem Set 12: Applications of Calculus

### 1. Problem 1

For each question, state whether the main tool is a derivative, a definite integral, optimisation, or a differential model:

- (a) find instantaneous velocity;
- (b) find total water discharged from a known flow rate;
- (c) choose dimensions giving the largest area;
- (d) model a population whose growth rate is proportional to its size.

### 2. Problem 2

A rectangle has perimeter 40. Express its area as a function of one side and find the dimensions that maximise the area.

### 3. Problem 3

A rectangular enclosure is built beside a straight wall, so only three sides require fencing. If 60 metres of fencing are available, find the dimensions that maximise the area.

### 4. Problem 4

Squares of side  $x$  are cut from the corners of a 20 cm  $\times$  30 cm sheet to form an open box. Build the volume function, determine the meaningful domain, and find the value of  $x$  that maximises the volume.

### 5. Problem 5

Find the area enclosed by

$$y = 4 - x^2$$

and the  $x$ -axis.

### 6. Problem 6

The curve

$$y = \sqrt{x}, \quad 0 \leq x \leq 9,$$

is rotated about the  $x$ -axis. Find the resulting volume.

### 7. Problem 7

Set up, but do not necessarily evaluate, the arc-length integral for

$$y = x^2, \quad 0 \leq x \leq 2.$$

### 8. Problem 8

A particle has position

$$s(t) = t^3 - 6t^2 + 9t, \quad t \geq 0.$$

Find its velocity and acceleration.

### 9. Problem 9

For the particle in the previous problem, determine when it is at rest and on which intervals it is speeding up or slowing down.

**10. Problem 10**

A particle has acceleration

$$a(t) = 6t - 4,$$

initial velocity  $v(0) = 3$ , and initial position  $s(0) = 2$ . Find  $v(t)$  and  $s(t)$ .

**11. Problem 11**

A spring obeys Hooke's law  $F(x) = 40x$  newtons. Find the work required to stretch it from  $x = 0.1$  m to  $x = 0.25$  m.

**12. Problem 12**

Water flows into a tank at the non-negative rate

$$r(t) = 12 - 2t$$

litres per minute for  $0 \leq t \leq 6$ . How much water enters during the first four minutes?

**13. Problem 13**

The temperature over a six-hour interval is modelled by

$$T(t) = 18 + 2t - \frac{t^2}{3}, \quad 0 \leq t \leq 6.$$

Find its average value on the interval.

**14. Problem 14**

A population satisfies

$$N'(t) = 0.04N(t), \quad N(0) = 500.$$

Find  $N(t)$  and the doubling time.

**15. Problem 15**

A radioactive substance satisfies

$$M'(t) = -0.08M(t), \quad M(0) = 120.$$

Find  $M(t)$  and its half-life.

**16. Problem 16**

A quantity  $Q(t)$  has rate of change

$$Q'(t) = 3t^2 - 4t + 1$$

and  $Q(0) = 7$ . Find  $Q(t)$  and explain the role of the initial condition.

**17. Problem 17**

A closed cylindrical can must have fixed volume  $V$ . Derive the relation between radius and height that minimises its surface area.

**18. Problem 18**

A factory is 400 m downstream on the opposite side of a river 80 m wide. Underwater cable costs three times as much per metre as land cable. Let  $x$  be the downstream distance from the point directly opposite the station to the landing point. Build the cost function and determine the optimal  $x$ .

**19. Problem 19**

A car's velocity is

$$v(t) = t^2 - 6t + 8, \quad 0 \leq t \leq 6.$$

Find the signed displacement and the total distance travelled on the interval.

**20. Problem 20**

A company models daily profit by

$$P(x) = -2x^3 + 30x^2 - 96x - 50, \quad 0 \leq x \leq 10,$$

where  $x$  is production in hundreds of units. Determine the production levels at which profit is locally maximal or minimal, compare all relevant endpoint and critical values, and interpret the economically best choice within the model.